| Main Category | Category | Terms |
|---|---|---|
| Disease | OMIM_DISEASE | Cowden syndrome 1 |
| Disease | OMIM_DISEASE | Lhermitte-Duclos disease |
| Disease | OMIM_DISEASE | Prostate cancer, somatic |
| Disease | OMIM_DISEASE | Macrocephaly/autism syndrome |
| Disease | OMIM_DISEASE | Meningioma |
| Disease | OMIM_DISEASE | Glioma susceptibility 2 |
| Disease | UP_KW_DISEASE | Tumor suppressor |
| Disease | UP_KW_DISEASE | Disease variant |
| Disease | UP_KW_DISEASE | Autism spectrum disorder |
| Functional Annotations | UP_KW_BIOLOGICAL_PROCESS | Apoptosis |
| Functional Annotations | UP_KW_BIOLOGICAL_PROCESS | Lipid metabolism |
| Functional Annotations | UP_KW_BIOLOGICAL_PROCESS | Neurogenesis |
| Functional Annotations | UP_KW_CELLULAR_COMPONENT | Nucleus |
| Functional Annotations | UP_KW_CELLULAR_COMPONENT | Synapse |
| Functional Annotations | UP_KW_CELLULAR_COMPONENT | Cytoplasm |
| Functional Annotations | UP_KW_CELLULAR_COMPONENT | Secreted |
| Functional Annotations | UP_KW_CELLULAR_COMPONENT | Cell projection |
| Functional Annotations | UP_KW_MOLECULAR_FUNCTION | Hydrolase |
| Functional Annotations | UP_KW_MOLECULAR_FUNCTION | Protein phosphatase |
| Functional Annotations | UP_KW_PTM | Acetylation |
| Functional Annotations | UP_KW_PTM | Phosphoprotein |
| Functional Annotations | UP_KW_PTM | Ubl conjugation |
| Functional Annotations | UP_KW_PTM | Isopeptide bond |
| Functional Annotations | UP_SEQ_FEATURE | ACT_SITE:Phosphocysteine intermediate |

| Main Category | Category | Terms |
|---|---|---|
| Functional Annotations | UP_SEQ_FEATURE | COMPBIAS:Acidic residues |
| Functional Annotations | UP_SEQ_FEATURE | COMPBIAS:Low complexity |
| Functional Annotations | UP_SEQ_FEATURE | CROSSLNK:Glycyl lysine isopeptide (Lys-Gly) (interchain with G-Cter in ubiquitin) |
| Functional Annotations | UP_SEQ_FEATURE | DOMAIN:C2 tensin-type |
| Functional Annotations | UP_SEQ_FEATURE | DOMAIN:Phosphatase tensin-type |
| Functional Annotations | UP_SEQ_FEATURE | DOMAIN:Tyrosine specific protein phosphatases |
| Functional Annotations | UP_SEQ_FEATURE | MOTIF:PDZ domain-binding |
| Functional Annotations | UP_SEQ_FEATURE | MUTAGEN:A->P: Does not reduce phosphatase activity towards PtdIns(3,4,5)P3 and PtdIns(3,4)P2. |
| Functional Annotations | UP_SEQ_FEATURE | MUTAGEN:A->S: Does not reduce phosphatase activity towards PtdIns(3,4,5)P3 and PtdIns(3,4)P2. |
| Functional Annotations | UP_SEQ_FEATURE | MUTAGEN:A->V: Does not reduce phosphatase activity towards PtdIns(3,4,5)P3 and PtdIns(3,4)P2. |
| Functional Annotations | UP_SEQ_FEATURE | MUTAGEN:C->A: Loss of protein phosphatase activity. Unable to inhibit focal adhesion formation. |
| Functional Annotations | UP_SEQ_FEATURE | MUTAGEN:D->A: 700-fold reduction in phosphatase activity towards PtdIns(3,4,5)P3. Loss of protein phosphatase activity. Unable to inhibit focal adhesion formation. |
| Functional Annotations | UP_SEQ_FEATURE | MUTAGEN:H->A: 75% reduction in phosphatase activity towards PtdIns(3,4,5)P3. Modest reduction in phosphatase activity towards PtdIns(3,4)P2. |
| Functional Annotations | UP_SEQ_FEATURE | MUTAGEN:K->A: No effect on MAGI2-, MAST2- and DLG1-binding. |
| Functional Annotations | UP_SEQ_FEATURE | MUTAGEN:K->E: Cytoplasmic; when associated with E-13. Shows less tumor suppressive ability; when associated with E-13. |
| Functional Annotations | UP_SEQ_FEATURE | MUTAGEN:K->E: Nuclear. Cytoplasmic; when associated with E-289. Shows less tumor suppressive ability; when associated with E-289. |
| Functional | UP_SEQ_FEATURE | MUTAGEN:K->M: 85% reduction in |

| Main Category | Category | Terms |
|---|---|---|
| Annotations | | phosphatase activity towards PtdIns(3,4,5)P3. |
| Functional Annotations | UP_SEQ_FEATURE | MUTAGEN:K->M: Reduced phosphatase activity towards PtdIns(3,4,5)P3, PtdIns(3,4)P2 and PtdIns(3)P. |
| Functional Annotations | UP_SEQ_FEATURE | MUTAGEN:K->R: Does not reduce phosphatase activity towards PtdIns(3,4,5)P3. |
| Functional Annotations | UP_SEQ_FEATURE | MUTAGEN:K->W: Loss of DLG1-, MAGI2-, MAGI3- and MAST2-binding. Decrease of protein stability. |
| Functional Annotations | UP_SEQ_FEATURE | MUTAGEN:KANKDKANR->AAGADAANA: Reduces growth suppression activity and promotes anchorage-independent growth. Reduces binding to phospholipid membranes in vitro; phosphatase activity towards PtdIns(3,4,5)P3 is not affected. |
| Functional Annotations | UP_SEQ_FEATURE | MUTAGEN:KMLKKDK->AAGAADA: Reduces the growth suppression activity and cells show anchorage-independent growth. Reduces binding to phospholipid membranes in vitro. Phosphatase activity towards PtdIns(3,4,5)P3 is not affected. |
| Functional Annotations | UP_SEQ_FEATURE | MUTAGEN:M->I: Prevents expression of isoform 1 and increased expression of isoform alpha. |
| Functional Annotations | UP_SEQ_FEATURE | MUTAGEN:Q->A,E: 75% reduction in phosphatase activity towards PtdIns(3,4,5)P3. |
| Functional Annotations | UP_SEQ_FEATURE | MUTAGEN:R->M: Does not affect the ability to inhibit AKT/PKB activation. |
| Functional Annotations | UP_SEQ_FEATURE | MUTAGEN:S->A: Decreased stability. |
| Functional Annotations | UP_SEQ_FEATURE | MUTAGEN:S->A: No effect on interaction with DLC1 or p85. |
| Functional Annotations | UP_SEQ_FEATURE | MUTAGEN:T->A,D: 60% reduction in phosphatase activity towards PtdIns(3,4,5)P3. |
| Functional Annotations | UP_SEQ_FEATURE | MUTAGEN:T->A: Decreased stability. |
| Functional Annotations | UP_SEQ_FEATURE | MUTAGEN:T->A: Loss of DLG1-binding. No effect on MAGI2- and MAST2-binding. |
| Functional Annotations | UP_SEQ_FEATURE | MUTAGEN:T->A: No effect on interaction with DLC1 or p85. |
| Functional Annotations | UP_SEQ_FEATURE | MUTAGEN:T->A: Reduces phosphorylation of the C2 tensin-like domain and abolishes interaction with DLC1 following EGF |

| Main Category | Category | Terms |
|---|---|---|
| | | stimulation. Abolishes phosphorylation of the C2 tensin-like domain; when associated with A-319. |
| Functional Annotations | UP_SEQ_FEATURE | MUTAGEN:T->A: Reduces phosphorylation of the C2 tensin-like domain and abolishes interaction with DLC1 following EGF stimulation. Abolishes phosphorylation of the C2 tensin-like domain; when associated with A-321. |
| Functional Annotations | UP_SEQ_FEATURE | MUTAGEN:T->E: Constitutive interaction with DLC1. |
| Functional Annotations | UP_SEQ_FEATURE | MUTAGEN:T->Y: Constitutive interaction with DLC1 and interaction with p85 following EGF stimulation. |
| Functional Annotations | UP_SEQ_FEATURE | MUTAGEN:V->A: Loss of DLG1-, MAGI2-, MAGI3-, MAST1-, MAST2- and MAST3-binding. |
| Functional Annotations | UP_SEQ_FEATURE | MUTAGEN:Y->F: Significantly lower phosphatase activity, reduced protein stability and decreased growth-inhibitory effect. |
| Functional Annotations | UP_SEQ_FEATURE | REGION:Disordered |
| Functional Annotations | UP_SEQ_FEATURE | REGION:Required for interaction with NOP53 |
| Gene Ontology | GOTERM_BP_DIRECT | osteoblast differentiation |
| Gene Ontology | GOTERM_BP_DIRECT | protein dephosphorylation |
| Gene Ontology | GOTERM_BP_DIRECT | lipid metabolic process |
| Gene Ontology | GOTERM_BP_DIRECT | phosphatidylinositol biosynthetic process |
| Gene Ontology | GOTERM_BP_DIRECT | apoptotic process |
| Gene Ontology | GOTERM_BP_DIRECT | spindle assembly involved in female meiosis |
| Gene Ontology | GOTERM_BP_DIRECT | neuron-neuron synaptic transmission |
| Gene Ontology | GOTERM_BP_DIRECT | nervous system development |
| Gene Ontology | GOTERM_BP_DIRECT | synapse assembly |
| Gene Ontology | GOTERM_BP_DIRECT | central nervous system development |

| Main Category | Category | Terms |
|---|---|---|
| Gene Ontology | GOTERM_BP_DIRECT | heart development |
| Gene Ontology | GOTERM_BP_DIRECT | learning or memory |
| Gene Ontology | GOTERM_BP_DIRECT | locomotory behavior |
| Gene Ontology | GOTERM_BP_DIRECT | positive regulation of cell population proliferation |
| Gene Ontology | GOTERM_BP_DIRECT | negative regulation of cell population proliferation |
| Gene Ontology | GOTERM_BP_DIRECT | regulation of signal transduction |
| Gene Ontology | GOTERM_BP_DIRECT | positive regulation of macromolecule metabolic process |
| Gene Ontology | GOTERM_BP_DIRECT | negative regulation of cell communication |
| Gene Ontology | GOTERM_BP_DIRECT | negative regulation of epithelial to mesenchymal transition |
| Gene Ontology | GOTERM_BP_DIRECT | regulation of neuron projection development |
| Gene Ontology | GOTERM_BP_DIRECT | negative regulation of neuron projection development |
| Gene Ontology | GOTERM_BP_DIRECT | cell migration |
| Gene Ontology | GOTERM_BP_DIRECT | dentate gyrus development |
| Gene Ontology | GOTERM_BP_DIRECT | central nervous system neuron axonogenesis |
| Gene Ontology | GOTERM_BP_DIRECT | negative regulation of signaling |
| Gene Ontology | GOTERM_BP_DIRECT | negative regulation of cell migration |
| Gene Ontology | GOTERM_BP_DIRECT | adult behavior |
| Gene Ontology | GOTERM_BP_DIRECT | regulation of protein stability |
| Gene Ontology | GOTERM_BP_DIRECT | central nervous system myelin maintenance |
| Gene Ontology | GOTERM_BP_DIRECT | negative regulation of peptidyl-serine phosphorylation |
| Gene Ontology | GOTERM_BP_DIRECT | multicellular organismal response to stress |

| Main Category | Category | Terms |
|---|---|---|
| Gene Ontology | GOTERM_BP_DIRECT | social behavior |
| Gene Ontology | GOTERM_BP_DIRECT | maternal behavior |
| Gene Ontology | GOTERM_BP_DIRECT | phosphatidylinositol 3-kinase/protein kinase B signal transduction |
| Gene Ontology | GOTERM_BP_DIRECT | locomotor rhythm |
| Gene Ontology | GOTERM_BP_DIRECT | regulation of cell differentiation |
| Gene Ontology | GOTERM_BP_DIRECT | negative regulation of osteoblast differentiation |
| Gene Ontology | GOTERM_BP_DIRECT | negative regulation of cell size |
| Gene Ontology | GOTERM_BP_DIRECT | positive regulation of transcription by RNA polymerase II |
| Gene Ontology | GOTERM_BP_DIRECT | negative regulation of organ growth |
| Gene Ontology | GOTERM_BP_DIRECT | phosphatidylinositol dephosphorylation |
| Gene Ontology | GOTERM_BP_DIRECT | forebrain morphogenesis |
| Gene Ontology | GOTERM_BP_DIRECT | cell motility |
| Gene Ontology | GOTERM_BP_DIRECT | negative regulation of axonogenesis |
| Gene Ontology | GOTERM_BP_DIRECT | regulation of developmental process |
| Gene Ontology | GOTERM_BP_DIRECT | protein stabilization |
| Gene Ontology | GOTERM_BP_DIRECT | negative regulation of developmental process |
| Gene Ontology | GOTERM_BP_DIRECT | negative regulation of cellular component organization |
| Gene Ontology | GOTERM_BP_DIRECT | negative regulation of multicellular organismal process |
| Gene Ontology | GOTERM_BP_DIRECT | negative regulation of keratinocyte migration |
| Gene Ontology | GOTERM_BP_DIRECT | regulation of cell cycle |
| Gene Ontology | GOTERM_BP_DIRECT | negative regulation of focal adhesion assembly |

| Main Category | Category | Terms |
|---|---|---|
| Gene Ontology | GOTERM_BP_DIRECT | regulation of phosphatidylinositol 3-kinase/ protein kinase B signal transduction |
| Gene Ontology | GOTERM_BP_DIRECT | negative regulation of phosphatidylinositol 3-kinase/protein kinase B signal transduction |
| Gene Ontology | GOTERM_BP_DIRECT | rhythmic synaptic transmission |
| Gene Ontology | GOTERM_BP_DIRECT | synapse maturation |
| Gene Ontology | GOTERM_BP_DIRECT | prepulse inhibition |
| Gene Ontology | GOTERM_BP_DIRECT | dendritic spine morphogenesis |
| Gene Ontology | GOTERM_BP_DIRECT | regulation of biological quality |
| Gene Ontology | GOTERM_BP_DIRECT | cellular response to electrical stimulus |
| Gene Ontology | GOTERM_BP_DIRECT | negative regulation of excitatory postsynaptic potential |
| Gene Ontology | GOTERM_BP_DIRECT | presynaptic membrane assembly |
| Gene Ontology | GOTERM_BP_DIRECT | postsynaptic density assembly |
| Gene Ontology | GOTERM_BP_DIRECT | positive regulation of intracellular signal transduction |
| Gene Ontology | GOTERM_BP_DIRECT | negative regulation of cell cycle G1/S phase transition |
| Gene Ontology | GOTERM_BP_DIRECT | negative regulation of wound healing, spreading of epidermal cells |
| Gene Ontology | GOTERM_BP_DIRECT | negative regulation of vascular associated smooth muscle cell proliferation |
| Gene Ontology | GOTERM_BP_DIRECT | positive regulation of ubiquitin-dependent protein catabolic process |
| Gene Ontology | GOTERM_BP_DIRECT | negative regulation of G1/S transition of mitotic cell cycle |
| Gene Ontology | GOTERM_BP_DIRECT | positive regulation of excitatory postsynaptic potential |
| Gene Ontology | GOTERM_BP_DIRECT | negative regulation of cellular senescence |
| Gene Ontology | GOTERM_BP_DIRECT | negative regulation of synaptic vesicle clustering |
| Gene Ontology | GOTERM_CC_DIRECT | extracellular region |

| Main Category | Category | Terms |
|---|---|---|
| Gene Ontology | GOTERM_CC_DIRECT | nucleus |
| Gene Ontology | GOTERM_CC_DIRECT | nucleoplasm |
| Gene Ontology | GOTERM_CC_DIRECT | cytoplasm |
| Gene Ontology | GOTERM_CC_DIRECT | cytosol |
| Gene Ontology | GOTERM_CC_DIRECT | plasma membrane |
| Gene Ontology | GOTERM_CC_DIRECT | cytoplasmic side of plasma membrane |
| Gene Ontology | GOTERM_CC_DIRECT | postsynaptic density |
| Gene Ontology | GOTERM_CC_DIRECT | apical plasma membrane |
| Gene Ontology | GOTERM_CC_DIRECT | PML body |
| Gene Ontology | GOTERM_CC_DIRECT | myelin sheath adaxonal region |
| Gene Ontology | GOTERM_CC_DIRECT | cell projection |
| Gene Ontology | GOTERM_CC_DIRECT | neuron projection |
| Gene Ontology | GOTERM_CC_DIRECT | dendritic spine |
| Gene Ontology | GOTERM_CC_DIRECT | Schmidt-Lanterman incisure |
| Gene Ontology | GOTERM_CC_DIRECT | synapse |
| Gene Ontology | GOTERM_MF_DIRECT | phosphatidylinositol-3-phosphate phosphatase activity |
| Gene Ontology | GOTERM_MF_DIRECT | phosphoprotein phosphatase activity |
| Gene Ontology | GOTERM_MF_DIRECT | protein serine/threonine phosphatase activity |
| Gene Ontology | GOTERM_MF_DIRECT | protein tyrosine phosphatase activity |
| Gene Ontology | GOTERM_MF_DIRECT | protein binding |
| Gene Ontology | GOTERM_MF_DIRECT | beta-catenin binding |

| Main Category | Category | Terms |
|---|---|---|
| Gene Ontology | GOTERM_MF_DIRECT | lipid binding |
| Gene Ontology | GOTERM_MF_DIRECT | anaphase-promoting complex binding |
| Gene Ontology | GOTERM_MF_DIRECT | phosphatidylinositol-3,4,5-trisphosphate 3-phosphatase activity |
| Gene Ontology | GOTERM_MF_DIRECT | hydrolase activity |
| Gene Ontology | GOTERM_MF_DIRECT | phosphatase activity |
| Gene Ontology | GOTERM_MF_DIRECT | enzyme binding |
| Gene Ontology | GOTERM_MF_DIRECT | PDZ domain binding |
| Gene Ontology | GOTERM_MF_DIRECT | inositol-1,3,4,5,6-pentakisphosphate 3-phosphatase activity |
| Gene Ontology | GOTERM_MF_DIRECT | identical protein binding |
| Gene Ontology | GOTERM_MF_DIRECT | inositol-1,3,4,5-tetrakisphosphate 3-phosphatase activity |
| Gene Ontology | GOTERM_MF_DIRECT | phosphatidylinositol-3,4-bisphosphate 3-phosphatase activity |
| Gene Ontology | GOTERM_MF_DIRECT | phosphatidylinositol phosphate phosphatase activity |
| Gene Ontology | GOTERM_MF_DIRECT | molecular function inhibitor activity |
| Gene Ontology | GOTERM_MF_DIRECT | ubiquitin-specific protease binding |
| Gene Ontology | GOTERM_MF_DIRECT | ubiquitin ligase activator activity |
| General Annotations | CHROMOSOME | 10 |
| General Annotations | CYTOGENETIC_LOCATION | 10q23.31 |
| General Annotations | ENTREZ_GENE_SUMMARY | This gene was identified as a tumor suppressor that is mutated in a large number of cancers at high frequency. The protein encoded by this gene is a phosphatidylinositol-3,4,5-trisphosphate 3-phosphatase. It contains a tensin like domain as well as a catalytic domain similar to that of the dual specificity protein tyrosine phosphatases. Unlike most of the protein tyrosine phosphatases, this protein preferentially dephosphorylates |

| Main Category | Category | Terms |
|---|---|---|
| | | phosphoinositide substrates. It negatively regulates intracellular levels of phosphatidylinositol-3,4,5-trisphosphate in cells and functions as a tumor suppressor by negatively regulating AKT/PKB signaling pathway. The use of a non-canonical (CUG) upstream initiation site produces a longer isoform that initiates translation with a leucine, and is thought to be preferentially associated with the mitochondrial inner membrane. This longer isoform may help regulate energy metabolism in the mitochondria. A pseudogene of this gene is found on chromosome 9. Alternative splicing and the use of multiple translation start codons results in multiple transcript variants encoding different isoforms. [provided by RefSeq, Feb 2015] |
| General Annotations | UP_COMMENT_ACTIVITY_REGULATION | Enzymatic activity is enhanced in the presence of phosphatidylserine. {ECO:0000269\|PubMed:16824732}. |
| General Annotations | UP_COMMENT_ALTERNATIVE_PRODUCTS | Event=Alternative splicing, Alternative initiation; Named isoforms=3; Name=1; Synonyms=55kDa; IsoId=P60484-1; Sequence=Displayed; Name=alpha; Synonyms=70kDa, PTEN-long; IsoId=P60484-2; Sequence=VSP_055420; Name=3; IsoId=P60484-3; Sequence=VSP_055421, VSP_055422, VSP_055423; |
| General Annotations | UP_COMMENT_BIOPHYSICOCHEMICAL_PROPERTIES | Kinetic parameters: KM=26 uM for PtdIns(3,4,5)P3 {ECO:0000269\|PubMed:11418101}; KM=11.5 uM for 1D-myo-inositol 1,3,4-trisphosphate {ECO:0000269\|PubMed:11418101}; KM=4.6 uM for 1D-myo-inositol 1,3,4,5-tetrakisphosphate {ECO:0000269\|PubMed:11418101}; KM=7.1 uM for 1D-myo-inositol 1,3,4,5,6-pentakisphosphate {ECO:0000269\|PubMed:11418101}; Vmax=60 nmol/min/mg enzyme with PtdIns(3,4,5)P3 as substrate {ECO:0000269\|PubMed:11418101}; Vmax=24 nmol/min/mg enzyme with 1D-myo-inositol 1,3,4-trisphosphate as substrate {ECO:0000269\|PubMed:11418101}; Vmax=112 nmol/min/mg enzyme with 1D-myo-inositol 1,3,4,5-tetrakisphosphate as substrate {ECO:0000269\|PubMed:11418101}; Vmax=67 nmol/min/mg enzyme with 1D-myo-inositol 1,3,4,5,6- pentakisphosphate as substrate {ECO:0000269\|PubMed:11418101}; |
| General Annotations | UP_COMMENT_CATALYTIC_ACTIVITY | Reaction=1,2-dihexadecanoyl-sn-glycero-3-phospho-(1D-myo- inositol-3,4,5-trisphosphate) + H2O = 1,2-dihexadecanoyl- |

| Main Category | Category | Terms |
|---|---|---|
| | | sn-glycero- 3-phospho-(1D-myo-inositol-4,5-bisphosphate) + phosphate; Xref=Rhea:RHEA:43560, ChEBI:CHEBI:15377, ChEBI:CHEBI:43474, ChEBI:CHEBI:83420, ChEBI:CHEBI:83423; Evidence={ECO:0000256\|ARBA:ARBA00034256}; PhysiologicalDirection=left-to-right; Xref=Rhea:RHEA:43561; Evidence={ECO:0000256\|ARBA:ARBA00034256}; |
| General Annotations | UP_COMMENT_CATALYTIC_ACTIVITY | Reaction=1,2-dihexadecanoyl-sn-glycero-3-phospho-(1D-myo-inositol- 3,4,5-trisphosphate) + H2O = 1,2-dihexadecanoyl-sn-glycero-3-phospho- (1D-myo-inositol-4,5-bisphosphate) + phosphate; Xref=Rhea:RHEA:43560, ChEBI:CHEBI:15377, ChEBI:CHEBI:43474, ChEBI:CHEBI:83420, ChEBI:CHEBI:83423; Evidence={ECO:0000269\|PubMed:16824732, ECO:0000269\|PubMed:9593664}; PhysiologicalDirection=left-to-right; Xref=Rhea:RHEA:43561; Evidence={ECO:0000305\|PubMed:16824732}; |
| General Annotations | UP_COMMENT_CATALYTIC_ACTIVITY | Reaction=1,2-dioctanoyl-sn-glycero-3-phospho-(1D-myo-inositol-3,4,5-trisphosphate) + H2O = 1,2-dioctanoyl-sn-glycero-3-phospho-(1D-myo- inositol-4,5-bisphosphate) + phosphate; Xref=Rhea:RHEA:43552, ChEBI:CHEBI:15377, ChEBI:CHEBI:43474, ChEBI:CHEBI:83416, ChEBI:CHEBI:83419; Evidence={ECO:0000256\|ARBA:ARBA00034268}; PhysiologicalDirection=left-to-right; Xref=Rhea:RHEA:43553; Evidence={ECO:0000256\|ARBA:ARBA00034268}; |
| General Annotations | UP_COMMENT_CATALYTIC_ACTIVITY | Reaction=1,2-dioctanoyl-sn-glycero-3-phospho-(1D-myo-inositol-3,4,5-trisphosphate) + H2O = 1,2-dioctanoyl-sn-glycero-3-phospho-(1D-myo- inositol-4,5-bisphosphate) + phosphate; Xref=Rhea:RHEA:43552, ChEBI:CHEBI:15377, ChEBI:CHEBI:43474, ChEBI:CHEBI:83416, ChEBI:CHEBI:83419; Evidence={ECO:0000269\|PubMed:16824732, ECO:0000269\|PubMed:9593664}; PhysiologicalDirection=left-to-right; Xref=Rhea:RHEA:43553; Evidence={ECO:0000305\|PubMed:16824732}; |
| General Annotations | UP_COMMENT_CATALYTIC_ACTIVITY | Reaction=1D-myo-inositol 1,3,4,5,6-pentakisphosphate + H2O = 1D-myo- |

| Main Category | Category | Terms |
|---|---|---|
| | | inositol 1,4,5,6-tetrakisphosphate + phosphate; Xref=Rhea:RHEA:77143, ChEBI:CHEBI:15377, ChEBI:CHEBI:43474, ChEBI:CHEBI:57627, ChEBI:CHEBI:57733; Evidence={ECO:0000256\|ARBA:ARBA00043762}; PhysiologicalDirection=left-to-right; Xref=Rhea:RHEA:77144; Evidence={ECO:0000256\|ARBA:ARBA00043762}; |
| General Annotations | UP_COMMENT_CATALYTIC_ACTIVITY | Reaction=1D-myo-inositol 1,3,4,5,6-pentakisphosphate + H2O = 1D-myo-inositol 1,4,5,6-tetrakisphosphate + phosphate; Xref=Rhea:RHEA:77143, ChEBI:CHEBI:15377, ChEBI:CHEBI:43474, ChEBI:CHEBI:57627, ChEBI:CHEBI:57733; Evidence={ECO:0000269\|PubMed:11418101}; PhysiologicalDirection=left-to-right; Xref=Rhea:RHEA:77144; Evidence={ECO:0000269\|PubMed:11418101}; |
| General Annotations | UP_COMMENT_CATALYTIC_ACTIVITY | Reaction=1D-myo-inositol 1,3,4,5-tetrakisphosphate + H2O = 1D-myo-inositol 1,4,5-trisphosphate + phosphate; Xref=Rhea:RHEA:77155, ChEBI:CHEBI:15377, ChEBI:CHEBI:43474, ChEBI:CHEBI:57895, ChEBI:CHEBI:203600; Evidence={ECO:0000256\|ARBA:ARBA00043734}; PhysiologicalDirection=left-to-right; Xref=Rhea:RHEA:77156; Evidence={ECO:0000256\|ARBA:ARBA00043734}; |
| General Annotations | UP_COMMENT_CATALYTIC_ACTIVITY | Reaction=1D-myo-inositol 1,3,4,5-tetrakisphosphate + H2O = 1D-myo-inositol 1,4,5-trisphosphate + phosphate; Xref=Rhea:RHEA:77155, ChEBI:CHEBI:15377, ChEBI:CHEBI:43474, ChEBI:CHEBI:57895, ChEBI:CHEBI:203600; Evidence={ECO:0000269\|PubMed:9593664}; PhysiologicalDirection=left-to-right; Xref=Rhea:RHEA:77156; Evidence={ECO:0000305\|PubMed:9593664}; |
| General Annotations | UP_COMMENT_CATALYTIC_ACTIVITY | Reaction=O-phospho-L-seryl-[protein] + H2O = L-seryl-[protein] + phosphate; Xref=Rhea:RHEA:20629, Rhea:RHEA-COMP:9863, Rhea:RHEA-COMP:11604, ChEBI:CHEBI:15377, ChEBI:CHEBI:29999, ChEBI:CHEBI:43474, ChEBI:CHEBI:83421; EC=3.1.3.16; Evidence={ECO:0000256\|ARBA:ARBA00047986}; PhysiologicalDirection=left-to-right; Xref=Rhea:RHEA:20630; Evidence={ECO:0000256\|ARBA:ARBA00047986}; |

| Main Category | Category | Terms |
|---|---|---|
| General Annotations | UP_COMMENT_CATALYTIC_ACTIVITY | Reaction=O-phospho-L-seryl-[protein] + H2O = L-seryl-[protein] + phosphate; Xref=Rhea:RHEA:20629, Rhea:RHEA-COMP:9863, Rhea:RHEA-COMP:11604, ChEBI:CHEBI:15377, ChEBI:CHEBI:29999, ChEBI:CHEBI:43474, ChEBI:CHEBI:83421; EC=3.1.3.16; Evidence={ECO:0000269\|PubMed:9256433}; PhysiologicalDirection=left-to-right; Xref=Rhea:RHEA:20630; Evidence={ECO:0000305\|PubMed:9256433}; |
| General Annotations | UP_COMMENT_CATALYTIC_ACTIVITY | Reaction=O-phospho-L-threonyl-[protein] + H2O = L-threonyl-[protein] + phosphate; Xref=Rhea:RHEA:47004, Rhea:RHEA-COMP:11060, Rhea:RHEA-COMP:11605, ChEBI:CHEBI:15377, ChEBI:CHEBI:30013, ChEBI:CHEBI:43474, ChEBI:CHEBI:61977; EC=3.1.3.16; Evidence={ECO:0000256\|ARBA:ARBA00048832}; PhysiologicalDirection=left-to-right; Xref=Rhea:RHEA:47005; Evidence={ECO:0000256\|ARBA:ARBA00048832}; |
| General Annotations | UP_COMMENT_CATALYTIC_ACTIVITY | Reaction=O-phospho-L-threonyl-[protein] + H2O = L-threonyl-[protein] + phosphate; Xref=Rhea:RHEA:47004, Rhea:RHEA-COMP:11060, Rhea:RHEA-COMP:11605, ChEBI:CHEBI:15377, ChEBI:CHEBI:30013, ChEBI:CHEBI:43474, ChEBI:CHEBI:61977; EC=3.1.3.16; Evidence={ECO:0000269\|PubMed:9256433}; PhysiologicalDirection=left-to-right; Xref=Rhea:RHEA:47005; Evidence={ECO:0000305\|PubMed:9256433}; |
| General Annotations | UP_COMMENT_CATALYTIC_ACTIVITY | Reaction=O-phospho-L-tyrosyl-[protein] + H2O = L-tyrosyl-[protein] + phosphate; Xref=Rhea:RHEA:10684, Rhea:RHEA-COMP:10136, Rhea:RHEA-COMP:20101, ChEBI:CHEBI:15377, ChEBI:CHEBI:43474, ChEBI:CHEBI:46858, ChEBI:CHEBI:61978; EC=3.1.3.48; Evidence={ECO:0000256\|ARBA:ARBA00051341}; PhysiologicalDirection=left-to-right; Xref=Rhea:RHEA:10685; Evidence={ECO:0000256\|ARBA:ARBA00051341}; |
| General Annotations | UP_COMMENT_CATALYTIC_ACTIVITY | Reaction=O-phospho-L-tyrosyl-[protein] + H2O = L-tyrosyl-[protein] + phosphate; Xref=Rhea:RHEA:10684, Rhea:RHEA-COMP:10136, Rhea:RHEA-COMP:20101, ChEBI:CHEBI:15377, ChEBI:CHEBI:43474, ChEBI:CHEBI:46858, ChEBI:CHEBI:61978; EC=3.1.3.48; Evidence={ECO:0000269\|PubMed:9187108, ECO:0000269\|PubMed:9256433}; PhysiologicalDirection=left-to-right; Xref=Rhea:RHEA:10685; |

| Main Category | Category | Terms |
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| | | Evidence={ECO:0000305\|PubMed:9256433}; |
| General Annotations | UP_COMMENT_CATALYTIC_ACTIVITY | Reaction=a 1,2-diacyl-sn-glycero-3-phospho-(1D-myo-inositol-3,4,5- trisphosphate) + H2O = a 1,2-diacyl-sn-glycero-3-phospho-(1D-myo- inositol-4,5-bisphosphate) + phosphate; Xref=Rhea:RHEA:25017, ChEBI:CHEBI:15377, ChEBI:CHEBI:43474, ChEBI:CHEBI:57836, ChEBI:CHEBI:58456; EC=3.1.3.67; Evidence={ECO:0000256\| ARBA:ARBA00043760}; PhysiologicalDirection=left-to-right; Xref=Rhea:RHEA:25018; Evidence={ECO:0000256\| ARBA:ARBA00043760}; |
| General Annotations | UP_COMMENT_CATALYTIC_ACTIVITY | Reaction=a 1,2-diacyl-sn-glycero-3-phospho-(1D-myo-inositol-3,4,5- trisphosphate) + H2O = a 1,2-diacyl-sn-glycero-3-phospho-(1D-myo- inositol-4,5-bisphosphate) + phosphate; Xref=Rhea:RHEA:25017, ChEBI:CHEBI:15377, ChEBI:CHEBI:43474, ChEBI:CHEBI:57836, ChEBI:CHEBI:58456; EC=3.1.3.67; Evidence={ECO:0000269\| PubMed:16824732, ECO:0000269\| PubMed:9811831}; PhysiologicalDirection=left-to-right; Xref=Rhea:RHEA:25018; Evidence={ECO:0000305\| PubMed:16824732}; |
| General Annotations | UP_COMMENT_COFACTOR | Name=Mg(2+); Xref=ChEBI:CHEBI:18420; |
| General Annotations | UP_COMMENT_COFACTOR | Name=Mg(2+); Xref=ChEBI:CHEBI:18420; Evidence={ECO:0000256\| ARBA:ARBA00001946}; |
| General Annotations | UP_COMMENT_DISEASE | Cowden syndrome 1 (CWS1) [MIM:158350]: An autosomal dominant hamartomatous polyposis syndrome with age-related penetrance. Cowden syndrome is characterized by hamartomatous lesions affecting derivatives of ectodermal, mesodermal and endodermal layers, macrocephaly, facial trichilemmomas (benign tumors of the hair follicle infundibulum), acral keratoses, papillomatous papules, and elevated risk for development of several types of malignancy, particularly breast carcinoma in women and thyroid carcinoma in both men and women. Colon cancer and renal cell carcinoma have also been reported. Hamartomas can be found in virtually every organ, but most commonly in the skin, gastrointestinal tract, breast and thyroid. {ECO:0000269\| PubMed:10051160, ECO:0000269\| PubMed:10234502, ECO:0000269\| PubMed:10400993, ECO:0000269\| PubMed:10772390, ECO:0000269\| |

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| | | PubMed:10866302, ECO:0000269\| PubMed:11230179, ECO:0000269\| PubMed:11494117, ECO:0000269\| PubMed:15355975, ECO:0000269\| PubMed:18716620, ECO:0000269\| PubMed:21659347, ECO:0000269\| PubMed:24778394, ECO:0000269\| PubMed:9140396, ECO:0000269\| PubMed:9241266, ECO:0000269\| PubMed:9259288, ECO:0000269\| PubMed:9345101, ECO:0000269\| PubMed:9399897, ECO:0000269\| PubMed:9425889, ECO:0000269\| PubMed:9467011, ECO:0000269\| PubMed:9593664, ECO:0000269\| PubMed:9600246, ECO:0000269\| PubMed:9735393, ECO:0000269\| PubMed:9797362, ECO:0000269\| PubMed:9811831, ECO:0000269\| PubMed:9832031, ECO:0000269\| PubMed:9915974}. Note=The disease is caused by variants affecting the gene represented in this entry. |
| General Annotations | UP_COMMENT_DISEASE | Endometrial cancer (ENDMC) [MIM:608089]: A malignancy of endometrium, the mucous lining of the uterus. Most endometrial cancers are adenocarcinomas, cancers that begin in cells that make and release mucus and other fluids. {ECO:0000269\| PubMed:16506206}. Note=Disease susceptibility is associated with variants affecting the gene represented in this entry. |
| General Annotations | UP_COMMENT_DISEASE | Glioma 2 (GLM2) [MIM:613028]: Gliomas are benign or malignant central nervous system neoplasms derived from glial cells. They comprise astrocytomas and glioblastoma multiforme that are derived from astrocytes, oligodendrogliomas derived from oligodendrocytes and ependymomas derived from ependymocytes. {ECO:0000269\| PubMed:12085208}. Note=Disease susceptibility is associated with variants affecting the gene represented in this entry. |
| General Annotations | UP_COMMENT_DISEASE | Lhermitte-Duclos disease (LDD) [MIM:158350]: A rare disease characterized by the occurrence of a slowly enlarging mass within the cerebellar cortex corresponding histologically to a cerebellar hamartoma. It manifests, most commonly in the third and fourth decades of life, with increased intracranial pressure, headache, nausea, cerebellar dysfunction, occlusive hydrocephalus, ataxia, visual disturbances and other cranial nerve palsies. Various associated abnormalities may be present such as megalencephaly, microgyria, hydromyelia, polydactyly, partial gigantism, |

| Main Category | Category | Terms |
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| | | macroglossia. LDD is part of the PTEN hamartoma tumor syndromes spectrum that also includes Cowden syndrome. Note=The disease is caused by variants affecting the gene represented in this entry. |
| General Annotations | UP_COMMENT_DISEASE | Macrocephaly/autism syndrome (MCEPHAS) [MIM:605309]: Patients have autism spectrum disorders and macrocephaly, with head circumferences ranging from +2.5 to +8 SD for age and sex (average head circumference +4.0 SD). {ECO:0000269\|PubMed:15805158, ECO:0000269\|PubMed:23160955, ECO:0000269\|PubMed:26637798}. Note=The disease is caused by variants affecting the gene represented in this entry. |
| General Annotations | UP_COMMENT_DISEASE | Note=A microdeletion of chromosome 10q23 involving BMPR1A and PTEN is a cause of chromosome 10q23 deletion syndrome, which shows overlapping features of the following three disorders: Bannayan-Zonana syndrome, Cowden disease and juvenile polyposis syndrome. |
| General Annotations | UP_COMMENT_DISEASE | Note=PTEN mutations are found in a subset of patients with Proteus syndrome, a genetically heterogeneous condition. The molecular diagnosis of PTEN mutation positive cases classifies Proteus syndrome patients as part of the PTEN hamartoma syndrome spectrum. As such, patients surviving the early years of Proteus syndrome are likely at a greater risk of developing malignancies. |
| General Annotations | UP_COMMENT_DISEASE | Prostate cancer (PC) [MIM:176807]: A malignancy originating in tissues of the prostate. Most prostate cancers are adenocarcinomas that develop in the acini of the prostatic ducts. Other rare histopathologic types of prostate cancer that occur in approximately 5% of patients include small cell carcinoma, mucinous carcinoma, prostatic ductal carcinoma, transitional cell carcinoma, squamous cell carcinoma, basal cell carcinoma, adenoid cystic carcinoma (basaloid), signet-ring cell carcinoma and neuroendocrine carcinoma. {ECO:0000269\|PubMed:26504226, ECO:0000269\|PubMed:9072974}. Note=Disease susceptibility is associated with variants affecting the gene represented in this entry. |
| General Annotations | UP_COMMENT_DISEASE | Squamous cell carcinoma of the head and neck (HNSCC) [MIM:275355]: A non-melanoma skin cancer affecting the head and neck. The hallmark of cutaneous SCC is malignant transformation of normal |

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| | | epidermal keratinocytes. {ECO:0000269\|PubMed:11801303}. Note=The disease is caused by variants affecting the gene represented in this entry. |
| General Annotations | UP_COMMENT_DOMAIN | The C2 domain binds phospholipid membranes in vitro in a Ca(2+)-independent manner; this binding is important for its tumor suppressor function. {ECO:0000269\|PubMed:10468583, ECO:0000269\|PubMed:10555148}. |
| General Annotations | UP_COMMENT_FUNCTION | Dual-specificity protein phosphatase, dephosphorylating tyrosine-, serine- and threonine-phosphorylated proteins (PubMed:9187108, PubMed:9256433, PubMed:9616126). Also functions as a lipid phosphatase, removing the phosphate in the D3 position of the inositol ring of PtdIns(3,4,5)P3/phosphatidylinositol 3,4,5-trisphosphate, PtdIns(3,4)P2/phosphatidylinositol 3,4-diphosphate and PtdIns3P/phosphatidylinositol 3-phosphate with a preference for PtdIns(3,4,5)P3 (PubMed:16824732, PubMed:26504226, PubMed:9593664, PubMed:9811831). Furthermore, this enzyme can also act as a cytosolic inositol 3-phosphatase acting on Ins(1,3,4,5,6)P5/inositol 1,3,4,5,6 pentakisphosphate and possibly Ins(1,3,4,5)P4/1D-myo-inositol 1,3,4,5-tetrakisphosphate (PubMed:11418101, PubMed:15979280). Antagonizes the PI3K-AKT/PKB signaling pathway by dephosphorylating phosphoinositides and thereby modulating cell cycle progression and cell survival (PubMed:31492966, PubMed:37279284). The unphosphorylated form cooperates with MAGI2 to suppress AKT1 activation (PubMed:11707428). In motile cells, suppresses the formation of lateral pseudopods and thereby promotes cell polarization and directed movement (PubMed:22279049). Dephosphorylates tyrosine-phosphorylated focal adhesion kinase and inhibits cell migration and integrin-mediated cell spreading and focal adhesion formation (PubMed:22279049). Required for growth factor-induced epithelial cell migration; growth factor stimulation induces PTEN phosphorylation which changes its binding preference from the p85 regulatory subunit of the PI3K kinase complex to DLC1 and results in translocation of the PTEN-DLC1 complex to the posterior of migrating cells to promote RHOA activation (PubMed:26166433). Meanwhile, TNS3 switches binding preference from DLC1 to p85 and the TNS3-p85 complex translocates to the leading edge of migrating cells to activate RAC1 |

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| | | activation (PubMed:26166433). Plays a role as a key modulator of the AKT-mTOR signaling pathway controlling the tempo of the process of newborn neurons integration during adult neurogenesis, including correct neuron positioning, dendritic development and synapse formation (By similarity). Involved in the regulation of synaptic function in excitatory hippocampal synapses. Recruited to the postsynaptic membrane upon NMDA receptor activation, is required for the modulation of synaptic activity during plasticity. Enhancement of lipid phosphatase activity is able to drive depression of AMPA receptor-mediated synaptic responses, activity required for NMDA receptor-dependent long-term depression (LTD) (By similarity). May be a negative regulator of insulin signaling and glucose metabolism in adipose tissue. The nuclear monoubiquitinated form possesses greater apoptotic potential, whereas the cytoplasmic nonubiquitinated form induces less tumor suppressive ability (PubMed:10468583, PubMed:18716620). {ECO:0000250\|UniProtKB:O08586, ECO:0000250\|UniProtKB:O54857, ECO:0000269\|PubMed:10468583, ECO:0000269\|PubMed:11418101, ECO:0000269\|PubMed:11707428, ECO:0000269\|PubMed:15979280, ECO:0000269\|PubMed:16824732, ECO:0000269\|PubMed:18716620, ECO:0000269\|PubMed:22279049, ECO:0000269\|PubMed:26166433, ECO:0000269\|PubMed:26504226, ECO:0000269\|PubMed:31492966, ECO:0000269\|PubMed:37279284, ECO:0000269\|PubMed:9187108, ECO:0000269\|PubMed:9256433, ECO:0000269\|PubMed:9593664, ECO:0000269\|PubMed:9616126, ECO:0000269\|PubMed:9811831}. |
| General Annotations | UP_COMMENT_FUNCTION | Dual-specificity protein phosphatase, dephosphorylating tyrosine-, serine- and threonine-phosphorylated proteins. Also functions as a lipid phosphatase, removing the phosphate in the D3 position of the inositol ring of PtdIns(3,4,5)P3/phosphatidylinositol 3,4,5-trisphosphate, PtdIns(3,4)P2/phosphatidylinositol 3,4-diphosphate and PtdIns3P/phosphatidylinositol 3-phosphate with a preference for PtdIns(3,4,5)P3. {ECO:0000256\|PIRNR:PIRNR038025}. |
| General Annotations | UP_COMMENT_FUNCTION | [Isoform alpha]: Functional kinase, like isoform 1 it antagonizes the PI3K-AKT/PKB signaling pathway. Plays a role in mitochondrial energetic metabolism by |

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| | | promoting COX activity and ATP production, via collaboration with isoform 1 in increasing protein levels of PINK1. {ECO:0000269\|PubMed:23744781}. |
| General Annotations | UP_COMMENT_INDUCTION | Down-regulated by TGFB1. {ECO:0000269\|PubMed:9187108}. |
| General Annotations | UP_COMMENT_INTERACTION | P60484; Q9ULX6: AKAP8L; NbExp=2; IntAct=EBI-696162, EBI-357530; P60484; P35226: BMI1; NbExp=7; IntAct=EBI-696162, EBI-2341576; P60484; P30260: CDC27; NbExp=7; IntAct=EBI-696162, EBI-994813; P60484; Q9P0U4: CXXC1; NbExp=2; IntAct=EBI-696162, EBI-949911; P60484; Q16643: DBN1; NbExp=5; IntAct=EBI-696162, EBI-351394; P60484; P42685: FRK; NbExp=7; IntAct=EBI-696162, EBI-1383583; P60484; O60307: MAST3; NbExp=3; IntAct=EBI-696162, EBI-311420; P60484; Q9Y6Q9: NCOA3; NbExp=2; IntAct=EBI-696162, EBI-81196; P60484; P46934: NEDD4; NbExp=4; IntAct=EBI-696162, EBI-726944; P60484; O14745: NHERF1; NbExp=7; IntAct=EBI-696162, EBI-349787; P60484; Q15599: NHERF2; NbExp=5; IntAct=EBI-696162, EBI-1149760; P60484; P09619: PDGFRB; NbExp=3; IntAct=EBI-696162, EBI-641237; P60484; Q9NRD5: PICK1; NbExp=2; IntAct=EBI-696162, EBI-79165; P60484; P62136: PPP1CA; NbExp=2; IntAct=EBI-696162, EBI-357253; P60484; Q06830: PRDX1; NbExp=7; IntAct=EBI-696162, EBI-353193; P60484; P60484: PTEN; NbExp=9; IntAct=EBI-696162, EBI-696162; P60484; O43791: SPOP; NbExp=4; IntAct=EBI-696162, EBI-743549; P60484; Q62696: Dlg1; Xeno; NbExp=2; IntAct=EBI-696162, EBI-389325; P60484; O88382: Magi2; Xeno; NbExp=3; IntAct=EBI-696162, EBI-696179; P60484; Q9JK71: Magi3; Xeno; NbExp=3; IntAct=EBI-696162, EBI-696226; P60484; Q9R1L5: Mast1; Xeno; NbExp=3; IntAct=EBI-696162, EBI-491771; P60484; Q60592: Mast2; Xeno; NbExp=4; IntAct=EBI-696162, EBI-493888; P60484; Q9JHL1: Nherf2; Xeno; NbExp=2; IntAct=EBI-696162, EBI-538451; P60484-1; Q19T08: ECSCR; NbExp=4; IntAct=EBI-15722967, EBI-15778214; P60484-1; Q5T2D3: OTUD3; NbExp=9; IntAct=EBI-15722967, EBI-16170539; P60484-1; P60484-1: PTEN; NbExp=3; IntAct=EBI-15722967, EBI-15722967; P60484-1; Q93009: USP7; NbExp=5; IntAct=EBI-15722967, EBI-302474; |

| Main Category | Category | Terms |
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| General Annotations | UP_COMMENT_MISCELLANEOUS | [Isoform alpha]: Produced by alternative initiation at an upstream CTG start codon. May contain a signal peptide at positions 1-21. {ECO:0000305}. |
| General Annotations | UP_COMMENT_PTM | Constitutively phosphorylated by CK2 under normal conditions. Phosphorylated in vitro by MAST1, MAST2, MAST3 and STK11. Phosphorylation results in an inhibited activity towards PIP3. Phosphorylation can both inhibit or promote PDZ-binding. Phosphorylation at Tyr-336 by FRK/PTK5 protects this protein from ubiquitin-mediated degradation probably by inhibiting its binding to NEDD4. Phosphorylation by ROCK1 is essential for its stability and activity. Phosphorylation by PLK3 promotes its stability and prevents its degradation by the proteasome. Phosphorylated on Thr-319 and Thr- 321 in the C2-type tensin domain following EGF stimulation which changes its binding preference from the p85 regulatory subunit of the PI3K kinase complex to DLC1 (PubMed:26166433). {ECO:0000269\|PubMed:10646847, ECO:0000269\|PubMed:11035045, ECO:0000269\|PubMed:11707428, ECO:0000269\|PubMed:12297295, ECO:0000269\|PubMed:15951562, ECO:0000269\|PubMed:15987703, ECO:0000269\|PubMed:18716620, ECO:0000269\|PubMed:19345329, ECO:0000269\|PubMed:20940307, ECO:0000269\|PubMed:26166433}. |
| General Annotations | UP_COMMENT_PTM | ISGylated. ISGylation promotes PTEN degradation. {ECO:0000269\|PubMed:37279284}. |
| General Annotations | UP_COMMENT_PTM | Monoubiquitinated; monoubiquitination is increased in presence of retinoic acid. Deubiquitinated by USP7; leading to its nuclear exclusion. Monoubiquitination of one of either Lys-13 and Lys-289 amino acid is sufficient to modulate PTEN compartmentalization. Ubiquitinated by XIAP/BIRC4. {ECO:0000269\|PubMed:18716620, ECO:0000269\|PubMed:19473982}. |
| General Annotations | UP_COMMENT_PTM | Ubiquitinated by the DCX(DCAF13) E3 ubiquitin ligase complex, leading to its degradation. {ECO:0000269\|PubMed:31492966}. |
| General Annotations | UP_COMMENT_SIMILARITY | Belongs to the PTEN phosphatase protein family. {ECO:0000256\|ARBA:ARBA00007881, ECO:0000256\|PIRNR:PIRNR038025}. |
| General | UP_COMMENT_SIMILARITY | Belongs to the PTEN phosphatase protein |

| Main Category | Category | Terms |
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| Annotations | | family. {ECO:0000256\|ARBA:ARBA00007881}. |
| General Annotations | UP_COMMENT_SIMILARITY | Belongs to the PTEN phosphatase protein family. {ECO:0000305}. |
| General Annotations | UP_COMMENT_SUBCELLULAR_LOCATION | Cell projection, dendritic spine {ECO:0000256\|ARBA:ARBA00004552}. Cytoplasm {ECO:0000256\|ARBA:ARBA00004496}. Nucleus, PML body {ECO:0000256\|ARBA:ARBA00004322}. Postsynaptic density {ECO:0000256\|ARBA:ARBA00034105}. |
| General Annotations | UP_COMMENT_SUBCELLULAR_LOCATION | Cell projection, dendritic spine {ECO:0000256\|ARBA:ARBA00004552}. Cytoplasm {ECO:0000256\|PIRNR:PIRNR038025}. Nucleus {ECO:0000256\|PIRNR:PIRNR038025}. Nucleus, PML body {ECO:0000256\|PIRNR:PIRNR038025}. Postsynaptic density {ECO:0000256\|ARBA:ARBA00034105}. |
| General Annotations | UP_COMMENT_SUBCELLULAR_LOCATION | Cytoplasm {ECO:0000269\|PubMed:15987703, ECO:0000269\|PubMed:18716620, ECO:0000269\|PubMed:19473982, ECO:0000269\|PubMed:25801959, ECO:0000269\|PubMed:9187108}. Nucleus {ECO:0000269\|PubMed:15987703, ECO:0000269\|PubMed:18716620, ECO:0000269\|PubMed:19473982, ECO:0000269\|PubMed:25801959}. Nucleus, PML body {ECO:0000269\|PubMed:18716620}. Cell projection, dendritic spine {ECO:0000250\|UniProtKB:O54857}. Postsynaptic density {ECO:0000250\|UniProtKB:O54857}. Note=Monoubiquitinated form is nuclear. Nonubiquitinated form is cytoplasmic. Colocalized with PML and USP7 in PML nuclear bodies (PubMed:18716620). XIAP/BIRC4 promotes its nuclear localization (PubMed:19473982). Associares with the postsynaptic density in response to NMDAR activation (By similarity). {ECO:0000250\|UniProtKB:O54857, ECO:0000269\|PubMed:18716620, ECO:0000269\|PubMed:19473982}. |
| General Annotations | UP_COMMENT_SUBCELLULAR_LOCATION | [Isoform alpha]: Secreted {ECO:0000269\|PubMed:23744781, ECO:0000269\|PubMed:24768297}. Note=May be secreted via a classical signal peptide and reenter into cells with the help of a poly-Arg motif. {ECO:0000269\|PubMed:23744781, ECO:0000269\|PubMed:24768297}. |
| General Annotations | UP_COMMENT_SUBUNIT | Monomer. The unphosphorylated form interacts with the second PDZ domain of MAGI2 and with DLG1 and MAST2 in vitro (PubMed:10646847, PubMed:10760291, |

| Main Category | Category | Terms |
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| | | PubMed:11707428). Interacts with MAGI2, MAGI3, MAST1 and MAST3, but neither with MAST4 nor with DLG5; interaction with MAGI2 increases protein stability (PubMed:10748157, PubMed:15951562). Interacts with NEDD4 (PubMed:17218260). Interacts with NDFIP1 and NDFIP2; in the presence of NEDD4 or ITCH, this interaction promotes PTEN ubiquitination (PubMed:20534535, PubMed:25801959). Interacts (via C2 domain) with FRK (PubMed:19345329). Interacts with USP7; the interaction is direct (PubMed:18716620). Interacts with ROCK1 (By similarity). Interacts with XIAP/BIRC4 (PubMed:19473982). Interacts with STK11; the interaction phosphorylates PTEN (PubMed:15987703). Interacts with PPP1R16B (PubMed:25007873). Interacts with NOP53; regulates PTEN phosphorylation and increases its stability (PubMed:15355975). Interacts (via PDZ domain-binding motif) with DLG4; the interaction is induced by NMDA and is required for PTEN location at postsynaptic density (By similarity). Interacts with the regulatory p85 subunit of the PI3K kinase complex and with Rho GTPase-activating protein DLC1; in resting cells, interacts (via C2 tensin-type domain) with p85 but, following growth factor stimulation, PTEN is phosphorylated which leads to weakened interaction with p85 and enhanced interaction (via C2 tensin-type domain) with DLC1 while p85 interaction with TNS3 increases (PubMed:26166433). {ECO:0000250\|UniProtKB:O08586, ECO:0000250\|UniProtKB:O54857, ECO:0000269\|PubMed:10555148, ECO:0000269\|PubMed:10646847, ECO:0000269\|PubMed:10748157, ECO:0000269\|PubMed:10760291, ECO:0000269\|PubMed:11707428, ECO:0000269\|PubMed:15355975, ECO:0000269\|PubMed:15951562, ECO:0000269\|PubMed:15987703, ECO:0000269\|PubMed:17218260, ECO:0000269\|PubMed:18716620, ECO:0000269\|PubMed:19345329, ECO:0000269\|PubMed:19473982, ECO:0000269\|PubMed:20534535, ECO:0000269\|PubMed:25007873, ECO:0000269\|PubMed:25801959, ECO:0000269\|PubMed:26166433}. |
| General Annotations | UP_COMMENT_TISSUE_SPECIFICITY | Expressed at a relatively high level in all adult tissues, including heart, brain, placenta, lung, liver, muscle, kidney and pancreas. {ECO:0000269\| PubMed:9090379}. |

| Main Category | Category | Terms |
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| Identifiers | ENTREZ_GENE_ID | 5728 |
| Identifiers | ENSEMBL_GENE_ID | ENSG00000171862 |
| Identifiers | UNIPROT_ID | PTEN_HUMAN |
| Identifiers | UNIPROT_ID | H6WA52_HUMAN |
| Identifiers | UNIPROT_ID | H6WA51_HUMAN |
| Identifiers | UNIPROT_ID | H6WA46_HUMAN |
| Identifiers | UNIPROT_ID | F6KD01_HUMAN |
| Identifiers | UNIPROT_ID | F6KCZ3_HUMAN |
| Identifiers | UNIPROT_ID | F6KCZ4_HUMAN |
| Identifiers | UNIPROT_ID | A0AAQ5BH13_HUMAN |
| Identifiers | UNIPROT_ID | A0AAQ5BGY7_HUMAN |
| Identifiers | UNIPROT_ID | A0AA34QVZ8_HUMAN |
| Identifiers | UNIPROT_ID | A0A9L9PYJ2_HUMAN |
| Identifiers | UNIPROT_ID | A0A8V8TQX2_HUMAN |
| Identifiers | UNIPROT_ID | A0A8V8TPK6_HUMAN |
| Identifiers | UNIPROT_ID | A0A8V8TP77_HUMAN |
| Identifiers | UNIPROT_ID | A0A8I5KSF9_HUMAN |
| Identifiers | UNIPROT_ID | A0A8I5KVQ8_HUMAN |
| Identifiers | UNIPROT_ID | A0A8I5KR36_HUMAN |
| Identifiers | UNIPROT_ID | A0A386NBZ2_HUMAN |
| Identifiers | UNIPROT_ID | A0A386NCS8_HUMAN |
| Identifiers | UNIPROT_ID | A0A2R4SE35_HUMAN |
| Identifiers | UNIPROT_ID | A0A2P0XNP6_HUMAN |
| Identifiers | UNIPROT_ID | A0A1V0DNR7_HUMAN |
| Identifiers | UNIPROT_ID | A0A142FGX1_HUMAN |
| Identifiers | UNIPROT_ID | A0A087X033_HUMAN |
| Identifiers | UNIPROT_ID | A0A024QYR6_HUMAN |
| Interactions | DRUGBANK | Phosphatidylethanolamine |
| Interactions | UP_KW_LIGAND | Lipid-binding |
| Literature | GENERIF_SUMMARY | PTEN mutations are uncommon in Proteus syndrome |
| Literature | GENERIF_SUMMARY | PTEN protects p53 from survival signals, permitting p53 to function as a guardian of the genome. By virtue of its capacity to |

| Main Category | Category | Terms |
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| | | protect p53, PTEN can sensitize tumor cells to chemotherapy that relies on p53 activity |
| Literature | GENERIF_SUMMARY | novel germline mutation of the PTEN gene in a patient with macrocephaly, ventricular dilatation, and features of VATER association |
| Literature | GENERIF_SUMMARY | modulates vascular endothelial growth factor-mediated signaling and angiogenic effects |
| Literature | GENERIF_SUMMARY | blockage of tumor necrosis factors-induced NF-kappa B-dependent transcription by inhibiting transactivation potential of the p65 subunit |
| Literature | GENERIF_SUMMARY | Germline PTEN mutations have not been identified nor has linkage to 10p21 been demonstrated in 8 subjects with Birt-Hogg-Dube (BHD) syndrome, suggesting that PTEN should be excluded as a candidate gene for BHD. |
| Literature | GENERIF_SUMMARY | PTEN mutations are associated with endometrial carcinomas and hyperplasia |
| Literature | GENERIF_SUMMARY | Mutations in endometrial cancer are the consequence of mismatch repair deficiencies in HNPCC-related cases, but may precede them in sporadic cases. |
| Literature | GENERIF_SUMMARY | expression reduced in pancreatic cancer may give tumor cells growth advantage and contribute to aggressive phenotype |
| Literature | GENERIF_SUMMARY | Review |
| Literature | GENERIF_SUMMARY | Abnormalities of the PTEN gene are associated with tumor progression, metastasis, and survival. |
| Literature | GENERIF_SUMMARY | Promoter analysis of PTEN |
| Literature | GENERIF_SUMMARY | MMAC1 may not play an important role in carcinogenesis in primary brain lymphoma, as in the systemic counterpart. |
| Literature | GENERIF_SUMMARY | reversible inactivation by hydrogen peroxide |
| Literature | GENERIF_SUMMARY | The phosphatidylinositol phosphatase PTEN is under control of costimulation and regulates proliferation in human T cells. PTEN negatively controls costimulatory signals by antagonizing PI3 K activity in the absence of TCR engagement. |
| Literature | GENERIF_SUMMARY | data suggest that PTEN expression is frequently reduced in advanced breast cancers |
| Literature | GENERIF_SUMMARY | Results show that 9 out of 10 carcinomas |

| Main Category | Category | Terms |
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| | | exhibited some degree of staining for Akt or PTEN, while normal tissue showed no staining. Akt and PTEN may be diagnostic markers for lung carcinoma. |
| Literature | GENERIF_SUMMARY | PTEN normally is subjected to a feedback mechanism of regulation aimed at maintaining homeostatic levels of D3-phosphoinositides, which are crucial for T cell survival and activation. |
| Literature | GENERIF_SUMMARY | The tumor-suppressive function of Pten depends on its lipid phosphatase activity which disrupts PI-3K/Akt signaling. Loss/ mutation of Pten is a marker for poor prognosis in gliomas. |
| Literature | GENERIF_SUMMARY | An Arg234Gln missense PTEN mutation from a patient with multiple brain tumors can't induce apoptosis and leads to high PKB/Akt activation & cell proliferation. |
| Literature | GENERIF_SUMMARY | down regulation of PTEN is associated with HCV-positive cirrhotic hepatocarcinogenesis and increased expression of iNOS and COXII |
| Literature | GENERIF_SUMMARY | role in the pathogenesis of epithelial thyroid neoplasias - review |
| Literature | GENERIF_SUMMARY | PTEN potently inhibited the growth and reduced the size of Jurkat cells. The growth-suppressive effect of PTEN was associated with its ability to induce apoptotic cell death with little or no effect on cell cycle. |
| Literature | GENERIF_SUMMARY | PTEN modulates angiogenesis in prostate cancer by regulating VEGF expression. |
| Literature | GENERIF_SUMMARY | PTEN alterations are late genetic events in the progression of low to high-grade astrocytomas. |
| Literature | GENERIF_SUMMARY | search for potential PTEN modulators through protein-protein interaction |
| Literature | GENERIF_SUMMARY | PTEN may play a critical role in regulating cellular signaling in prostate cancer cells |
| Literature | GENERIF_SUMMARY | PTEN mutations were detected in ILC (truncating mutations) in around 2% of the tumors |
| Literature | GENERIF_SUMMARY | PTEN silencing in anaplastic thyroid cancer is a result of a wide variety of epigenetic and/or structural silencing mechanisms rather than a consequence of structural biallelic inactivation of the classical type. |
| Literature | GENERIF_SUMMARY | PTEN expression in endometrial neoplasms |
| Literature | GENERIF_SUMMARY | These results suggest that PTEN controls mammary gland development and, |

| Main Category | Category | Terms |
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| | | consequently, lactation. |
| Literature | GENERIF_SUMMARY | used mass spectrometric methods to identify Ser(370) and Ser(385) as in vivo phosphorylation sites of PTEN |
| Literature | GENERIF_SUMMARY | cell growth control by PTEN and it's ability to regulate known cell size regulators involved in protein translation (review) |
| Literature | GENERIF_SUMMARY | there are high frequencies of somatic mutations in PTEN (encoding phosphate and tensin homolog) in breast neoplastic epithelium and stroma |
| Literature | GENERIF_SUMMARY | PTEN suppresses hyaluronic acid-induced matrix metalloproteinase-9 expression in U87MG glioblastoma cells through focal adhesion kinase dephosphorylation. |
| Literature | GENERIF_SUMMARY | A patient with Cowden Disease and all family members exhibited the same 4-bp deletion in exon 8 of the PTEN gene. |
| Literature | GENERIF_SUMMARY | induction of Skp2 may be causally linked with decreased levels of p27 in prostate cancer and implicate PTEN in the regulation of Skp2 expression |
| Literature | GENERIF_SUMMARY | PTEN gene was deleted or weakly expressed in primary renal significant cell carcinoma, which is probably related to tumorigenesis and development of renal cell carcinoma. |
| Literature | GENERIF_SUMMARY | PTEN inactivation may be important for the propagation of melanoma cells in culture, and that another chromosome 10 tumour suppressor gene may be important for melanoma pathogenesis. |
| Literature | GENERIF_SUMMARY | REVIEW |
| Literature | GENERIF_SUMMARY | loss or reduced expression of PTEN protein occures commonly in tumorigenesis and progression of gastric carcinoma |
| Literature | GENERIF_SUMMARY | restoration of PTEN function in gliomas may induce therapeutic effect by downregulating VEGF. |
| Literature | GENERIF_SUMMARY | Mutation and epidermal growth factor receptor activation regulate vascular endothelial growth factor mRNA expression in human glioblastoma cells by transactivating the proximal VEGF promoter. |
| Literature | GENERIF_SUMMARY | Results suggest that activation of peroxisome proliferative activated receptor gamma (PPARgamma) may represent a novel approach for the treatment of pancreatic cancer by increasing PTEN levels |

| Main Category | Category | Terms |
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| | | and inhibiting PI3K activity. |
| Literature | GENERIF_SUMMARY | Nuclear PTEN varies throughout the cell cycle. Higher nuclear PTEN levels were associated with G0-G1 phase, and lower nuclear PTEN levels were associated with S phase. |
| Literature | GENERIF_SUMMARY | Frequent monoallelic deletion of PTEN and its reciprocal associatioin with PIK3CA amplification is associated with gastric carcinoma |
| Literature | GENERIF_SUMMARY | PTEN/MMAC1 gene mutation is a rare event in soft tissue sarcomas without specific balanced translocations. |
| Literature | GENERIF_SUMMARY | Dysregulated PTEN-PKB and negative receptor status in human breast cancer |
| Literature | GENERIF_SUMMARY | BMP2 exposure can regulate PTEN protein levels by decreasing PTEN's association with the degradative pathway |
| Literature | GENERIF_SUMMARY | Loss of PTEN expression and increased levels of HIF-1alpha and VEGF may play an important role in carcinogenesis and progression of colorectal adenocarcinoma. |
| Literature | GENERIF_SUMMARY | PTEN and p27Kip1 have roles in tumor cell proliferation, and increased risk of recurrence |
| Literature | GENERIF_SUMMARY | overexpression of RRM1 in human and mouse lung cancer cell lines induced PTEN expression, reduced phosphorylation of focal adhesion kinase (FAK), suppressed migration, invasion, and metastasis formation, and increased survival in an animal model. |
| Literature | GENERIF_SUMMARY | study suggests mutations of the PTEN/MMAC1 gene do not occur at a significant rate in advanced gastric carcinoma, but the rare clustered mutation site (exons 2-6) suggests that PTEN/MMAC1 might contribute to gastric carcinogenesis and its progression |
| Literature | GENERIF_SUMMARY | In certain tumor subtypes, PTEN/MMAC1 gene alteration may be useful clinically in the evaluation of the tumor. |
| Literature | GENERIF_SUMMARY | functional role of PTEN in the molecular pathogenesis of prostatic disease [review] |
| Literature | GENERIF_SUMMARY | exposure to Zn2+ ions causes PTEN degradation and loss of function, which is mediated by an ubiquitin-associated proteolytic process in the airway epithelium |
| Literature | GENERIF_SUMMARY | Loss of this protein in EGF receptor- |

| Main Category | Category | Terms |
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| | | expressing tumor cells counteracts the antitumor action of EGFR tyrosine kinase inhibitors. |
| Literature | GENERIF_SUMMARY | Data suggest that PTEN tumor suppressor gene malfunction seems to be involved in metastasing capacity of hepatocellular carcinoma. |
| Literature | GENERIF_SUMMARY | existence of a regulatory mechanism of protein stability and PTEN-protein interactions during apoptosis, executed by caspase-3 in a PTEN phosphorylation-regulated manner. |
| Literature | GENERIF_SUMMARY | PTEN modulates IGF-2-mediated signaling. The phosphoprotein phosphatase activity of PTEN downregulates IGF-2 expression in hepatoma cells. |
| Literature | GENERIF_SUMMARY | phosphorylation/dephosphorylation of the C-terminal region of PTEN serves as an electrostatic switch that controls the membrane translocation of the protein |
| Literature | GENERIF_SUMMARY | PTEN has a novel germ-line mutation in a PHTS patient with a granular cell tumor. |
| Literature | GENERIF_SUMMARY | Hepatitis B X protein in liver cells down-regulates the expression of PTEN and activates AKT. |
| Literature | GENERIF_SUMMARY | Germline PTEN promoter mutations and deletions in Cowden/Bannayan-Riley-Ruvalcaba syndrome result in aberrant PTEN protein and dysregulation of the phosphoinositol-3-kinase/Akt pathway. |
| Literature | GENERIF_SUMMARY | a PI(4,5)P2 monomer binds to PTEN, initiates an allosteric conformational change and, thereby, activates PTEN independent of membrane binding |
| Literature | GENERIF_SUMMARY | activated Src inhibits PTEN function leading to alterations in signaling through the PI3K/AKT pathway |
| Literature | GENERIF_SUMMARY | The data are consistent with an oncogenesis model in which a lack of PTEN fuels the cell cycle by increasing the nuclear availability of cyclin D1 through the Akt/PKB pathway. |
| Literature | GENERIF_SUMMARY | Down-regulated expression of PTEN plays important role in tumorigenesis, progression, growth, differentiation, and angiogenesis of gastric cancer. Low expression of PTEN can decrease expression of caspase-3 to disorder apoptosis of tumor cells. |
| Literature | GENERIF_SUMMARY | tumor suppressor PTEN, inhibits the |

| Main Category | Category | Terms |
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| | | interaction of CH-ILKBP with ILK |
| Literature | GENERIF_SUMMARY | Down-regulated PTEN expression and up-regulated MMP-7 expression were greatly implicated in tumorigenesis and progression of gastric carcinoma. |
| Literature | GENERIF_SUMMARY | The survival data on 129 glioblasstoma patients were correlated with the results of a detailed analysis of this gene. |
| Literature | GENERIF_SUMMARY | Oxidative stress activates PI 3-kinase-dependent signalling via the inactivation of PTEN. |
| Literature | GENERIF_SUMMARY | ovarian cancer PTEN does not play a major role in disease progression and is not involved in the alteration of p27Kip1 expression |
| Literature | GENERIF_SUMMARY | ovarian steroids regulate the endometrial PTEN pool |
| Literature | GENERIF_SUMMARY | PTEN inhibits MDM2 and protects p53 through both p13k/Akt-dependent and -independent pathways in ALL. |
| Literature | GENERIF_SUMMARY | PTEN has a role in anchorage-independent colony formation and anchorage-dependent proliferation |
| Literature | GENERIF_SUMMARY | loss of PTEN function is sufficient to cause Lhermitte-Duclos disease |
| Literature | GENERIF_SUMMARY | Mutation in PTEN is associated with Bannayan-Riley-Ruvalcaba syndrome |
| Literature | GENERIF_SUMMARY | TNF-alpha-induced down-regulation of PTEN is mediated through a TNF-alpha/NIK/NF-kappaB pathway. |
| Literature | GENERIF_SUMMARY | The results indicate that the evaluation of PTEN mutation are useful to predict dissemination and prognosis of glioblastomas. |
| Literature | GENERIF_SUMMARY | These data also suggest that location of intragenic PTEN mutations and their coexistence with the CMYC amplification may play a crucial part in the development of various subtypes of endometrial carcinoma. |
| Literature | GENERIF_SUMMARY | Loss of PTEN expression is common and correlates with tumor progression and lymph node metastasis in breast carcinoma. |
| Literature | GENERIF_SUMMARY | Lipid rafts(LR) may play important role in determining function of PI 3-K/Akt2 signaling, including stimulation of intestinal Na absorption. LR-associated Akt2 may be involved in enterocyte differentiation. |

| Main Category | Category | Terms |
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| Literature | GENERIF_SUMMARY | A conserved N-terminal motif of PTEN is needed for correct membrane orientation, cellular activity and tumour-suppressor function. |
| Literature | GENERIF_SUMMARY | importance of PTEN in primary sporadic colorectal cancer |
| Literature | GENERIF_SUMMARY | regulates cell proliferation and apoptosis through inhibition of IGF-IR synthesis |
| Literature | GENERIF_SUMMARY | results show that PTEN can inhibit cell migration through its C2 domain, independent of its lipid phosphatase activity; ability of PTEN to control cell migration through its C2 domain is likely to be an important feature of its tumor suppressor activity |
| Literature | GENERIF_SUMMARY | Loss of PTEN gene not uncommon in NSCLC, but level of PTEN protein expression is not an independent prognostic marker in early-stage NSCLC. |
| Literature | GENERIF_SUMMARY | possible cooperation between BRAF activation and PTEN loss in melanoma development. |
| Literature | GENERIF_SUMMARY | PTEN downregulation is implied in GIST progression. |
| Literature | GENERIF_SUMMARY | PTEN is regulated by anionic lipids and reactive oxygen species [review] |
| Literature | GENERIF_SUMMARY | a tumour suppressor candidate protein binds to extreme C-terminal region of PTEN and regulates PTEN protein turnover [review] |
| Literature | GENERIF_SUMMARY | PTEN mutations were detected in 7 cases (14.3%) of synovial sarcoma, and all of these were monophasic tumors |
| Literature | GENERIF_SUMMARY | Hypermethylation of the PTEN promoter correlated significantly with either decreased or complete loss of PTEN protein expression in colorectal cancer. |
| Literature | GENERIF_SUMMARY | a role of these tumor suppressor and metastasis suppressor genes in the evolution and progression of NSCLC |
| Literature | GENERIF_SUMMARY | PTEN mutation is associated with HPV-negative adenocarcinoma of the uterine cervix |
| Literature | GENERIF_SUMMARY | PTEN methylation and loss of PTEN expression are early events in the development of cervical cancer and may have prognostic significance. |
| Literature | GENERIF_SUMMARY | PTEN, via distinct mechanisms, differentially regulates androgen receptors in various |

| Main Category | Category | Terms |
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| | | stages of prostate cancers. |
| Literature | GENERIF_SUMMARY | One common way of stimulating the 1-Phosphatidylinositol 3-Kinase kinase pathway occurs through inactivation of the PTEN tumor suppressor |
| Literature | GENERIF_SUMMARY | Modification of the Egr-1-dependent mechanisms may play a role in the silencing of PTEN gene expression occurring during thyroid cell transformation. |
| Literature | GENERIF_SUMMARY | We found that indomethacin and NS-398 treatment significantly upregulated expression of the tumor suppressor gene, PTEN. |
| Literature | GENERIF_SUMMARY | Inhibitory effects of PTEN on cell motility translate into suppression of in vivo invasion. |
| Literature | GENERIF_SUMMARY | PTEN promoter hypermethylation is a common event in sporadic breast cancer, correlating with other well-established prognostic factors of this malignancy |
| Literature | GENERIF_SUMMARY | PICT-1 (protein interacting with carboxyl terminus 1) binds to the C terminus of PTEN and regulates its phosphorylation and turnover. |
| Literature | GENERIF_SUMMARY | Methylation of the PTEN promoter leads to PTEN inactivation in a subset of human breast cancers |
| Literature | GENERIF_SUMMARY | PTEN mRNA and protein expression as well as PTEN-related cell growth inhibition in endometrial cancer cells. |
| Literature | GENERIF_SUMMARY | inactivation of PTEN gene and over-expression of VEGF contribute to the neovascularization and progression of gastric cancer |
| Literature | GENERIF_SUMMARY | PTEN acts in T cells primarily to control basal PI-3,4,5-P3 levels, rather than opposing PI3K acutely during TCR stimulation. |
| Literature | GENERIF_SUMMARY | PTEN expression was a strong predictor of both, shorter relapse-free survival and shorter disease-specific survival. |
| Literature | GENERIF_SUMMARY | Data show that PTEN and MXI1 were two candidate tumor suppressor genes on 10q23 and 10q24-q25 and may be potentially involved in the initiation and progression of prostate carcinoma. |
| Literature | GENERIF_SUMMARY | in contrast to prostate cancer, mutations in the PTEN gene seem not to affect cellular distribution of the beta-catenin protein in |

| Main Category | Category | Terms |
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| | | endometrial carcinomas. |
| Literature | GENERIF_SUMMARY | The Tumor suppressor PTEN was involved in control of VEGF/VEGFR-2 stimulated prostate cancer cell adhesion as well as proliferation. |
| Literature | GENERIF_SUMMARY | findings expand our understanding of tumor promoter/suppressor inter-relationships and downstream transcriptional effects of PTEN loss and c-Met overexpression in malignant gliomas |
| Literature | GENERIF_SUMMARY | PTEN expression in the initial steps of the apoptotic pathway is regulated by EGR-1 |
| Literature | GENERIF_SUMMARY | the function of the PTEN gene in hepatocarcinomas may be impaired mainly through point mutations and expression deficiency; and the defect of PTEN in tumor cells may alter the phosphorylation of FAK. |
| Literature | GENERIF_SUMMARY | reactive oxygen species generated by insulin stimulation contributes to the inactivation of PTEN and not to the activation of PI-3 kinase in the PI-3 kinase/Akt pathway |
| Literature | GENERIF_SUMMARY | reduced PTEN expression may be an independent prognostic indicator in patients with invasive ductal carcinoma. |
| Literature | GENERIF_SUMMARY | PTEN has a role in regulating actin polymerization but not directionality during mammalian cell chemotaxis |
| Literature | GENERIF_SUMMARY | Overexpression of human PTEN in rat vascular smooth muscle cells inhibits the cellular processes necessary for neointimal hyperplasia. PTEN is a critical regulator of cell growth in the vasculature. |
| Literature | GENERIF_SUMMARY | Review. PTEN removal of the phosphate from PtdIns(3,4,5)P(3) inhibits the PI3-kinase/Akt pathway, preventing localisation of pleckstrin homology domain proteins to the cell membrane. Alterations of PTEN are associated with cancer & other diseases. |
| Literature | GENERIF_SUMMARY | Increase of PTEN expression in ethanol-exposed cells is the main causative factor in altering the balance between prosurvival and prodeath signals initiated by TNF. |
| Literature | GENERIF_SUMMARY | PTEN mutations are restricted to advanced gastric cancer, loss of heterozygosity and mutation of PTEN gene are closely related to the infiltration and metastasis of gastric cancer. |
| Literature | GENERIF_SUMMARY | Alterations in beta-catenin and PTEN genes, as well as MSI, are frequent in low-stage ovarian carcinomas of endometrioid type |

| Main Category | Category | Terms |
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| | | that have a favorable prognosis. |
| Literature | GENERIF_SUMMARY | Intratubular germ cell neoplasias intensely expressed PTEN, indicating that loss of PTEN expression is not an early event in testicular tumor development. |
| Literature | GENERIF_SUMMARY | Sod2 can serve as an alternative physiological source of the potent signaling molecule, H2O2 via PTEN oxidation |
| Literature | GENERIF_SUMMARY | Transient and near-complete loss of PTEN expression induces loss of adhesion of the cells. |
| Literature | GENERIF_SUMMARY | loss of PTEN and subsequent activation of AKT impair CHK1 through phosphorylation, ubiquitination, and reduced nuclear localization to promote genomic instability in tumor cells |
| Literature | GENERIF_SUMMARY | May be susceptible to dimethyl sulfoxide, shich might therefore be an antineoplastic agent. |
| Literature | GENERIF_SUMMARY | Dimethylsulfoxide induced upregulation of PTEN in HL-60 leukemia cells. |
| Literature | GENERIF_SUMMARY | S1P2R receptor actively regulates the PTEN phosphatase by a Rho GTPase-dependent pathway to inhibit cell migration. |
| Literature | GENERIF_SUMMARY | Data show that DJ-1 is a key negative regulator of PTEN that may be a useful prognostic marker for cancer. |
| Literature | GENERIF_SUMMARY | the regulation of PTEN expression may play an important role in the development of the early gestational trophoblast and in the pathogenesis of hydatidiform mole, but not in its malignant transformation |
| Literature | GENERIF_SUMMARY | The PTEN is a phosphatase that dephosphorylates both protein and phosphoinositide substrates. |
| Literature | GENERIF_SUMMARY | EGF and hypoxia induce CXCR4 in non-small cell lung cancer, a process regulated by the PI3-kinase/PTEN/AKT/mTOR signaling pathway and activation of HIF-1alpha |
| Literature | GENERIF_SUMMARY | The PTEN is a phosphatase that dephosphorylates both protein and phosphoinositide substrates. |
| Literature | GENERIF_SUMMARY | Subset of individuals with autism spectrum disorders and extreme macrocephaly associated with germline PTEN tumour suppressor gene mutations |
| Literature | GENERIF_SUMMARY | protection of p53 from MDM2 by PTEN and the damage-induced activation of PTEN by |

| Main Category | Category | Terms |
|---|---|---|
| | | phosphorylated p53 leads to the formation of an apoptotic amplification cycle in which p53 and PTEN coordinately increase cellular apoptosis |
| Literature | GENERIF_SUMMARY | expression of PTEN increases during terminal follicular growth |
| Literature | GENERIF_SUMMARY | the present results suggest that PTEN expression may be a useful indicator of cell proliferation in patients with STS. |
| Literature | GENERIF_SUMMARY | We demonstrate that the expression of the tumor suppressor gene PTEN has a significant inverse correlation with fatty acid synthase (FAS) expression in prostate cancer, and inhibition of the PTEN gene leads to the overexpression of FAS in vitro. |
| Literature | GENERIF_SUMMARY | Activated AKT was associated with increased resistance to multiple chemotherapeutic agents in gastric cancer. |
| Literature | GENERIF_SUMMARY | loss of PTEN expression is the first biomarker in endometrial intraepithelial neoplasia that increases the accuracy of the prognostic morphometric D score to predict cancer progression risk. |
| Literature | GENERIF_SUMMARY | the C-terminal region of PTEN in is important for the selective association with scaffolding and/or regulatory molecules and PDZ domain binding stabilizes PTEN and targets it for phosphorylation by microtubule-associated serine/threonine kinases |
| Literature | GENERIF_SUMMARY | The colocalization of sphingomyelinases, ceramide, polyphosphoinositides, and PTEN in the raft fraction further suggests that the association of these lipids is critical for regulating cell death |
| Literature | GENERIF_SUMMARY | PTEN protein could inhibit cell invasion even in the presence of the constitutively active epidermal growth factor receptor(EGFR) |
| Literature | GENERIF_SUMMARY | We identified the dual phosphatase and tumor suppressor protein PTEN as an LKB1-interacting protein.PTEN is a substrate of kinase LKB1 in vitro. |
| Literature | GENERIF_SUMMARY | PTEN contributes to G1 growth arrest through an Akt-independent signaling pathway. |
| Literature | GENERIF_SUMMARY | PTEN expression negatively regulates chemotaxis of lymphoid mammalian cells via its lipid phosphatase activity |
| Literature | GENERIF_SUMMARY | Study demonstrated that PTEN-positive and phosphorylated-Akt-negative expression |

| Main Category | Category | Terms |
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| | | was a predictor of survival for patients with advanced endometrial carcinoma. |
| Literature | GENERIF_SUMMARY | Reduced expression of PTEN protein correlated with lymph node metastases in the patients with breast cancer |
| Literature | GENERIF_SUMMARY | PTEN expression is down-regulated in HCC cell lines probably due to loss of activity of PTEN promoter. |
| Literature | GENERIF_SUMMARY | PTEN represses RNA Pol I transcription through a novel mechanism that involves disruption of the SL1 complex |
| Literature | GENERIF_SUMMARY | sequestration of PTEN in the cytoplasm likely limits PTEN nuclear translocation |
| Literature | GENERIF_SUMMARY | Notch homolog 1 can participate in cross-talk with other signaling pathways such as Ras/Raf/MEK/ERK through the regulation of the PTEN tumor suppressor |
| Literature | GENERIF_SUMMARY | multiple kinases, including CK2 and GSK3beta, participate in PTEN phosphorylation and GSK3beta may provide feedback regulation of PTEN |
| Literature | GENERIF_SUMMARY | Data on loss of heterozygosity at specific loci on chromosomes 10q, 9p, and 17p and PTEN mutations on exons 5 and 6 indicated their role in glioma progression in Malay population. |
| Literature | GENERIF_SUMMARY | the clinical data linking IGFBP-2 expression to poor prognosis may arise, at least in part, because high levels of IGFBP-2 expression correlate with loss of function of PTEN |
| Literature | GENERIF_SUMMARY | Inhibition of PTEN expression in fibroblasts may contribute to the pathogenesis of fibrotic lung disease. |
| Literature | GENERIF_SUMMARY | loss of Pten is more frequent in anaplastic large cell lymphoma as compared to other mature T-/NK-cell lymphomas, which strongly correlates with the loss of cyclin-dependent kinase inhibitor 1B expression |
| Literature | GENERIF_SUMMARY | The TGFbeta(1)-induced destabilisation of E-cadherin-mediated cell-cell adhesion involves phosphorylation of beta-catenin, which is regulated by E-cadherin adhesion complex-associated PI3-kinase and PTEN. |
| Literature | GENERIF_SUMMARY | The alteration of the PTEN gene may be associated with malignant transformation of ovarian epithelial tumors. The PTEN gene seems to be a negative regulator of cell proliferation in ovarian adenocarcinomas. |
| Literature | GENERIF_SUMMARY | CKIepsilon-induced down-regulation of |

| Main Category | Category | Terms |
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| | | PI3K/Akt signaling through PTEN leads to amplified sensitivity to apoptosis. |
| Literature | GENERIF_SUMMARY | Together, these studies demonstrate a novel signal transduction pathway involving p38MAPK-NF-kappaB-PTEN in IL-18-mediated HCMEC death, and identify IL-18 as potential therapeutic target to inhibit or reduce myocardial inflammation and injury. |
| Literature | GENERIF_SUMMARY | Results suggest that the PTEN gene may be involved in the etiology of both smoking initiation and nicotine dependence. |
| Literature | GENERIF_SUMMARY | SEL1L modulates the expression of the matrix metalloproteinase inhibitors TIMP1, TIMP2 and the PTEN gene |
| Literature | GENERIF_SUMMARY | alterations in ROS levels contribute to macrophage homeostasis by altering the balance between PI 3-kinase/Akt and the phosphatase, PTEN |
| Literature | GENERIF_SUMMARY | SPRY2 mediates its anti-proliferative actions by altering PTEN content and activity |
| Literature | GENERIF_SUMMARY | Resistin induces PTEN expression by activating stress signaling p38 pathway, which may activate target transcription factor ATF-2, which in turn induces PTEN expression |
| Literature | GENERIF_SUMMARY | PTEN loss is correlated with substantial increases in Akt(Ser473) and integrin-linked kinase expression, both of which promote Ser(9) phospho-inhibition of GSK3beta and inactivation of apoptotic factors |
| Literature | GENERIF_SUMMARY | data provide clues to understand regulation of PTEN expression and possible mechanisms of pathogenesis of subset of Cowden syndrome individuals with germ line promoter variation and who lack mutations in PTEN coding region and splice sites |
| Literature | GENERIF_SUMMARY | Data suggest that PTEN may regulate the expression of p27 by negatively regulating S phase kinase-associated protein 2 expression. |
| Literature | GENERIF_SUMMARY | Pten has a role in prostate cancer progression |
| Literature | GENERIF_SUMMARY | The differential expression of PTEN and its splice variants could play a role in the pathogenesis of sporadic breast neoplasms and Cowden disease. |
| Literature | GENERIF_SUMMARY | Our observations suggest potential prognostic significance of p185(HER2) overexpression with PTEN loss in gastric |

| Main Category | Category | Terms |
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| | | adenocarcinoma patients. This opens up the possibility of considering p185(HER2)and PTEN as a therapeutic target in gastric cancer. |
| Literature | GENERIF_SUMMARY | these results show a high frequency of PTEN promoter hypermethylation, especially in follicular tumors, suggesting its possible role in thyroid tumorigenesis |
| Literature | GENERIF_SUMMARY | PTEN inactivation may play a role in progression to an androgen independence (AI) stage of advanced prostate cancer. |
| Literature | GENERIF_SUMMARY | Multiple PTEN mutations in the same tumor were more frequent in tumors with microsatellite instability. |
| Literature | GENERIF_SUMMARY | Results suggest that the altered PTEN expression and its inverse correlation with survivin may be involved in the development and progression of ovarian tumors. |
| Literature | GENERIF_SUMMARY | Survivin is positively correlated with PTEN expression in gastric cancer; PTEN expression is a molecular marker of advanced gastric cancer |
| Literature | GENERIF_SUMMARY | The dynamic membrane association could be modulated temporally or spatially to alter PTEN activity in specific physiological situations and could have important implications for tumor suppressor function. |
| Literature | GENERIF_SUMMARY | OPN acts downstream of PI3K in melanoma and PTEN loss contributes to melanoma development |
| Literature | GENERIF_SUMMARY | PTEN functions as a key regulator of mast cell homeostasis and FcepsilonRI-responsiveness |
| Literature | GENERIF_SUMMARY | PTEN induction confers androgen independent CaP cells enhanced responsiveness to the anti-proliferative effects of anti-androgens and this action may involve non-AR mediated effects. |
| Literature | GENERIF_SUMMARY | eukaryotic translation initiation factor 4E binding protein 1(4E-BP1) overexpression is associated with prostate cancer, especially when combined with phosphatase and tensin homolog(PTEN) and mammalian target of rapamycin(mTOR) expression data |
| Literature | GENERIF_SUMMARY | The data demonstrate that c-Jun contributes to the promotion of cellular survival by regulating the expression of PTEN. |
| Literature | GENERIF_SUMMARY | The negative expression of PTEN in ovarian endometriosis without atypia (EN), ovarian |

| Main Category | Category | Terms |
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| | | endometriosis with atypia (EA), and endometriosis with adjacent ovarian carcinoma (ET) is a manifestation of the inactivation of this gene. |
| Literature | GENERIF_SUMMARY | PTEN loss causes reduced NKX3.1 expression in both murine and human prostate cancers. |
| Literature | GENERIF_SUMMARY | These genetic and translational observations establish a cooperative role of Pten and Ink4a/Arf in the development of HS and provide mechanistic insights into the pathogenesis of human HS. |
| Literature | GENERIF_SUMMARY | Loss of heterozygosity is an independent prognostic factor and PTEN is a candidate as a haploinsufficient tumor suppressor in gastric cancers. |
| Literature | GENERIF_SUMMARY | activation of the PI3K/AKT pathway constitutes an important step in the molecular pathogenesis of medulloblastoma and that dysregulation of PTEN may play a significant role in this context |
| Literature | GENERIF_SUMMARY | Hypermethylation of PTEN gene promoter is associated with loss of its transcription in laryngeal squamous cell carcinoma, and may be related to lymph node metastasis. |
| Literature | GENERIF_SUMMARY | modulation of PTEN inactivation may also occur at the transcription level influencing the specific phenotypes seen in Cowden and Bannayan-Riley-Ruvalcaba syndromes |
| Literature | GENERIF_SUMMARY | Multiple nuclear exclusion motifs and a nuclear localization domain control PTEN nuclear localization by a Ran-dependent mechanism and suggest a proapoptotic role for PTEN in the cell nucleus. |
| Literature | GENERIF_SUMMARY | Microsatellite instability (MSI) leading to the functional inactivation of the PTEN gene has also been reported for ovarian adenocarcinomas with frequencies varying from 6 to 37%. |
| Literature | GENERIF_SUMMARY | Epigenetic PTEN silencing seems to be a relevant mechanism of inactivating this tumor suppressor gene in melanoma that may promote melanoma development by derepression of the AKT pathway. |
| Literature | GENERIF_SUMMARY | PC12 and U251 cells overexpressing PTEN had decreases in adrenomedullin mRNA levels. Cellular and secreted adrenomedullin peptide was similarly reduced. U251 cells overexpressing PTEN did not respond to exogenous adrenomedullin. |

| Main Category | Category | Terms |
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| Literature | GENERIF_SUMMARY | SHIP2 substitutes for PTEN in the acute regulation of PKB in PC3 cells but not other prostate cell lines, where PTEN may share this role with further PIP3-degrading mechanisms. |
| Literature | GENERIF_SUMMARY | REVIEW |
| Literature | GENERIF_SUMMARY | findings demonstrate that PTEN potently modulates hepatitis B virus-X protein-mediated signaling |
| Literature | GENERIF_SUMMARY | PTEN gene plays an important role in the occurrence and development of laryngeal carcinoma. |
| Literature | GENERIF_SUMMARY | Combination therapy provides added benefit in promoting cell death in PTEN-deficient tumor cells. |
| Literature | GENERIF_SUMMARY | Acetylation promotes p53 tetramerization, which, in turn, is required for the PTEN-p53 interaction and subsequent maintenance of high p53 acetylation. |
| Literature | GENERIF_SUMMARY | The high frequency of PTEN deletion observed in CaP versus precursor lesions implicates a pivotal role for PTEN haploinsufficiency in the transition from preneoplastic PIN to CaP. |
| Literature | GENERIF_SUMMARY | results confirm that PIK3CA mutations are frequent in endometrial carcinoma and support the hypothesis that PIK3CA mutations may have an additive effect to PTEN monoallelic inactivation in endometrial carcinoma |
| Literature | GENERIF_SUMMARY | In advanced bladder cancers, PTEN protein was significantly reduced by initiating transitional cell carcinoma and promoting progresssion. |
| Literature | GENERIF_SUMMARY | collagen matrix contraction activates PTEN by a mechanism involving cytoskeletal disassembly |
| Literature | GENERIF_SUMMARY | Data show that UVB irradiation increases PTEN/Akt phosphorylation in dermal fibroblasts, and inhibition of PTEN and activation of Akt by phosphorylation are involved in UVB-induced MMP-1 and -3 secretions through upregulation of AP-1 activity. |
| Literature | GENERIF_SUMMARY | AdPTEN strongly inhibits the growth of human prostate tumors, especially when combined with radiation therapy, and that this effect is mediated by the induction of apoptosis, and by the inhibition of angiogenesis and cellular proliferation. |

| Main Category | Category | Terms |
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| Literature | GENERIF_SUMMARY | PTEN mRNA levels were significantly lower in the glioma tissues than in the benign brain tumors and tumor-adjacent normal tissues, whereas there were no statistical differences between benign brain tumor and the tumor-adjacent normal tissues. |
| Literature | GENERIF_SUMMARY | Akt activ correlates with HER2 overexpr or LOH at the PTEN gene locus while inversely correlating with PR expression. When LOH at the PTEN gene locus and HER2 overexpr occurred simultaneously, incidence of Akt activ and reduced PR expr was significant |
| Literature | GENERIF_SUMMARY | report on the first somatic mutation of a tumor-suppressor gene, PTEN, in Focal cortical dysplasias with Taylor-type balloon cells |
| Literature | GENERIF_SUMMARY | took a haplotype-based approach and investigated the association of specific genomic regions of the PTEN locus with PTEN hamartoma tumor syndrome |
| Literature | GENERIF_SUMMARY | PTEN nonsense mutation in a Bannayan-Riley-Ruvalcaba syndrome family disrupts a potential exonic splicing enhancer and causes exon skipping |
| Literature | GENERIF_SUMMARY | Activation of PI3K signaling by mutations in PTEN or PIK3CA can lead to activation of p53-mediated growth suppression. phosphatidylinositol (3,4,5)-triphosphate-induced mitogenesis during human cancer pathogenesis. |
| Literature | GENERIF_SUMMARY | PTEN mutations were significantly correlated with uterine endometrioid carcinoma compared with atypical hyperplasia. |
| Literature | GENERIF_SUMMARY | PTEN may down-regulate the expression of PIP3, and then down-regulate the expression of cyclin D1, which leads to the suppression of cell growth. |
| Literature | GENERIF_SUMMARY | Altered phosphorylation of PTEN protein and the consequent activation of survival signals may contribute to the pathomechanism of leiomyoma. |
| Literature | GENERIF_SUMMARY | Cooperation between this gene and BMPR1A gene is deleted on chromoome 10 in juvenile polyposis coli. |
| Literature | GENERIF_SUMMARY | Our results indicate that Ad-PTEN exerts its tumor suppressive effect on bladder cancer cells through inhibiting survivin and upregulating caspase-related proteins. Thus Ad-PTEN may be potentially therapeutic for the treatment of bladder cancers. |

| Main Category | Category | Terms |
|---|---|---|
| Literature | GENERIF_SUMMARY | In conclusion, using a biologically relevant model system to dissect PTEN tumor suppressor function in human bladder cancer, we identified three molecules important for many cellular functions in complex with PTEN. |
| Literature | GENERIF_SUMMARY | determinatin of types and frequencies of PTEN gene alterations in Turkish patients with glioblastoma multiforme & relationship between alterations of the gene & histopathological properties; mutations may have an effect on aggressiveness of GBM tumors |
| Literature | GENERIF_SUMMARY | PTEN reportedly interacts in vitro with the EF hand-like motif of major vault protein in a Ca(2+)-dependent manner in a breast tumor cell line. |
| Literature | GENERIF_SUMMARY | NEDD4-1 is a potential proto-oncogene that negatively regulates PTEN via ubiquitination, a paradigm analogous to that of Mdm2 and p53. |
| Literature | GENERIF_SUMMARY | Nuclear PTEN is essential for tumor suppression and that PTEN nuclear import is mediated by its monoubiquitination. |
| Literature | GENERIF_SUMMARY | These results demonstrate that PTEN plays a fundamental role in the maintenance of chromosomal stability through the physical interaction with centromeres and control of DNA repair. |
| Literature | GENERIF_SUMMARY | We conclude that most frequently occurring mutations in the PTEN gene may be a key event for the tumorigenesis of endometrioid endometrial carcinomas. |
| Literature | GENERIF_SUMMARY | These data indicate that the aberrant expression of PTEN contributes to the activation of the PI3kinase/Akt pathway and its transcription factor mediators in glioma. |
| Literature | GENERIF_SUMMARY | To detect and analysis epidermal growth factor receptor (EGFR) and phosphatase and tensin homolog deleted on chromosome ten (PTEN) in different malignancy brain tumors, and to evaluate their prognostic significance. |
| Literature | GENERIF_SUMMARY | PTEN tumor-suppressor protein has a role in response to epidermal growth factor receptor tyrosine kinase inhibitors in glioblastoma |
| Literature | GENERIF_SUMMARY | PTEN acts as an anti-inflammatory protein to prevent neointima formation in an animal model of arterial injury. |
| Literature | GENERIF_SUMMARY | p21 plays a key role in mechanisms used by |

| Main Category | Category | Terms |
|---|---|---|
| | | PTEN-deficient tumors to escape chemotherapy |
| Literature | GENERIF_SUMMARY | Alterations in PTEN and amplification of EGFR are uncommon in pediatric malignant gliomas, in contrast to adult malignant gliomas. |
| Literature | GENERIF_SUMMARY | TNF-alpha induces upregulation of PTEN expression through NF-kappaB activation in human leukemic cells |
| Literature | GENERIF_SUMMARY | Results further affirmed the role of this element in PTEN's regulation and deregulation, and its contribution to the pathogenesis of Cowden syndrome. |
| Literature | GENERIF_SUMMARY | The high expression of Skp2 was negatively correlated with lower PTEN expression in laryngeal carcinoma, which suggested that PTEN may regulate the expression of Skp2. |
| Literature | GENERIF_SUMMARY | loss of PTEN function in human prostate cancer may specifically facilitate bone rather than other organ metastasis and suggest that Rac1, as a PTEN effector, may contribute to this metastatic tropism. |
| Literature | GENERIF_SUMMARY | PTEN could be a good prognostic factor for lung carcinomas, regardless of the histological types. |
| Literature | GENERIF_SUMMARY | The abnormal expression of Survivin and PTEN gene may be related to carcinogenesis, tumor progress and prognosis in laryngeal squamous cell carcinoma. |
| Literature | GENERIF_SUMMARY | IGF-II and IGFBP-2 differentially regulate PTEN in human breast cancer cells |
| Literature | GENERIF_SUMMARY | PTEN and IGFBP-2 expression are inversely correlated in human brain and prostate cancers |
| Literature | GENERIF_SUMMARY | Comparative genomic hybridization and immunohistochemical assessment of EGFR, PTEN, p53, and MIB-1 expression in 13 oligodendrogliomas, one oligoastrocytoma and 23 high-grade astrocytomas is reported. |
| Literature | GENERIF_SUMMARY | bi-allelic inactivation of PTEN can lead to developmental anomalies instead of malignant transformation, thus raising the question of the limitations of the tumor suppressive function in this gene. |
| Literature | GENERIF_SUMMARY | Exogenous PTEN gene induces apoptosis in breast carcinoma cell line MDA468. |
| Literature | GENERIF_SUMMARY | Increased PTEN expression in unstimulated MCF-7 breast cancer cells results in a 51% |

| Main Category | Category | Terms |
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| | | increase in phosphatidic acid, with a decrease in phosphatidylcholine, suggesting that PTEN may regulate phospholipase D and phospholipase C. |
| Literature | GENERIF_SUMMARY | GAL-3 mediates TRAIL signaling by regulating PTEN in human breast carcinoma cells |
| Literature | GENERIF_SUMMARY | PTEN mutations are a relatively infrequent cause of autism spectrum disorders with macrocephaly. |
| Literature | GENERIF_SUMMARY | PTEN/PI 3-kinase signaling and cyclin D1 control a novel pathway that regulates assembly of the SCF(SKP2) complex. |
| Literature | GENERIF_SUMMARY | The protein expression of PTEN was correlated with clinical stage, tumor differentiation grade, and cervical lymph node metastasis in laryngeal squamous cell carcinoma |
| Literature | GENERIF_SUMMARY | A model in which the phosphorylation of Thr366 plays a role in destabilizing the PTEN protein is reported. |
| Literature | GENERIF_SUMMARY | Up-regulation of PTEN at the transcriptional level is an advers prognostic factor in female lung adenocarcinomas. |
| Literature | GENERIF_SUMMARY | aberrant PI3K pathway signaling is strongly associated with metastasis and poor survival across carcinoma types |
| Literature | GENERIF_SUMMARY | findings show that unconjugated bilirubin modulates a signaling pathway involving APE1/Ref-1, Egr-1, and PTEN |
| Literature | GENERIF_SUMMARY | PPARgamma and RAR play an important role in controlling the growth of leukemia cells via the up-regulation of PTEN |
| Literature | GENERIF_SUMMARY | In Ishikawa H cells that model type I endometrial cancer in the loss of PTEN and RB1, re-expressing PTEN and RB1 increased the apoptotic and G1 phases and decreased the S and G2-M phases, which further sensitize the cells to gefitinib. |
| Literature | GENERIF_SUMMARY | Loss of PTEN may also be associated with a worse prognosis in patients with early-stage endometrial carcinoma. |
| Literature | GENERIF_SUMMARY | methylation of the PTEN promoter may represent an alternate mechanism by which PI3K signaling is increased in grade II and III gliomas as well as secondary GBMs |
| Literature | GENERIF_SUMMARY | Vascular anomalies in patients with a PTEN mutation are typically multifocal intramuscular combinations of fast-flow |

| Main Category | Category | Terms |
|---|---|---|
| | | channels and ectopic fat; cerebral developmental venous anomalies are very common |
| Literature | GENERIF_SUMMARY | It is unlikely that the EMX2 or PTEN gene variants investigated contribute to risk for initiation and/or development of endometriosis. |
| Literature | GENERIF_SUMMARY | the COOH-terminal tail can act as an autoinhibitory domain to control both PTEN membrane recruitment and phosphatase activity |
| Literature | GENERIF_SUMMARY | PIK3CA mutations and PTEN loss were not mutually exclusive events and associated with similar prognostic factors |
| Literature | GENERIF_SUMMARY | identifies the RhoA/ROCK pathway as a major target of p110delta-mediated PI 3-kinase signalling |
| Literature | GENERIF_SUMMARY | Metastasis of renal cell carcinoma is correlated with inactivation of PTEN. |
| Literature | GENERIF_SUMMARY | loss of PTEN expression, together with increased Akt phosphorylation, contributes to progression and recurrence of prostate cancer |
| Literature | GENERIF_SUMMARY | PTEN expression in leukemic blasts may be correlated with risk of relapse and survival |
| Literature | GENERIF_SUMMARY | IKK alpha controls mTOR kinase activity in Akt-active, PTEN-null prostate cancer cells, with less involvement by IKK beta |
| Literature | GENERIF_SUMMARY | PTEN germline mutations are rare |
| Literature | GENERIF_SUMMARY | Pten inactivation has a role in emergence of androgen-independent prostate cancer [review] |
| Literature | GENERIF_SUMMARY | transcriptional down-regulation of the tumor suppressor PTEN in pancreatic adenocarcinoma if facilitated by effect of K-RAS/ERK on TGFbeta |
| Literature | GENERIF_SUMMARY | Mutations in PTEN is asociated with clear cell sarcoma of the kidney |
| Literature | GENERIF_SUMMARY | PIK3CA mutations are mutually exclusive with PTEN loss in diffuse large B-cell lymphoma |
| Literature | GENERIF_SUMMARY | results suggest that the induction of PTEN-modulated apoptosis is one of the putative mechanisms of tumor suppressive activity by GLTSCR2 |
| Literature | GENERIF_SUMMARY | Increased promoter methylation of PTEN was present in renal metastasis (of breast |

| Main Category | Category | Terms |
|---|---|---|
| | | cancer), coinciding with the decrease in the level of normal PTEN transcript |
| Literature | GENERIF_SUMMARY | The functional down-regulation of PTEN contributes to cervical tumorigenesis through HR-HPV-independent mechanism. |
| Literature | GENERIF_SUMMARY | Aberrant expression of miR-21 can contribute to hepatocellular carcinoma growth and spread by modulating PTEN expression and PTEN-dependent pathways. |
| Literature | GENERIF_SUMMARY | PTEN dominantly inhibits Akt activation, the coexistence of high levels of the PTEN protein with enhanced Akt activation suggests the existence of novel mechanisms which attenuate PTEN |
| Literature | GENERIF_SUMMARY | The up-regulation of PTEN inhibited Akt and MDM2, which enhanced the level of p53, thereby inducing G(2)/M arrest and apoptosis. |
| Literature | GENERIF_SUMMARY | Findings suggest that the JNK/PTEN and NF-kappaB/PTEN pathways play a critical role in normal intestinal homeostasis and colon carcinogenesis. |
| Literature | GENERIF_SUMMARY | Data indicates that haploinsufficiency or PTEN genomic loss is an indicator of more advanced disease at surgery, and is predictive of a shorter time to biochemical recurrence of disease. |
| Literature | GENERIF_SUMMARY | PTEN expression suppresses the growth and proliferation of gastric carcinoma cell SGC7901, possibly through an inhibition of the expressions of VEGF and MMP-9. |
| Literature | GENERIF_SUMMARY | The aberrant expressions of PTEN and phosphorylated Smad2 and their interaction may play an important role in the pathogenesis of hepatocellular carcinoma. |
| Literature | GENERIF_SUMMARY | decreased protein expressions due to increased MMAC gene mutation in advanced endometriosis and ovarian cancers were not observed; a significant difference in MMAC immunostaining between endometriosis and ovarian endometriod adenocarcinoma was seen |
| Literature | GENERIF_SUMMARY | REsults describe PTEN and NDUFB8 aberrations in cervical cancer tissue. |
| Literature | GENERIF_SUMMARY | Aberrant location of expression/staining intensity of PTEN, PPM1A and P-Smad2 in hepatocellular carcinoma may impact disease progression. |
| Literature | GENERIF_SUMMARY | These data show that PTEN plays a non-redundant role in EGF-induced chemotaxis |

| Main Category | Category | Terms |
|---|---|---|
| | | of human breast cancer cells, and an optimal level of PTEN is required in these responses. |
| Literature | GENERIF_SUMMARY | Genetic alterations of chromosome 10q23, including the PTEN gene, could be important in hepatocarcinogenesis |
| Literature | GENERIF_SUMMARY | Inhibition of PI3K signaling by PTEN inhibits tumor angiogenesis and growth. AKT is the downstream target of PI3K in controlling angiogenesis and tumor growth. |
| Literature | GENERIF_SUMMARY | Loss of PTEN expression in Jurkat T cells does not impact on the PDK-1/PKC pathway and that only a subset of kinases, such as PKB/Akt, are perturbed as a consequence PTEN loss. |
| Literature | GENERIF_SUMMARY | (PTEN) functions as a phosphoinositide 3-phosphatase, that antagonizes phosphatidylinositol 3-kinase --{REVIEW} |
| Literature | GENERIF_SUMMARY | Altogether, our results suggested that E-cadherin mediated cell-cell adhesion was essential for preventing the proteasome degradation of PTEN. |
| Literature | GENERIF_SUMMARY | examined the downstream effect of five PTEN promoter variants (-861G/T, -853C/G, -834C/T, -798G/C, and -764G/A) that are not within any known cis-acting regulatory elements |
| Literature | GENERIF_SUMMARY | TLX recruits histone deacetylases to repress transcription of p21(CIP1/WAF1) and pten and regulate neural stem cell proliferation |
| Literature | GENERIF_SUMMARY | 2 point mutations in PTEN (697C>T & 253 +1G>T)were associated with juvenile polyposis syndrome. |
| Literature | GENERIF_SUMMARY | Mutational loss of PTEN induces resistance to NOTCH1 inhibition in T-cell leukemia. |
| Literature | GENERIF_SUMMARY | PTEN plays a critical role in MAGI-2-induced inhibition of cell migration and proliferation in human hepatocarcinoma cells |
| Literature | GENERIF_SUMMARY | A reduction in PTEN expression appears to be an early step in renal cell carcinogenesis. However, the PTEN expression pattern of renal cell carcinomas apparently is not prognostic for patient survival. |
| Literature | GENERIF_SUMMARY | In patients with mycosis fungoides, several loci associated with the tumor suppressor gene PTEN on chromosome 10 appear to be associated with progression from plaque to tumor stage. |
| Literature | GENERIF_SUMMARY | PTEN deficiency has a role in trastuzumab |

| Main Category | Category | Terms |
|---|---|---|
| | | resistance, which can be reversed by combinations of trastuzumab with triciribine or RAD001 |
| Literature | GENERIF_SUMMARY | In breast cancer stem-like cells, the STAT3 pathway was found to be positively regulated by mTOR signaling, whereas PTEN served as a negative regulator of both STAT3 and mTOR signaling. |
| Literature | GENERIF_SUMMARY | Western blot analysis showed that six out of 25 (24%) lung cancer cell lines displayed low expression of PTEN protein. |
| Literature | GENERIF_SUMMARY | loss of PTEN expression was significantly associated with sentinel lymph node micro-metastasis |
| Literature | GENERIF_SUMMARY | High levels of p-AKT expression occurred independently of the presence of PTEN or PIK3CA mutations in endometrial cancer. |
| Literature | GENERIF_SUMMARY | Wild-type PTEN gene with dual-specific phosphatase activity can inhibit the proliferation and metastasis of ZR-75-1 cells. |
| Literature | GENERIF_SUMMARY | The expression of Survivin and PTEN proteins are closely related to the prognosis of adrenal tumors. |
| Literature | GENERIF_SUMMARY | Cell cycle progression from the G(1) to the S phase is arrested in cells expressing wild-type PTEN, suggesting a role for PTEN as a tumor suppressor gene. |
| Literature | GENERIF_SUMMARY | In trastuzumab-treated breast cancer patients combined analysis of PTEN and PIK3CA identified twice as many patients at increased risk for progression compared to PTEN alone. |
| Literature | GENERIF_SUMMARY | Expression of TF is inhibited by PTEN gene via inactivating PI3K/AKT pathway. |
| Literature | GENERIF_SUMMARY | the loss of PTEN protein expression is associated with nonresponsiveness to cetuximab. |
| Literature | GENERIF_SUMMARY | PTEN expression is infrequent in ovarian carcinoma |
| Literature | GENERIF_SUMMARY | Analysis of PTEN gene mutations may be useful for characterization of the molecular event in hepatic angiosarcoma and cancer predisposition. |
| Literature | GENERIF_SUMMARY | PTEN may play an important role in carcinogenesis and the progression of esophageal squamous cell carcinoma in a high incidence area of northern China, and PTEN could serve as an important factor to |

| Main Category | Category | Terms |
|---|---|---|
| | | predict clinical outcome and prognosis |
| Literature | GENERIF_SUMMARY | The PTEN is crucial onvolement in neuronal injury as well as in neurological and psychiatric disorders. |
| Literature | GENERIF_SUMMARY | These findings imply that oncogenic Ras suppresses the apoptotic gene PTEN via the Raf-MEK-ERK-c-Jun pathway to induce antiapoptosis and cellular transformation. |
| Literature | GENERIF_SUMMARY | Combination of PTEN/p53/PCNA represent an independent prognostic factor for tumor recurrence and disease-specific survival in hepatocellular carcinoma after surgery. |
| Literature | GENERIF_SUMMARY | The positive rate of PTEN was significantly lower in gastric cancer than in normal gastric tissue. |
| Literature | GENERIF_SUMMARY | there is a reciprocal regulation between PI3K and PTEN that defines a novel negative-feedback loop in cell cycle progression |
| Literature | GENERIF_SUMMARY | BMP2 downregulates PTEN via RAS/ERK in a SMAD4-null environment that contributes to cell growth, and constitutes a SMAD4-independent but BMP-responsive signaling pathway. |
| Literature | GENERIF_SUMMARY | Upregulation of sterol response element-binding protein (SREBP), known to induce PPARgamma expression, can increase PTEN expression. |
| Literature | GENERIF_SUMMARY | Study shows that loss of PTEN expression is significantly associated with the basal-like breast cancer subtype in human sporadic and BRCA1-associated hereditary breast cancers. |
| Literature | GENERIF_SUMMARY | Loss of PTEN gene expression is one of the important factors of p-Akt high expression in myelodysplastic syndrome. |
| Literature | GENERIF_SUMMARY | Down-regulated expression of FHIT and PTEN may contribute to gastric carcinogenesis possibly by involving in the imbalance between apoptosis and proliferation of cells. |
| Literature | GENERIF_SUMMARY | Large genomic deletions of SMAD4, BMPR1A and PTEN are a common cause of JPS. |
| Literature | GENERIF_SUMMARY | combined analysis of Cdx2 and nuclear PTEN expression can have significant value in distinguishing histological types of gastric cancer and assessing prognosis in patients with gastric cancer. |

| Main Category | Category | Terms |
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| Literature | GENERIF_SUMMARY | miR-214 induces cell survival and cisplatin resistance primarily through targeting the PTEN/Akt pathway. |
| Literature | GENERIF_SUMMARY | Mammalian target of rapamycin is the key effector of phosphatidylinositol-3-OH-initiated proliferative signals in the thyroid follicular epithelium in PTEN mutant mice. |
| Literature | GENERIF_SUMMARY | Nuclear PTEN plays a role in chromosome stability, DNA repair, cell cycle arrest and cellular stability. The balance between these functions is an important factor in determining whether a cell remains benign or becomes neoplastic. |
| Literature | GENERIF_SUMMARY | PTEN binds to membranes through multiple sites, but only PI(4,5)P2 binding to the N-terminal domain triggers a conformational change with increased alpha-helicity |
| Literature | GENERIF_SUMMARY | Overexpressions of survivin mRNA and p27 protein and reduced expression of PTEN protein might be valuable markers to predict the presence of hepatocellular carcinoma. |
| Literature | GENERIF_SUMMARY | frequency of the occurrence of BRAF mutation and/or RET/PTC in H4-PTEN positive tumors was extremely high (75%) in papillary thyroid carcinoma |
| Literature | GENERIF_SUMMARY | PTEN gene plays an important role in ovarian cancer multidrug resistance. |
| Literature | GENERIF_SUMMARY | In pancreatic tumor cells calcium-dependent PKC-alpha mediates TGF-beta-induced transcriptional downregulation of PTEN, and this pathway promotes cell migration in a SMAD4-null environment. |
| Literature | GENERIF_SUMMARY | The loss of PTEN was shown to lead to increased levels of ARF4L protein but no change in transcript levels. The ARF4L transcript preferentially localized to the polysomal compartment after PTEN loss in glioma. |
| Literature | GENERIF_SUMMARY | The population of tumor cells with monosomy of chromosome 10 and PTEN locus correlated significantly with histological grade. High-grade astrocytomas (AAs and GBMs) had elevated fractions of tumor cells with monosomy of chromosome 10 and PTEN locus. |
| Literature | GENERIF_SUMMARY | The mRNA expression of PTEN was higher than that of the protein expression in precancerous lesion and laryngeal carcinoma tissue. |
| Literature | GENERIF_SUMMARY | The mitochondrial localization of tumor suppressor PTEN promotes apoptosis in |

| Main Category | Category | Terms |
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| | | A431 cells. |
| Literature | GENERIF_SUMMARY | Results discuss the role of PTEN upon the E3 ubquitin ligase Nedd4 as a negative feedback regulator as well as a substrate. |
| Literature | GENERIF_SUMMARY | the regulation of PTEN,phosphatase and tensin homolog gene expression . |
| Literature | GENERIF_SUMMARY | APE1 regulated PTEN expression is mediated by Egr-1. |
| Literature | GENERIF_SUMMARY | Mutations in the key residues which control PTEN lipid and protein phosphatase may act as dominant-negative mutants to suppress endogenous PTEN and alter the sensitivity of breast cancer patients to chemo- and targeted therapies. |
| Literature | GENERIF_SUMMARY | the expression manner of PTEN, beta-catenin, and p53 immunocytochemistry was observed in the normal endometrium (proliferative, secretory, and atrophic, and endometrial glandular and stromal breakdown |
| Literature | GENERIF_SUMMARY | H4-PTEN rearrangement can occur simultaneously with RET/PTC rearrangement in papillary thyroid carcinoma. |
| Literature | GENERIF_SUMMARY | PTEN acquires unexpected properties by enhancing gain-of-function mutant p53 (mut-p53) protein levels. PTEN restoration to cells harboring mut-p53 leads to induction of G(1)-S cell cycle progression and cell proliferation and to inhibition of cell death. |
| Literature | GENERIF_SUMMARY | constitutive activation of Akt correlates with the expression of the phosphorylated, inactive form of PTEN |
| Literature | GENERIF_SUMMARY | Data show that early inactivation of PTEN in saphenous vein smooth muscle cells leads to temporally increased activity of the pro-hyperplasia PI3K-AKT/PKB pathway leading to intimal hyperplasia-induced vein graft occlusion. |
| Literature | GENERIF_SUMMARY | Development of endometrial carcinoma is associated with an overexpression of NDRG1 and the loss of PTEN expression. |
| Literature | GENERIF_SUMMARY | Increased PTEN expression was associated with invasive adenocarcinoma of the prostate |
| Literature | GENERIF_SUMMARY | PTEN represses RNA polymerase III-dependent transcription by targeting the TFIIIB complex |

| Main Category | Category | Terms |
|---|---|---|
| Literature | GENERIF_SUMMARY | Enhanced insulin signaling in Pten deficiency suppresses autophagy at the formation and maturation steps of autophagosomes. |
| Literature | GENERIF_SUMMARY | PTEN and p27Kip1 are not downregulated in the majority of renal cell carcinomas |
| Literature | GENERIF_SUMMARY | germline PTEN mutation is associated with Hamartoma Tumor Syndrome |
| Literature | GENERIF_SUMMARY | Gene deletions in the PTEN gene are associated with Cowden disease. |
| Literature | GENERIF_SUMMARY | differential expression of PTEN-targeting miR-19a and miR-21 modulates the PTEN protein levels and the CS and CSL phenotypes, irrespective of the patient's mutation status, and support their roles as genetic modifiers in CS and CSL. |
| Literature | GENERIF_SUMMARY | VHL mutation alone is insufficient for tumor formation; study shows that epididymal cystadenomas from VHL patients frequently also lack expression of the PTEN tumor suppressor and display activation of phosphatidylinositol 3-kinase pathway signaling. |
| Literature | GENERIF_SUMMARY | PTEN abrogates TGF-beta-induced Smad2/3 phosphorylation. This study establishes a novel role for nuclear PTEN in the stabilization of PPM1A. |
| Literature | GENERIF_SUMMARY | Down-regulation of PTEN expression via methylation is associated with oral squamous cell carcinoma |
| Literature | GENERIF_SUMMARY | patients with 10q23 microdeletions involving the PTEN and BMPR1A genes have variable clinical phenotypes, which cannot be explained merely by the deletion sizes, and are not restricted to severe infantile juvenile polyposis |
| Literature | GENERIF_SUMMARY | Deficient PTEN expression is associated with worse overall survival in high-grade gliomas |
| Literature | GENERIF_SUMMARY | These results demonstrate that the PTEN plays a critical role in regulating cell cycle entry. |
| Literature | GENERIF_SUMMARY | Loss of PTEN function is insufficient to adequately predict responsiveness to mTOR inhibitors in glioblastoma multiforme. |
| Literature | GENERIF_SUMMARY | mTOR signaling pathway is activated in two endometrial carcinoma cell strains and the status of activation is related with PTEN expression of the cells |

| Main Category | Category | Terms |
|---|---|---|
| Literature | GENERIF_SUMMARY | ATP regulates PTEN subcellular localization in breast, colon and thyroid cancer cell lines. |
| Literature | GENERIF_SUMMARY | 1) PAK plays a required role in hyperosmotic signaling through the PI3K/pTEN/Cdc42/PP2Calpha/p38 pathway, and 2) PAK and PP2Calpha modulate the effects of this pathway on focal adhesion dynamics. |
| Literature | GENERIF_SUMMARY | First report of Cowden syndrome presenting with ovarian dysgerminoma, which implicates PTEN in the molecular pathogenesis of dysgerminoma and adds it to the phenotypic manifestations of Cowden syndrome. |
| Literature | GENERIF_SUMMARY | Cellular resistance to TRAIL could be developed through phosphorylation (activation) of Akt and phosphorylation (inactivation) of PTEN in acute lymphoblastic leukemia cells. |
| Literature | GENERIF_SUMMARY | PTEN gene mutation and expression may play an important role in the occurrence and development of gastric cancer. |
| Literature | GENERIF_SUMMARY | no association between methylation and loss of expression could be established for PTEN in gliomas and glioma cell lines |
| Literature | GENERIF_SUMMARY | Necrosis and hemorrhage were associated with PTEN expression |
| Literature | GENERIF_SUMMARY | PTEN is expressed in endometrium as it progresses to endometrial carcinoma |
| Literature | GENERIF_SUMMARY | Adiponectin blocks interleukin-18-mediated endothelial cell death via APPL1-dependent AMP-activated protein kinase (AMPK) activation and IKK/NF-kappaB/PTEN suppression. |
| Literature | GENERIF_SUMMARY | These findings indicate that PTEN plays an important role in the coordinated induction of apoptosis and G(1) arrest by carnosic acid and arsenic trioxide. |
| Literature | GENERIF_SUMMARY | Data suggest that alteration of PTEN may upregulate NDRG1, which may be an important gene in facilitating endometrium carcinogenesis. |
| Literature | GENERIF_SUMMARY | results demonstrated that PTEN had played an important role in the cell proliferation, cell migration and invasion dependent on its phosphatase activity |
| Literature | GENERIF_SUMMARY | KRAS mutations and PI3KCA/PTEN deregulation significantly correlate with resistance to cetuximab. |
| Literature | GENERIF_SUMMARY | PIK3CA mutations, in comparison with |

| Main Category | Category | Terms |
|---|---|---|
| | | PTEN loss and AKT1 mutations, were associated with significantly less and inconsistent activation of AKT and of downstream PI3K/AKT signaling in tumors and cell lines |
| Literature | GENERIF_SUMMARY | CSIG acts as a novel regulatory component of replicative senescence, which requires PTEN as a mediator and involves in a translational regulatory mechanism. |
| Literature | GENERIF_SUMMARY | Lack of PTEN gene amplification was associated with more responses to cetuximab and longer time to progression in coloretal cancers. |
| Literature | GENERIF_SUMMARY | in prostate cancer cells at least two mechanisms of drug resistance are interconnected. PTEN and mTOR signaling were shown |
| Literature | GENERIF_SUMMARY | coexpression of PTEN and AR should be undertaken to validate this pilot study and the utility of these biomarkers in routine histopathologic workup of patients with PC |
| Literature | GENERIF_SUMMARY | results delineate a previously unknown PML-DAXX-HAUSP molecular network controlling PTEN deubiquitinylation and trafficking, which is perturbed by oncogenic cues in human cancer |
| Literature | GENERIF_SUMMARY | The expression of metastasis inhibitor genes PTEN and thrombospondin 2 was down-regulated in the supraglottic carcinoma tissue with lymph node metastasis. |
| Literature | GENERIF_SUMMARY | Chk1 is phosphorylated at Ser(317) by ATR resulting in stabilization of CKII, which in turn leads to phosphorylation of PTEN at Thr(383). |
| Literature | GENERIF_SUMMARY | Loss of PTEN function may protect against resistance to PI3-kinase inhibitors in human prostate and bladder cancer cell lines. |
| Literature | GENERIF_SUMMARY | report a statistically significant lower expression intensity of PTEN and HePTP and higher nuclear SHP2 expression |
| Literature | GENERIF_SUMMARY | Although p110alpha activation is required to sustain the proliferation of established PIK3CA-mutant tumors, PTEN-deficient tumors are dependent instead on p110beta signaling. |
| Literature | GENERIF_SUMMARY | PTEN mutations were seen concomitantly with FGFR2 mutations |
| Literature | GENERIF_SUMMARY | Lys(402) acetylation modulates PTEN interaction with PDZ domain-containing |

| Main Category | Category | Terms |
|---|---|---|
| | | proteins, indicating a potential role of acetylation in regulating PTEN function |
| Literature | GENERIF_SUMMARY | Our results provide evidence that PTEN germline mutations may sustain a more wide phenotypical spectrum than previously suggested. |
| Literature | GENERIF_SUMMARY | impairment of function in senescent ECs in culture is mediated by an increase in S1P signaling through S1P(2)-mediated activation of the lipid phosphatase PTEN |
| Literature | GENERIF_SUMMARY | TGFbeta activates PI3K to downregulate PTEN for enhancement of cell proliferation that is independent of SMAD proteins |
| Literature | GENERIF_SUMMARY | PTEN hamartoma tumor syndromes |
| Literature | GENERIF_SUMMARY | PTEN is mutated in a significant subgroup of colorectal carcinomas, and all novel mutations found were in tumors harboring wild-type TP53. |
| Literature | GENERIF_SUMMARY | findings show mTOR is targeted for ubiquitination & degradation by binding to FBXW7; breast cancer cell lines & primary tumors showed a reciprocal relation between loss of FBXW7 & deletion or mutation of PTEN, which also activates mTOR |
| Literature | GENERIF_SUMMARY | SHH-dependent and -independent brain tumor growth require phosphoinositide 3-kinase-mammalian target of rapamycin signaling; PTEN has a role in shorter survival time |
| Literature | GENERIF_SUMMARY | The characterization of PTEN and PTEN-related pathways from a multidisciplinary perspective underscores the importance of incorporating data from different -omics, which is crucial for the advancement of personalized medicine. |
| Literature | GENERIF_SUMMARY | Mechanisms by which PTEN maintains genomic stability. |
| Literature | GENERIF_SUMMARY | The diverse ways in which PTEN signaling is modified in cancer. |
| Literature | GENERIF_SUMMARY | The reported poorer response to chemotherapeutic agents in patients with Metaplastic breast carcinomas may stem from downregulated DNA damage response pathways, PTEN and TOP2A. |
| Literature | GENERIF_SUMMARY | Total PTEN was absent in 33.3% of ameloblastomas, while its stabilized, phosphorylated(ser380 / thr382 / thr383) form was absent in 83.3% of tumors. |

| Main Category | Category | Terms |
|---|---|---|
| Literature | GENERIF_SUMMARY | PTEN posttranslational inactivation and hyperacivation of the PI3K/Akt pathway sustain primary T cell leukemia. |
| Literature | GENERIF_SUMMARY | Aberrant methylation and hence silencing of the PTEN gene coexisting with activating genetic alterations of the PI3K/AKT pathway is associated with thyroid tumors. |
| Literature | GENERIF_SUMMARY | Hypermethylation in the promoter region is one of major mechanisms responsible for transcriptional suppression of PTEN. |
| Literature | GENERIF_SUMMARY | a role for both cytoplasmic and nuclear PTEN in progression of prostate cancer to the hormone-refractory state |
| Literature | GENERIF_SUMMARY | PTEN protein expression in both myelodysplastic syndrome and acute myeloid leukemia was lower than in controls. |
| Literature | GENERIF_SUMMARY | PTEN upregulation is responsible for Pfn1-dependent attenuation of AKT activation in MDA-MB-231 cells. |
| Literature | GENERIF_SUMMARY | Myc as an important target for cooperative actions of p53 and Pten in the regulation of normal and malignant stem/progenitor cell differentiation, self-renewal and tumorigenic potential |
| Literature | GENERIF_SUMMARY | High expression of Skp2 was negatively correlated with lower PTEN expression in glottic carcinoma. |
| Literature | GENERIF_SUMMARY | DHA & EPA treatment increased PTEN mRNA & protein expression in MDA MB-231 cells. PTEN expression significantly inhibited NFkappaB-dependent transcription of Bcl-2 and Bcl-XL genes. |
| Literature | GENERIF_SUMMARY | Reduced PTEN expression was detected in more than one third of ovarian clear cell adenocarcinoma cases. Neither PTEN promoter methylation nor LOH at 10q23 locus is significantly related to PTEN inactivation and is not an adverse prognostic factor in OCCA. |
| Literature | GENERIF_SUMMARY | PTEN deficiency is very common in JMML and is in part due to hypermethylation of the PTEN gene promoter |
| Literature | GENERIF_SUMMARY | The present study suggested that neither the immunohistochemical expression of PTEN nor the expression of pAkt was associated with the clinical outcome of trastuzumab-containing neo-adjuvant chemotherapy. |
| Literature | GENERIF_SUMMARY | Loss of PTEN expression is associated with |

| Main Category | Category | Terms |
|---|---|---|
| | | colorectal cancer liver metastasis and poor patient survival. |
| Literature | GENERIF_SUMMARY | Frequent PTEN genomic alterations and activated phosphatidylinositol 3-kinase pathway is associated with basal-like breast cancer. |
| Literature | GENERIF_SUMMARY | Unsaturated fatty acids inhibit PTEN expression in hepatocytes by up-regulating microRNA-21 synthesis via an mTOR/NF-kappaB-dependent mechanism. |
| Literature | GENERIF_SUMMARY | the consequence of PTEN loss and Akt2 overexpression function synergistically to promote metastasis |
| Literature | GENERIF_SUMMARY | Id-1 is a novel PTEN inhibitor that could activate the Akt pathway and its downstream effectors, the Wnt/TCF pathway and p27(Kip1) phosphorylation |
| Literature | GENERIF_SUMMARY | Loss of PTEN results in the activation of JNK pathway in MCF-7 cells, and improves sensitivity of MCF-7 cells to JNK inhibitor. |
| Literature | GENERIF_SUMMARY | PTEN influences Fas signaling, at least in part, by regulating PEA-15 phosphorylation and activity that, in turn, regulate the ability of Bcl-2 to suppress Fas-induced apoptosis. |
| Literature | GENERIF_SUMMARY | A hypothetical model whereby PTEN loss upregulates cell cycle genes such as cdc6 and cyclin E2 that in turn promote metastatic colonization at distant sites. |
| Literature | GENERIF_SUMMARY | PTEN membrane binding requires PIP(2) and phosphorylation regulates an intramolecular interaction. |
| Literature | GENERIF_SUMMARY | the PTEN/PI3K/Akt pathways are critical for prostate cancer stem-like cell maintenance |
| Literature | GENERIF_SUMMARY | PTEN polymorphism may effect the risk of developing esophageal and gastric cardia neoplasms in Chinese population. |
| Literature | GENERIF_SUMMARY | PTEN, PTEN/p-AKT and PTEN/p-mTOR have roles in extrahepatic cholangiocarcinoma |
| Literature | GENERIF_SUMMARY | status of PTEN determines chromatin modifications leading to antioxidant responsive element activation |
| Literature | GENERIF_SUMMARY | genetic variations in PTEN were associated with survival in esophageal cancer |
| Literature | GENERIF_SUMMARY | These results reveal an unknown function for Pten signaling in the commitment of multipotential-secretory progenitor cells and suggest that epithelial Pten functions as a modifier gene in intestinal neoplasia. |

| Main Category | Category | Terms |
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| Literature | GENERIF_SUMMARY | a novel p73 regulated mechanism for PTEN expression in thyroid cancer cells. |
| Literature | GENERIF_SUMMARY | Results show that transformed human prostate epithelial cells lacking PTEN require mTORC2 to form tumors when injected into nude mice. |
| Literature | GENERIF_SUMMARY | PTEN mutation in malignancies; functions of PTEN; restoring PTEN functions in cancer cells directly or indirectly holds great promise for cancer therapy [review] |
| Literature | GENERIF_SUMMARY | PTEN loss amplifies c-Met-induced glioblastoma malignancy. |
| Literature | GENERIF_SUMMARY | PTEN was overexpressed in diagnostic ALL, Childhood samples |
| Literature | GENERIF_SUMMARY | Data show that TGF-beta1-mediated down-regulation of PTEN is post-transcriptional and post-translational, depending on its coding sequence, and that Smad and p38-MAPK signal pathways are involved in this down-regulation. |
| Literature | GENERIF_SUMMARY | Deletion of Pten transgene in adult neural stem cells (NSCs) is not tumorigenic but promotes NSC expansion; perturbation of PTEN-phosphatidylinositol 3-kinase (PI3K) pathway alone in NSCs is not tumorigenic. |
| Literature | GENERIF_SUMMARY | PTEN is possibly one of the targets of miR-21 in breast cancer |
| Literature | GENERIF_SUMMARY | PTEN controls the cellular response to cetuximab in KRAS wild-type prostate carcinoma PC3 cells |
| Literature | GENERIF_SUMMARY | The integration of PTEN and p53 into a common pathway for the induction of another tumor suppressor, Maspin, constitutes a tumor suppressor network of PTEN/p53/Mapsin that is operational under limited oxygen conditions. |
| Literature | GENERIF_SUMMARY | Loss of PTEN expression was an early event in endometrial tumorigenesis. |
| Literature | GENERIF_SUMMARY | IGF1-induced AKT phosphorylation and cell proliferation are suppressed with the increase in PTEN during luteinization in human granulosa cells. |
| Literature | GENERIF_SUMMARY | PTEN alterations are more common in pure endometrioid adenocarcinomas compared with that in non-endometrioid adenocarcinomas and mixed tumors. |
| Literature | GENERIF_SUMMARY | Report expression of FAK and PTEN in bronchioloalveolar carcinoma and lung adenocarcinoma. |

| Main Category | Category | Terms |
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| Literature | GENERIF_SUMMARY | combined deletion of p53 and Pten in bladder epithelium leads to invasive cancer in a novel mouse model. Inactivation of p53 and PTEN promotes tumorigenesis in human bladder cells and is correlated with poor survival in human tumors. |
| Literature | GENERIF_SUMMARY | The prevalence of PTEN mutations in a clinical pediatric cohort with autism spectrum disorders, developmental delay, and macrocephaly |
| Literature | GENERIF_SUMMARY | loss of PTEN as well as other genetic alterations that result in constitutive activation of Akt/mTORC1 signaling may contribute to the development of sporadic chordomas. |
| Literature | GENERIF_SUMMARY | Restoration of PTEN expression in PTEN-deficient GBM cells diminished EGFR-induced TF expression by inhibiting JunD/AP-1 transcriptional activity. PTEN antagonizes phosphatidylinositol 3-kinase activity, which attenuates both Akt and JNK activities. |
| Literature | GENERIF_SUMMARY | Overexpression of PTEN by the synthetic compound HO-3867 effectively suppressed cell cycle progression in human vascular smooth-muscle cells and inhibited neointimal hyperplasia in a rat carotid artery injury model. |
| Literature | GENERIF_SUMMARY | Inactivation of PTEN/p27(kip1) proteins is a specific feature in the progression of endometrial carcinoma in obese patients. |
| Literature | GENERIF_SUMMARY | PTEN may influence the proliferation of human granulosa cells as well as disturbance of follicular growth in PCOS patients. |
| Literature | GENERIF_SUMMARY | EBV latent membrane protein 2A (LMP2A) up-regulated DNMT1, leading to an increase in methylation of the PTEN promoter in stomach cancer. |
| Literature | GENERIF_SUMMARY | Authors observed homogeneity in ERG gene rearrangement status in CTCs in contrast to significant heterogeneity of AR copy number gain and PTEN loss, suggesting that rearrangement of ERG may be an earlier event in prostate carcinogenesis. |
| Literature | GENERIF_SUMMARY | In T-ALL, PTEN mutations were detected in 27 of 43 specimens, 22 of which would result in truncated proteins. 2 of these contained a mutation in the ATG translation start. There was no correlation with treatment response. |
| Literature | GENERIF_SUMMARY | Reduced MMAC/PTEN expression by |

| Main Category | Category | Terms |
|---|---|---|
| | | genetic and/or epigenetic mechanisms in low grade gliomas might be associated with glioma tumorigenesis. |
| Literature | GENERIF_SUMMARY | Results indicate that key regulators in BmK venom-induced apoptosis are PTEN, acting through downregulation of the PI3K/Akt signal pathway. |
| Literature | GENERIF_SUMMARY | PTEN loss in metastases may be predictive of resistance to cetuximab plus irinotecan in metastatic colorectal neoplasms. |
| Literature | GENERIF_SUMMARY | Activation of PI3K signaling through PI3K/PTEN mutation or altered expression is not associated with ovarian granulosa cell tumors. |
| Literature | GENERIF_SUMMARY | PTEN deletion and ERG rearrangement may cooperate, but contribute at different stages, in prostate cancer progression. |
| Literature | GENERIF_SUMMARY | E-cadherin regulates both the PTEN protein levels and its recruitment to cell-cell junctions, which integrates proliferative and morphogenetic signaling in breast epithelial cells. |
| Literature | GENERIF_SUMMARY | dysregulation of the phosphoinositide 3-kinase/Akt/PTEN pathway is relevant to the prognosis in node-negative breast carcinoma |
| Literature | GENERIF_SUMMARY | PTEN levels inversely correlates with levels of tyrosine-phosphorylated IGF-I receptor in tissue lysate arrays of primary breast cancers and deletion promotes resistance to estrogen antagonists. |
| Literature | GENERIF_SUMMARY | PTEN modulated VEGF-mediated signaling and affected tumor angiogenesis through PI3K/Akt/VEGF/eNOS pathway. |
| Literature | GENERIF_SUMMARY | Data demonstrate that stem cell protein SALL4 represses its target genes, PTEN and SALL1, through the epigenetic repressor Mi-2/NuRD complex. |
| Literature | GENERIF_SUMMARY | results suggest that genetic alterations in the PI3K pathway are common in prostate cancer, and occur mainly through PIK3CA amplification and PTEN hemizygous or homozygous deletion |
| Literature | GENERIF_SUMMARY | Pten functions downstream of ErbB-2 to restrict prostate epithelial transformation by blocking full activation of the PDK1 signaling cascade. |
| Literature | GENERIF_SUMMARY | The causative mutation of the PTEN gene is rare in Japanese women with idiopathic POF. |

| Main Category | Category | Terms |
|---|---|---|
| Literature | GENERIF_SUMMARY | Germline and somatic mutations within PTEN's ATP-binding domain influence its subcellular localization and tumor suppressive function. |
| Literature | GENERIF_SUMMARY | clustering of PTEN mutations in exon 7 in 12 of 44 cases of T-cell acute lymphoblastic leukemia |
| Literature | GENERIF_SUMMARY | Germline PTEN mutations are found in 85% of patients with CS and in more than 60% of patients with BRRS. |
| Literature | GENERIF_SUMMARY | cooperation between Par-4 and PTEN as relevant for the development of prostate cancer and implicate the NF-kappaB pathway as a critical event in prostate tumorigenesis. |
| Literature | GENERIF_SUMMARY | Conditional deletion of PTEN in the mouse prostate causes deregulated induction of survivin before full-blown transformation in vivo, whereas expression of survivin and PTEN is inversely correlated in cancer patients |
| Literature | GENERIF_SUMMARY | Loss of PTEN function in breast cancer can occur either by mutation or reduction of PTEN expression in almost half of sporadic breast tumors. |
| Literature | GENERIF_SUMMARY | XIAP acts as an E3 ubiquitin ligase for PTEN and promotes Akt activity by regulating PTEN content and compartmentalization. |
| Literature | GENERIF_SUMMARY | In melanoma cell lines, increased nuclear localization of PTEN correlates with G2 rather than with G1 arrest. Further, the specific cell cycle phase regulated by nuclear PTEN is cell-type dependent. |
| Literature | GENERIF_SUMMARY | Defects in Notch-1 signaling may play a role in human prostate tumor formation in part via a mechanism that involves regulation of the PTEN tumor suppressor gene. |
| Literature | GENERIF_SUMMARY | Results suggest that the concerted local inactivation of signaling pathways downstream of PTEN and DLC1, respectively, is required for the tight control of cell migration. |
| Literature | GENERIF_SUMMARY | PTEN induces HIF-2alpha transcriptional activity by inhibiting expression of Yin Yang 1 (YY1), which acts as a novel corepressor of HIF-2alpha. |
| Literature | GENERIF_SUMMARY | miR-26a as a direct regulator of PTEN expression and miR-26a is frequently amplified at the DNA level in human glioma, most often in association with monoallelic PTEN loss |

| Main Category | Category | Terms |
|---|---|---|
| Literature | GENERIF_SUMMARY | These studies demonstrate an important role of the PTEN/Akt/beta-catenin signaling pathway in regulation of normal and malignant mammary stem/progenitor cell populations. |
| Literature | GENERIF_SUMMARY | PTEN protein deficiency alters multiple cell cycle checkpoints possibly leaving less time for DNA damage repair and/or chromosome segregation. |
| Literature | GENERIF_SUMMARY | Tumor-infiltrating lymphocyte density in colorectal cancers with microsatellite instability was associated with the overall number of frameshift mutations and with the presence of frameshift mutations within two target genes, namely ASTE1/HT001 and PTEN |
| Literature | GENERIF_SUMMARY | p16INK4A, DAPK1, PTEN and MT1G genes were not frequently methylated in the stage I non-small cell lung cancer in China. |
| Literature | GENERIF_SUMMARY | analysis of PTEN expression in peripheral T-cell lymphoma |
| Literature | GENERIF_SUMMARY | Decreasing or depletion of PTEN might be a sign of nasal inverted papilloma development and unfavorable prognosis. |
| Literature | GENERIF_SUMMARY | Upregulation of PTEN by CKD712 inhibits VCAM-1 expression, but not ICAM-1 expression, in lipopolysaccharide-treated endothelial cells. |
| Literature | GENERIF_SUMMARY | The IGF-1/PI3K/PTEN/Akt/NF-small ka, CyrillicB cascade may be a key pathway stimulating metastasis of pancreatic cancer cells. |
| Literature | GENERIF_SUMMARY | PTEN gene might inhibit the proliferation, metastasis and invasive ability of leukemia cells via down-regulating FAK expression. |
| Literature | GENERIF_SUMMARY | Data provide a basis for rational combination of MEK and PI3K inhibitors in basal-like cancers with both intact and deleted PTEN. |
| Literature | GENERIF_SUMMARY | Results suggested that PTEN/AKT pathway was involved in trichostatin A induced cell growth inhibition and apoptosis of oral squamous cell carcinoma cells. |
| Literature | GENERIF_SUMMARY | These results suggest that abnormally low levels of PTEN, MGMT and DNA-PKcs protein in gynaecomastia breast tissue may play a role in the development of gynaecomastia. |
| Literature | GENERIF_SUMMARY | Data show although detected at a higher frequency in primary effusion lymphoma cell |

| Main Category | Category | Terms |
|---|---|---|
| | | lines than in primary primary effusion lymphoma tumors, TP53 and/or PTEN mutations, as well as deletion of CDKN2A-ARF locus are uncommon in primary effusion lymphoma. |
| Literature | GENERIF_SUMMARY | PTEN is the master regulator of the process whereby WEB-2170 induces apoptosis in acute myeloid leukemia cells, including upregulation of cyclin G2, p21/WAF1, NIX, TNF-alpha, and PTEN expression, and downregulation of cyclin D2 and BCL2 expression. |
| Literature | GENERIF_SUMMARY | genetic variation in PTEN may be an important risk factor of chronic obstructive pulmonary disease in Xuanwei. |
| Literature | GENERIF_SUMMARY | PTEN-p85 association involves the unphosphorylated form of PTEN engaged within the PTEN-associated complex and also includes the p110beta isoform of PI3K. |
| Literature | GENERIF_SUMMARY | PTEN expression predicts sensitivity to anti-HER2 therapy. |
| Literature | GENERIF_SUMMARY | Reducing expression of miR-21 or miR-155 led to up-regulation of phosphatase and tensin homologue (PTEN), programmed cell death 4 (PDCD4), or Src homology-2 domain-containing inositol 5-phosphatase 1 (SHIP1). |
| Literature | GENERIF_SUMMARY | Results reveal a unique function of HIF1alpha in up-regulation of PTEN and provide a new mechanism of reduced Akt phosphorylation in Tsc2 null cells. |
| Literature | GENERIF_SUMMARY | The combined measurement of PTEN and MMP9 transcripts might represent a novel reliable tool for the differential diagnosis of high-grade gliomas. |
| Literature | GENERIF_SUMMARY | Mutations of TP53, STK11, and PTEN were identified in 20, 21, and 20% of tumors, respectively. |
| Literature | GENERIF_SUMMARY | results indicate that LOH of the PTEN gene and of D12S1051 is the molecular pathogenesis of the gastric adenocarcinoma, and the mutation of the TP53 gene is an additional hit for the oncogenesis of choriocarcinoma arising from gastric adenocarcinoma |
| Literature | GENERIF_SUMMARY | Identified up-regulation of microRNA-21 (hsa-miR-21) concurrent with down-regulation of potent tumor suppressor proteins PTEN and programmed cell death 4 in cholesteatoma as compared with normal skin. |

| Main Category | Category | Terms |
|---|---|---|
| Literature | GENERIF_SUMMARY | Results do not support PTEN as a colon cancer susceptibility gene. |
| Literature | GENERIF_SUMMARY | Studies indicate that drugs are effective in targeting essential molecular pathways of BRAF, PTEN, Akt and mammalian target of rapamycin. |
| Literature | GENERIF_SUMMARY | Pin1 overexpression in tamoxifen-resistant - MCF-7 cells is involved in the epithelial-mesenchymal transition process via PTEN-PI3-kinase-Akt-GSK-3beta and/or GSK-3beta-NF-kappaB-dependent Snail activation |
| Literature | GENERIF_SUMMARY | Presence of PTEN NULL phenotype in Surface epithelial changes and associated endometrial hyperplasia does not necessarily predict an increased incidence of endometrioid carcinoma |
| Literature | GENERIF_SUMMARY | PTEN expression may be a good marker for the prognosis of colorectal carcinoma. |
| Literature | GENERIF_SUMMARY | identified P-REX2a as a PTEN-interacting protein; P-REX2a inhibited PTEN lipid phosphatase activity & stimulated the PI3K pathway only in the presence of PTEN; P-REX2a is a component of the PI3K pathway that can antagonize PTEN in cancer cells |
| Literature | GENERIF_SUMMARY | Data show that Protein levels of tumor suppressor targets of the miRNAs were increased by antisense to miR-21 (PTEN and RECK) and miR-221 (p27). |
| Literature | GENERIF_SUMMARY | propose a model that the relative amount/ activity of Pten between different ovarian compartments (follicular, granulosa and epithelial cells) is spatiotemporally programmed, creating a ""menopause tone""" |
| Literature | GENERIF_SUMMARY | endogenous hnRNP C and PTEN interact and co-localize within the nucleus |
| Literature | GENERIF_SUMMARY | Loss of function of PTEN alters the relationship between glucose concentration and cell proliferation, increases glycolysis, and sensitizes cells to 2-deoxyglucose. |
| Literature | GENERIF_SUMMARY | Reduced PTEN protein expression levels were observed in 34% of NSCLC cases. PTEN expression level was reduced in 26% of cases that showed increased EGFR expression. |
| Literature | GENERIF_SUMMARY | The over-expression of AKT accompanied by the loss of PTEN involves carcinogenesis and development of colon carcinoma. |
| Literature | GENERIF_SUMMARY | Over expression of PTEN gene can inhibit K562 cells proliferation and promote cell |

| Main Category | Category | Terms |
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| | | apoptosis. |
| Literature | GENERIF_SUMMARY | PTEN deletion rates were 50.0%, 50.0%, 66.7% and 81.8% in proliferative endometrium, benign hyperplasia, EIN and endometrial carcinoma, respectively |
| Literature | GENERIF_SUMMARY | The development of cervical adenocarcionomas is correlated with the mutation and absence of the protein expression of PTEN. |
| Literature | GENERIF_SUMMARY | PTEN dependent gene expression is important in cell-cycle regulation. |
| Literature | GENERIF_SUMMARY | antiandrogenic therapy reduced PTEN expression by epigenetic mechanisms |
| Literature | GENERIF_SUMMARY | Hypermethylation of PTEN was significantly associated with shorter time to recurrence of nonsmall cell lung cancer. |
| Literature | GENERIF_SUMMARY | PTEN has a role in malignant melanoma of the skin |
| Literature | GENERIF_SUMMARY | Phosphatase and tensin homologue on chromosome 10 (PTEN) directs prostaglandin E2-mediated fibroblast responses via regulation of E prostanoid 2 receptor expression |
| Literature | GENERIF_SUMMARY | Negative prognostic impact of PTEN mutation in pediatric T-cell acute lymphoblastic leukemia. |
| Literature | GENERIF_SUMMARY | Deletion of PTEN promotes tumorigenic signaling, resistance to anoikis, and altered response to chemotherapeutic agents in human mammary epithelial cells. |
| Literature | GENERIF_SUMMARY | PTEN dosage and its controlled signaling pathways are critical for transformation of PTEN dosage and its controlled signaling pathways are critical for transformation of NFs to malignant peripheral nerve sheath tumor. |
| Literature | GENERIF_SUMMARY | suggests that aberrant control of PTEN by P-REX2a may represent a key tumorigenic mechanism, in agreement with recent studies supporting the pathological relevance of several other proposed PTEN regulators |
| Literature | GENERIF_SUMMARY | The PTEN-mediated down-regulation of phosphatidylinositol 3-kinase activity may be involved in the regulation of Rap1B-dependent inhibition of Rac1 activity. |
| Literature | GENERIF_SUMMARY | report here that the combination of ERK and AKT activation is crucial for PTEN suppression in surviving cells following |

| Main Category | Category | Terms |
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| | | ultraviolet B irradiation. |
| Literature | GENERIF_SUMMARY | BRAF status, EGFR amplification, and cytoplasmic expression of PTEN were associated with outcome measures in KRAS wild-type patients treated with a cetuximab-based regimen. |
| Literature | GENERIF_SUMMARY | ablation of PTEN expression partially rescued the migratory/invasive phenotype of Bmi-1-silenced cells, indicating that PTEN might be a major mediator of Bmi-1-induced epithelial-mesenchymal transition |
| Literature | GENERIF_SUMMARY | Nuclear PTEN reduces BMI1 function independently of its phosphatase activity. |
| Literature | GENERIF_SUMMARY | Interaction between PTEN to FABP4 suggests a role for this phosphatase in the regulation of lipid metabolism and adipocyte differentiation. |
| Literature | GENERIF_SUMMARY | PTEN's lipid and protein phosphatase activities proliferation and together suppress invasion. |
| Literature | GENERIF_SUMMARY | Data reveal PTPN2, PTPRJ and PTEN as potent regulators of Akt signalling which contribute to ameliorating the consequences of oncogenic K-Ras activity. |
| Literature | GENERIF_SUMMARY | pathogenic role of germline mutations in PTEN in hereditary breast and ovarian cancer syndrome |
| Literature | GENERIF_SUMMARY | 17beta-estradiol can promote cell proliferation in endometriosis by activating PI3K/AKt pathway via an NFkappaB/PTEN-dependent pathway. |
| Literature | GENERIF_SUMMARY | The TGF-beta/NF-kappaB/PTEN cascade may be a critical pathway for pancreatic cancer cells to proliferate and metastasize |
| Literature | GENERIF_SUMMARY | The loss of PTEN expression on metastatic sites was negatively associated with cetuximab response |
| Literature | GENERIF_SUMMARY | These results suggest that bust autophagic activity target at PTEN might be a potential therapy target for glioma therapy. |
| Literature | GENERIF_SUMMARY | Tumor suppression by PTEN requires the activation of the PKR-eIF2alpha phosphorylation pathway. |
| Literature | GENERIF_SUMMARY | Study supports clinical studies identifying an association of PTEN loss with late stage cancer. Cellular factors secreted from the surrounding tumor milieu act in concert with genetic changes in the tumor cells and contribute to enhanced tumor invasion. |

| Main Category | Category | Terms |
|---|---|---|
| Literature | GENERIF_SUMMARY | Quantitative methylation analysis identified ABCB1, FOXC1, PPP2R2B and PTEN as novel genes to be methylated in ductal carcinoma in situ. |
| Literature | GENERIF_SUMMARY | Patients with ulcerative colitis higher PTEN expression and lower AKT expression than those with colorectal cancer, but lower PTEN and higher Akt expressions than in normal tissue. |
| Literature | GENERIF_SUMMARY | Loss of PTEN in endometrial carcinoma cells may inhibit the cell proliferation and increase apoptosis upon exposure to signal transduction inhibitors. |
| Literature | GENERIF_SUMMARY | Lack of PTEN expression may be the earliest event in endometrial carcinogenesis. |
| Literature | GENERIF_SUMMARY | This study provides the first demonstration that TSPYL5 modulates cell growth and sensitization of cells to the detrimental effects of damaging agents via regulation of p21(WAF1/Cip1) and PTEN/AKT pathway. |
| Literature | GENERIF_SUMMARY | The PTEN-deficient endometrial cancer cells exhibit higher p38MAPK activity, and genetic studies demonstrate that p38MAPK functions dependently of AKT. |
| Literature | GENERIF_SUMMARY | loss of nuclear PTEN expression was a marker of poor clinical outcome in patients with stage II colorectal cancer |
| Literature | GENERIF_SUMMARY | Caffeine confers radiosensitization of PTEN-deficient malignant glioma cells by enhancing ionizing radiation-induced G1 arrest and negatively regulating Akt phosphorylation |
| Literature | GENERIF_SUMMARY | ERG/ETV1 gene rearrangements and PTEN gene loss status should now prospectively be incorporated into a predictive model to establish whether predictive performance is improved. |
| Literature | GENERIF_SUMMARY | miR-21 inhibitor might interrupt the activity of EGFR pathways, independently of PTEN status |
| Literature | GENERIF_SUMMARY | the PTEN/AKT pathway has a role as a target of CsA in epidermal keratinocytes, suggesting a previously unknown mechanism in CsA-enhanced skin carcinogenesis |
| Literature | GENERIF_SUMMARY | TP53 gene mutations occur in some of endometrioid endometrial cancers in the presence of PTEN gene mutations. |
| Literature | GENERIF_SUMMARY | Characterizing the expression of PTEN, AKT |

| Main Category | Category | Terms |
|---|---|---|
| | | and pS6 Ser235/236 might be a useful predictive tool for deciding whether to use mTOR inhibitors to treat conjunctiva melanomas. |
| Literature | GENERIF_SUMMARY | promoter methylation is not an important mechanism for PTEN deregulation in non-small-cell lung cancers (NSCLCs) from Koreans. |
| Literature | GENERIF_SUMMARY | ubiquitination may represent a regulated mechanism of direct reversible control over the PTEN enzyme. |
| Literature | GENERIF_SUMMARY | We evaluated the presence of mutations in PIK3CA, AKT1, AKT2, AKT3, PTEN, and PDPK1 genes in 83 papillary thyroid carcinomas |
| Literature | GENERIF_SUMMARY | This is the first demonstration of increased PTEN expression in human microvascular endothelial cells derived from aging tissues and that elevated PTEN is a major factor responsible for aging-related impairment of in vitro angiogenesis. |
| Literature | GENERIF_SUMMARY | The radiosensitivity shown in H460 cells may be due to the higher expression of PTEN through p53 signaling pathway. |
| Literature | GENERIF_SUMMARY | PTEN and PPARgamma are up-regulated in leiomyoma tissues. Estradiol stimulates expression of PTEN and PPARgamma in leiomyoma cells, but not in normal cells. |
| Literature | GENERIF_SUMMARY | Alterations in the EGFR-PTEN signaling pathway are present in a third of prostate adenocarcinomas, particularly affecting the more advanced cases. |
| Literature | GENERIF_SUMMARY | Compared with their parental cells, the expressions of CDA, RRM1, PTEN and ERCC1 increase in human gemcitabine-resistant non-small cell lung cancer cell lines. |
| Literature | GENERIF_SUMMARY | PTEN, MMP-2 and MMP-9 may play an important role in the occurrence and development of thymoma. The expression of PTEN and MMP-2 correlates with the malignance and the aggression of thymoma. |
| Literature | GENERIF_SUMMARY | p85alpha regulatory protein binds directly to and enhances PTEN lipid phosphatase activity |
| Literature | GENERIF_SUMMARY | Data show that the coexpression of HSP90 with PI3K-p110alpha or expression of HSP90 along with PTEN loss has a potential as a molecular prognostic marker to predict early relapse in patients with invasive breast |

| Main Category | Category | Terms |
|---|---|---|
| | | cancers. |
| Literature | GENERIF_SUMMARY | findings show that basal PI3K activity in PTEN-deficient PCa cells is RTK-independent and can be mediated by p110beta and p110delta |
| Literature | GENERIF_SUMMARY | A novel pathway involving PTEN inhibition promotes GluR1 trafficking to hippocampal synapses, a process that has important implications for the role of leptin in hippocampal synaptic function. |
| Literature | GENERIF_SUMMARY | These findings suggested that promoter hypermethylation of PTEN might contribute to the progression of sporadic breast cancer in human. |
| Literature | GENERIF_SUMMARY | The PTEN mutations have significant effect on the expression silencing of PTEN protein indicating the important role of PTEN mutation in carcinogenesis of breast cancers. |
| Literature | GENERIF_SUMMARY | Finding provides important information regarding the efficacy of Nutlin-3 and indicates that patients with tumors deficient in p53 function due to p73 or PTEN loss may benefit from Nutlin-3 treatment. |
| Literature | GENERIF_SUMMARY | The antifibrotic effect of dhS1P in scleroderma fibroblasts is mediated through normalization of PTEN expression. |
| Literature | GENERIF_SUMMARY | We investigated methylation patterns in the promoter regions of ABCB1, ATM, BRCA1, CDH3, CDKN2A, CXCR4, ESR1, FBXW7, FOXC1, GSTP1, IGF2, HMLH1, PPP2R2B, and PTEN75 in well-described pre-treatment samples from locally advanced breast cancer |
| Literature | GENERIF_SUMMARY | Head circumference is a useful clinical feature, but is insufficient as a singular screening tool for detection of PTEN hamartoma tumor syndrome in a clinic population at high-risk of breast cancer. |
| Literature | GENERIF_SUMMARY | This article focuses on the latest data about the role of the gene for phosphatase and tensin homologue located on chromosome 10 (PTEN) in malignant gliomas. |
| Literature | GENERIF_SUMMARY | An evolutionarily conserved PTEN-C/EBPalpha-CTNNA1 axis controls myeloid development and transformation. |
| Literature | GENERIF_SUMMARY | Observed a reduction of Akt signaling proteins in prostate cancer tissue compared to benign controls with significantly lower expression of p27kip1, PTEN and p-Akt. |

| Main Category | Category | Terms |
|---|---|---|
| Literature | GENERIF_SUMMARY | Our results establish that inactivation of PTEN is a rare and late event in the pathogenesis of nephroblastomas. |
| Literature | GENERIF_SUMMARY | uncovered a proto-oncogenic miRNA-dependent network for PTEN regulation and defined the MCM7 locus as a critical factor in initiating prostate tumorigenesis |
| Literature | GENERIF_SUMMARY | In HER2-positive breast cancer, basal co-expression of PTEN and p-Akt might identify those patients who are more likely to benefit from trastuzumab-based therapies. |
| Literature | GENERIF_SUMMARY | Most patients with glioma had three or more mutations in the PTEN gene, with exons 2, 3, 4, 5, 6 and 7 as hot mutation regions. 68.4% of mutations were missense, 26.3% same-sense and 5.3% nonsense mutations. |
| Literature | GENERIF_SUMMARY | Data demonstrate that caveolin-1 expression is a determinant of membrane PTEN levels and show that PTEN interacts with caveolin-1 via its caveolin-1-binding sequence. |
| Literature | GENERIF_SUMMARY | variations in Pten dose determine cancer susceptibility |
| Literature | GENERIF_SUMMARY | PTEN gene was significantly linked with the progression and metastasis of gastrointestinal stromal tumors. |
| Literature | GENERIF_SUMMARY | inactivation of functional PTEN and the consequent activation of the Akt pathway prevents TRAIL-induced apoptosis, leading to TRAIL resistance |
| Literature | GENERIF_SUMMARY | PTEN suppresses angiogenesis and VEGF expression in Hepatocellular carcinoma through both phosphatase-dependent and -independent mechanisms. |
| Literature | GENERIF_SUMMARY | PTEN instability-mediated constitutive Akt activation is involved in acquired resistance mechanisms to cetuximab and also induces de novo resistance to gefitinib in non-small cell lung cancer cells. |
| Literature | GENERIF_SUMMARY | germline mutations, c.825_840del, resp. c.438delT, in 2 patients. Both patients fulfilled strict diagnostic criteria. |
| Literature | GENERIF_SUMMARY | Microarray data revealed that 153 genes were upregulated in DNMT3A knockdown cells; 13 of them including PTEN, a crucial tumor suppressor gene in HCC, are genes involved in cell cycle and cell proliferation. |
| Literature | GENERIF_SUMMARY | IMP3 immunoexpression and the IMP3+/PTEN+ pattern are the best independent and combination markers, respectively, to |

| Main Category | Category | Terms |
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| | | predict uterine serous carcinomas. |
| Literature | GENERIF_SUMMARY | Improved phagocytosis by transgenic alveolar macrophages lacking PTEN ultimately results in unaltered lung colony forming units (CFU)following Streptococcus pneumoniae infection. |
| Literature | GENERIF_SUMMARY | Results characterize a novel immunosuppressant, which shows potential anti-tumor effects on gastric cancer by PTEN activation through p53-independent mechanism, especially in combination with Cisplatin. |
| Literature | GENERIF_SUMMARY | Decreased expression of PTEN in gastric cancer contributes to disease progression. |
| Literature | GENERIF_SUMMARY | the PTEN phosphatase is active against the PI3K p85beta subunit and dephosphorylates a protein involved in insulin signaling where known downstream consequences are increased cell migration, motility, and invasion. |
| Literature | GENERIF_SUMMARY | miR-22 acts to fine-tune the dynamics of PTEN/AKT/FoxO1 pathway. |
| Literature | GENERIF_SUMMARY | PTEN gene can inhibit the effects of Triptorelin or GnRH-II on human endometrial carcinoma cells. |
| Literature | GENERIF_SUMMARY | Combined with data from our previous cohort, we have found PTEN mutations in 7/99 (7.1%) of individuals with ASD and 8/100 (8.0%) of individuals with MR/DD, all of whom had macrocephaly. |
| Literature | GENERIF_SUMMARY | PTEN/Akt and MAP kinase signaling pathways are regulated by the ubiquitin ligase activators Ndfip1 and Ndfip2 |
| Literature | GENERIF_SUMMARY | PTEN expression was not related to VEGF expression and clinicopathological features of squamous cell carcinoma of the larynx. |
| Literature | GENERIF_SUMMARY | The study provides evidence that the interaction between chronic lymphocytic leukemia cells (CLL) and bone marrow stromal cells leads to the activation of PI3-K/Akt cascade, inactivation of PTEN, and inhibition of apoptosis in CLL cells. |
| Literature | GENERIF_SUMMARY | We present a review of Cowden syndrome with particular emphasis on the diagnostic criteria, clinical features, genetics, management and surveillance. |
| Literature | GENERIF_SUMMARY | Results highlight the co-operation and importance of ERG gene rearrangements and PTEN deletions as significant drivers of prostate cancer development and |

| Main Category | Category | Terms |
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| | | progression. |
| Literature | GENERIF_SUMMARY | PTEN-associated Cowden syndrome should be considered a mixed polyp syndrome, with hyperplastic polyps most prevalent, with a risk of early onset colorectal cancer |
| Literature | GENERIF_SUMMARY | Laminarin sulfate can inhibit the proliferation, arrest the cell cycle in the S phase and induce apoptosis of prostate cancer PC-3 cells by increased expressions of PTEN and P27kip1. |
| Literature | GENERIF_SUMMARY | Studies indicate that targeting the deregulated PI3K/PTEN-Akt signaling axis has currently emerged as one of the major tenets in anticancer drug development. |
| Literature | GENERIF_SUMMARY | PAX2 and PTEN protein loss occurs independently and accumulates with increasing age in latent precancers of normal premenopausal endometrium. Loss of both characterizes the emergence of a premalignant lesion which carries forward to carcinoma. |
| Literature | GENERIF_SUMMARY | Oxidized PTEN was reduced by glutathione in a concentration- and time-dependent manner. Glutathionylated PTEN was detected. |
| Literature | GENERIF_SUMMARY | The expression of PTEN at the mRNA and protein level in the gastric cancer xenografts were significantly increased after administration of doxorubicin. |
| Literature | GENERIF_SUMMARY | Blockade of PI3K/Akt signaling by PTEN may render breast cancer cells more dependent on the MEK/Erk pathway for their proliferation and survival. |
| Literature | GENERIF_SUMMARY | Data suggest that the absent or decreased expression of PTEN and the actived mTOR may play important roles in carcinogenesis and metastasis of NSCLC. |
| Literature | GENERIF_SUMMARY | PTEN, through residues 394-403, serves as a scaffold for TRPC6, enabling cell surface expression of the channel. |
| Literature | GENERIF_SUMMARY | Pten inactivation can drive the genesis and promote the metastatic progression of RAS activated Ink4a/Arf deficient melanomas |
| Literature | GENERIF_SUMMARY | Data suggest that activation of the ERBB2-PI3K-AKT pathway by loss of PTEN at early stages of tumorigenesis promotes the formation of breast tumors with apocrine features. |
| Literature | GENERIF_SUMMARY | Data show that binding to PI(4,5)P(2) model membranes is 2.5-fold decreased for the |

| Main Category | Category | Terms |
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| | | mutant PTEN in comparison to wild-type PTEN. |
| Literature | GENERIF_SUMMARY | Sodium butyrate increased PTEN expression, promoted the expression of MUC2 and induced the differentiation of gastric cancer cells through the PTEN/PI3K signalling pathway. |
| Literature | GENERIF_SUMMARY | Findings define a tumor suppressor role for NHERF1 at the plasma membrane, and reveal a novel mechanism for PI3K/Akt activation through PTEN inactivation caused by a loss of membrane-localized NHERF1. |
| Literature | GENERIF_SUMMARY | findings showed, for the first time, overexpression of DJ-1 at both transcriptional and protein levels in gestational trophoblastic disease |
| Literature | GENERIF_SUMMARY | STAT3 is not only a downstream target of IL-6 but, with miR-21, miR-181b-1, PTEN, and CYLD, is part of the positive feedback loop that underlies the epigenetic switch that links inflammation to cancer. |
| Literature | GENERIF_SUMMARY | HBx suppresses adriamycin-induced apoptosis of HepG2 cells and PTEN expression. |
| Literature | GENERIF_SUMMARY | PTEN loss is associated with poor trastuzumab response in HER2-positive metastatic breast cancer. |
| Literature | GENERIF_SUMMARY | activation of Wnt signalling pathway in the absence of PTEN potently promotes urothelial cell carcinoma in vivo. |
| Literature | GENERIF_SUMMARY | PTEN, in combination with NEDD4, provides signals for axonal branching in humans. |
| Literature | GENERIF_SUMMARY | we demonstrate that loss of the PTEN tumor suppressor gene, with associated activation of the PI3K/Akt/mTOR pathway, leads to a human glioma phenotype that induces autologous T-cell apoptosis upon contact |
| Literature | GENERIF_SUMMARY | Polyps from PTEN(+/-) patients lose PTEN expression, but loss is not a universal early feature of all hamartomatous polyposis syndrome. |
| Literature | GENERIF_SUMMARY | COX-2 is constitutively expressed in osteoblasts in the dynamic bone growth area, which facilitates osteoblast proliferation via PTEN/Akt/p27(Kip1) signaling. |
| Literature | GENERIF_SUMMARY | Promoter sequence of the PTEN gene differ among normal Japanese individuals. Promoter polymorphisms in this gene can |

| Main Category | Category | Terms |
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| | | influence gene transcription. |
| Literature | GENERIF_SUMMARY | PTEN selectively inhibits expression of VCAM-1 but not ICAM-1 through modulation of PI3K/Akt/GSK-3beta/GATA-6 signaling cascade in TNF-alpha-treated endothelial cells |
| Literature | GENERIF_SUMMARY | PTEN expression at both mRNA and protein levels were not significantly affected by erbB2 silencing (P>0.05). |
| Literature | GENERIF_SUMMARY | Mutations in PTEN is associated with squamous cell carcinomas of the lung. |
| Literature | GENERIF_SUMMARY | Data indicate that NEDD4-1 may play an important role in thedevelopment of NSCLC, and that NEDD4-1 overexpression may modulate the malignant phenotype of NSCLC cells, likely by regulating the intracellular levels of PTEN. |
| Literature | GENERIF_SUMMARY | the functional consequences of nuclearcytoplasmic mislocalization of these PTEN ATP-binding mutants in breast carcinogenesis were analyzed. |
| Literature | GENERIF_SUMMARY | Relevance of phosphatase and tensin homolog on chromosome 10 in the etiopathophysiology of diabetic nephropathy |
| Literature | GENERIF_SUMMARY | Plk3 as a new player in the regulation of the PI3K/PDK1/Akt signaling axis by phosphorylation and stabilization of PTEN. |
| Literature | GENERIF_SUMMARY | Hepatitis B virus X-protein induced reactive oxygen species activate hepatocellular carcinogenesis via dysregulation of PTEN/Akt pathway. |
| Literature | GENERIF_SUMMARY | PTEN and SHIP act cooperatively to suppress B cell lymphoma and provides the first direct evidence that SHIP is a tumor suppressor. |
| Literature | GENERIF_SUMMARY | loss of PTEN expression contributes to gefitinib and erlotinib resistance in non-small cell lung cancer. |
| Literature | GENERIF_SUMMARY | Alkylation of PTEN by alpha,beta-enals/enones and interference with its restraint of cellular PKB/Akt signaling may accentuate hyperplastic and neoplastic disorders associated with chronic inflammation, oxidative stress, or aging. |
| Literature | GENERIF_SUMMARY | The overexpression of bFGF and underexpression of PTEN are closely related to the invasion and growth of cervical carcinoma. |
| Literature | GENERIF_SUMMARY | Results reveal that up-regulation of the key |

| Main Category | Category | Terms |
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| | | upstream signaling factor, PTEN, in MCF-7/DOX cells inhibited Akt phosphorylation, which ultimately causes increase in p53 levels affecting the induction of G2/M cell cycle arrest and apoptosis. |
| Literature | GENERIF_SUMMARY | Loss of PTEN expression does not lead to a loss of cetuximab efficacy in inhibiting EGFR-downstream signaling pathways. |
| Literature | GENERIF_SUMMARY | a known tumor suppressor in ovarian cancer. Confirmed that miR-21 was significantly overexpressed in human EOC tissues and cell lines. Demonstrated knockdown of miR-21 significantly increased PTEN expression, a known tumor suppressor in ovarian cancer. |
| Literature | GENERIF_SUMMARY | This study demonstrates that PTEN overexpression inhibits the proliferation and migration of human ASMCs by down-regulating the activity of the Akt and FAK signaling pathways |
| Literature | GENERIF_SUMMARY | Various studies have revealed the essential roles of these PTEN-targeting noncoding RNAs during tumor development, thus providing a paradigm to explore the molecular mechanisms underlying the dosage-dependent effects of key oncogenes and tumor suppressors |
| Literature | GENERIF_SUMMARY | PTEN acts as a pivotal determinant of cell fate, regarding senescence and apoptosis in ionizing radiation-exposed glioma cells. |
| Literature | GENERIF_SUMMARY | Data suggest that loss of PTEN permits CXCR4-mediated functions in prostate cancer cells through the ERK1/2 pathway. |
| Literature | GENERIF_SUMMARY | Deletion of the gene encoding PTEN enables retinal ganglion cells to regenerate axons the full length of the optic nerve in mature transgenic mice. |
| Literature | GENERIF_SUMMARY | Common genetic variation in PTEN, PIK3CA, AKT1, MLH1, or MSH2 was not statistically significantly associated with endometrial cancer. |
| Literature | GENERIF_SUMMARY | Data show that among 181 CRC patients, stratified by microsatellite instability status, DNA sequence changes were identified in KRAS (32%), BRAF (16%), PIK3CA (4%), PTEN (14%) and TP53 (51%). |
| Literature | GENERIF_SUMMARY | Data show that MiR-21 was overexpressed in vitro and esophageal squamous cell carcinoma, and promoted the cell proliferation, might target PTEN at post-transcriptional level, and regulated the cancer invasion in Kazakh's ESCC. |

| Main Category | Category | Terms |
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| Literature | GENERIF_SUMMARY | PTEN expression is related to cell proliferation and self-renewal independent of CD133 phenotype in glioma-initiating cells |
| Literature | GENERIF_SUMMARY | Over-expression of DJ-1 suppresses PTEN expression which promotes SWO-38 cell migration and invasion. |
| Literature | GENERIF_SUMMARY | The findings of the present study demonstrate that there are mutations in the PTEN gene of the esophageal squamous cell carcinoma (ESCC) cells and that the wild type PTEN gene has important effects on the ESCC cells in vitro and in vivo. |
| Literature | GENERIF_SUMMARY | Data show that miR-183 has a potential oncogenic role through the regulation of 2 tumor suppressor genes, EGR1 and PTEN, and the deregulation of this fundamental miRNA regulatory network may be central to many tumor types. |
| Literature | GENERIF_SUMMARY | Lower expression of PTEN protein and higher expression of P-AKT protein may play an important role in leukemia pathogenesis. |
| Literature | GENERIF_SUMMARY | Our results prove that PTEN haplotypes may be associated with susceptibility to hepatocellular carcinoma in a Chinese Han population. |
| Literature | GENERIF_SUMMARY | Relevance of the combinatorial inactivation of PTEN in non-small cell lung carcinoma progession. |
| Literature | GENERIF_SUMMARY | PTEN expression is regulated by (nuclear factor of activated T cell) NFAT signaling. |
| Literature | GENERIF_SUMMARY | The combined deregulation of the pVHL/HIF axis and PTEN alone appear to be important for multilocular cystic renal cell carcinoma development but not sufficient to drive metastatic processes. |
| Literature | GENERIF_SUMMARY | Data show that KRAS mutation and PTEN protein expression are significantly correlated with the response rate and survival time of Chinese metastatic colorectal cancer patients treated with cetuximab. |
| Literature | GENERIF_SUMMARY | The PTEN mRNA and protein expression in rectal cancer tissues on neoadjuvant chemoradiotherapy were significantly higher than those before chemoradiotherapy. |
| Literature | GENERIF_SUMMARY | methylation of the PTEN promoter related to both SAM increase and DNMT1 activation contributes to persistent Akt activation and are potential therapeutic targets for reversing TAM resistance in breast cancer. |

| Main Category | Category | Terms |
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| Literature | GENERIF_SUMMARY | Germline KILLIN methylation is common among patients with Cowden syndrome or Cowden-like syndrome and is associated with increased risks of breast and renal cancer over PTEN mutation-positive individuals |
| Literature | GENERIF_SUMMARY | The differences of ERG and /or PTEN genetic aberrations in transition zone tumors may point toward a subset of tumors with adverse behavior. |
| Literature | GENERIF_SUMMARY | high expression of PTTG1 and low expression of PTEN may be involved in pathogenesis and development of endometrial carcinoma |
| Literature | GENERIF_SUMMARY | the mechanism of miR-21 antineoplastic drug resistance involves the PI3K/Akt pathway and changes following PTEN protein expression. |
| Literature | GENERIF_SUMMARY | Cell lines with a high expression of PTEN are sensitive to chemotherapy with 5-FU and oxaliplatin |
| Literature | GENERIF_SUMMARY | PTEN controls cellular polarity, establishment of cell-cell junctions, paracellular permeability, migration and tumorigenic/metastatic potential of human colorectal cancer cells |
| Literature | GENERIF_SUMMARY | Migration of renal tumor cells depends on dephosphorylation of Shc by PTEN. |
| Literature | GENERIF_SUMMARY | EGFR, PI3Kp110 and PTEN expression is a common feature of neuroblastoma |
| Literature | GENERIF_SUMMARY | Compared with normal monocytes, advanced glycosylation end product (AGE)-treated monocytes and monocytes from chronic renal failure patients exhibit lower PTEN levels and delayed apoptosis. |
| Literature | GENERIF_SUMMARY | Study shows that nuclear PTEN interacts with APC/C, promotes APC/C association with CDH1, and thereby enhances the tumor-suppressive activity of the APC-CDH1 complex. |
| Literature | GENERIF_SUMMARY | study identified five PTEN mutations, two indels and three missense in pancreatic neuroendocrine tumors |
| Literature | GENERIF_SUMMARY | Dependence of Evi1-expressing leukemic cells on AKT/mTOR signaling provides the first example of targeted therapeutic modalities that suppress the leukemogenic activity of Evi1. |
| Literature | GENERIF_SUMMARY | The PTEN and MLH1 immunohistochemical analysis broadens the microscopic |

| Main Category | Category | Terms |
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| | | diagnosis of the endometrial carcinomas. |
| Literature | GENERIF_SUMMARY | loss of heterozygosity at the PTEN locus with retention of the mutated alleles, and greatly reduced or absent PTEN expression. |
| Literature | GENERIF_SUMMARY | found that E-cadherin negatively regulates tumor cell growth, in part, by positively regulating PTEN expression via beta-catenin-mediated Egr1 regulation, thus influencing PI3K/Akt signaling |
| Literature | GENERIF_SUMMARY | Role of Pten in the breast tumor microenvironment. |
| Literature | GENERIF_SUMMARY | Presence of the PTEN gene product in the tumor cells is associated with good prognosis in patients with triple-negative breast cancer. |
| Literature | GENERIF_SUMMARY | A role for PTEN in the regulation of PLX4720-mediated BIM expression was confirmed by siRNA knockdown of PTEN and through reintroduction of PTEN into cells that were PTEN-. |
| Literature | GENERIF_SUMMARY | augments invasiveness and growth of lung cancer cells through the PI3K/AKT/NFkB pathway |
| Literature | GENERIF_SUMMARY | role of PTEN germline mutations in macrocephaly |
| Literature | GENERIF_SUMMARY | mPGES-1 promotes experimental cholangiocarcinogenesis and tumor progression by inhibiting PTEN. |
| Literature | GENERIF_SUMMARY | demonstrate here that breast cancer cells harbouring PIK3CA mutations are selectively sensitive to mTOR allosteric and kinase inhibitors. However, cells with PTEN loss of function are not sensitive to these drugs |
| Literature | GENERIF_SUMMARY | These results suggest that the reduction of oxidized hPTEN is regulated by Grx5 and Ycp4. |
| Literature | GENERIF_SUMMARY | Expression rates of FHIT and PTEN in prostate cancer were significantly lower than in benign prostatic nodular hyperplasia. |
| Literature | GENERIF_SUMMARY | 148 Penile squamous cell carcinoma were microarrayed and immunostained for phosphorylated EGFR (pEGFR), HER2, HER3, HER4, phosphorylated Akt (pAkt), Akt1 and PTEN proteins. |
| Literature | GENERIF_SUMMARY | Single nucleotide polymorphisms in PTEN gene is not associated with colorectal cancer. |

| Main Category | Category | Terms |
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| Literature | GENERIF_SUMMARY | germline PTEN mutations have a role in differentiated thyroid cancer |
| Literature | GENERIF_SUMMARY | Upregulation of PTEN gene can effectively inhibit airway smooth muscle cell migration. |
| Literature | GENERIF_SUMMARY | The COOH-terminal truncated form of p17 (p17Delta36) induced activation of the PI3K/Akt pathway by maintaining PTEN in an inactive phosphorylated form. |
| Literature | GENERIF_SUMMARY | Protein interactions of phosphatase and tensin homologue (PTEN) and its cancer-associated G20E mutant compared by using stable isotope labeling by amino acids in cell culture-based parallel affinity purification. |
| Literature | GENERIF_SUMMARY | BMP-4 induces PTEN in E6 and BPH-1 prostate cells and reduces cell proliferation. Knockdown of PTEN attenuated the growth-inhibiting effects of BMP-4. BMP-4 also increased PTEN protein stability. |
| Literature | GENERIF_SUMMARY | The down-regulation of PTEN and IRS1 is a critical event leading to the HCV genotype 3a-induced formation of large lipid droplets in hepatocytes. |
| Literature | GENERIF_SUMMARY | PTEN, rather than mTOR, was found to be involved in side population cell regulation in primary esophageal squamous cell carcinoma cells |
| Literature | GENERIF_SUMMARY | These results suggest that ERalpha induces the down-regulation of PTEN through PI3K activation in breast cancer cells. |
| Literature | GENERIF_SUMMARY | frequent occurrence of loss of expression in colorectal cancers |
| Literature | GENERIF_SUMMARY | role of post-translational modification of PTEN in host tumor suppressor functions; possibility of antineoplastic therapy using drugs that regulate post-translational modification of PTEN [REVIEW] |
| Literature | GENERIF_SUMMARY | a new mechanism is proposed where glucocorticoids treat asthma partly through up-regulation of PTEN expression. The in vitro studies also suggest that the PTEN pathway may be involved in human asthma |
| Literature | GENERIF_SUMMARY | PTEN was lost in 14 (30%) primary tumors and 13 (25%) metastases |
| Literature | GENERIF_SUMMARY | Up-regulation of PTEN may be involved in oleanolic acid-induced Jurkat cell apoptosis. |
| Literature | GENERIF_SUMMARY | Results suggest that prosurvival PTEN-Akt signalling is impaired in the diseased diabetic myocardium. |

| Main Category | Category | Terms |
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| Literature | GENERIF_SUMMARY | Posttranscriptional deregulation of MYC via PTEN constitutes a major alternative pathway of MYC activation in T-cell acute lymphoblastic leukemia. |
| Literature | GENERIF_SUMMARY | Our results provide new insight into the role of PTEN in cAMP-induced apoptosis in glial cells. |
| Literature | GENERIF_SUMMARY | PIK3CA mutations were significantly associated with PTEN mutations in endometrial carcinomas. |
| Literature | GENERIF_SUMMARY | results reveal a functional E3 ubiquitin ligase for PTEN that plays a vital role in tumour-cell survival |
| Literature | GENERIF_SUMMARY | It was shown for the first time that androgen receptor inhibits PTEN transcription in prostate cancer cells, whereas androgen receptor upregulates PTEN transcription in breast cancer cells. |
| Literature | GENERIF_SUMMARY | these results provide further support for the existence of a DNA damage-inducible size checkpoint that is regulated by a major tumor suppressor, and they provide a novel Akt-independent mechanism by which PTEN controls cell size. |
| Literature | GENERIF_SUMMARY | MAN2C1 binds PTEN thereby inhibiting its phosphatidylinositol 3,4,5 triphosphate phosphatase activity |
| Literature | GENERIF_SUMMARY | PTEN protein has different conformations within the nucleus and cytoplasm of neurons |
| Literature | GENERIF_SUMMARY | PTEN may suppress the invasive and migration ability of the ovarian cancer cell line A2780, possibly by downregulating MMP9. |
| Literature | GENERIF_SUMMARY | Report reciprocal feedback regulation of PI3K and androgen receptor signaling in PTEN-deficient prostate cancer. |
| Literature | GENERIF_SUMMARY | Deficiency of CDKN2A and PTEN expression, although shared across many other different types of tumors, likely represents a key aspect of chordoma pathogenesis |
| Literature | GENERIF_SUMMARY | a previously unrecognized function of miR-21 that is the reciprocal regulation of PTEN levels and Akt/TORC1 activity that mediate critical pathologic features of diabetic kidney disease |
| Literature | GENERIF_SUMMARY | show that castration-resistant growth is an intrinsic property of Pten null prostate cancer (CaP) cells, independent of cancer |

| Main Category | Category | Terms |
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| | | development stage |
| Literature | GENERIF_SUMMARY | Complex endometrial hyperplasia was associated with a reduction in number of strongly PTEN positive glands, with an increase in null glands, seen in clusters. |
| Literature | GENERIF_SUMMARY | these studies add to the body of knowledge on the regulatory role of PTEN in CXCR4-mediated cancer progression. |
| Literature | GENERIF_SUMMARY | that germ-line variants in PTEN do not have an important role in prostate cancer susceptibility. |
| Literature | GENERIF_SUMMARY | deltaNp63alpha negatively regulates PTEN, thereby providing a feedback loop between PTEN, Akt and deltaNp63alpha, which has an integral role in skin cancer development |
| Literature | GENERIF_SUMMARY | beta-arrestins can thus differentially control distinct functional outputs of PTEN important for cell proliferation and migration. |
| Literature | GENERIF_SUMMARY | Results identify unique sites of PTEN and Akt regulation by means of S-nitrosylation, resulting in an ""on-off"" pattern of control of Akt/PI3K signaling." |
| Literature | GENERIF_SUMMARY | Overexpression of PTEN reduces the overexpression of type I collagen and CCN2 by dSSc fibroblasts |
| Literature | GENERIF_SUMMARY | A mutation prediction model has been developed which can help identify patients appropriate for PTEN testing in clinical practice. |
| Literature | GENERIF_SUMMARY | Results suggest that the abnormal expression and localization of PTEN during cervical carcinogenesis may be a consequence of modifications in the expression patterns of hDlg. |
| Literature | GENERIF_SUMMARY | MAGI1 may inhibit the cancer cell migration and invasion of hepatocellular carcinoma via regulating PTEN. |
| Literature | GENERIF_SUMMARY | PTEN, one of the target genes of miRNA-214 was detected by using of immunocytochemistry and Western blotting. |
| Literature | GENERIF_SUMMARY | Metastases in most subcellular compartments showed comparable and stronger expression for pAKT, pmTOR, and PTEN than primary renal cell carcinomas (PRCCs). |
| Literature | GENERIF_SUMMARY | Promoter methylation and loss of PTEN expression occur frequently in carcinoma of uterine cervix. Our results suggest that |

| Main Category | Category | Terms |
|---|---|---|
| | | PTEN plays an important role in the carcinogenesis of cervical cancer. |
| Literature | GENERIF_SUMMARY | overexpression of EZH2 and loss of PTEN expression might be closely related to the carcinogenesis, progression, clinical biological behavior, and prognosis of gallbladder adenocarcinoma. |
| Literature | GENERIF_SUMMARY | identified concurrent mutational inactivation of the PTEN and RB1 tumor suppressors as a mechanism for loss of BRAF/MEK dependence in melanomas harboring (V600E)BRAF mutations |
| Literature | GENERIF_SUMMARY | UV damage downregulates skin PTEN during carcinogenesis. This impaired global genomic nucleotide excision repair and increased the activation of the Chk1 DNA damage pathway in an AKT-independent manner. |
| Literature | GENERIF_SUMMARY | show for the first time a significant disruption of all three members of the PTEN-NHERF1-PHLPP1 tumor suppressor network in high-grade tumors, correlating with Akt activation and patient's abysmal survival |
| Literature | GENERIF_SUMMARY | Host genetic aberrations in PTEN contribute to the required activity of the PI3K-Akt-mTOR pathway in primary effusion lymphoma and Kaposi's sarcoma. |
| Literature | GENERIF_SUMMARY | Data suggest that dysregulation of miR-21 plays a critical role in resistance of breast cancer to doxorubicin, at least in part via targeting PTEN. |
| Literature | GENERIF_SUMMARY | Overexpression of DJ-1 decreased the expression of PTEN, promoted the activation of Akt. |
| Literature | GENERIF_SUMMARY | absence of mutations in AKT1, PTEN, and RAS genes in the present study is in accordance with previous studies confirming that these genes are rarely mutated in oral squamouos cell carcinoma |
| Literature | GENERIF_SUMMARY | These findings reveal distinctive functional patterns among PTEN mutations found in tumors and in the germline of PTEN hamartoma tumor-related syndromes and autism spectrum disorders patients. |
| Literature | GENERIF_SUMMARY | data establish a conceptual framework for progression of PTEN mutant prostate cancer to life-threatening disease |
| Literature | GENERIF_SUMMARY | Reduction in PTEN protein was found in 90% of breast tumor samples when compared to the adjacent normal breast tissue. All of the |

| Main Category | Category | Terms |
|---|---|---|
| | | tumor samples had decreased levels of PTEN transcripts. 20% of tumor tissues were hypermethylated. |
| Literature | GENERIF_SUMMARY | the results provide strong evidence that CTNNB1 is the driver in these contexts and that KRAS(G12D) and Pten loss promote the program set in motion by the CTNNB1. |
| Literature | GENERIF_SUMMARY | Induction of apoptotic genes by a p73-phosphatase and tensin homolog (p73-PTEN) protein complex in response to genotoxic stress. |
| Literature | GENERIF_SUMMARY | Aberrant expressions of p-AKT1 and p-MTOR in intrahepatic cholangiocarcinoma were associated with a favorable prognosis, possibly in a PTEN-independent manner. |
| Literature | GENERIF_SUMMARY | PTEN loss seems to be an early event during tumorigenesis in primary and metastatic clear cell renal cell carcinomas. |
| Literature | GENERIF_SUMMARY | a role of Plk1 in facilitating loss of Pten-induced prostate cancer formation |
| Literature | GENERIF_SUMMARY | Allelic loss of 10q23.3, the PTEN gene locus is associated with cervical carcinoma. |
| Literature | GENERIF_SUMMARY | Loss of PTEN expression in adenomatous thyroid nodules appears to be both sensitive and specific for Cowden syndrome. |
| Literature | GENERIF_SUMMARY | data suggest that stratification of CaP patients for HER2/3 and PTEN status could identify patients with aggressive CaP who may respond favorably to MEK inhibition |
| Literature | GENERIF_SUMMARY | Data show that Rac1 was highly activated in PTEN deficient and IGF-IR overexpressing Trastuzumab-resistant cells in a HER2-independent manner. |
| Literature | GENERIF_SUMMARY | PTEN regulates levels of pluripotency network genes in human embryonic stem cells. |
| Literature | GENERIF_SUMMARY | PTEN regulates colorectal epithelial apoptosis through Cdc42 signalling |
| Literature | GENERIF_SUMMARY | Cowden syndrome patients have elevated risks of follicular thyroid cancer due to PTEN pathogenic mutations |
| Literature | GENERIF_SUMMARY | This study described that PTEN mutaion with autism,macrocephaly. |
| Literature | GENERIF_SUMMARY | Multiple PIK3R1 and PIK3R2 mutations demonstrate gain of function, including disruption of a novel mechanism of pathway regulation wherein p85alpha dimers bind and stabilize PTEN. |

| Main Category | Category | Terms |
|---|---|---|
| Literature | GENERIF_SUMMARY | these data suggest that PTEN inhibits proliferation and function of hypertrophic scar fibroblasts through the AKT pathway |
| Literature | GENERIF_SUMMARY | Upon competition with neural precursor cell expressed, developmentally down-regulated 4, NHERF1 then interacts with PTEN to inhibit PTEN degradation, through an ubiquitination-dependent pathway. |
| Literature | GENERIF_SUMMARY | Study identified and validated endogenous protein-coding transcripts that regulate PTEN, antagonize PI3K/AKT signaling, and possess growth- and tumor-suppressive properties. |
| Literature | GENERIF_SUMMARY | Tumors with homozygotic deletion of PTEN expressed little or no mRNA or protein. Positive p53, high Ki67, and basal-like tumors expressed significant lower MGMT and PTEN. |
| Literature | GENERIF_SUMMARY | Transient silencing of PTEN in CD34(+) cells promoted their entry into cell cycle, and increased their expansion in vitro compared with control siRNA-treated CD34(+) cells. |
| Literature | GENERIF_SUMMARY | activating mutation of PIK3CA, loss of PTEN, or IGF1R expression have a role in round cell transformation |
| Literature | GENERIF_SUMMARY | Discoverery of a novel reverse regulation within the traditional PTEN/AKT signaling pathway, whereby AKT induces GATA2 with consequent decreased PTEN transcription in breast cancer. |
| Literature | GENERIF_SUMMARY | Study confirms the contribution of loss of PTEN expression in Akt phosphorylation and spontaneous apoptosis suppression in the specimens of oral cancer. |
| Literature | GENERIF_SUMMARY | results suggest that a high level of PTEN expression is associated with low-grade liver metastasis and satisfactory patient survival in pancreatic cancer |
| Literature | GENERIF_SUMMARY | Pten-deficient mice developed full-penetrance HNSCC in combination with type I TGF-beta receptor (Tgfbr1) deletion |
| Literature | GENERIF_SUMMARY | unstable segments of repeated DNA (SD17 and SD18) can lead to the recurrent deletions of the entire PTEN gene in prostate cancer |
| Literature | GENERIF_SUMMARY | Two out of 86 cases revealed a 130th codon G>A and a 334th codon A>T missense mutation in exon 8 of PTEN. |
| Literature | GENERIF_SUMMARY | ligand-activated PPARdelta confers resistance to Ang II-induced senescence by |

| Main Category | Category | Terms |
|---|---|---|
| | | up-regulation of PTEN and ensuing modulation of the PI3K/Akt signaling to reduce ROS generation in vascular cells. |
| Literature | GENERIF_SUMMARY | Persistent activation of signaling by the AKT-survivin pathway induced by PTEN loss underlies a mechanism of resistance to erlotinib-induced apoptosis in EGFR mutation-positive non-small cell lung cancer. |
| Literature | GENERIF_SUMMARY | The rate of PTEN alterations in melanoma cell lines, primary melanoma, and metastatic melanoma is 27.6, 7.3, and 15.2%, respectively. Review. |
| Literature | GENERIF_SUMMARY | we assessed PTEN genomic level and protein expression in this large series of sarcomas with complex genomics |
| Literature | GENERIF_SUMMARY | Concomitant CD26-positive and PTEN-negative status in cases of follicular adenoma suggests a state close to follicular carcinoma or further disease progression thus warranting careful follow-up. |
| Literature | GENERIF_SUMMARY | The mutation in cysteine 124 present in the catalytic domain of PTEN abolished cis-GS-induced HO-1 expression as well as Akt phosphorylation. |
| Literature | GENERIF_SUMMARY | combination of low CEN7 copy number, PTEN loss, and PI3KCA gain may be useful for identifying NSCLC patients unlikely to benefit from treatment with EGFR (TKIs), specifically in wild-type EGFR cases |
| Literature | GENERIF_SUMMARY | Decreased PTEN expression correlates closely with the cytological atypia in endometrial cells. |
| Literature | GENERIF_SUMMARY | novel mutation of PTEN in breast neoplasms |
| Literature | GENERIF_SUMMARY | These results strongly implicate that PTEN might be helpful as a new prognostic factor. |
| Literature | GENERIF_SUMMARY | Loss of PTEN is associated with non-seminomatous testicular germ cell tumors. |
| Literature | GENERIF_SUMMARY | The absence of and/or a low expression of PTEN and activated mTOR may play an important role in the development of non-small cell lung cancer. |
| Literature | GENERIF_SUMMARY | Abnormal expression of PTEN and FAK in patients with multiple myeloma may be associated with disease progression and extramedullary infiltration. |
| Literature | GENERIF_SUMMARY | PTEN-mediated miR-21 regulation is achieved by inhibiting the interaction |

| Main Category | Category | Terms |
|---|---|---|
| | | between the Drosha complex and RNH1, revealing previously unidentified role of PTEN in the oncogenic miR-21 biogenesis |
| Literature | GENERIF_SUMMARY | these results indicate that miR-22 is an important player in the cellular stress response upon UV radiation, which may promote cell survival via the repression of PTEN expression. |
| Literature | GENERIF_SUMMARY | Inherited variation in the PTEN promoter region affects the tumor progression and gene expression profile in breast cancer. |
| Literature | GENERIF_SUMMARY | No significant correlations were observed between the clinicopathological variables and PIK3CA, PTEN, and pHER2 status in breast cnacer. |
| Literature | GENERIF_SUMMARY | unrecognized cell-autonomous function of TORC1 and TORC2 in the up-regulation of PTEN, which prevents phosphorylation of Akt and may shield against the development of malignancy in TSC patients |
| Literature | GENERIF_SUMMARY | Data suggest that wild-type PTEN gene transfection might increase drug sensitivity or reverse drug resistance via inhibiting the PI3K/Akt pathway and regulating downstream molecules of the cell signaling transduction pathway in K562/ADM cells. |
| Literature | GENERIF_SUMMARY | PTEN plays a key role in the progression of pulmonary hypertension |
| Literature | GENERIF_SUMMARY | Oleic acid activates GPR40-phospholipase C-calcium pathway to increase the expression of PPAR-delta and PPAR-delta further decreased the expression of PTEN to regulate insulin sensitivity in hepatic steatosis. |
| Literature | GENERIF_SUMMARY | these results suggest a mechanism by which loss of PTEN may promote prostate cancer progression via PDGF D/beta-PDGFR signal transduction. |
| Literature | GENERIF_SUMMARY | Ars2 is overexpressed in human cholangiocarcinoma and may be a diagnostic marker. Ars2 depletion increases PTEN and PDCD4 protein levels via the reduction of miR-21. |
| Literature | GENERIF_SUMMARY | Loss of PTEN expression was present in 59 out of 273 carcinomas and was significantly correlated to genomic PTEN deletion, high-grade malignancy, increased tumour size (P=0.036), lymph node metastases (P=0.007) and worse disease-specific survival |
| Literature | GENERIF_SUMMARY | investigated the direct role of YY1 in the |

| Main Category | Category | Terms |
|---|---|---|
| | | regulation of EMT. This review discusses the molecular regulation of EMT in cancer cells through the activity of the dysregulated NF-kappaB/Snail/ YY1/PTEN/RKIP circuitry |
| Literature | GENERIF_SUMMARY | PTEN promoter mutations were associated with breast cancer, wheras nonsense mutations were associated with colorectal neoplasms. |
| Literature | GENERIF_SUMMARY | In hepatocellular carcinoma miR-519d is up-regulated by p53 and DNA hypomethylation and targets CDKN1A/p21, PTEN, AKT3 and TIMP2. |
| Literature | GENERIF_SUMMARY | Reduced activity of PTEN therefore affects homeostasis of human germinal center B cells by increasing PI3K-AKT signaling via mammalian target of rapamycin as well as antiapoptotic signals |
| Literature | GENERIF_SUMMARY | microRNA-21 promotes tumor proliferation and invasion in gastric cancer by targeting PTEN. |
| Literature | GENERIF_SUMMARY | IGF2BP1 promotes the velocity and directionality of tumor-derived cell migration by determining the cytoplasmic fate of two novel target mRNAs |
| Literature | GENERIF_SUMMARY | We report the first description of a pigmented choroidal schwannoma in PTEN hamartoma tumor syndrome (PHTS). This rare tumor showed a unique combination of reduction of PTEN and absence of NF2 expression. |
| Literature | GENERIF_SUMMARY | We demonstrate that c-MYC gain alone, or combined c-MYC gain and PTEN loss, were increasingly prognostic for relapse |
| Literature | GENERIF_SUMMARY | loss of PTEN expression (by immunohistochemistry) and PIK3CA mutation is likely to be predictive of a lack of benefit to anti-EGFR therapy in metastatic colorectal cancer |
| Literature | GENERIF_SUMMARY | miR-29b regulates PTEN gene expression through downregulating Dnmts expression and subsequently induces hypomethylation in PTEN promoter. |
| Literature | GENERIF_SUMMARY | PTEN up-regulates superoxide dismutase, glutathione peroxidase and catalase activity by inhibition of PI3K/AKT pathway in non-small cell lung cancer cell lines. |
| Literature | GENERIF_SUMMARY | data suggest that lamin A/C proteins are positively involved in malignant behavior of PC cells through the PI3K/AKT/PTEN pathway. Lamin A/C may represent a new oncogenic factor and a novel therapeutic |

| Main Category | Category | Terms |
|---|---|---|
| | | target for PC |
| Literature | GENERIF_SUMMARY | PTEN loss defines a TGF-beta-induced tubule phenotype of failed differentiation and JNK signaling during renal fibrosis. |
| Literature | GENERIF_SUMMARY | Immunohistochemistry appears to be a preferable technique for identifying endometrial tumors with loss of PTEN function. |
| Literature | GENERIF_SUMMARY | miR-21 is up-regulated and PTEN is down-regulated, and their expressions are reversely correlated in human laryngeal squamous cell carcinomas and hypopharyngeal squamous cell carcinomas. |
| Literature | GENERIF_SUMMARY | A wide range of germline PTEN mutations may play a role in the pathogenesis of breast cancer in Pakistan. |
| Literature | GENERIF_SUMMARY | Cortical dysplasia associated with the PTEN mutation in Bannayan Riley Ruvalcaba syndrome |
| Literature | GENERIF_SUMMARY | The over-expression of PTEN gene may inhibit the proliferation of K562 cells and promote cell apoptosis via the regulation of Survivin, Xiap and Smac genes. |
| Literature | GENERIF_SUMMARY | Endometrial carcinoma patients with loss of PTEN and p-AKT positivity have a favorable prognosis. |
| Literature | GENERIF_SUMMARY | PTEN rs2735343 might be a susceptibility factor for esophageal squamous cell carcinoma. |
| Literature | GENERIF_SUMMARY | PTEN overexpression attenuates angiogenic processes of endothelial cells by blockade of endothelin-1/endothelin B receptor signaling, contributing to thrombosis of arteriovenous grafts in hemodialysis patients. |
| Literature | GENERIF_SUMMARY | transcription factor activities may be used as ""signatures"" for predicting PTEN functional status and elucidate the transcriptional architectures underlying human cancers caused by PTEN loss" |
| Literature | GENERIF_SUMMARY | Pten loss and RAS/MAPK activation cooperate to promote EMT and metastasis initiated from prostate cancer stem/ progenitor cells. |
| Literature | GENERIF_SUMMARY | Data conclude that miR-205 contributes to radioresistance of nasopharyngeal carcinoma by directly targeting PTEN. |
| Literature | GENERIF_SUMMARY | PTEN protein phosphatase activity correlates with control of gene expression |

| Main Category | Category | Terms |
|---|---|---|
| | | and invasion, a tumor-suppressing phenotype, but not with AKT activity. |
| Literature | GENERIF_SUMMARY | Combined application of spiral CT and PTEN/COX-2 expression improves the accuracy of diagnosis and staging in pancreatic cancer. |
| Literature | GENERIF_SUMMARY | PDPN expression in human glioblastoma multiforme is caused by loss of PTEN function and activation of the PI3K-AKT-AP-1 signaling pathway, accompanied by epigenetic regulation of PDPN promoter activity |
| Literature | GENERIF_SUMMARY | The loss of PTEN function through mutation and other mechanisms is evident in as many as a third of all human tumors. (Review) |
| Literature | GENERIF_SUMMARY | PTEN may serve as a potent prognostic and predictive marker in postoperative radiotherapy for high-risk squamous cell cancer of the head and neck. |
| Literature | GENERIF_SUMMARY | PTEN mutations play a role in cancer. |
| Literature | GENERIF_SUMMARY | Data show that CHIP interacts with PTEN and promotes its proteasomal degradation. |
| Literature | GENERIF_SUMMARY | PTEN mRNA expression was significantly downregulated in brain metastases compared with primary breast tumors, and PTEN mutations were frequently found in brain metastases. |
| Literature | GENERIF_SUMMARY | Data suggest that hDlg may serve as a platform to bring in proximity APC and PTEN tumor suppressor activities. |
| Literature | GENERIF_SUMMARY | Gross rearrangements of the PTEN locus occur in prostate cancer and can be detected by a 'break-apart' FISH assay. |
| Literature | GENERIF_SUMMARY | miR-93 can directly target PTEN, and participates in the regulation of the AKT signaling pathway |
| Literature | GENERIF_SUMMARY | The repertoire of PTEN functions has recently been expanded to include phosphatase-independent activities and crucial functions within the nucleus. [Review] |
| Literature | GENERIF_SUMMARY | Ursolic acid may elicit its strong antitumor effects via upregulation of the PTEN gene and inhibition of the PI3K/Akt pathway. |
| Literature | GENERIF_SUMMARY | Loss of expression of PTEN does not confer poor overall survival in the temozolomide era. |
| Literature | GENERIF_SUMMARY | PTEN IHC in TLs may be helpful in screening |

| Main Category | Category | Terms |
|---|---|---|
| | | TL for association with CS, but should be used in context with other established clinical criteria, and possibly germline PTEN genotyping to confirm a diagnosis of CS. |
| Literature | GENERIF_SUMMARY | PTEN mutations are associated with pediatric T-cell acute lymphoblastic leukemia. |
| Literature | GENERIF_SUMMARY | PI3K/PTEN pathway plays an important role in the translocation of TRPC6 to the plasma membrane and may thus have a significant impact on Ca(2+) signaling in cells that endogenously express TRPC6. |
| Literature | GENERIF_SUMMARY | At least 1 mutation was identified in 48 of 87 (55%) specimens tested for PTEN, making it the most common abnormality in this study of patients with endometrial cancer. |
| Literature | GENERIF_SUMMARY | Neutron reflection measurements show that the PTEN phosphatase ""scoots"" along the membrane surface." |
| Literature | GENERIF_SUMMARY | PTEN germ-line alteration and beta-catenin accumulation, observed in peripheral blood cells of PHTS patients, could determine a cell survival and pro-inflammatory stimulation mediated by the expression of molecules such as COX2, CCND1 and TNFa receptors. |
| Literature | GENERIF_SUMMARY | aberrant expression of PTEN and p-PTEN at residue Ser380 was early event that could contribute to gastric carcinogenesis, and that PTEN phosphorylation at residue Ser380 could be a mechanism for PTEN inactivation |
| Literature | GENERIF_SUMMARY | This study concludes that the abnormal methylation of PTEN gene promoter CpG islands exists in the colorectal tumors and metastases. |
| Literature | GENERIF_SUMMARY | Although there is an association of PI3 kinase signaling and chemoresistance in advanced ovarian cancer, There is no clear evidence that this is driven specifically by deletion of PTEN. |
| Literature | GENERIF_SUMMARY | CXCL8 and CXCR1 are involved in the pathogenesis of endometriosis by up-regulating proliferation and growth and restricting apoptosis in endometrial stromal cells by activating the PTEN/Akt pathway and mediating the expression of survivin and Bcl-2 |
| Literature | GENERIF_SUMMARY | In the present study, we did not find any significant correlations between KRAS, BRAF and PIK3CA mutations and the loss of |

| Main Category | Category | Terms |
|---|---|---|
| | | PTEN expression and various clinicopathological features in Chinese patients with colorectal cancer. |
| Literature | GENERIF_SUMMARY | Multiple de novo PTEN mutations are responsible for the PTEN hamartoma tumor syndrome. |
| Literature | GENERIF_SUMMARY | PTEN signaling regulates infection of both noncancerous and cancerous cells by multiple intracellular mycobacterial pathogens and that pharmacological modulation of PTEN signaling can affect mycobacterial infection. |
| Literature | GENERIF_SUMMARY | miR-26b and miR-128 as central regulators of pituitary somatomammotrophic tumor growth through activation of the PTEN-AKT pathway. |
| Literature | GENERIF_SUMMARY | This paper describes the analysis of BRCA1, PTEN, and p53 at the single cell level in 55 BRCA1-associated breast tumors and computational methods to predict the relative temporal order of somatic events |
| Literature | GENERIF_SUMMARY | Single nucleotide polymorphism in PTEN is associated with synchronous endometrial and ovarian cancer. |
| Literature | GENERIF_SUMMARY | Genomic PTEN deletion is a rare event in gastric adenocarcinoma but correlates with metastatic disease. The homogeneous distribution pattern indicates that this alteration occurs early in tumour development. |
| Literature | GENERIF_SUMMARY | This review offer insight into the role of PTEN in a subset of autism spectrum disorders patients and highlight the complexity of the pathogenesis of autism spectrum disorders. |
| Literature | GENERIF_SUMMARY | Benign gastrointestinal lesions are common in PTEN mutation carriers, and a three- to four-fold increased lifetime risk of colorectal cancer compared with the general population may exist. |
| Literature | GENERIF_SUMMARY | PTEN loss, high MYC expression, and low Ki67 expression in primary tumor samples are associated with response to therapy in prostate cancer. |
| Literature | GENERIF_SUMMARY | Our results reveal a novel mechanism for the therapeutic function of fish oil diet that blocks miR-21, thereby increasing PTEN levels to prevent expression of CSF-1 in breast cancer. |
| Literature | GENERIF_SUMMARY | Multiple nucleotides in the PTEN are altered in normal Japanese individuals. |

| Main Category | Category | Terms |
|---|---|---|
| Literature | GENERIF_SUMMARY | the expression levels of EGFR, pAkt, and PTEN differ between oropharyngeal and oral cavity cancer and it may be attributed to HPV-related molecular pathogenesis |
| Literature | GENERIF_SUMMARY | In patients with clinically localized prostate cancer treated by prostatectomy, decreased PTEN expression was associated with an increased risk of recurrence, independent of known clinicopathological factors. |
| Literature | GENERIF_SUMMARY | Data indicate that a signature of seven biomarkers (Bax, Bcl-X, PTEN, COX-2, beta-Catenin, MTAP and CD20) was found to be an independent negative predictor for overall and recurrence-free survival in cutaneous malignant melanoma (MM). |
| Literature | GENERIF_SUMMARY | PTEN loss activates cofilin and induces cytoskeletal changes via a PI3K-independent mechanism. |
| Literature | GENERIF_SUMMARY | Differential expression of microRNAs miR-21, miR-31, miR-203, miR-125a-5p and miR-125b and proteins PTEN and p63 in verrucous carcinoma of the head and neck. |
| Literature | GENERIF_SUMMARY | Haploinsufficiency or complete loss of Pten dramatically accelerated neurofibroma development and led to the development of higher grade PNSTs in the context of Nf1 loss |
| Literature | GENERIF_SUMMARY | Metastasis-associated protein 1/histone deacetylase 4-nucleosome remodeling and deacetylase complex regulates phosphatase and tensin homolog gene expression and function. |
| Literature | GENERIF_SUMMARY | Deletion of PTEN is associated with tumor progression and early PSA recurrence in ERG fusion-positive and fusion-negative prostate cancer. |
| Literature | GENERIF_SUMMARY | SUMO1 modification is required for PTEN tumour suppressor function by controlling PTEN membrane association and regulation of the phosphatidylinositol-3 kinase/AKT pathway. |
| Literature | GENERIF_SUMMARY | No association was found between PTEN expression and disease persistance/ progression in endometrial hyperplasia treated with progestins. |
| Literature | GENERIF_SUMMARY | this study evaluated whether activating mutations of PI3KCA or complete loss of PTEN is associated with response to anti-human Her2 treatment in breast cancer |
| Literature | GENERIF_SUMMARY | PTEN promoter methylation and activation of the PI3K/Akt/mTOR pathway is |

| Main Category | Category | Terms |
|---|---|---|
| | | associated with pediatric gliomas. |
| Literature | GENERIF_SUMMARY | Combinatorial immunophenotypic analyses showed that loss of PTEN expression with concomitant IGFR-1 expression correlated with poor disease-free survival |
| Literature | GENERIF_SUMMARY | Report miRNA and PTEN expression in normal, hyperplastic and malignant endometrial tissues. |
| Literature | GENERIF_SUMMARY | microRNA-21 expression is up-regulated in human cholangiocarcinoma and PTEN, PDCD4 are direct effectors of microRNA-21. |
| Literature | GENERIF_SUMMARY | Upregulating PTEN expression increases the chemosensitivity of K562/ADM cells. |
| Literature | GENERIF_SUMMARY | study reveals a unique mechanism through which ER-beta/PTEN signaling induces cell death in TCAM2 seminoma cell line by autophagy and necroptosis |
| Literature | GENERIF_SUMMARY | The association between single nucleotide polymorphisms (SNPs) of PTEN/AKT/mTOR genes and prostate cancer (PCa) in Chinese population, was examined. |
| Literature | GENERIF_SUMMARY | aberrant p16 alterations directly correlated with altered those in PTEN in non-small cell lung cancer. |
| Literature | GENERIF_SUMMARY | Loss of PTEN staining was associated with a coexistent of an endometrial adenocarcinoma in the presence of atypical complex hyperplasia. |
| Literature | GENERIF_SUMMARY | These results uncover an essential role of TGFbeta-induced expression of miR-21, which targets PTEN to initiate a non-canonical signaling circuit involving Akt/mTORC1 axis for mesangial cell hypertrophy and matrix protein synthesis. |
| Literature | GENERIF_SUMMARY | Our study demonstrated that miR-26a enhanced lung cancer cell metastasis potential via modulation of metastasis-related gene expression, and activation of AKT pathway by PTEN suppression |
| Literature | GENERIF_SUMMARY | our results imply that LOH of PTEN could be used as a good prognostic characteristic for the outcome of breast cancer patients treated with tamoxifen. |
| Literature | GENERIF_SUMMARY | This review will summarize the new points on the regulation of PTEN and briefly discuss the potential therapeutic role of PTEN in some diseases [review] |
| Literature | GENERIF_SUMMARY | The phosphatase activity of PTEN seems to be active because p-Akt levels were |

| Main Category | Category | Terms |
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| | | reduced. |
| Literature | GENERIF_SUMMARY | Data indicate that mutation frequency in malanoma patients was found witih BRAF(V600) in 51%, NRAS in 19%, PI3K pathway in 41% and PTEN in 22%. |
| Literature | GENERIF_SUMMARY | Based on these results we have formulated a hypothesis about the different types of endometrial hyperplasia morphogenesis and its possible transfer to cervical cancer associated with features of tumor suppressor gene PTEN. |
| Literature | GENERIF_SUMMARY | study concludes PTEN and Livin are important in renal clear cell carcinoma development |
| Literature | GENERIF_SUMMARY | PTEN loss induces a selective upregulation of CXCL8 signalling that sustains the growth and survival of PTEN-deficient prostate epithelium. |
| Literature | GENERIF_SUMMARY | Reduced PTEN expression and MMP-7 over-expression may play important roles in the pathogenesis of colorectal cancer. |
| Literature | GENERIF_SUMMARY | PTEN loss is associated with hepatocellular carcinoma. |
| Literature | GENERIF_SUMMARY | Mammalian target of rapamycin, PTEN, and beta-catenin are independently involved in different molecular subtypes of endometrial carcinoma with diverse patients' prognosis and support their distinctive treatment based on targeted drugs. |
| Literature | GENERIF_SUMMARY | The tumorigenic role of miR-21 in non-small cell lung cancer development through modulation of the PTEN signaling pathway. |
| Literature | GENERIF_SUMMARY | data suggest that EZH2 is overexpressed in B-ALL and promotes the progression of B-ALL by directly mediating the inactivation of tumor suppressor genes p21 and PTEN |
| Literature | GENERIF_SUMMARY | PTEN expression is increased in radioresistant esophageal squamous cell carcionoma cells. |
| Literature | GENERIF_SUMMARY | We demonstrate an apparently divergent effect of PTEN mutations |
| Literature | GENERIF_SUMMARY | Loss of phosphatase and tensin homolog protein expression is associated with lung adenocarcinoma. |
| Literature | GENERIF_SUMMARY | Bcl-2 was positive in 18.18% of the cases, in distal colorectal cancer, only in men, 63.5 +/- 13.2-year-old. |
| Literature | GENERIF_SUMMARY | PTEN was identified to be directly regulated |

| Main Category | Category | Terms |
|---|---|---|
| | | by miR-1297 in laryngeal squamous cell carcinoma. |
| Literature | GENERIF_SUMMARY | Our observations indicate that the R280T mutation of p53 regulates the proliferation of human glioma cells related to the GSK-3beta/PTEN pathway. |
| Literature | GENERIF_SUMMARY | low expression of PTEN was associated with shorter time to first treatment (TTFT) (p = 0.040). These results indicate that PTEN might be a new prognostic marker in patients with CLL. |
| Literature | GENERIF_SUMMARY | PTEN is modified by SUMO1 and SUMO2, and reveal the existence of a crosstalk between SUMOylation and ubiquitination of PTEN. |
| Literature | GENERIF_SUMMARY | Report changes in PTEN cell signaling associated with ischemic/idiopathic cardiomyopathies. |
| Literature | GENERIF_SUMMARY | PTEN losses exhibit heterogeneity in multifocal prostatic adenocarcinoma and are associated with higher Gleason grade. |
| Literature | GENERIF_SUMMARY | Loss of heterozygosity of PTEN in thirty percent of human melanomas implicate the PI3K-PTEN-AKT signal pathway in this cancer.[Review] |
| Literature | GENERIF_SUMMARY | Studied the cytotoxicity and anti-tumour activity of novel MEK inhibitor, E6201, in a panel of melanoma cell lines. Most melanoma cell lines were sensitive or hypersensitive to E6201; the sensitivity correlated with wildtype PTEN and mutant BRAF status. |
| Literature | GENERIF_SUMMARY | Our results suggest that miR-153 play an important role in promoting proliferation of human prostate cancer cells and present a novel mechanism of microRNA-mediated direct suppression of PTEN expression in prostate cancer cells. |
| Literature | GENERIF_SUMMARY | The (-/-) genotype of PTEN IVS4 gene might be associated with increased risk for development of CRC in a Turkish population |
| Literature | GENERIF_SUMMARY | High PTEN expression is associated with low response to HSP90 inhibitor NVP-AUY922 in esophageal cancer. |
| Literature | GENERIF_SUMMARY | PTEN protein expression could serve as a screening molecular correlate to predict for germline PTEN mutation in Cowden syndrome (CS) and CS-like presentations of thyroid cancer. |
| Literature | GENERIF_SUMMARY | Our study provides the first evidence of an |

| Main Category | Category | Terms |
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| | | association between a PTEN intronic variant and the risk of atherosclerotic cerebral infarction |
| Literature | GENERIF_SUMMARY | this study was focused on aberrations of the tumor suppressor genes TP53 and PTEN in cancer stromal fibroblasts of breast cancer. |
| Literature | GENERIF_SUMMARY | Data show that Tbx3 is up-regulated in tissue samples of head and neck squamous cell cancer (HNSCC) patients and that Tbx3 represses PTEN transcription. |
| Literature | GENERIF_SUMMARY | either mutant PIK3CA expression or PTEN loss, but not mutant AKT1 E17K, cause disrupted epithelial architecture. |
| Literature | GENERIF_SUMMARY | Polymorphisms in the HER2 gene (Ile655Val) and an ATCTT insertion at intron 4 of PTEN, are associated with breast cancer susceptibility. |
| Literature | GENERIF_SUMMARY | these results suggest that Sp1 is involved in the development and invasiveness of cancer through inhibition of PTEN |
| Literature | GENERIF_SUMMARY | Overexpression of EGFR or EGFRvIII, accompanied by a loss of PTEN, contributed to the activation of multiple intracellular signalling pathways in GBM cells. |
| Literature | GENERIF_SUMMARY | Analysis of the PTEN gene revealed the presence of a constitutional mutation (c.510 T>A; p.Ser170Arg) which had been associated with the Cowden phenotype. |
| Literature | GENERIF_SUMMARY | our studies reveal PNUTS as a novel PTEN regulator and a likely oncogene. |
| Literature | GENERIF_SUMMARY | Overexpression of bcl2 and loss of PTEN expression were correlated to poor differentiation, lymph node involvement and late stages in oral squamous cell carcinoma. |
| Literature | GENERIF_SUMMARY | Novel PTEN germline mutation in a family with mild phenotype |
| Literature | GENERIF_SUMMARY | Low PTEN expression is associated with nasopharyngeal carcinoma. |
| Literature | GENERIF_SUMMARY | PTEN downregulation as a critical component of UVA's molecular impact during keratinocyte transformation. |
| Literature | GENERIF_SUMMARY | model of tumor progression in NSCLC and intrinsic resistance to TKI in loss of PTEN expression |
| Literature | GENERIF_SUMMARY | MYC acts via the PTEN tumor suppressor to elicit autoregulation and genome-wide gene repression by activation of the Ezh2 |

| Main Category | Category | Terms |
|---|---|---|
| | | methyltransferase. |
| Literature | GENERIF_SUMMARY | PTEN promoter methylation may be a mechanism for PTEN downregulation or silencing in Metabolic syndrome |
| Literature | GENERIF_SUMMARY | The role of Pten, a key regulator of the Pi3k/Akt/mTOR pathway, in human malignant peripheral nerve sheath tumours and benign neurofibromas. |
| Literature | GENERIF_SUMMARY | Data suggest the potential of targeting of Nemo-Like Kinase (NLK) to treat a range of tumourigenic conditions characterised by PTEN deficiency. |
| Literature | GENERIF_SUMMARY | PTEN (phosphatase and tensin homolog deleted from chromosome 10), a cytosolic enzyme, can be transferred between cells in exosomes to suppress signaling and proliferation in target cells. |
| Literature | GENERIF_SUMMARY | PTEN expression was detected in 46.2% of supraglottic squamous cell carcinoma and in 90.5% of adjacent non-cancerous tissues |
| Literature | GENERIF_SUMMARY | loss of PTEN can be detected in a significant subset of esophageal adenocarcinoma and is associated with an aggressive phenotype. |
| Literature | GENERIF_SUMMARY | The identification of H61D mutation is imparted maximum functional effect on PTEN function. |
| Literature | GENERIF_SUMMARY | The positive expression rates of survivin and PTEN in vocal cord polyps were obviously lower than in papilloma of larynx and laryngeal squamous cell carcinoma. |
| Literature | GENERIF_SUMMARY | PTEN deletion was also linked to diminished expression of PTEN (P |
| Literature | GENERIF_SUMMARY | Low PTEN expression is associated with colorectal cancer. |
| Literature | GENERIF_SUMMARY | IL-18 deactivated phosphatase and tensin homolog deleted on chromosome 10 (PTEN) in normal human foreskin-derived epidermal melanocytes. |
| Literature | GENERIF_SUMMARY | Letter |
| Literature | GENERIF_SUMMARY | delineate a new NPM1-HAUSP molecular interaction controlling PTEN deubiquitination and trafficking |
| Literature | GENERIF_SUMMARY | FAK expression was positively correlated with ILK expression but negatively correlated with PTEN expression in SACC tissues. FAK and ILK expression was positively associated with advanced stage |

| Main Category | Category | Terms |
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| Literature | GENERIF_SUMMARY | miR-23b-3p is an oncogenic miRNA and inhibits PTEN tumor suppressor gene in renal cell carcinoma. |
| Literature | GENERIF_SUMMARY | In this review PTEN (and SOCS3) are shown to act as negative regulators of major signaling pathways, such as protein translation and transcription. |
| Literature | GENERIF_SUMMARY | in primary endometrial carcinoma, mutations in PIK3CA, TP53, PTEN and FBXW7 correlate with high tumor grade, endometrial cancer type and lymph node status |
| Literature | GENERIF_SUMMARY | These data also implicate PPARgamma and PTEN inactivation in the development of keratinizing squamous metaplasia. |
| Literature | GENERIF_SUMMARY | Our study identifies PTEN loss as a potentially useful marker to distinguish intraductal carcinoma from high-grade prostatic intraepithelial neoplasia |
| Literature | GENERIF_SUMMARY | to understand the molecular basis of contextual regulation of PTEN is crucial in order to understand the biological function of PTEN and, subsequently, to develop appropriate approaches to therapeutically target PTEN signaling |
| Literature | GENERIF_SUMMARY | Aberrations of MET are associated with EGFR and PTEN signalling and might possess relevance for targeted therapies of salivary gland carcinomas in the future. |
| Literature | GENERIF_SUMMARY | Loss of PTEN protein expression is associated with malignant pleural mesothelioma. |
| Literature | GENERIF_SUMMARY | Data indicate that the PTEN loss and activation of AKT signaling pathway contributed to erlotinib resistance in EGFR-mutant non-small cell lung cancer (NSCLC) cell line H1650. |
| Literature | GENERIF_SUMMARY | loss of PTEN expression was an independent prognostic factor for poor overall survival in patients with stage II pancreatic ductal adenocarcinoma. |
| Literature | GENERIF_SUMMARY | Vandetanib might be effective in non small cell lung carcinoma with EGFR mutations that lack PTEN expression. |
| Literature | GENERIF_SUMMARY | Overexpression of PTEN inhibited proliferation of human airway smooth muscle cells and was associated with downregulation of Akt phosphorylation levels. |
| Literature | GENERIF_SUMMARY | Alterations of EGFR, p53 and PTEN that |

| Main Category | Category | Terms |
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| | | mimic changes found in basal-like breast cancer promote transformation of human mammary epithelial cells. |
| Literature | GENERIF_SUMMARY | Results showed that PTEN expression progressively decreased with the continuum from normal epithelium to squamous cell carcinoma (P |
| Literature | GENERIF_SUMMARY | PTEN appears to function as a crucial inhibitor of glioblastoma stem cells and as an inducer of senescence, suggesting that functional enhancement of the PTEN pathway will be useful to provide a therapeutic strategy for targeting glioblastoma. |
| Literature | GENERIF_SUMMARY | analysis of PTEN status in 135 sporadic invasive ductal carcinomas reveals that relapsed tumors are associated with a specific, complex DNA profile at the PTEN locus with a strikingly similar highly rearranged pan-genomic profile |
| Literature | GENERIF_SUMMARY | Transfection with PTEN antisense oligonucleotide dose- and time-dependently enhanced cell proliferation and inhibited apoptosis in both EC9706 and EC1 cells. |
| Literature | GENERIF_SUMMARY | we report convergent cellular and behavioral data that the interaction between the 5-HT(2C)R and PTEN serves as a regulatory mechanism to control 5-HT(2C)R-mediated biology |
| Literature | GENERIF_SUMMARY | Automated dual ERG-PTEN and ERG-SPINK1 immunohistochemisty assays are simple, reliable and portable across study sites for the simultaneous assessment of these proteins in prostate cancer. |
| Literature | GENERIF_SUMMARY | NOTCH1 activation clinically antagonizes the unfavorable effect of PTEN inactivation in BFM-treated children with precursor T-cell acute lymphoblastic leukemia. |
| Literature | GENERIF_SUMMARY | PTEN loss of expression is associated with colorectal cancer. |
| Literature | GENERIF_SUMMARY | Our findings suggest that altered PTEN, ATRX, CHGA, and CHGB expression are associated with aggressive PNET phenotype in VHL and may serve as useful adjunct prognostic markers to Ki-67 in PNETs. |
| Literature | GENERIF_SUMMARY | Cases of extreme macrocephaly have been correlated to mutations in the gene phosphatase and tensin homolog. |
| Literature | GENERIF_SUMMARY | This suggests that PI3K/AKT signaling contributes to both metabolic and cognitive activities. |

| Main Category | Category | Terms |
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| Literature | GENERIF_SUMMARY | Our patients present with a unique constellation of features that have not yet been described in humans with PTEN germline mutations,severe hypoglycemia, growth hormone deficiency, Von Willebrand disease, and dyslipidemia. |
| Literature | GENERIF_SUMMARY | Glioblastoma cell miR-17 prolonged tumor cell survival, induced angiogenesis and promoted stem-like cell aggregation by repressing expression of MDM2 and PTEN. |
| Literature | GENERIF_SUMMARY | Seventy-seven patients (13%) were found to have colorectal polyposis-associated mutations, including 13 in PTEN (2.2%) |
| Literature | GENERIF_SUMMARY | the PTEN IVS4 polymorphism might contribute to the development of gastric cancer in a Turkish population |
| Literature | GENERIF_SUMMARY | Gene expression analyses highlighted the association between xenograftability and a unique gene expression signature, including down-regulated PTEN tumor-suppressor gene expression and a progenitor-like phenotype. |
| Literature | GENERIF_SUMMARY | RFP-mediated ubiquitination of PTEN modulates its effect on AKT activation. |
| Literature | GENERIF_SUMMARY | study demonstrates that the Tgfbr1/Pten 2cKO mouse model of human HNSCC is a useful model for assessing antitumor activity of new cancer therapeutic agents, and that IL-13-PE has therapeutic potential to treat human head and neck cancer |
| Literature | GENERIF_SUMMARY | Transforming growth factor beta-regulated microRNA-29a promotes angiogenesis through targeting the phosphatase and tensin homolog in endothelium. |
| Literature | GENERIF_SUMMARY | GATA3 antagonizes prostate tumor formation mediated by Pten-deficiency. |
| Literature | GENERIF_SUMMARY | SPRY2, PTEN, and PP2A status is an important determinant of prostate cancer progression. |
| Literature | GENERIF_SUMMARY | Disruption of this asRNA-regulated network induces cell-cycle arrest and sensitizes cells to doxorubicin, which suggests a biological function for the respective PTENpg1 expressed asRNAs |
| Literature | GENERIF_SUMMARY | HER2 overexpression was associated with an increasing risk of developing metastases. |
| Literature | GENERIF_SUMMARY | PTEN, VEGF, HER2, p53 expressions were compared with bevacizumab response. |

| Main Category | Category | Terms |
| --- | --- | --- |
| Literature | GENERIF_SUMMARY | genomic PTEN gene alterations caused by deletion or gene disruption characterize a fraction of rectal cancers with particularly poor outcome. |
| Literature | GENERIF_SUMMARY | results provide novel insights into the cellular functions of PTEN and reveal molecular mechanisms whereby PTEN mutations increase proteasome activity and lead to neurologic phenotypes |
| Literature | GENERIF_SUMMARY | Treatment of transplanted laryngeal squamous cell carcinoma with Gold Throat Tablets reduced expression of survivin and PTEN. |
| Literature | GENERIF_SUMMARY | PTEN might act as a negative regulator of AIB1 whereby the association of PTEN with both AIB1 and Fbw7alpha could lead to the downregulation of AIB1 transcriptional activity. |
| Literature | GENERIF_SUMMARY | The oxidative stress-induced up regulation of PTEN in VSMC of injured arteries augments the sensitivity of VSMC to apoptotic stimuli in the early phase following vascular injury, augmenting the initial injury and cell loss of the injured vessel wall |
| Literature | GENERIF_SUMMARY | PTEN loss tends to occur as a subclonal event within a given established prostatic carcinoma clone after ERG gene fusion. |
| Literature | GENERIF_SUMMARY | miRNA-200c inhibited Akt signaling through its effects on E-cadherin and PTEN, resulting in the inhibition of doxorubicin resistance in breast cancer cells. |
| Literature | GENERIF_SUMMARY | The coexpression levels of PTEN and Cdx2 in GC tumors correlated with prognosis in GC patients. Cdx2 may play a role in the upregulation of PTEN by triggering PI3K/Akt inactivation in GC cells. |
| Literature | GENERIF_SUMMARY | The combined detection of PTEN and miR-92 may provide critical clinical evidence for the early diagnosis and prognosis of hepatocellular carcinoma |
| Literature | GENERIF_SUMMARY | In HER2-positive recurrent or metastatic breast cancer patients PTEN loss might indicate resistance to trastuzumab-based salvage treatment. |
| Literature | GENERIF_SUMMARY | The findings presented in this study demonstrate a novel role for PTEN and Akt in controlling centrosome composition and integrity during mitosis. |
| Literature | GENERIF_SUMMARY | DeltaNp63alpha-mediated suppression of nuclear PTEN in basal layer keratinocytes |

| Main Category | Category | Terms |
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| | | occurs through repression of NEDD4-1. |
| Literature | GENERIF_SUMMARY | Studies indicate ETS transcription factors, PTEN protein and androgen receptor as tumor markers for diagnosis and therapy of prostate cancer. |
| Literature | GENERIF_SUMMARY | These findings suggest that hepatitis B virus X protein disrupts stress fiber formation and triggers apoptosis via an MLCK and a PTEN-dependent pathway. |
| Literature | GENERIF_SUMMARY | PTEN knockdown-mediated radioresistance is accompanied by repression of radiation-induced cytotoxic autophagy. |
| Literature | GENERIF_SUMMARY | PTEN was involved in CTinduced SWO-38 cell differentiation via its lipid phosphatase activity, which inhibits the PI3K/AKT pathway. |
| Literature | GENERIF_SUMMARY | Angiosarcomas are a heterogeneous group of vascular malignancies in which active TGF-beta signaling and decreased expression of PTEN separates angiosarcoma of bone from its soft tissue counterpart. |
| Literature | GENERIF_SUMMARY | PTEN has a role in regulating ABCG2 expression and the PI3K/Akt pathway in adult acute leukemia |
| Literature | GENERIF_SUMMARY | inactivation of TGF-beta signaling and loss of PTEN cooperate to drive intestinal cancer formation and progression by suppressing cell cycle inhibitors |
| Literature | GENERIF_SUMMARY | tumor cells with inactivated PTEN are responsive to metformin, phenformin and AICAR |
| Literature | GENERIF_SUMMARY | No statistically significant correlation between trastuzumab resistance and the expression of PTEN was noted in breast cancer. |
| Literature | GENERIF_SUMMARY | PTEN and p53 immunoexpression helps both in accurate diagnosis and proper therapeutic approach of the various endometrial carcinomas. |
| Literature | GENERIF_SUMMARY | article will summarize recent laboratory and clinical investigations relating to PTEN, highlighting the overgrowth aspects of this syndrome and the molecular drivers behind these key phenotypes |
| Literature | GENERIF_SUMMARY | MicroRNA-32 (miR-32) regulates phosphatase and tensin homologue (PTEN) expression and promotes growth, migration, and invasion in colorectal carcinoma cells. |

| Main Category | Category | Terms |
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| Literature | GENERIF_SUMMARY | Reactivation of PTEN tumor suppressor pathway leads to a 50% reduction in colorectal cancer metastasis without affecting primary tumor formation. |
| Literature | GENERIF_SUMMARY | Downregulation of PTEN protein phosphatase activity by HCV affects cholesterol metabolism, thereby inducing the appearance of large lipid droplets and increasing virion egress. |
| Literature | GENERIF_SUMMARY | PIK3CA exon 20 and PTEN mutations associated with features of colon cancer tumorigenesis |
| Literature | GENERIF_SUMMARY | The role reduced PTEN expression and/or inactivation in head and neck squamous cell carcinoma progression. |
| Literature | GENERIF_SUMMARY | DKK1 and PTEN serve as competitive endogenous RNA, affecting the expression of each other via competition for miRNAs binding in the diabetic heart. |
| Literature | GENERIF_SUMMARY | A SNP within PTEN is associated with increased risk of testicular germ cell tumor. |
| Literature | GENERIF_SUMMARY | PTEN deletion is associated with capsular penetration in prostate cancer. |
| Literature | GENERIF_SUMMARY | Aberrations in PIK3CA and PTEN did not add significant prognostic information on colorectal carcinoma patients. |
| Literature | GENERIF_SUMMARY | Patients with mtp53 tumors were older at diagnosis, had more incidence of liver metastasis, and more frequent PTEN loss. |
| Literature | GENERIF_SUMMARY | miR-21-mediated suppression of both hSulf-1 and PTEN led to activation of AKT/ERK pathways and epithelial-mesenchymal transition in hepatocellular carcinoma, promoting tumor growth. |
| Literature | GENERIF_SUMMARY | PTEN and MTOR expression correlate with clinicopathological features in pancreatic neuroendocrine tumors. |
| Literature | GENERIF_SUMMARY | PTEN hypermethylation is a common event in Soft tissue sarcomas suggesting that the inactivation of PTEN may be due to hypermethylation in the promoter of PTEN. |
| Literature | GENERIF_SUMMARY | In low-risk patients, PTEN immunohistochemistry loss adds prognostic value to Gleason score, PSA, Ki-67 and extent of disease. |
| Literature | GENERIF_SUMMARY | Association between PTEN mutations and clinical markers in individuals with autism spectrum disorders and macrocephaly. |

| Main Category | Category | Terms |
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| Literature | GENERIF_SUMMARY | Progression of mycosis fungoides and diminished PTEN expression seems to correlate with PTEN damage at the molecular level. |
| Literature | GENERIF_SUMMARY | The direct correlation of PTEN in Cowden syndrome and sporadic trichilemmoma remains unclear. |
| Literature | GENERIF_SUMMARY | IGF-1 may exert its proliferative effects by negatively regulating the PTEN/PHLPP2 signaling pathway in CM cells. |
| Literature | GENERIF_SUMMARY | We also examined 61 of our cases for the presence of PIK3CA, AKT1, PTEN and K-RAS mutations. |
| Literature | GENERIF_SUMMARY | These results identify casein kinase 2 as a new target of PD-1 and reveal an unexpected mechanism by which PD-1 decreases PTEN protein expression while increasing PTEN activity, thereby inhibiting the PI3K/Akt signaling axis. |
| Literature | GENERIF_SUMMARY | identified a 576-amino acid translational variant of PTEN, PTEN-Long, that arises from an alternative translation start site 519 base pairs upstream of the ATG initiation sequence; PTEN-Long is a membrane-permeable lipid phosphatase that is secreted from cells and can enter other cells |
| Literature | GENERIF_SUMMARY | MAGI2 enhances the sensitivity of BEL-7404 human hepatocellular carcinoma cells to staurosporine-induced apoptosis by increasing PTEN stability. |
| Literature | GENERIF_SUMMARY | Because loss of PTEN can activate mTOR and mTOR inhibitors are Food and Drug Administration approved to treat renal cell carcinoma, these agents have clinical potential in renal cell carcinoma associated with Cowden syndrome |
| Literature | GENERIF_SUMMARY | PTEN mutations are more frequent in low-grade endometrial endometrioid carcinomas (67%) compared with low-grade ovarian endometrioid carcinomas (17%) (P |
| Literature | GENERIF_SUMMARY | Id4 regulates NKX3.1, Sox9 and PTEN. |
| Literature | GENERIF_SUMMARY | Of the total cases examined for PTEN promoter methylation we found that 35% were positive and 65% were negative. When evaluated in connection with tumor differentiation it was found that 82% of poorly differentiated, 24% of moderately differentiated and 32% of well differentiated OSCC samples were methylated. |
| Literature | GENERIF_SUMMARY | Notch1 represses AKT independent of its PTEN transcriptional effects |

| Main Category | Category | Terms |
|---|---|---|
| Literature | GENERIF_SUMMARY | The CG genotype of PTEN -9C>G polymorphism was not associated with MetS and some of its components |
| Literature | GENERIF_SUMMARY | Ectopic expression of wild type PTEN reduced the frequency of etoposide-induced multinucleated HepG2 cells & restored their etoposide sensitivity.The Akt/PTEN cellular axis is a major determinant of etoposide resistance in these cells. |
| Literature | GENERIF_SUMMARY | MITF, PTEN, and CDKN2A germline mutations associated with multiple primary melanomas. |
| Literature | GENERIF_SUMMARY | The c.80-96A.G (rs1903858) polymorphism was identified in one individual. In a single SPS individual, we also identified a novel variant in intron 2 (c.164+223T.C) of PTEN, which was predicted by in silico analysis to have no functional consequences. |
| Literature | GENERIF_SUMMARY | Endoplasmic reticulum (ER)-localized PTEN regulates Ca(2+) release from the ER in a protein phosphatase-dependent manner. |
| Literature | GENERIF_SUMMARY | PIK3CA mutation and copy number gain were detected in 3 (4.9%) and 8 (13.1%), respectively, of 61 gastric cancer samples while PTEN loss was detected in 24 (39%) of the tumours. Two tumours had both PTEN loss and PIK3CA copy number gain. |
| Literature | GENERIF_SUMMARY | Expression of ERG and ETV1 correlated with higher AR transcriptional output in PTEN-deficient prostate cancer specimens. |
| Literature | GENERIF_SUMMARY | PTEN and MMP-2 immunoexpression in ectopic endometrial foci cannot be used to identify women with an increased risk of neoplastic transformation. |
| Literature | GENERIF_SUMMARY | PTEN gene expression is down regulated by oxidative stress through the mediation of reactive oxygen species |
| Literature | GENERIF_SUMMARY | analysis of clinical datasets from liver tissues showed a negative correlation between expressions of ERRalpha and PTEN in patients with liver cancer. |
| Literature | GENERIF_SUMMARY | results indicate that PTEN loss defines a PI3K/AKT-dependent GCB DLBCL subtype that is addicted to PI3K and MYC signaling and suggest that pharmacologic inhibition of PI3K might represent a promising therapeutic approach in these lymphomas |
| Literature | GENERIF_SUMMARY | the implication of PTEN in genesis of plexiform neurofibromas and malignant plexiform neurofibromas |

| Main Category | Category | Terms |
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| Literature | GENERIF_SUMMARY | PIK3CA mutations and/or PTEN loss were identified in the wide morphologic spectrum of salivary duct carcinomas. |
| Literature | GENERIF_SUMMARY | Low PTEN expression is associated with malignant phenotypes in non-small cell lung cancer. |
| Literature | GENERIF_SUMMARY | This study shows that PI3K inhibitors only transiently inhibit AKT which limits their antitumor activities. |
| Literature | GENERIF_SUMMARY | we show that modest inhibition of PNKP in a PTEN knockout background enhances cellular radiosensitivity, suggesting that such a ""synthetic sickness"" approach involving the combination of PNKP inhibition with radiotherapy" |
| Literature | GENERIF_SUMMARY | study demonstrates that SUMOylation of PTEN controls its nuclear localization; cells lacking nuclear PTEN were hypersensitive to DNA damage, whereas PTEN-deficient cells were susceptible to killing by a combination of genotoxic stress and a small-molecule PI3K inhibitor |
| Literature | GENERIF_SUMMARY | activation of CXCL12 signaling through CXCR4 in prostate cancer driven by the loss of PTEN and subsequent activation of Akt, which in turn promotes tumor growth |
| Literature | GENERIF_SUMMARY | PTEN loss leads to the phosphorylation and inactivation of Lgl by atypical protein kinase C in glioblastoma cells. |
| Literature | GENERIF_SUMMARY | Data indicate that knockdown of any component of the novel mut-p53/c-Myc/Bcl-XL axis and complex reversed the oncogenic effects of PTEN. |
| Literature | GENERIF_SUMMARY | Two different groups have demonstrated that PTEN is secreted/exported into the extracellular environment for uptake by recipient cells, and functions as a tumor suppressor in a cell non-autonomous manner. |
| Literature | GENERIF_SUMMARY | This study showed that female PTEN mutation carriers have an approximately eight times increased risk of developing cancer and/Lhermitte-Duclos disease, compared to the general population. |
| Literature | GENERIF_SUMMARY | In prostate neoplasms patients, PTEN is incorporated in the cargo of exosomes that circulate in their blood. |
| Literature | GENERIF_SUMMARY | PTEN-knockdown increases p-Akt/pS166Mdm2 over basal levels and p53 levels are significantly lowered. |

| Main Category | Category | Terms |
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| Literature | GENERIF_SUMMARY | Data indicate that loss of Pten in the lung epithelial cells in vivo activated adipokine signaling and induced leptin synthesis. |
| Literature | GENERIF_SUMMARY | PTEN promoter methylation was significantly higher in esphageal squamous cell carcinoma patients than controls. PTEN methylation levels were also higher in male patients with ESCC than in male controls. |
| Literature | GENERIF_SUMMARY | Low PTEN protein expression significantly increases the risk of lethal prostate cancer, particularly when the IGF-IR expression remains at normal level. |
| Literature | GENERIF_SUMMARY | These data support the use of both PI3K and mTOR inhibitors to treat urothelial carcinoma, in particular in the absence of functional PTEN. |
| Literature | GENERIF_SUMMARY | The loss of expression of PTEN protein was observed in 54.5% of Malaysian patients with colorectal neoplasms. |
| Literature | GENERIF_SUMMARY | PTEN is mutated in a small subset of acral lentiginous melanoma tumors of Swedish patients |
| Literature | GENERIF_SUMMARY | Multivariate analyses revealed that PTEN expression (HR, 0.48; p = 0.02) and absence of PI3K expression (HR, 0.2; p = 0.001) were independent prognostic factors for increased PFS |
| Literature | GENERIF_SUMMARY | PTEN is linked to aggressive tumour phenotype and to unfavourable outcome in early bladder cancer. Heterozygous PTEN loss, i.e. reduced PTEN gene dosage, might be sufficient to cause aggressive tumour behaviour in bladder cancer cells. |
| Literature | GENERIF_SUMMARY | TOPK phosphorylates and inactivates PTEN, which in turn activates Akt that leads to proper G2/M progression. |
| Literature | GENERIF_SUMMARY | nherf1 Mutations (K172N and D301V) caused complete or partial loss of NHERF1 functions by affecting the PTEN/NHERF1/PDGFRbeta complex formation. |
| Literature | GENERIF_SUMMARY | Overexpression of microRNA-92a in colorectal cancer cells decreases PTEN expression at the translational level |
| Literature | GENERIF_SUMMARY | Presence of ER and PgR and decreased expression of PTEN gene. |
| Literature | GENERIF_SUMMARY | Initial risk assessment by a genetics specialist was significantly associated with immediate PTEN testing in patients also meeting hereditary breast and ovarian cancer testing criteria. |

| Main Category | Category | Terms |
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| Literature | GENERIF_SUMMARY | Taken together, our data reveal that PTEN can convey the death signal by preventing MADD phosphorylation by Akt. |
| Literature | GENERIF_SUMMARY | Nuclear/chromatin PTEN mediates DNA damage repair through interacting with and modulating the activity of Rad52. |
| Literature | GENERIF_SUMMARY | Genistein inhibited the Akt activation by upregulation of PTEN in SGC-7901 cells and BGC-823 cells. |
| Literature | GENERIF_SUMMARY | a human glioma PTEN mutation generates aberrant cell growth in primary astrocytes, even in the presence of wt PTEN, by recruiting through PKA an autocrine IGF-1 loop that generates a constitutive NFkappaB activity |
| Literature | GENERIF_SUMMARY | PTEN is deleted in 36% of Sezary syndrome cases & downregulated in almost all samples. miR-21, miR-106b, and miR-486 were able to control PTEN abundance either in vitro or in vivo. |
| Literature | GENERIF_SUMMARY | PTEN phosphorylation and nuclear export mediate free fatty acid-induced oxidative stress. |
| Literature | GENERIF_SUMMARY | the SHH and PI3K pathways synergize to promote tumor growth and viability in PTEN-deficient glioblastomas |
| Literature | GENERIF_SUMMARY | this review questions the cancer theory of the two hits because the functions of both PTEN and TP53 are not fully explained by this theory. |
| Literature | GENERIF_SUMMARY | Our results demonstrated that miR-205 is involved in the tumorigenesis of Non-small cell lung cancer through modulation of the PTEN signaling pathway. |
| Literature | GENERIF_SUMMARY | Our data suggest that the LYN/PTEN expression ratio may be a sensitive monitor of disease progression in unmutated CML patients under imatinib or nilotinib treatment. |
| Literature | GENERIF_SUMMARY | Acid ceramidase promotes nuclear export of PTEN through sphingosine 1-phosphate mediated Akt signaling. |
| Literature | GENERIF_SUMMARY | 4 microRNAs can effectively suppress PTEN expression by directly interacting with its 3' UTR in prostate epithelial and cancer cells. |
| Literature | GENERIF_SUMMARY | this study further extends the biological role of miR-92b in non-small cell lung cancer A549 cells and for the first time identifies PTEN as a novel target of miR-92b. |

| Main Category | Category | Terms |
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| Literature | GENERIF_SUMMARY | A pathogenicheterozygous PTEN |
| Literature | GENERIF_SUMMARY | TGF-beta induces miR10a/10b expression and TGF-beta-induced miR10a/10b expression promotes cell migration through the suppression of PTEN. |
| Literature | GENERIF_SUMMARY | the differential PTEN mutation status with corresponding LDD phenotypes suggests a potential correlation between germline or somatic mutation and coexisting LDD/CS or isolated LDD, respectively. |
| Literature | GENERIF_SUMMARY | A novel AKT1/NF-kappaB/Notch1/PTEN axis has an important role in the development of chemoresistance in gastric cancer. |
| Literature | GENERIF_SUMMARY | our finding suggest that molecular testing for PTEN gene mutations in patients with extreme macrocephaly, developmental delay, and hamartomatous lesions, even in the absence of other CS/BRRS related clinical features, should be considered. |
| Literature | GENERIF_SUMMARY | The incorporation of PTEN expression status in making surgical decisions may also benefit IPMN patients and should warrant further investigation |
| Literature | GENERIF_SUMMARY | Found progressive loss of PTEN and increase in EGFR, TGF-alpha, P-AKT expression from benign samples to non small cell lung cancer. Changes were correlated to differentiation extent of cancer tissue, metastasis of lymph nodes and histological classification. |
| Literature | GENERIF_SUMMARY | The effect of PTEN on neural stem-cell development contributes significantly to the pathophysiology of autism spectrum disorders. |
| Literature | GENERIF_SUMMARY | c-Myc modulates genes associated with oncogenesis in glioblastoma multiforme through deregulation of miRNAs via the c-Myc-miR-26a-PTEN signaling pathway. |
| Literature | GENERIF_SUMMARY | PTEN functions as a melanoma tumor suppressor in part by regulating the host immune response. |
| Literature | GENERIF_SUMMARY | functional upregulation of secreted-MMP7 is associated to the loss of PTEN |
| Literature | GENERIF_SUMMARY | Akt-p53-miR-365-cyclin D1/cdc25A axis contributes to gastric tumorigenesis induced by PTEN deficiency. |
| Literature | GENERIF_SUMMARY | PTEN and the PI3K/Akt-Bad axis may have a role in the pathogenesis of endometriosis |

| Main Category | Category | Terms |
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| Literature | GENERIF_SUMMARY | PTEN might play a role in degeneration of osteoarthritic chondrocytes by regulating the PI3K/Akt pathway. It might be possible to increase matrix synthesis in OA chondrocytes by inhibiting PTEN to improve balance between the PI3K/Akt and MEK/ERK pathways. |
| Literature | GENERIF_SUMMARY | the present study indicates that DJ-1 and PTEN may play their roles in progression of GC in a cooperating pattern. |
| Literature | GENERIF_SUMMARY | PTEN expression is consistent in colorectal cancer primaries and metastases and associates with patient survival. |
| Literature | GENERIF_SUMMARY | The data in this study suggest that 2ME2 may be more efficacious in patients whose GBM tumors express a wild type PTEN tumor suppressor protein. |
| Literature | GENERIF_SUMMARY | Report role of PTEN oncogenetic risk classification in the prognosis of adult T-cell acute lymphoblastic leukemia. |
| Literature | GENERIF_SUMMARY | TNBC was significantly associated with PTEN loss, high Ki67 and the CD44+/CD24- phenotype. The loss of PTEN impacts DFS significantly in African Americans. |
| Literature | GENERIF_SUMMARY | The downregulation of PTEN expressions may play a role in the overactivation of the PI3K/Akt pathway in psoriatic lesions and correlate with the hyperproliferation of psoriatic keratinocytes. |
| Literature | GENERIF_SUMMARY | Loss of PTEN expression is associated with poor disease-free survival and overall survival in patients with ampullary adenocarcinoma after pancreaticoduodenectomy. |
| Literature | GENERIF_SUMMARY | Simultaneous deletion of Lkb1 and Pten, however, led to development of high-grade ovarian serous histotype tumors with 100% penetrance that expressed WT1, ERalpha, PAX8, TP53 and cytokeratin 8, markers used in differential diagnosis of serous ovarian cancer |
| Literature | GENERIF_SUMMARY | Curcumin shows enhanced cytotoxicity toward PTEN-deficient colorectal cancer cells. |
| Literature | GENERIF_SUMMARY | These results imply that curcumin inhibits Ab-induced tau hyperphosphorylation involving PTEN/Akt/GSK-3b pathway. |
| Literature | GENERIF_SUMMARY | In chemotherapy-refractory colorectal cancer, neither PIK3CA mutation status nor PTEN expression were prognostic, nor were they predictive of benefit from cetuximab |

| Main Category | Category | Terms |
|---|---|---|
| Literature | GENERIF_SUMMARY | MicroRNA-21 significantly increases expression of PTEN protein. |
| Literature | GENERIF_SUMMARY | USP13 is a tumour-suppressing protein that functions through deubiquitylation and stabilization of PTEN. |
| Literature | GENERIF_SUMMARY | PTEN loss was greater in lung squamous cell carcinomas than in adenocarcinom |
| Literature | GENERIF_SUMMARY | we found Kae-induced apoptosis in EJ cells to be associated with phosphatase and the tensin homolog deleted on the chromosome 10 (PTEN)/PI3K/Akt pathway. |
| Literature | GENERIF_SUMMARY | miR-221 may promote trastuzumab resistance and metastasis of HER2-positive breast cancers by targeting PTEN, suggesting its role as a potential biomarker for progression and poor prognosis |
| Literature | GENERIF_SUMMARY | In ATXN3-depleted cells, under conditions of transcriptional inhibition, PTEN and PTENP1 mRNAs rapidly decay. |
| Literature | GENERIF_SUMMARY | Membrane recruitment of PTEN is required for PTEN function in cells. |
| Literature | GENERIF_SUMMARY | miR-221/222 confers radioresistance in glioblastoma cells through activating Akt independent of PTEN status. |
| Literature | GENERIF_SUMMARY | Data indicate that seven genes showed statistical enrichment for mutation |
| Literature | GENERIF_SUMMARY | PIK3CA mutations/amplification, HRAS mutation, or PTEN loss that likely lead to PI3K pathway activation were identified in 34 of 75 cases (45%) and did not correlate with disease specific survival. |
| Literature | GENERIF_SUMMARY | PIASxalpha is a novel SUMO E3 ligase for PTEN, and it positively regulates PTEN protein level in tumor suppression. |
| Literature | GENERIF_SUMMARY | Patients who had a PTEN score >20 (78.1%) showed longer PFS. |
| Literature | GENERIF_SUMMARY | The membrane-binding function of the catalytically inert PTEN C2 domain influences cdc42/aPKC-dependent AM dynamics and gland formation in a highly relevant 3D CRC morphogenesis model system. |
| Literature | GENERIF_SUMMARY | results suggest a new mTOR-independent signaling pathway by which PTEN can regulate in opposite directions the main mechanisms of intracellular protein degradation. |
| Literature | GENERIF_SUMMARY | Data indicate that NVP-BEZ235-induced |

| Main Category | Category | Terms |
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| | | androgen receptor (AR) suppression resulted in decreased expression of both tumor suppressor proteins PTEN and KLLN in AR+/ER+ breast cancer cells. |
| Literature | GENERIF_SUMMARY | We hypothesize that the loss of PTEN expression as well as the subcellular localization could be of interest as a predictive marker of recurrence in head and neck squamous cell carcinoma . |
| Literature | GENERIF_SUMMARY | P-REX2 PH-domain-mediated inhibition of PTEN has a role in regulating insulin sensitivity and glucose homeostasis |
| Literature | GENERIF_SUMMARY | The tight link to ERG activation and PTEN deletions suggest interaction with key molecular pathways in prostate cancer. |
| Literature | GENERIF_SUMMARY | The polymorphisms of PTEN rs701848 T/C and rs2735343 C/G might represent crucial modifying factors for development of esophageal squamous cell carcinoma. |
| Literature | GENERIF_SUMMARY | Results suggest Pten is a developmental rheostat regulating interneuron and oligodendroglial differentiation. |
| Literature | GENERIF_SUMMARY | Anti-miR-221 can enhance the radiosensitivity of colorectal carcinoma cells cells by upregulating PTEN expression. |
| Literature | GENERIF_SUMMARY | miR-200a functions as an oncogene, affecting proliferation and apoptosis by regulating the expression of the tumor suppressor PTEN at the translational level. |
| Literature | GENERIF_SUMMARY | The expression of PTEN protein does not correlate with the abnormalities of the loss of heterozygosity of the gene. Interestingly, patients with glioblastomas presenting either LOH of 10q23 or heterogeneous PTEN expression have a poorer prognosis. |
| Literature | GENERIF_SUMMARY | Decreased cytoplasmic PTEN was associated with mesothelioma. |
| Literature | GENERIF_SUMMARY | Incidence of PTEN mutations in Chinese patients with primary small cell carcinoma of the esophagus is higher than that in previouis reports of other histological subtypes of esophageal cancer. |
| Literature | GENERIF_SUMMARY | Data indicate that KRT10 is a downstream molecule of PTEN which improves cisplatin-resistance of ovarian cancer cells. |
| Literature | GENERIF_SUMMARY | Data indicate that phosphatase and tensin homolog (PTEN) mRNA level was inversely correlated with miR-17 level in osteosarcoma (OS) tissues. |

| Main Category | Category | Terms |
|---|---|---|
| Literature | GENERIF_SUMMARY | mutations may confer resistance to epidermal growth factor receptor-tyrosine kinase inhibitors in lung adenocarcinoma patients |
| Literature | GENERIF_SUMMARY | MiR-20b, -21, and -130b inhibit PTEN expression resulting in B7-H1 over-expression in advanced colorectal cancer. |
| Literature | GENERIF_SUMMARY | Study reveals a critical role for the PTEN/PI3K pathway in both SCLC and lung adenocarcinoma. |
| Literature | GENERIF_SUMMARY | DNA sequencing of 18 benign and atypical nevi and 75 melanomas showed that nevi had a similar high frequency of PTEN mutations as melanomas. Both had a very high proportion of UV-type mutations (occurring at adjacent pyrimidines). |
| Literature | GENERIF_SUMMARY | PTEN inhibited PI3K/NF-kappaB pathway and inhibited the DNA binding of NF-kappaB on FAK promoter. |
| Literature | GENERIF_SUMMARY | show how CBP and PTEN interact to mediate tumor suppression in the prostate, establishing a central role for histone modification in the etiology of prostate cancer |
| Literature | GENERIF_SUMMARY | Data introduce hitherto unappreciated paradigms whereby reactive oxygen species can reciprocally regulate the cellular localization of pro- and anti-migratory signaling molecules, p130cas and PTEN, respectively. |
| Literature | GENERIF_SUMMARY | Data indicate that cell lines carrying mutations in both p53 and PTEN showed the highest capacitance C m values of 3.7-4.0 microF/cm(2). |
| Literature | GENERIF_SUMMARY | Case Report |
| Literature | GENERIF_SUMMARY | Study identifies an association of NBS1 expression with surrogates of genomic instability in prostate cancer including TMPRSS2-ERG rearrangements and PTEN deletion. |
| Literature | GENERIF_SUMMARY | PTEN loss is a frequent event in triple-negative breast cancers and is significantly associated with younger age at onset of breast cancer, late stage, and IGFBP2 expression. |
| Literature | GENERIF_SUMMARY | We investigated the role of SOCS3 in p53- and PTEN-mediated activation of inflammatory feedback loop and malignant transformation |
| Literature | GENERIF_SUMMARY | Lipopolysaccharide induced miR-181a |

| Main Category | Category | Terms |
|---|---|---|
| | | promotes pancreatic cancer cell migration via targeting PTEN and MAP2K4. |
| Literature | GENERIF_SUMMARY | PTEN and mTOR might play an important role in the local invasiveness of sacral chordoma |
| Literature | GENERIF_SUMMARY | Novel function of Pnck is revealed in the regulation of ERK1/2 and p38 MAP kinase activity and cell proliferation, which is mediated by paradoxical PTEN functions. |
| Literature | GENERIF_SUMMARY | In endometrioid endometrial carcinoma, 43% showed loss of PTEN expression vs. 16.6% in non-endometrioid endometrial carcinoma. |
| Literature | GENERIF_SUMMARY | Understanding the genetic mechanisms of PTEN loss and implementing improved and standardized methodologies of PTEN assessment are required to clarify the role of PTEN as a biomarker in colorectal cancer. |
| Literature | GENERIF_SUMMARY | Data show the expression of N-myc downstream-regulated gene 2 (NDRG2) is frequently downregulated in adult T-cell leukaemia-lymphoma, resulting in enhanced phosphorylation of PTEN and activation of the phosphatidylinositol 3-kinase (PI3K)-AKT pathway. |
| Literature | GENERIF_SUMMARY | our data support a critical role for NF-kappaB-dependent upregulation of miR-425, which represents a new pathway for the repression of phosphatase and tensin homolog activation and the promotion of cell survival upon IL-1beta induction. |
| Literature | GENERIF_SUMMARY | Deletion of the PTEN gene in RCC was detected with a frequency of approximately 40% and its presence was not determinant of lower survival rates |
| Literature | GENERIF_SUMMARY | PTEN depletion decreases AMPA receptor expression and AMPA-mediated whole-cell currents, resulting in inhibition of AMPA-induced neuronal death in primary cultured and iPS-derived motor neurons. |
| Literature | GENERIF_SUMMARY | A novel frameshift variant of PTEN in a patient with extreme macrocephaly, autism, intellectual disability and seizures. |
| Literature | GENERIF_SUMMARY | Data identify autocrine activation of FGF signaling as an essential mechanism in promoting Pten-deficient skin tumors. |
| Literature | GENERIF_SUMMARY | PTEN hamartoma tumor syndrome has a high prevalence of colon polyposis with multiple histologic types in hamartoma tumor syndrome |

| Main Category | Category | Terms |
|---|---|---|
| Literature | GENERIF_SUMMARY | The phosphatase and tensin homologue (PTEN) and the Ciona intestinalis voltage sensitive phosphatase (Ci-VSP) are both phosphatidylinositol phosphate (PIP) phosphatases that contain a C2 domain. |
| Literature | GENERIF_SUMMARY | In this cetuximab-treated colorectal cancer population, EGFR gain was associated with better outcome and PTEN protein expression with longer TTP in KRAS WT, KRAS WT/AREG high and KRAS/BRAF WT subpopulations. |
| Literature | GENERIF_SUMMARY | Loss of PTEN expression is asssociated with breast cancer. |
| Literature | GENERIF_SUMMARY | Data suggest that the expression of ATP-binding cassette sub-family G member 2 (ABCG2) and the side population may be regulated by phosphatase and tensin homologue deleted on chromosome -10 (PTEN) through the PI3K/Akt pathway. |
| Literature | GENERIF_SUMMARY | Data indicate the potential of miR-21 and the PTEN/Akt pathway as therapeutic targets in intervertebral disc degeneration. |
| Literature | GENERIF_SUMMARY | These results first reveal an epigenetic alteration, aberrant methylation of PTEN, in nasopharyngeal carcinoma (NPC), which is probably an early event and may be regarded as a novel candidate biomarker for early stage of NPC detection and prevention. |
| Literature | GENERIF_SUMMARY | morgana causes PTEN destabilization, by inhibiting ROCK activity |
| Literature | GENERIF_SUMMARY | Suggest role for PTEN/mTOR/B7-H1 pathway in immune-resistance, immune escape and progression of pancreatic cancer. |
| Literature | GENERIF_SUMMARY | genetic variants in the PTEN, AKT1, MDM2 and p53 tumor suppressor-oncoprotein network may play roles in mediating the susceptibility to NPC in Chinese populations |
| Literature | GENERIF_SUMMARY | HIF-1alpha/PTEN/CD44v6/Survivin, as measured by immunohistochemistry, can be used to predict the prognosis and potential for invasion and metastasis in Asian patients with gastric cancer. |
| Literature | GENERIF_SUMMARY | PTEN downregulation is a fairly frequent event in colorectal carcinoma, but this apparently is not usually caused by two hits on the gene |
| Literature | GENERIF_SUMMARY | miR-21 is overexpressed in gastric cancer and its aberrant expression may have important role in gastric cancer growth and |

| Main Category | Category | Terms |
|---|---|---|
| | | dissemination by modulating the expression of the tumor suppressors PTEN and PDCD4 |
| Literature | GENERIF_SUMMARY | Mutations in PTEN predict resistance to anti-EGFR therapies. |
| Literature | GENERIF_SUMMARY | MiR-181a expression is increased and thus induces a metabolic shift in colon cancer cells by inhibiting the expression of PTEN. |
| Literature | GENERIF_SUMMARY | PTEN biallelic loss may be a prognostic marker for clear cell renal cell carcinoma |
| Literature | GENERIF_SUMMARY | PTEN was upregulated by BCA treatment or EGR1 overexpression. |
| Literature | GENERIF_SUMMARY | Plk1 as an important regulator of PTEN during the cell cycle |
| Literature | GENERIF_SUMMARY | Low expression of PTEN is associated with gastrointestinal bleeding in primary gastrointestinal stromal tumors. |
| Literature | GENERIF_SUMMARY | Findings suggest that PTEN as well as p53 plays a critical role in DNA damage response. There cooperation might be an essential blockage in development of mammary tumors.[Review] |
| Literature | GENERIF_SUMMARY | Data proposed a novel signaling mechanism by which Sema 3A regulates PTEN, FOXO 3a and MelCAM in a coordinated manner that leads to suppression of breast cancer growth and angiogenesis. |
| Literature | GENERIF_SUMMARY | Human fibroblast reprogramming to pluripotent stem cells is regulated by the miR19a/b-PTEN axis. |
| Literature | GENERIF_SUMMARY | Depleted PTEN expression associated with mono-allelic PTEN loss occurs frequently in imatinib-resistant GIST and might serve as a biomarker for stratifying patients for optimal treatment. |
| Literature | GENERIF_SUMMARY | we show that Pten expression in Rip(+) neurons has a critical role in diabetes pathogenesis through mediating the anti-inflammatory reflex. |
| Literature | GENERIF_SUMMARY | The observation that PTEN alterations develop subsequent to ERG fusion strongly suggests that ERG expression may directly drive development of PTEN aberrations in prostate cancer |
| Literature | GENERIF_SUMMARY | The presence of individual and multiple alterations of p53, p16 and PTEN were compared in in non-small cell lung carcinoma (NSCLC), glioma and breast carcinoma. |

| Main Category | Category | Terms |
|---|---|---|
| Literature | GENERIF_SUMMARY | Patients with CS with germline PTEN mutations are at higher risk for SMNs compared with the general population. |
| Literature | GENERIF_SUMMARY | inhibitory of cell growth by melittin might be led by HDAC2-mediated PTEN upregulation, Akt inactivation, and inhibition of the PI3K/Akt signaling pathways |
| Literature | GENERIF_SUMMARY | PTP-1B, SHP-2, and PTEN all play a functional role in Rb/E2F-associated apoptotic signal transduction |
| Literature | GENERIF_SUMMARY | studies identify Cdh1 as an important regulator of nuclear/chromatin PTEN during mitosis. |
| Literature | GENERIF_SUMMARY | we explored a synthetic lethal relationship between PTEN and APE1 in melanoma |
| Literature | GENERIF_SUMMARY | Suggest PI3 kinase inhibitor BENC-511 as a drug for the treatment of PTEN-deficient and androgen-independent prostate cancers. |
| Literature | GENERIF_SUMMARY | The specific role of AKT2 in tumor maintenance provides a rationale for the development of isoform-specific inhibitors for patients with PTEN-deficient cancers. |
| Literature | GENERIF_SUMMARY | In Graves disease orbital fibroblasts, relatively high basal PTEN levels prevent secreted IL-1RA expression or release. |
| Literature | GENERIF_SUMMARY | the PTEN tumor suppressor pathway has a role in carcinoma in situ of the bladder |
| Literature | GENERIF_SUMMARY | cancer associated fibroblasts with down-regulated miR-106b could significantly inhibit gastric cancer cell migration and invasion by targeting PTEN |
| Literature | GENERIF_SUMMARY | PTEN positivity strongly argues against a diagnosis of pancreatic ductal adenocarcinoma |
| Literature | GENERIF_SUMMARY | Study demonstrated that PTEN loss and MCT-1 induction synergistically promoted the neoplastic multinucleation via the Src/p190B signaling activation. |
| Literature | GENERIF_SUMMARY | We identified differential therapeutic vulnerabilities that emerge upon the loss of both PTEN and p53, and observed that combined inhibition of PARP and PI3K provides increased efficacy in hormone-insensitive advanced prostate cancer. |
| Literature | GENERIF_SUMMARY | These results demonstrate that the NF-kappab/miR-802/PTEN signaling pathway plays an important role in the development of cholesteatoma. |

| Main Category | Category | Terms |
|---|---|---|
| Literature | GENERIF_SUMMARY | Increased activity of NRF2 due to loss of PTEN is instrumental in human carcinogenesis and represents a novel therapeutic target |
| Literature | GENERIF_SUMMARY | proliferation of VSMC was inhibited in PTEN transfected VSMCs induced in PDGF conditioned medium |
| Literature | GENERIF_SUMMARY | PTEN IVS4 (-/-) genotype was significantly associated with increased risk of cancer especially for digestive tract cancer compared with (+/+) genotype. |
| Literature | GENERIF_SUMMARY | These findings suggest that resistance to TRAIL could induce epithelial-mesenchymal transition and enhance invasiveness by suppressing PTEN via miR-221. |
| Literature | GENERIF_SUMMARY | miR-185 inhibits Hepatocellular carcinoma cell growth by targeting DNMT1, leading to PTEN induction and Akt inhibition. |
| Literature | GENERIF_SUMMARY | MiR-17-5p is a predictive factor for chemotherapy response and a prognostic factor for overall survival in CRC, which is due to its regulation of PTEN expression. |
| Literature | GENERIF_SUMMARY | The study demonstrates that the expression of PTEN is directly regulated by miR-205 in endometrial cancer cells and leads to the inhibition of cellular apoptosis. |
| Literature | GENERIF_SUMMARY | Data indicate that among eight genotyped single-nucleotide polymorphisms (SNPs) in tensin homolog deleted on chromosome 10 (PTEN), rs11202607 was found associated with increased colorectal cancer (CRC) risk. |
| Literature | GENERIF_SUMMARY | results show that PI3Ks dimerize in vivo and that PI3K and PTEN activities modulate each other in a complex that controls cell PI(3,4,5)P3 levels. |
| Literature | GENERIF_SUMMARY | PTEN regulates leukemia stem cells and hematopoietic stem cells through different mechanisms. [review] |
| Literature | GENERIF_SUMMARY | Data indicate that chemokine CXCL8 signaling regulates chemokines CCL2 and CXCL12 expression and secretion in PTEN phosphohydrolase-deficient prostate cancer cells. |
| Literature | GENERIF_SUMMARY | TRX1 physically interacts with and reactivates PTP1B and PTEN through a direct thiol disulfide exchange. |
| Literature | GENERIF_SUMMARY | An enhanced engineered PTEN dramatically represses PIP3 signaling with increased tumor suppressor activities. |

| Main Category | Category | Terms |
|---|---|---|
| Literature | GENERIF_SUMMARY | Mutations in PTEN occurred in areas of chromosomal copy loss in chordoma tumor samples. |
| Literature | GENERIF_SUMMARY | A unified nomenclature and amino acid numbering for human PTEN. |
| Literature | GENERIF_SUMMARY | PTEN loss in Gleason score 6 biopsies identifies a subset of prostate tumors at increased risk of upgrading at radical prostatectomy |
| Literature | GENERIF_SUMMARY | PTEN status in Ewing sarcoma affects cellular responses to IGFI and mTOR-directed therapy. |
| Literature | GENERIF_SUMMARY | These results indicated that PTEN mutations and activation of the PI3K/Akt/mTOR signaling pathway may play a role in the biology of papillary tumors of the pineal region. |
| Literature | GENERIF_SUMMARY | Our findings demonstrate that the PTEN protein expression is significantly diminished in CSCC |
| Literature | GENERIF_SUMMARY | PTEN expression did not significantly affect survival outcomes of the present study |
| Literature | GENERIF_SUMMARY | the frequency of detected cytogenetic aberrations MYC/MYCN, EGFR and PDGRFA, homozygous deletion of the CDKN2A gene, and deletion of the PTEN gene, pediatric glioblastomas bear analogy to the subgroup of IDH1-mutant glioblastomas in adults. |
| Literature | GENERIF_SUMMARY | found miR-106b directly targeted PTEN at transcriptional and translational levels |
| Literature | GENERIF_SUMMARY | Results show that PTEN expression is modulated by Plk1. Its phosphorylation results in its inactivation promotimg the warburg effect. |
| Literature | GENERIF_SUMMARY | Studies indicate the importance of the EGFR/PI3K/PTEN/Akt/mTORC1/GSK-3 pathway in breast cancer. |
| Literature | GENERIF_SUMMARY | PTEN was neither a significant prognostic nor predictive factor. PIK3CA mutations were an adverse prognostic factor for survival but not predictive for lapatinib benefit. |
| Literature | GENERIF_SUMMARY | Results show that up-regulation of miR-21 and down-regulation of Pten in patients with non-small cell lung cancer negatively correlate with shorter disease-free survival and poor sensitivity to treatment with tyrosine kinase inhibitors. |
| Literature | GENERIF_SUMMARY | Upregulation of PTEN may attenuate the |

| Main Category | Category | Terms |
|---|---|---|
| | | adhesion, migration and invasion capabilities of osteosarcoma cells. This may occur via decreased FAK and MMP9 expression. |
| Literature | GENERIF_SUMMARY | Data suggest that combined low tumor expression of PTEN (phosphatase and tensin homologue) and PR/PGR (progesterone receptor) may be used as prognostic biomarkers in subjects with pancreatic neuroendocrine tumors following surgery. |
| Literature | GENERIF_SUMMARY | Results provide evidence that p62 contributes to HER2-induced mammary tumorigenesis through multiple signaling pathways and provide insights into its potential role in the regulation of the tumor suppressor PTEN. |
| Literature | GENERIF_SUMMARY | defective miR-7 regulation of PTEN contributes to B cell hyperresponsiveness in systemic lupus erythematosus. |
| Literature | GENERIF_SUMMARY | The knockdown of Notch1 in both hepatocellular carcinoma cell lines increased both the total expression of PTEN and its phosphorylated form but FAK and phospho-FAK expression was decreased following Notch1 depletion. |
| Literature | GENERIF_SUMMARY | Silencing Bcl-2 in PTEN-mutated prostate cancer cells enhances the apoptotic effects of combined starvation and taxol treatments, indicating that inhibition of Bcl-2 may be of significant value in PTEN-mutant tumor therapy. |
| Literature | GENERIF_SUMMARY | knocking-down of PTEN rescued the extent of suppressed cell migration and proliferation induced by miR-200a and miR-141 |
| Literature | GENERIF_SUMMARY | The regulatory mechanism of miR-218 in human hepatocellular carcinoma cells was acting through PTEN/AKT/PI3K pathway and possibly associated with HoxA10. |
| Literature | GENERIF_SUMMARY | Low PTEN expression is associated with testicular germ cell tumor. |
| Literature | GENERIF_SUMMARY | PTEN regulates plasma membrane expression of glucose transporter 1 and glucose uptake in thyroid cancer cells. |
| Literature | GENERIF_SUMMARY | miR-29b functions as a tumor suppressor in TSCC, and the miR-29b/Sp1/PTEN/AKT axis might represent a potential therapeutic target for TSCC intervention |
| Literature | GENERIF_SUMMARY | Due to the Akt activation, PTEN loss may play an important role in prostate cancer by |

| Main Category | Category | Terms |
|---|---|---|
| | | promoting cancer stemness through a mechanism that involves enhanced NF-kappaB signaling. |
| Literature | GENERIF_SUMMARY | Studies indicate that the PI3K/PTEN/Akt/mTOR pathway is in the pathogenesis of malignancies and development of resistance to anticancer therapies, and inhibitors are a promising therapeutic option. |
| Literature | GENERIF_SUMMARY | Data indicate that microRNA miR-181b could activate HSCs, at least in part, via tensin homologs deleted on chromosome 10 (PTEN)/Akt proto-oncogene protein pathway. |
| Literature | GENERIF_SUMMARY | Data suggest a novel mechanism whereby SDHD germline variants induce thyroid tumorigenesis mediated by PTEN accumulation in the nucleus. |
| Literature | GENERIF_SUMMARY | The data demonstrate large differences in invasion capacities of 5 tested glioblastoma cell lines differing in p53 and PTEN status, although histologically all cell lines belong to the glioblastoma grade IV. |
| Literature | GENERIF_SUMMARY | SIPL1 binds PTEN and enhances PTEN polyubiquitination. |
| Literature | GENERIF_SUMMARY | Data suggest that expression of PTEN (phosphatase and tensin homolog) and PP2A is down-regulated in adenomyosis as compared to normal endometrium; expression of survivin appears to be up-regulated. |
| Literature | GENERIF_SUMMARY | PTEN loss was significantly associated with OS and time to MBM in patients with BRAF(V600) mutations. |
| Literature | GENERIF_SUMMARY | Circulating MIR-21 and PTEN might represent new complementary tumour markers for oral squamous cell carcinoma. |
| Literature | GENERIF_SUMMARY | PTEN expression did not correlate with an inhibition of cellular proliferation or control of AKT phosphorylation |
| Literature | GENERIF_SUMMARY | Loss of PTEN expression was more common in endometrial endometrioid adenocarcinomas than in ovarian endometrioid adenocarcinomas. |
| Literature | GENERIF_SUMMARY | The OHPg/PR-B dependent induction of PTEN causes the down-regulation of PI3K/AKT signal, switching on the autophagy process |
| Literature | GENERIF_SUMMARY | Data suggest that anionic lipids (such as phosphatidylinositol phosphates) play |

| Main Category | Category | Terms |
|---|---|---|
| | | crucial role in localization of peripheral membrane proteins (such as PTEN, talin-1, auxilin-1, and Ci-VSP [voltage-sensor containing phosphatase]). [review-like article] |
| Literature | GENERIF_SUMMARY | A novel IL-6/Stat3-miR-17-92 cluster-PTEN signaling axis is crucial for cholangiocarcinogenesis and tumor progression. |
| Literature | GENERIF_SUMMARY | Results identified a new class of cancer-associated PTEN mutations that specifically inhibit the recruitment of PTEN to the plasma membrane as a result of impaired interactions with phospholipids. |
| Literature | GENERIF_SUMMARY | PTEN and PI3K at the protein and at the DNA level as well as the presence of AKT1 mutations in skin lesions from 50 patients with mycosis fungoides |
| Literature | GENERIF_SUMMARY | we provided convincing evidence that upregulation of miR-494 was associated with tumor aggressiveness and tumor metastasis and promoted cell migration and invasion by targeting PTEN gene in colorectal cancer. |
| Literature | GENERIF_SUMMARY | In established stably transfected HCT-8 cell lines expressing siRNA targeting the PTEN gene, constitutive knockdown of the PTEN gene significantly promoted cell proliferation and migration. |
| Literature | GENERIF_SUMMARY | The present study examined molecular and phenotypic characteristics of individuals with germline heterozygous PTEN mutations and autism spectrum disorder (ASD) |
| Literature | GENERIF_SUMMARY | Most notably, WWP1 downregulation both inactivated PTEN-Akt signaling pathway in MKN-45 and AGS cells.  our findings suggest that WWP1 acts as an oncogenic factor and should be considered as a novel interfering molecular target for gastric carcinoma |
| Literature | GENERIF_SUMMARY | The proliferationpromoting effect of PTEN knockdown was similar to that of miR214 overexpression. |
| Literature | GENERIF_SUMMARY | PTEN mutation is associated with neoplasms. |
| Literature | GENERIF_SUMMARY | Deletion of Pten and p53 in mammary epithelium accelerates triple-negative breast cancer with dependency on eEF2K activity. |
| Literature | GENERIF_SUMMARY | The activity and function of PTEN are regulated by coordinated epigenetic, |

| Main Category | Category | Terms |
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| | | transcriptional, posttranscriptional, and posttranslational modifications. |
| Literature | GENERIF_SUMMARY | BCR-ABL promotes PTEN downregulation through a MEK dependent pathway. |
| Literature | GENERIF_SUMMARY | mechanistic studies disclosed that EBV-miR-BART7-3p targeted a major human tumor suppressor PTEN, modulating PI3K/Akt/GSK-3beta signaling |
| Literature | GENERIF_SUMMARY | In a population of Chinese patients with gastric cancer and poor prognosis, PTEN protein was down-regulated and IGF1R was up-regulated. |
| Literature | GENERIF_SUMMARY | Data show that microRNAs miR-26a and miR-214 directly targeted PTEN phosphohydrolase. |
| Literature | GENERIF_SUMMARY | Activation of miR-21 by STAT3 induces proliferation and suppresses apoptosis in nasopharyngeal carcinoma by targeting PTEN gene. |
| Literature | GENERIF_SUMMARY | Enterolactone alters the expression of the miR-106b cluster leading to increased PTEN expression in prostate cancer cells. |
| Literature | GENERIF_SUMMARY | Transnitrosylation from DJ-1 to PTEN decreases PTEN phosphatase activity, thus promoting neuron survival in parkinson's disease model. |
| Literature | GENERIF_SUMMARY | HCV NS5A down-regulates PTEN expression through a cooperation of ROS-dependent and -independent pathways that subsequently drives a PTEN-PI3K/Akt feedback loop to support cell survival. |
| Literature | GENERIF_SUMMARY | these results suggest that PTEN is required for apoptosis-mediated cavitation during epithelial morphogenesis by regulating the expression of HIF-2alpha and Bnip3 |
| Literature | GENERIF_SUMMARY | parallel genetic evolution of separate metastatic sites with different PTEN genomic alterations leads to a convergent PTEN-null phenotype resistant to PI(3)Kalpha inhibition |
| Literature | GENERIF_SUMMARY | The PTEN/AKT signaling axis was defined as a critical pathway regulated by betaIII-tubulin in NSCLC cells. |
| Literature | GENERIF_SUMMARY | analysis of the PTEN binding site and mechanism of phosphatidylinositol 4,5-bisphosphate activation |
| Literature | GENERIF_SUMMARY | the endogenous S-sulfhydration of PTEN via CBS/H2S plays a role in preventing the S-nitrosylation that would inhibition its |

| Main Category | Category | Terms |
|---|---|---|
| | | enzymatic activity under physiological condition |
| Literature | GENERIF_SUMMARY | In particular, the bisperoxovanadate compounds were found to inhibit PTEN poorly in the presence of reducing agents including the cellular redox buffer glutathione. |
| Literature | GENERIF_SUMMARY | With potential relevance for tumor suppression, a PDZ-mediated interaction of PTEN and PTPN13 is described. |
| Literature | GENERIF_SUMMARY | EGFR activity in this PTEN null background also results in Wnt/beta-catenin signaling and activation of NF-kappaB. |
| Literature | GENERIF_SUMMARY | Study shows that although HER2 expression itself was not a prognostic factor, the combination of HER2 expression and loss of heterozygosity of PTEN exacerbates the malignant potential of gastric cancer through its proliferative function. |
| Literature | GENERIF_SUMMARY | alleles causing incomplete loss of PTEN function are more commonly linked to autism than to severe PTEN hamartoma tumour cases. |
| Literature | GENERIF_SUMMARY | Combined analysis of PIK3CA mutation and PTEN expression may serve as critical indicators to identify patients unlikely to respond to trastuzumab. |
| Literature | GENERIF_SUMMARY | SOX2, a predictor of survival in gastric cancer, inhibits cell proliferation and metastasis by regulating PTEN. |
| Literature | GENERIF_SUMMARY | PTEN testing should be considered in any child presenting with severe macrocephaly and another feature of PTEN hamartoma tumor syndromes |
| Literature | GENERIF_SUMMARY | Hepatitis C virus (HCV) proteins could modulate PTEN by activating NF-kappaB. |
| Literature | GENERIF_SUMMARY | Data show PTEN protein frameshift mutation and a SMAD7 protein missense mutation occurring in a father and son who had a syndrome of gastrointestinal hamartomatous and ganglioneuromatous polyposis, and who both developed esophageal adenocarcinoma. |
| Literature | GENERIF_SUMMARY | Expression of PTEN in several HCC cell lines was lower than in normal cell lines |
| Literature | GENERIF_SUMMARY | PTEN is a good prognostic marker for vulvar cancer, correlating with higher rates of patient survival. |
| Literature | GENERIF_SUMMARY | The enhanced antitumor activity generated |

| Main Category | Category | Terms |
|---|---|---|
| | | by Ad.RGD-ING4-PTEN was closely associated with activation of the intrinsic and extrinsic apoptotic pathways and additive inhibition of tumor angiogenesis both in vitro and in vivo. |
| Literature | GENERIF_SUMMARY | PTEN mutation is associated with glioma. |
| Literature | GENERIF_SUMMARY | our data suggest that PTEN is among the active regulatory components prior to human NK cells transitioning from the noncytolytic CD56(bright) NK cell to the cytolytic CD56(dim) NK cells. |
| Literature | GENERIF_SUMMARY | data indicate that miRNA-21 regulates hTERT expression via the PTEN/ERK1/2 signaling pathway, therefore controlling CRC cell line growth. MiRNA-21 may serve as a novel therapeutic target in CRC |
| Literature | GENERIF_SUMMARY | Results show that PTEN activity depends on the concentration of salt and PIP2 product while PTEN-L, an alternative mRNA translation product is constitutively active. This activation is abrogated in the K13A variant disrupting PIP2 putative binding site. |
| Literature | GENERIF_SUMMARY | Results show that PTEN loss is a common event in high-grade serous ovarian cancer and defines a subgroup with significantly worse prognosis. |
| Literature | GENERIF_SUMMARY | Results showed that loss of PTEN function can be a driver event and prognostic factor in high-grade serous ovarian cancer. |
| Literature | GENERIF_SUMMARY | lung cancer patients with Oct4 high, PTEN low and TNC high expression profile significantly correlated with poor disease-free survival |
| Literature | GENERIF_SUMMARY | A significantly negative correlation between PTEN and hTERT gene expression indicates that hTERT activation and upregulation may be conferred by the loss of PTEN gene expression in hepatocellular carcinoma (HCC). |
| Literature | GENERIF_SUMMARY | we detected higher activity of AKT (p = 0.011) and FAK (p = 0.054), higher integrin alpha5 expression (p = 0.052) and lower PTEN expression in primary cells from patients with bone metastasis compared to nonmetastasizing cells. |
| Literature | GENERIF_SUMMARY | This study suggests that PTEN knockout potentiates the invasiveness of colorectal cancer spheroidal cells through a 3D extracellular matrix. |
| Literature | GENERIF_SUMMARY | These results illuminate the complex interplay of AR, ETV1, and PTEN pathways in |

| Main Category | Category | Terms |
|---|---|---|
| | | pre-cancerous neoplasia and early tumorigenesis |
| Literature | GENERIF_SUMMARY | miR-93 stimulated cell proliferation, migration, and invasion through the oncogenic c-Met/PI3K/Akt pathway and also inhibited apoptosis by directly inhibiting PTEN and CDKN1A expression in human hepatocellular carcinoma. |
| Literature | GENERIF_SUMMARY | the oncogenic miR-17-92 cluster and its paralogs are involved in cutaneous B-cell lymphoma progression, and the downregulation of the target gene PTEN is associated with shorter disease-free survival |
| Literature | GENERIF_SUMMARY | miR-130b plays an oncogenic role in ESCC cells by repressing PTEN expression and Akt phosphorylation, which would be helpful in developing miRNA-based treatments for ESCC. |
| Literature | GENERIF_SUMMARY | we propose that RRM1 protein can inhibit CRC invasion and metastasis at the advanced stage by regulating PTEN transactivation and its downstream pathways in addition to forming an RR holoenzyme for supporting cancer proliferation. |
| Literature | GENERIF_SUMMARY | CpG hypermethylation of the PTEN gene contributes to the decreased expression of PTEN during acquired resistance to gefitinib or erlotinib. |
| Literature | GENERIF_SUMMARY | PTEN is involved in the regulation of cytoskeletal rearrangement in podocytes and that loss of PTEN predisposes to the development of proteinuria and diabetic nephropathy |
| Literature | GENERIF_SUMMARY | Hepatitis C virus infection suppresses the insulin signaling pathway and promotes insulin resistance by repressing PTEN, subsequently leading to decreased levels of IRS-1 and increased levels of Ser307-phosphorylated IRS-1. |
| Literature | GENERIF_SUMMARY | These findings suggest that prenylation inhibition may be able to target melanoma cells with mutant NRAS or with mutant BRAF and PTEN. |
| Literature | GENERIF_SUMMARY | Variants of human PTEN linked to somatic cancers and disorders on the autism spectrum are shown to be impaired in their conformational stability, catalytic activity, or both. |
| Literature | GENERIF_SUMMARY | Down-expression of PTEN could attenuate the inhibitory effects of knockdown miR-32 |

| Main Category | Category | Terms |
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| | | on the proliferation, migration, and invasion of liver cancer cells, suggesting that miR-32 could be a potential target for hepatocellular carcinoma treatment. |
| Literature | GENERIF_SUMMARY | Increased plasma levels of FABP4 and PTEN is associated with more severe insulin resistance in women with gestational diabetes mellitus. |
| Literature | GENERIF_SUMMARY | miR-21 is a key regulator of fibrogenic epithelial-mesenchymal transition in hepatocytes via PTEN/Akt pathway. |
| Literature | GENERIF_SUMMARY | AKR1C3 mediated doxorubicin resistance might be resulted from the activation of anti-apoptosis PTEN/Akt pathway via PTEN loss. |
| Literature | GENERIF_SUMMARY | miR-494-3p inhibitor could prevent migration, invasion, proliferation, and promote apotosis in gliomas through PTEN/AKT pathway. |
| Literature | GENERIF_SUMMARY | Germline promoter hypermethylation of KLLN is associated with particular malignant and benign Cowden syndrome features, which is dependent on the PTEN mutation status. |
| Literature | GENERIF_SUMMARY | Results suggest that the loss of the susceptible region on chromosome 10q, which implicates PTEN, FAS and PAPSS2 may serve as genetic predictors of PSA recurrence after radical prostatectomy. |
| Literature | GENERIF_SUMMARY | Glioma cell VEGFR-2 confers resistance to chemotherapeutic and antiangiogenic treatments in PTEN-deficient glioblastoma. |
| Literature | GENERIF_SUMMARY | Thus, targeting miR-181c and restoration of PTEN can be used in conjunction with other therapies to prevent progression of IBC. |
| Literature | GENERIF_SUMMARY | PTEN hypermethylation is a rare event and did not play an important role in the prognosis and loss of PTEN expression in esophageal squamous cell carcinoma |
| Literature | GENERIF_SUMMARY | heterogeneity of ERG and PTEN is critical for developing optimal yet cost-effective strategies to identify these underlying molecular abnormalities of prostate cancer |
| Literature | GENERIF_SUMMARY | High Tau-protein and low PTEN expression showed a significant association with poor response to H + P chemotherapy in patients with HER2-positive advanced breast cancer. |
| Literature | GENERIF_SUMMARY | SLUG regulates PTEN expression in prostate cancer cell lines. |
| Literature | GENERIF_SUMMARY | PTEN IVS4 (rs3830675)gene polymorphism |

| Main Category | Category | Terms |
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| | | is associated with reduced risk of cancer. |
| Literature | GENERIF_SUMMARY | miR-494 upregulation was associated with PTEN downregulation. Inhibition of miR-494 suppressed cell proliferation and growth by directly targeting the 3'-UTR of PTEN mRNA. |
| Literature | GENERIF_SUMMARY | detected expression of a tumor-suppressor gene, phosphatase and tensin homolog (PTEN) |
| Literature | GENERIF_SUMMARY | the activation of PTEN and NHERF-1 may impede the evolution of macrophages beyond the M1 into M2 phenotype in tumor microenvironment. |
| Literature | GENERIF_SUMMARY | DNA damage induced phosphorylation of C-terminal STT motif of PTEN is necessary for DNA repair. |
| Literature | GENERIF_SUMMARY | Data show that functional loss of PTEN has been detected in a variety of bone malignancies. PTEN loss leads to aberrant activation of oncogenic PI3K/Akt/mTOR pathway which is instrumental in promoting tumor development and progression. [review] |
| Literature | GENERIF_SUMMARY | the downregulation of muscle PTEN may explain the retention of insulin sensitivity with higher adiposity in women compared to men. |
| Literature | GENERIF_SUMMARY | Inactivation of the FAK signaling pathway by either genetic or pharmacologic methods significantly sensitizes both murine and human PTEN-null T-ALL cells to PI3K/AKT/mTOR inhibition when cultured in vitro on feeder layer cells or a matrix and in vivo |
| Literature | GENERIF_SUMMARY | PTPN14 and PTEN were identified as direct and functional targets of miR-21. |
| Literature | GENERIF_SUMMARY | PTEN loss initiates tubular dysfunction via SMAD3- and p53-mediated fibrotic gene induction, with accompanying PAI-1-dependent proliferative arrest, and cooperates with TGFbeta1 to induce the expression of profibrotic genes and tubular apoptosis. |
| Literature | GENERIF_SUMMARY | reveal a positive feedback loop that is activated in response to PTEN loss and sustains BMI-1 |
| Literature | GENERIF_SUMMARY | Activation of nuclear PTEN by inhibition of Notch signaling induces G2/M cell cycle arrest in gastric cancer. |
| Literature | GENERIF_SUMMARY | Our data identify IL6 detection as a potential causal biomarker for MYC-driven metastasis after loss of PTEN and p53. |

| Main Category | Category | Terms |
|---|---|---|
| Literature | GENERIF_SUMMARY | PTEN inhibits PREX2-catalyzed activation of RAC1 to restrain tumor cell invasion. |
| Literature | GENERIF_SUMMARY | NUAK2 silencing and inactivation of the PI3K pathway efficiently controlled CDK2 expression, whereas CDK2 inactivation specifically abrogated the growth of NUAK2-amplified and PTEN-deficient melanoma cells. |
| Literature | GENERIF_SUMMARY | Data suggest that PTEN protein could be a biomarker for predicting response to focal Adhesion Kinase (FAK)-targeted therapy during clinical development. |
| Literature | GENERIF_SUMMARY | Notably, miR-21 regulated the potential anticancer effects of icariin on cell proliferation and apoptosis by targeting PTEN, RECK and Bcl-2 in the ovarian cancer A2780 cells |
| Literature | GENERIF_SUMMARY | Significant downregulation of LKB1 and PTEN mRNA expression levels were detected in tumour tissues compared to non-cancerous tissues. |
| Literature | GENERIF_SUMMARY | Loss of PTEN is associated with breast cancer. |
| Literature | GENERIF_SUMMARY | Combined detection of PTEN and miR-494 can aid in determining malignancy degree and the prognosis of patients with non-small cell lung cancer |
| Literature | GENERIF_SUMMARY | PAR1 plays a role in thrombin-mediated reduction of PTEN expression. |
| Literature | GENERIF_SUMMARY | We observed decreased Bcl-2 expression and increased Bax and Pten expression induced by TIEG1 in 4 different leukemia cell lines. |
| Literature | GENERIF_SUMMARY | The association of MVP amplification and PTEN deletion could be a potential biomarker in glioblastoma. |
| Literature | GENERIF_SUMMARY | Basal cell carcinoma of the prostate is an aggressive tumor with frequent loss of PTEN expression and overexpression of Epidermal growth factor receptor |
| Literature | GENERIF_SUMMARY | our results offer a mechanistic rationale for clinical evaluation of ATM inhibitors in PTEN-deficient tumors |
| Literature | GENERIF_SUMMARY | findings reveal a novel mechanism by which PTEN inhibits the progression of colon cancer by inhibiting paxillin expression downstream of PI3K/AKT/NF-kappaB pathway |
| Literature | GENERIF_SUMMARY | Our functional dissection of the PTEN N- |

| Main Category | Category | Terms |
|---|---|---|
| | | terminus provides the basis for a systematic analysis of tumor-related and experimentally engineered PTEN mutations. |
| Literature | GENERIF_SUMMARY | Data show that silencing of PMEPA1 protein facilitates the growth of prostate cancer cells and modulates androgen receptor (AR) through NEDD4 ubiquitin protein ligase and PTEN protein. |
| Literature | GENERIF_SUMMARY | Data shows that PTEN protein is expressed in both nucleus and cytosol of lung neuroendocrine tumors (NET) and suggests that cellular ubiquitinylation and sumoylation likely influence the functional PTEN loss in high grade lung NETs. |
| Literature | GENERIF_SUMMARY | The DJ-1-induced PTEN downregulation is associated with proliferative and invasive activity of laryngeal cancer cells. |
| Literature | GENERIF_SUMMARY | KLF5 loss enhances tumor angiogenesis by attenuating PI3K/AKT signaling and subsequent accumulation of HIF1alpha in PTEN deficient prostate tumors. |
| Literature | GENERIF_SUMMARY | ERG overexpression was associated with favorable recurrence-free survival after radical prostatectomy for prostate cancer, whereas PTEN loss was associated with unfavorable outcomes. |
| Literature | GENERIF_SUMMARY | PTEN is a potent tumor suppressor in human salivary gland tumors, and targeting PI3K/mTOR pathway may be effective in the targeted therapy for human SGT patients with loss of PTEN expression. |
| Literature | GENERIF_SUMMARY | PTEN expression regulates extracellular trap formation in neutrophils. |
| Literature | GENERIF_SUMMARY | SMYD2-mediated methylation negatively regulates PTEN tumor suppressor activity and results in activation of the phosphatidylinositol 3-kinase-AKT pathway. |
| Literature | GENERIF_SUMMARY | In HEK 293 cells, transfected with human uPAR, uPAR overexpression down-regulates PTEN and activates the PI3K/Akt-pathway. |
| Literature | GENERIF_SUMMARY | The PTEN autism spectrum disorder alleles are hypomorphic in regulating cell size and the parvalbumin/somatostatin ratio in comparison to WT PTEN. |
| Literature | GENERIF_SUMMARY | PTEN deletion is strongly associated with seminal vesicle involvement and extracapsular extension in Prostate cancer. |
| Literature | GENERIF_SUMMARY | Low expression of PTEN in clearcell renal cell carcinoma contributes to chemoresistance through activating the |

| Main Category | Category | Terms |
|---|---|---|
| | | Akt/HDM2 signaling pathway. |
| Literature | GENERIF_SUMMARY | negative expression of PTEN could be a useful biomarker to predict high risk for lymphatic-related metastasis in esophageal squamous cell carcinoma |
| Literature | GENERIF_SUMMARY | The present study showed the upregulation of miR-21 in invasive cervical cancers, and confirmed the promotion of miR-21 with regard to the proliferation, migration and invasion in cervical cancer cells via inhibiting the PTEN expression. |
| Literature | GENERIF_SUMMARY | results suggest that PTENP1 functions as a novel tumor suppressor in gastric cancer and its suppressive ability may be involved in the modulation of PTEN |
| Literature | GENERIF_SUMMARY | PTEN is a direct target of MIR205 in lung cancer tissue. MIR205 might inhibit lung cancer cell proliferation by regulating PTEN expression. |
| Literature | GENERIF_SUMMARY | PTEN signaling pathway might be used as potential therapeutic target in triple-negative breast cancers since loss of its expression is shown to be significantly associated with this aggressive subtype of breast cancer. |
| Literature | GENERIF_SUMMARY | PTEN alterations were found in up to one-third of colorectal cancers but did not impact CRC patients' prognosis. |
| Literature | GENERIF_SUMMARY | Study confirmed that PTEN might have a role in pathogenesis in a subset of PTC. |
| Literature | GENERIF_SUMMARY | miR-377 downregulated p53, PTEN and TIMP1 expression by directly targeting the 3'-untranslated region of these target genes. Collectively, miR-377 potentially served as a new molecular predictive biomarker of GC tumorigenesis and prognosis |
| Literature | GENERIF_SUMMARY | Results show that the deletion of 16q and PTEN cooperatively drives prostate cancer progression, which could be of clinical value for preoperative risk assessment of the clinically most challenging group of prostate cancers. |
| Literature | GENERIF_SUMMARY | Short-Term PTEN Inhibition Improves In Vitro Activation of Primordial Follicles, Preserves Follicular Viability, and Restores AMH Levels in Cryopreserved Ovarian Tissue From Cancer Patients. |
| Literature | GENERIF_SUMMARY | specific PTEN/10q haplotypes are significantly different between European American(EA)and African-American (AA) individuals (p |

| Main Category | Category | Terms |
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| Literature | GENERIF_SUMMARY | PSGR overexpression synergizes with loss of PTEN to accelerate prostate cancer development, and present a novel bigenic mouse model that mimics the human condition |
| Literature | GENERIF_SUMMARY | results of our study argue against a tumor-promoting role of SOX9 in prostate cancer, but demonstrate that loss of SOX9 expression characterizes a particularly aggressive subset of ERG positive cancers harboring PTEN deletion |
| Literature | GENERIF_SUMMARY | The impact of PTEN loss on tumor biology and clinical outcomes must be interpreted in the context of body mass index. |
| Literature | GENERIF_SUMMARY | Overexpression of miR-196a promoted proliferation of MG63 and U2OS cells by modulating the PTEN/PI3K/Akt/FOXO1 signaling pathway. |
| Literature | GENERIF_SUMMARY | cyclin D1 expression cooperates with KRAS and PTEN alterations in pathogenesis of NSCLC, and they could serve as potential multiple molecular markers for specific subgroups of NSCLC patients as well as prognostic markers for this type of cancer. |
| Literature | GENERIF_SUMMARY | results suggested that Piwil1 caused the loss of PTEN expression through DNMT1-mediated PTEN hypermethylation |
| Literature | GENERIF_SUMMARY | impaired functions of the FOXO1/3/PTEN pathways lead to dramatic changes in the molecular program within granulosa cells. |
| Literature | GENERIF_SUMMARY | Bioinformatics analysis combined with tumor metastasis PCR array showed that matrix metalloproteinase 2 (MMP2) and PTEN could be important target genes of miR-29b. |
| Literature | GENERIF_SUMMARY | the experimental results and the molecular modeling data from the interactions of AFP and HSA with PTEN will help us to identify targets for designing drugs and vaccines against human hepatocellular carcinoma. |
| Literature | GENERIF_SUMMARY | this study explored the relationship between polymorphisms in PTEN and primary ovarian insufficiency in a Han Chinese population and suggests that polymorphisms in PTEN may not be associated with a risk of POI. |
| Literature | GENERIF_SUMMARY | Berberine reverses drug resistance by sensitizing human ovarian cancer cells to cisplatin through miR-93/PTEN/Akt signaling pathway. |
| Literature | GENERIF_SUMMARY | identified a distinctive colorectal cancer |

| Main Category | Category | Terms |
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| | | (CSC)-like pre-invasive transition zone in PTEN-haploinsufficient CRC that shows convergent on-off regulation of the PTEN/AKT, TGF-beta/SMAD and Wnt/beta-catenin pathways |
| Literature | GENERIF_SUMMARY | The newly identified pathway HMGB1/RAGE/miR221/222 may represent an effective way of tumor escape from immune surveillance that could be used to develop new therapeutic strategies against anaplastic tumors. |
| Literature | GENERIF_SUMMARY | Exon 7 PTEN gene mutation is not associated with Endometrial Carcinoma. |
| Literature | GENERIF_SUMMARY | Loss of Expression of PTEN is Associated with Cancer. |
| Literature | GENERIF_SUMMARY | PTEN regulates IGF-1R-mediated therapy resistance in melanoma |
| Literature | GENERIF_SUMMARY | An association between the morphological appearance of endometrial hyperplasia and endometrial carcinoma and the degree of PTEN alterations, was investigated. |
| Literature | GENERIF_SUMMARY | Case Report |
| Literature | GENERIF_SUMMARY | It also inhibited extrinsic pathway protein caspase 8, growth-related protein PI3K and further activated PTEN in combination with cisplatin. |
| Literature | GENERIF_SUMMARY | recombinant adenovirus-mediated overexpression of KRT10 and PTEN may improve the cisplatin resistance of ovarian cancer in vitro and in vivo |
| Literature | GENERIF_SUMMARY | Germ-line mutations in PTEN cause a familial tumor syndrome, the PTEN hamartoma tumor syndrome (also known as Cowden syndrome), characterized by an increased risk to develop malignant tumors in several organs, including the thyroid. |
| Literature | GENERIF_SUMMARY | Data highlight the interplay of PTEN with Rad51 in promoting stalled fork restart. Loss of PTEN may initiate a replication stress cascade that progressively deteriorates through the cell cycle. |
| Literature | GENERIF_SUMMARY | A phosphorylation-mediated molecular switch comprising DLC), TNS3, PTEN and PI3K controls the spatiotemporal activation of Rac1 and RhoA, thereby initiating directional cell migration induced by growth factors. |
| Literature | GENERIF_SUMMARY | High levels of PTEN expression pointed to a favorable prognostic marker in non-small cell lung cancer. |

| Main Category | Category | Terms |
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| Literature | GENERIF_SUMMARY | we demonstrate that PTEN protein expression is a significant indicator of prognosis in ovarian/peritoneal high-grade serous carcinoma |
| Literature | GENERIF_SUMMARY | PTEN loss was enriched among ERG positive prostatic ductal adenocarcinomas. |
| Literature | GENERIF_SUMMARY | 8% of the ovarian tumors showed loss of PTEN, most commonly among endometrioid tumors. |
| Literature | GENERIF_SUMMARY | recurrence rate of p27 and/or PTEN-negative patients was higher than that of the positive ones,that should be followed up closely after treatment |
| Literature | GENERIF_SUMMARY | HER2 and PTEN expression appears as early as the 9-12 weeks period in the digestive tract, but HER2 expression decreases in the 21-24 weeks period and then disappears. EGFR appears only during the 13-16 weeks period and p53 is strong until week 21. |
| Literature | GENERIF_SUMMARY | We show that the decrease in PI(4,5)P2 level under non-stimulated conditions inhibits PTEN activity leading to the aberrant activation of the oncoprotein Akt. As well as defining a novel mechanism of Akt phosphorylation with important therapeutic consequences, we also demonstrate that differential expression levels of FGFR2, Plc11 and Grb2 correlate with patient survival |
| Literature | GENERIF_SUMMARY | for pediatric patients treated according to current protocols, routine screening for PTEN or RAS abnormalities at diagnosis is not warranted to further refine risk stratification |
| Literature | GENERIF_SUMMARY | As a consequence, homodimeric but not monomeric p85alpha suppresses the phosphatidylinositol 3-kinase pathway by protecting PTEN from E3 ligase WWP2-mediated proteasomal degradation. |
| Literature | GENERIF_SUMMARY | Potential compensatory indels were identified in two tumor suppressor genes, TP53 and PTEN, where compensatory indels are composed of frameshift indels that can recover the original reading frame. |
| Literature | GENERIF_SUMMARY | PTEN overexpression and lithium treatment separately reduce cell migration, colony formation, and invasion. |
| Literature | GENERIF_SUMMARY | MicroRNA-21 cross talk with PTEN and p53 in bladder cancer. |
| Literature | GENERIF_SUMMARY | there were no significant differences in the |

| Main Category | Category | Terms |
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| | | distribution of the genotype and allele frequencies of polymorphisms of the PTEN gene between the OSCC patients and controls in a Chinese Han population. |
| Literature | GENERIF_SUMMARY | Breast cancer patients with strong PTEN expression survived 3 years without metastases after adjuvant trastuzumab therapy. |
| Literature | GENERIF_SUMMARY | Expression levels of MAGI3 and PTEN were significantly downregulated in gliomas. |
| Literature | GENERIF_SUMMARY | work identifies FAK as a target of p110delta PI3K that links RhoA with PTEN and establishes for the first time that PTEN is a substrate of FAK-mediated Tyr phosphorylation |
| Literature | GENERIF_SUMMARY | miR-214 is negatively correlated with PTEN in osteosarcoma tissues. |
| Literature | GENERIF_SUMMARY | WM130 could inhibit cell proliferation, invasion, migration and induced apoptosis in HCC cells by suppressing EGFR/ERK/ MMP-2 and PTEN/AKT signaling pathways. |
| Literature | GENERIF_SUMMARY | These findings suggest that PTEN is implicated in multiple pathways and plays a crucial role in the pathogenesis of idiopathic pulmonary fibrosis. |
| Literature | GENERIF_SUMMARY | miRNA-1908 functions as an oncogene in glioblastoma by repressing the PTEN pathway. MiR-1908 is a potential new molecular marker for predicting the risk of recurrence and prognosis of glioblastoma. |
| Literature | GENERIF_SUMMARY | the down-regulation of NDRG2 expression promotes the inactive phosphorylated form of PTEN, which results in constitutively active PI3K/AKT signaling in various cancer cell types. |
| Literature | GENERIF_SUMMARY | Down-regulation of the PTEN protein is associated with breast carcinoma. |
| Literature | GENERIF_SUMMARY | OTUD3 is an essential regulator of PTEN and the OTUD3-PTEN signalling axis plays a critical role in tumour suppression. |
| Literature | GENERIF_SUMMARY | miR-21 has a role in upregulating PTEN, RECK and PDCD4 in glioma |
| Literature | GENERIF_SUMMARY | PTEN inactivation could be a rare event, but with a poor prognosis in advanced gallbladder cancer |
| Literature | GENERIF_SUMMARY | PPARgamma activation can activate PTEN expression, thereby suppressing the expression of MMP2 and hence inhibiting the invasion and metastasis of pancreatic |

| Main Category | Category | Terms |
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| | | cancer cells. |
| Literature | GENERIF_SUMMARY | The C-terminal PTEN tail functions in stabilizing the homodimer; tail phosphorylation interferes with this stabilization. |
| Literature | GENERIF_SUMMARY | We conclude that one mechanism by which lncRNAs function in in tumorigenesis is as ceRNAs for tumor suppressor mRNAs. |
| Literature | GENERIF_SUMMARY | Hepatocellular carcinoma miR-21 participates in the acquired resistance of sorafenib by suppresing autophagy through the Akt/PTEN pathway. |
| Literature | GENERIF_SUMMARY | PTEN insufficiency modulates estrogen receptor positive breast cancer cell cycle progression and increases cell growth in vitro and in vivo. |
| Literature | GENERIF_SUMMARY | MiRNA-10a is upregulated in non-small cell lung carcinoma and activates PTEN/AKT/ERK pathway. |
| Literature | GENERIF_SUMMARY | ectopically expressed miR-17, miR-20a and miR-106b directly target PTEN 3'UTR to reduce its expression, an effect rescued upon treatment with resveratrol and pterostilbene |
| Literature | GENERIF_SUMMARY | Study shows that miR-1908 is overexpressed in osteosarcoma cells and associated with poor patient survival. There is evidence indicating that miR-1908 has oncogenic and invasion regulatory effects mediated by PTEN in osteosarcoma cells. |
| Literature | GENERIF_SUMMARY | MTOR-low or PTEN-high patients had better 3-year rates of OS. |
| Literature | GENERIF_SUMMARY | PTEN loss is detectable in endometrial tissues that are benign and malignant, with substantially more frequent loss in endometrial cancer compared with benign endometrium |
| Literature | GENERIF_SUMMARY | The prognostic and predictive value of PTEN in CTCs warrants investigation in CRPC clinical trials of PI3K/AKT-targeted therapies. |
| Literature | GENERIF_SUMMARY | reduced AZGP1 expression is strongly related to adverse prostate cancer prognosis, independently of established clinic-pathological variables and PTEN deletions. |
| Literature | GENERIF_SUMMARY | a decrease in PTEN facilitates ROS/SHP2 signaling, causing lung cancer cells to become unresponsive to IFN-gamma |

| Main Category | Category | Terms |
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| Literature | GENERIF_SUMMARY | The upregulation of miR-221 significantly reduced the expression of PTEN. |
| Literature | GENERIF_SUMMARY | PTEN plays an important role in multicilia formation and cilia disassembly by controlling the phosphorylation of Dishevelled. |
| Literature | GENERIF_SUMMARY | PTEN maintains genome stability through regulation of DNA replication. |
| Literature | GENERIF_SUMMARY | Results identify PTEN as the substrate of AIF oxidoreductase and reveal a novel function for AIF in controlling tumor metastasis. |
| Literature | GENERIF_SUMMARY | PTEN IVS4 (-/-) genotype was significantly associated with increased risk of cancer especially for digestive tract cancer compared to the (+/+) genotype; a similar phenomenon was observed in the (-) allele of PTEN IVS4 polymorphism compared to the (+) allele and the recessive effect model [meta-analysis] |
| Literature | GENERIF_SUMMARY | Results indicate that PTEN loss occurs in cooperation with TMPRSS2-ERG fusion in prostate cancer and the majority of the samples harbor TMPRSS2-ERG fusion as well as PTEN gene deletion. |
| Literature | GENERIF_SUMMARY | A novel homozygous variant in PTEN identified in two adolescent siblings with severe macrocephaly and mild intellectual disability. The absence of tumours in the two homozygous siblings as well as lack of symptoms of PHTS in the heterozygous carriers of the family suggest that this particular variant is functionally hypomorphic rather than deleterious. |
| Literature | GENERIF_SUMMARY | miR-130a plays a critical role in attenuation of cardiac dysfunction and remodeling after myocardial infarction. The mechanisms involve activation of PI3K/Akt signaling via suppression of PTEN expression. |
| Literature | GENERIF_SUMMARY | there is remarkable plasticity of PTEN expression in metastatic tumour cells in response to different organ microenvironments; essential role of co-evolution between the metastatic cells and their microenvironment during the adaptive metastatic outgrowth |
| Literature | GENERIF_SUMMARY | Data show that microRNA miR-221 targets PTEN protein, leading to activation of the PI3K/Akt pathway. |
| Literature | GENERIF_SUMMARY | differing Metformin sensitivity in RCC cells is associated with and mediated through the regulation of miR-21/PTEN expression with |

| Main Category | Category | Terms |
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| | | an impact on subsequent AKT signaling |
| Literature | GENERIF_SUMMARY | Study shows that PTEN is a downstream target of miR-616 in hepatocellular carcinoma (HCC) and miR-616 exerts its functional significance on HCC by suppressing PTEN. |
| Literature | GENERIF_SUMMARY | Results indicate that p110alpha primarily mediates PI3K/AKT signaling in HER2+ PTEN deficient breast tumors whereas p110beta conditionally mediates HER2/PI3K signaling in the absence of p110alpha. |
| Literature | GENERIF_SUMMARY | A point mutation (p.A126G) produces an enzymatic gain-of-function in PTEN, shifting its function from a phosphoinositide (PI) 3-phosphatase to a phosphoinositide (PI) 5-phosphatase. |
| Literature | GENERIF_SUMMARY | GAS5 acts as an tumor suppressor lncRNA in endometrial cancer. Through inhibiting the expression of miR-103, GAS5 significantly enhanced the expression of PTEN to promote cancer cell apoptosis. |
| Literature | GENERIF_SUMMARY | Our results suggest that Fatty Acid Synthase upregulation and PTEN downregualtion may be involved in peritoneal dissemination for gastric cancer progression. |
| Literature | GENERIF_SUMMARY | Report detection of PTEN mRNA/protein in prostate neoplasms. |
| Literature | GENERIF_SUMMARY | Continued use of trastuzumab in HER2(+) PTEN(-) cells increased the frequency of cancer stem cells. |
| Literature | GENERIF_SUMMARY | CAIX and PTEN had prognostic importance for metastatic renal cell carcinoma patients receiving first-line VEGFR TKI. Future validation and mechanistic studies are required. |
| Literature | GENERIF_SUMMARY | PTEN is possibly involved in the pathogenesis of oncocytic follicular carcinoma. |
| Literature | GENERIF_SUMMARY | Results showed that EBP50 specifically scaffolds the interaction of PTEN with EGFR and enhances the inhibition of PTEN on EGF-induced AKT activation. These results elucidated a novel mechanism regulating EGF-induced AKT signaling. |
| Literature | GENERIF_SUMMARY | The carriers of homozygous mutant of rs2735343 and rs701848 polymorphism revealed increased esophageal squamous cell carcinoma risk. We also found a potential gene-gene interaction between rs2735343 and rs701848 and a potential gene-gene interaction among all five SNPs |

| Main Category | Category | Terms |
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| | | (rs2735343, rs555895, rs2299939, rs17431184 and rs701848). |
| Literature | GENERIF_SUMMARY | Data suggest that patients with IgA nephropathy exhibit higher microRNA-374b in B cells compared to controls; microRNA-374b appears to target PTEN (phosphatase and tensin homolog) and Cosmc (C1GALT1 specific chaperone 1) proteins. |
| Literature | GENERIF_SUMMARY | we identified that phosphatase and tensin homolog (PTEN) was a target of miR-141-3p. Inhibiting PTEN could alter the mitochondrial function, too. Our study suggested that mitochondria-related miR141-3p induced mitochondrial dysfunction by inhibiting PTEN |
| Literature | GENERIF_SUMMARY | PTEN regulates DNA replication through MCM2 and loss of PTEN function leads to replication defects and genomic instability. |
| Literature | GENERIF_SUMMARY | Data show that combined MYC activation and Pten loss driven by the Hoxb13 regulatory locus synergize to induce genomic instability and aggressive prostate cancer that phenocopies the human disease at the histologic and genomic levels. |
| Literature | GENERIF_SUMMARY | redox status of PTEN causes a functional variation in the PTEN interactome |
| Literature | GENERIF_SUMMARY | Data indicate that mutations in PTEN phosphatase A167T and NRAS protein Q61L, and proto-oncogene protein BRAF V600E were detected in melanoma cell line. |
| Literature | GENERIF_SUMMARY | these findings were reversed with the LC3I/II-shPTEN treatment. Therefore, the loss of PTEN may promote the development of primary resistance to trastuzumab in breast cancer via autophagy defects |
| Literature | GENERIF_SUMMARY | Data suggest a critical role for the I2PP2A protein (SET)-protein phosphatase-2A (PP2A) signaling axis in Pten protein |
| Literature | GENERIF_SUMMARY | Findings indicate that Epstein-Barr virus(EBV)-encoded latent membrane protein 1 (LMP1), PI3K/AKT, miR-21 and PTEN constitute a positive feedback loop and have a key role in LMP1-induced cancer stem cells (CSCs) in nasopharyngeal carcinoma (NPC). |
| Literature | GENERIF_SUMMARY | Studied the effects of alterations of the phosphatase and tensin homologue (PTEN) gene on the occurrence and development of gallbladder cancer, which has not been previously reported. |

| Main Category | Category | Terms |
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| Literature | GENERIF_SUMMARY | PTEN point mutation was present in 5% of clinically significant prostate tumors. PTEN loss is infrequent in clinically insignificant prostate cancer, and is associated with higher grade tumors. |
| Literature | GENERIF_SUMMARY | PTEN loss is independently associated with increased risk of lethal progression in prostate cancer, particularly in the ERG fusion-negative subgroup. |
| Literature | GENERIF_SUMMARY | Low expression of PTEN and high expression of NF-kappaB are significant risk factors for chemotherapy resistance of ovarian cancer patients |
| Literature | GENERIF_SUMMARY | identified a high frequency of mutations in PTEN (50%) and genes involved in the endometrial cancer-related molecular pathway, which involves the IL-7 signaling pathway |
| Literature | GENERIF_SUMMARY | disruption of PTEN leads to neoplastic transformation of human neural stem cells. |
| Literature | GENERIF_SUMMARY | Identify miR-382 as a promoter for hepatocyte proliferation and cell growth via targeting PTEN-Akt axis |
| Literature | GENERIF_SUMMARY | Results demonstrate that loss of PTEN promotes resistance to immunotherapy in melanoma. |
| Literature | GENERIF_SUMMARY | These data demonstrated that PTEN upregulation could reduce invasion by inhibiting the process of EMT in Tca8113 cells, which might be related to the Hedgehog signaling pathway. |
| Literature | GENERIF_SUMMARY | MiR-21 modulates human airway smooth muscle cell proliferation and migration in asthma through regulation of PTEN expression. |
| Literature | GENERIF_SUMMARY | Deletion or deficiency of PTEN leads to down regulation of TOP2A, dysfunction of the decatenation checkpoint and incomplete DNA decatenation in G2 and M phases. |
| Literature | GENERIF_SUMMARY | proliferation and induces apoptosis of hepatic stellate cells |
| Literature | GENERIF_SUMMARY | PTEN, a direct target gene of miR-21, was significantly downregulated in gemcitabine-resistant breast cancer cells and restoration of PTEN expression blocked miR-21-induced EMT and gemcitabine resistance. |
| Literature | GENERIF_SUMMARY | PTEN deletion occurs in a relevant fraction of breast cancers, and is linked to aggressive tumor behavior |

| Main Category | Category | Terms |
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| Literature | GENERIF_SUMMARY | a new ERK2/AP-1/miR-494/PTEN pathway that is responsible for the tumor-suppressive role of NFkappaB2 p100 in cellular transformation. |
| Literature | GENERIF_SUMMARY | TBX2 is a central component of the PTEN/PI3K/AKT signaling pathway deregulation in RMS cells and that targeting TBX2 in RMS tumors may offer a novel therapeutic approach for RMS |
| Literature | GENERIF_SUMMARY | Among oral squamous cell carcinoma in south Indian subjects, the PTEN gene was methylated in 24%. PTEN and p16 promoter methylation may be the main mechanism leading to the low expression of PTEN and p16 genes indicating the progress of tumor development. |
| Literature | GENERIF_SUMMARY | the association between the down-regulation of PTEN and NKX3.1 genes contributed to prostate tumorigenesis. |
| Literature | GENERIF_SUMMARY | Study identifies crosstalk between PTEN and IRF3 in tumor suppression and innate immunity. |
| Literature | GENERIF_SUMMARY | Among advanced lung adenocarcinoma patients treated with platinum-based chemotherapy, the carriers of homozygous PTEN SNPS rs701848 and rs11202592 had significantly worse overall survival than heterozygotes or wild-type homozygotes, especially in stage III. Haplotype analysis revealed a protective effect of the haplotype G-A-A (in the order of rs11202592, rs701848, and rs11202607) on chemotherapy efficacy and survival. |
| Literature | GENERIF_SUMMARY | SOX4 expression was induced by Pten loss as a result of the activation of PI3K-AKT-mTOR signaling |
| Literature | GENERIF_SUMMARY | miR-519a promotes tumor growth of hepatocellular carcinoma by targeting PTEN-mediated PI3K/AKT pathway |
| Literature | GENERIF_SUMMARY | Hypermethylation status of PTEN tumor suppressor gene is associated with breast cancer. |
| Literature | GENERIF_SUMMARY | 3 fold, whereas other genes (p53, PTEN, TIMP-1) were upregulated by |
| Literature | GENERIF_SUMMARY | PPAR-gamma and PTEN expressions are related to the clinical parameters and prognosis of renal cell carcinoma |
| Literature | GENERIF_SUMMARY | rs701848 in the PTEN gene might be a candidate pharmacogenomic factor to assess the susceptibility and prognosis in colorectal cancer patients. |

| Main Category | Category | Terms |
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| Literature | GENERIF_SUMMARY | Data indicate that SPINK1 expression status shows a lower rate of concordance between primary prostate cancer (PCA) site and nodule metastases, than those of ERG and PTEN. |
| Literature | GENERIF_SUMMARY | the markers Ets-1 and APAF-1 relative to p53, Ki-67 and PTEN expression in colon adenomas/polyps, were investigated. |
| Literature | GENERIF_SUMMARY | our results indicate that miR-181A expression is associated with CRC patients with late liver metastases through PTEN/ AKT signaling. |
| Literature | GENERIF_SUMMARY | PTEN promoter methylation is a mechanism partially responsible for PTEN silencing in sporadic breast cancer for North Indian women. |
| Literature | GENERIF_SUMMARY | PTEN analyzed as continuous variable was associated with Ki67 and hormone receptor status in breast cancer patients. |
| Literature | GENERIF_SUMMARY | The higher levels of pten mRNA expression post treatment confirmed its role in effective clearance of Leishmania. |
| Literature | GENERIF_SUMMARY | Low PTEN expression is associated with osteosarcoma. |
| Literature | GENERIF_SUMMARY | These results suggest that in the progression of renal aging, high expression of TIMP-1 up-regulates PTEN expression indicating that PTEN and TIMP-1 are involved in the aging-associated impairment of renal angiogenesis |
| Literature | GENERIF_SUMMARY | This study suggests a high frequency of miR-21 overexpression and aberrant promoter methylation in down-regulation of PTEN expression in colorectal carcinoma. Loss of PTEN may be a prognostic factor for patients with CRC. |
| Literature | GENERIF_SUMMARY | Profilin interacts with a protein phosphatase, phosphatase and tension homologue (PTEN), and protects it from degradation. |
| Literature | GENERIF_SUMMARY | MiR-148a-3p could directly bind to the PTEN 3'-UTR. |
| Literature | GENERIF_SUMMARY | Our findings, hence, provide proof-of-principle evidence that pharmacological inhibition of PTEN may represent a promising approach for HCC therapy in a subclass of patients with a low PTEN expression. |
| Literature | GENERIF_SUMMARY | 35 SNPs of the human phosphatase and tensin homolog (PTEN) gene were analyzed |

| Main Category | Category | Terms |
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| | | by mutational analysis for functional significance, and five were found to be deleterious based on different computational tools, including molecular dynamics simulations. |
| Literature | GENERIF_SUMMARY | Study shows that women with polycystic ovarian syndrome and endometrial cancer have an increased endometrial expression of genes (IGF1, IGFBP1 and PTEN) involved in the insulin signaling pathway compared with control women. |
| Literature | GENERIF_SUMMARY | DNA content in combination with PTEN deletions provide strong and independent prognostic information in prostate cancer. |
| Literature | GENERIF_SUMMARY | The effects of PTEN downregulation on miR-29-mediated high glucose-induced retinal pigment epithelial cell apoptosis were also examined. |
| Literature | GENERIF_SUMMARY | PTEN gene polymorphisms might relate to liver cancer risk. |
| Literature | GENERIF_SUMMARY | PTEN may inhibit TGFbeta1mediated myofibroblast differentiation of fibroblasts by attenuating signaling via the phosphatidylinositol 3kinase/Akt and TGFbeta/Smad3 pathways. |
| Literature | GENERIF_SUMMARY | Results provided evidence that the expression levels of miR-301a in Ewing's sarcoma (ES) cells were significantly increased and inversely correlated with PTEN expression levels, and that miR-301a played important roles in ES cell proliferation and tumorigenesis by targeting PTEN mRNA. |
| Literature | GENERIF_SUMMARY | upregulation of miR21 expression levels suppressed the effect of triptolide on cell viability and PTEN protein expression levels in PC9 cells. |
| Literature | GENERIF_SUMMARY | silencing HDAC7 can reset the tumor suppressor activity of STAT3, independently of the EGFR/PTEN/TP53 background of the glioblastoma. |
| Literature | GENERIF_SUMMARY | PTEN promoter methylation is an independent predictor for impaired survival with malignant melanoma. |
| Literature | GENERIF_SUMMARY | PTEN is a direct functional target of miR-130b. In hepatocellular carcinoma, miR-130b may promote proliferation and EMT-induced metastasis via PTEN/p-AKT/HIF-1alpha signaling. |
| Literature | GENERIF_SUMMARY | The everolimus plus CHOP was effective for peripheral T-cell lymphoma patients, and its |

| Main Category | Category | Terms |
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| | | efficacy might be related with the preservation of PTEN. |
| Literature | GENERIF_SUMMARY | miR-21 can confer drug resistance to 5-FU in pancreatic cancer cells by regulating the expression of tumor suppressor genes, as the target genes of miR-21, PTEN and PDCD4 can rescue 5-FU sensitivity and the phenotypic characteristics disrupted by miR-21. |
| Literature | GENERIF_SUMMARY | the transcriptional control of gene expression programs related to anabolic and catabolic pathways by NOTCH1 and Pten is yet to be linked with specific transcription factors and epigenetic regulators |
| Literature | GENERIF_SUMMARY | PTEN-dependent endosome maturation through phosphoregulation of Rab7 as an important route of controlling EGFR signalling. |
| Literature | GENERIF_SUMMARY | MiR-205 promotes proliferation, migration and invasion of nasopharyngeal carcinoma cells by activation of AKT signaling and down-regulation of PTEN. |
| Literature | GENERIF_SUMMARY | Data suggest that microRNA-92b up-regulates cell proliferation/cell migration/ tumor invasiveness and down-regulates apoptosis of glioma cells via PTEN/AKT (phosphatase and tensin homolog protein/ protein kinase b alpha) signaling pathway. |
| Literature | GENERIF_SUMMARY | High expression of PTEN was positively correlated with myometrial invasion in endometrial cancer. |
| Literature | GENERIF_SUMMARY | High PTEN expression is associated with drug resistance in ovarian cancer. |
| Literature | GENERIF_SUMMARY | 3,5-triazine derivatives exhibited generally low or medium interaction with PTEN gene promoter. However, the 1,3,5-triazine derivative 11 with the para-bromo substituent showed the highest affinity against H4 R with Ki value of 520 nm and may be considered as a new lead structure |
| Literature | GENERIF_SUMMARY | High PTEN expression is associated with breast cancer. |
| Literature | GENERIF_SUMMARY | We conclude that miR-21 post-transcriptionally down-regulates the expression of PTEN to promote cell proliferation and cervical cancer cell survival. Therefore, it may be a potential therapeutic target in gene therapy for cervical cancer |
| Literature | GENERIF_SUMMARY | Cowden syndrome, the principal PTEN- |

| Main Category | Category | Terms |
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| | | related disorder is characterized by multiple neoplasms and hamartomas, mucosal papillomatosis, and skin lesions, trichilemmomas[review] |
| Literature | GENERIF_SUMMARY | PTEN is required for the proper activation of checkpoints in response to DNA damage in MCF-7 cells. |
| Literature | GENERIF_SUMMARY | findings demonstrate that the BRD7/miR-141/PTEN/AKT axis has critical roles in the progression of NPC and provide some promising targets for the diagnosis and treatment of NPC. |
| Literature | GENERIF_SUMMARY | MiR-130a exerted promoting effects on metastatic behavior and epithelial-mesenchymal transition of osteosarcoma cells through suppressing PTEN expression. |
| Literature | GENERIF_SUMMARY | findings shed new light on the NF-kappaB/miR-130a/PTEN pathway in cervical cancer and add new insight regarding the carcinogenesis of cervical cancer. |
| Literature | GENERIF_SUMMARY | The proposed quantitative method of PTEN expression assessment creates an alternative to currently available subjective methods and forms the basis for inter-case and inter-tissue comparisons. |
| Literature | GENERIF_SUMMARY | Mutations of PIK3CA were found in our cohort in 21 cases (28 %), 10 (13 %) in exon 9 and 11(15 %) in exon 20. Mutation frequency of AKT1 and PTEN genes was 4 and 3 %, respectively |
| Literature | GENERIF_SUMMARY | Genetic testing confirmed that the proband and his daughter were heterozygous for a mutation in exon 8 of the PTEN gene. |
| Literature | GENERIF_SUMMARY | A common PTEN variation is associated with peripheral insulin resistance and subsequent risk of developing T2 diabetes. |
| Literature | GENERIF_SUMMARY | protein expression is downregulated early in non-alcoholic liver disease, but not in alcoholic liver disease. |
| Literature | GENERIF_SUMMARY | the current study observed overexpression of miR-337 in EC tissues. As a target of miR-337, PTEN regulates expression at the transcriptional level and the inhibition of PTEN promotes the proliferation of EC cells in vitro. |
| Literature | GENERIF_SUMMARY | PTEN is a direct target gene of miR-22 in 786-O and A498 renal cell carcinoma. |
| Literature | GENERIF_SUMMARY | Loss of PTEN, expression and cyclin D1 overexpression was seen in a significant number of well differentiated endometrial |

| Main Category | Category | Terms |
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| | | adenocarcinomas and complex hyperplasias with atypia, suggesting both as an early event in endometrial carcinogenesis. |
| Literature | GENERIF_SUMMARY | Upon arsenite exposure of cells, over-expression of miR-21 inhibits PTEN and activates ERK, which is involved in arsenite-induced autophagy. |
| Literature | GENERIF_SUMMARY | miR-10b regulates the self-renewal of the breast CSC phenotype by inhibiting PTEN and maintaining AKT pathway activation. |
| Literature | GENERIF_SUMMARY | Data show that ten-eleven translocation 1 (TET1) suppresses tumor cell growth, migration and invasion through demethylation of CpG island in PTEN promoter by increasing 5-hmC content. |
| Literature | GENERIF_SUMMARY | The endogenous siRNA (PTEN-sh-3p21) cleaved from PTEN-sh within PTEN mRNA 3'UTR modulates protein phosphatase (PPP2CA) and PTEN at the post-transcriptional level in liver cells. |
| Literature | GENERIF_SUMMARY | Several germline variants in Hamartomatous Polyposis Syndrome genes were detected, among them three in ENG, two in BMPR1A, one in PTEN, and one in SMAD4. Although some of the detected variants have been reported previously none could be definitely pathogenic or likely pathogenic. |
| Literature | GENERIF_SUMMARY | miR-543 was constitutively up-regulated in colorectal cancer tissues and its over-expression promoted tumor cell migration and invasion. Moreover, PTEN was identified as a direct target of miR-543. |
| Literature | GENERIF_SUMMARY | Underexpression of PTEN is associated with Breast Cancer. |
| Literature | GENERIF_SUMMARY | Overexpression of PTEN can effectively inhibit proliferation of liver cancer cells and promote their apoptosis. |
| Literature | GENERIF_SUMMARY | Immunohistochemistry assay for PTEN loss shows the highest specificity and sensitivity for PTEN gene deletion in prostate neoplasms. |
| Literature | GENERIF_SUMMARY | MiR-21 targets key proteins in PTEN/PI3K/AKT signal pathway, promoting proliferation, migration, invasion, and cell cycle, and inhibiting apoptosis of human esophageal cancer cells. |
| Literature | GENERIF_SUMMARY | ERG rearrangement, but not PTEN loss, is present in a minority of potential neoplastic precursor lesions in the prostate. |

| Main Category | Category | Terms |
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| Literature | GENERIF_SUMMARY | there is a HIF-1alpha-miR-21 positive feedback loop through the PTEN/Akt/HIF-1alpha pathway in cervical cancer cells. |
| Literature | GENERIF_SUMMARY | Deleterious Non-Synonymous Single Nucleotide Polymorphisms in the PTEN Gene is associated with breast cancer. |
| Literature | GENERIF_SUMMARY | LMP1-mediated NF-kappaB can up-regulate DNMT3b transcription, thereby leading to relatively higher methylation intensity at PTEN CpG islands, and ultimately silencing major tumor suppressor PTEN |
| Literature | GENERIF_SUMMARY | photo-cross-linking suggested a direct interaction between the PTEN C-tail and a segment in the N-terminal region of the catalytic domain |
| Literature | GENERIF_SUMMARY | PTEN nuclear expression increased with increasing mitotic rate in gastrointestinal stromal tumors. |
| Literature | GENERIF_SUMMARY | The protein level of PTEN in Capan-2 pancreatic ductal adenocarcinoma cells was downregulated by overexpressing miR-221, while upregulated by inhibiting miR-221. |
| Literature | GENERIF_SUMMARY | that Pten, through the Dlg1-binding ability of its PDZ-binding domain , accumulates phosphorylated Eg5 at duplicated centrosomes to establish symmetrical bipolar spindles that properly segregate chromosomes |
| Literature | GENERIF_SUMMARY | The study of biopsies from metastatic sites from patients with advanced prostate cancer has revealed potentially actionable or prognostic genomic alterations. Common genomic alterations observed in castration-resistant are |
| Literature | GENERIF_SUMMARY | Results presented strong evidence for the existence of three (or perhaps four) non-AUG-initiated proteoforms of PTEN that are expressed in human cells in addition to, and at lower levels than, the well-studied canonical AUG-initiated PTEN. All N-terminally extended PTEN proteoforms tested retain the ability to downregulate the PI3K pathway. |
| Literature | GENERIF_SUMMARY | High levels of PTEN protein expression are associated with better prognosis in NSCLC patients. |
| Literature | GENERIF_SUMMARY | PTEN was confirmed as a direct target of miR-454. |
| Literature | GENERIF_SUMMARY | IL-1beta/NF-kb signaling has a role in promoting colorectal cancer cell growth |

| Main Category | Category | Terms |
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| | | through miR-181a/PTEN axis |
| Literature | GENERIF_SUMMARY | These results may provide a linkage between PCa chemoresistance and exosome regulatory networks and thus lead us to propose that AR, PTEN and TCF4 genes may be the important genes which are regulated by exosome miRNAs in chemoresistance cancer cells. |
| Literature | GENERIF_SUMMARY | PI3K-PTEN upregulated-mTORC1 and mTORC2 complex plays a critical role in controlling BMAL1, establishing a connection between PI3K signaling and the regulation of circadian rhythm, ultimately resulting in deregulated BMAL1 in tumor cells with disrupted PI3K signaling |
| Literature | GENERIF_SUMMARY | Results identified PTEN as a target of PRMT5 in inducing senescence of glioblastoma cells. |
| Literature | GENERIF_SUMMARY | PTEN mutation is associated with response to chemotherapy in neoplasms. |
| Literature | GENERIF_SUMMARY | PTEN mRNA expression differs between left- and right-sided colorectal cancers. |
| Literature | GENERIF_SUMMARY | The endometrial tissue showed significantly lower levels of miR-200b, miR-15a-5p, miR-19b-1-5p, miR-146a-5p, and miR-200c, and higher levels of miR-16-5p, miR-106b-5p, and miR-145-5p. VEGFA was significantly upregulated, whereas EGFR2, PTEN, and CXCR4 were markedly downregulated, in the endometriotic tissues compared to that in the normal endometrial tissues |
| Literature | GENERIF_SUMMARY | These data suggest the involvement of IGF-2, IGF-1R, IGF-2R and PTEN in temporo-spatial patterning of basic cellular processes (proliferation, differentiation) during normal tooth development. |
| Literature | GENERIF_SUMMARY | Reduced PTEN expression may be an indicator for poor prognosis in patients with lung cancer. The presence of PTEN expression may define a subset of patients with lung cancer appropriate for investigational therapeutic strategies |
| Literature | GENERIF_SUMMARY | Our findings indicate that miR-214 may promote peritoneal metastasis of gastric cancer cells via down-regulation of PTEN, thus leading to the progression of gastric cancer. |
| Literature | GENERIF_SUMMARY | We show that elevated levels of PPARG strongly correlate with elevation of FASN in human prostate cancer (CaP) and that high levels of PPARG/FASN and PI3K/pAKT pathway activation confer a poor |

| Main Category | Category | Terms |
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| | | prognosis.These data suggest that CaP patients could be stratified in terms of PPARG/FASN and PTEN levels to identify patients with aggressive CaP who may respond favorably to PPARG/FASN inhibition. |
| Literature | GENERIF_SUMMARY | TUG1 could serve as a miR-26a sponge in human glioma cells, contributing to the upregulation of PTEN. This study revealed a new TUG1/miR-26a/PTEN regulatory mechanism and provided a further understanding of the tumor-suppressive role of TUG1 in glioma development. |
| Literature | GENERIF_SUMMARY | Data suggest that reactive oxygen species (ROS)-induced DNA damage results in hypomethylation of the PTEN promoter, upregulation of PTEN mRNA and protein, and a subsequent increase in apoptosis in irradiated lung tissue. |
| Literature | GENERIF_SUMMARY | FER1L4 not only showed downregulation in endometrial carcinoma tissues and cells, but also regulated PTEN expression and Akt signaling, which might contribute to its inhibition on cell proliferation. |
| Literature | GENERIF_SUMMARY | Mutations in PIK3CA, AKT1, and PTEN also were less frequently found in cancers with mutations in the DNA guardian, TP53. This observation might seem counterintuitive because increased genome instability might facilitate development and selection of advantageous mutations. |
| Literature | GENERIF_SUMMARY | PIK3CA, PTEN, and AKT1 mutations occurred more frequently in the presence of hormone receptor overexpression (androgen, progesterone, or estrogen receptor). PIK3CA mutations were also more common in the HER2-positive than in the HER2-negative group; the opposite pattern was seen for PTEN mutation or PTEN loss |
| Literature | GENERIF_SUMMARY | Inhibition of PLK1 activity or overexpression of a non-phosphorylatable PLK1 mutant reduces the polyploid cell population. |
| Literature | GENERIF_SUMMARY | High GFRalpha2 expression level leads to PTEN inactivation via Mir-17-5p up-regulation, promoting pancreatic tumor cell growth and chemoresistance. |
| Literature | GENERIF_SUMMARY | The present work identifies a key role for Lys(66) in the regulation of PTEN expression and provides both an opportunity to improve the stability of PTEN as a protein therapy and a mechanistic basis for efforts to stabilize endogenous PTEN. |

| Main Category | Category | Terms |
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| Literature | GENERIF_SUMMARY | Data suggest that phosphatase and tensin homolog deleted on chromosome 10 (PTEN) mutations might not be correlated with activated protein kinase B (Akt). |
| Literature | GENERIF_SUMMARY | results indicate that the combination of ING4 and PTEN may provide an effective therapeutic strategy for HCC |
| Literature | GENERIF_SUMMARY | we conclude that miR-1273g-3p might affect the activation and apoptosis of hepatitis C virus (HCV)- by directly targeting PTEN in HCV-related liver fibrosis. |
| Literature | GENERIF_SUMMARY | this study shows that PTEN mutations can cause increased AKT/mTOR/S6 signaling in lymphocytes and result in Activated phosphatidylinositol 3-kinase delta syndrome |
| Literature | GENERIF_SUMMARY | Low PTEN expression is associated with low radiosensitivity in neuroblastoma. |
| Literature | GENERIF_SUMMARY | UNDerexpression of PTEN is associated with pituitary neoplasms. |
| Literature | GENERIF_SUMMARY | Underexpression of PTEN is associated with breast neoplasms. |
| Literature | GENERIF_SUMMARY | We further showed that PTEN knockdown in MCF-7 cells could increase cell migration in the xenograft zebrafish model. Our findings reveal new therapeutic targets for breast cancer patients with PTEN insufficiency. |
| Literature | GENERIF_SUMMARY | Data suggest a possible molecular mechanism might be that NEDD4 silence led to an increase in PTEN (phosphatase and tensin homologue) expression. |
| Literature | GENERIF_SUMMARY | This study indicated that patients with PTEN deletion were associated with higher pathological stage or Gleason score and a higher risk in prostate cancer. |
| Literature | GENERIF_SUMMARY | this study shows that heterozygous loss of function of PTEN in human subjects has a significant effect on T- and B-cell immunity |
| Literature | GENERIF_SUMMARY | Structural mutation analysis of PTEN and its genotype-phenotype correlations in endometriosis and cancer have been presented. |
| Literature | GENERIF_SUMMARY | miR-101 could bind 3'-UTR of WT-PTEN with reduced fluorescent intensity, suggesting that PTEN was the target gene. SMMC-7721, HepG2 and Huh7 were eligible cell lines for miR-101 studies. miR-101 was an applicable molecular marker of HCC |
| Literature | GENERIF_SUMMARY | High PTEN expression is associated with |

| Main Category | Category | Terms |
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| | | gastric cancer. |
| Literature | GENERIF_SUMMARY | Study provide evidence that PTEN deletion and TMPRSS2-ERG gene fusion were mutually exclusive in patients with prostate neoplasm. TMPRSS2-ERG gene fusion was rare compared to peripheral zone tumors and to PTEN inactivation in T1a transition zone tumors. |
| Literature | GENERIF_SUMMARY | DNA copy number variation in PTEN gene is associated with glioblastoma. |
| Literature | GENERIF_SUMMARY | decreased expression of PTEN correlated with poor overall survival in non-small cell lung cancer patients and was indicative of a poor prognosis for disease-free survival and progression-free survival in patients with NSCLC [review and meta-analysis] |
| Literature | GENERIF_SUMMARY | The miR-19b-PTEN signaling pathway may serve as a novel therapeutic target in the context of ventilator-associated lung fibrosis. |
| Literature | GENERIF_SUMMARY | miR-106a promoted the growth and invasion of SKOV3 cells by targeting phosphatase and tensin homolog (PTEN). Furthermore, the present study revealed that IL-6 inhibited miR-106a expression by activating STAT3. Tocilizumab, a humanized anti-human IL-6R antibody, that competitively inhibits IL-6/IL-6R signaling, did not inhibit the proliferation and invasion of SKOV3 cells. |
| Literature | GENERIF_SUMMARY | PTEN loss was frequently found in HER2-overexpressing gastroesophageal adenocarcinoma. |
| Literature | GENERIF_SUMMARY | PTEN and GSK3beta are targeted by miR-29a, and miR-29a may contribute to ADR resistance through inhibition of the PTEN/AKT/GSK3beta pathway in breast cancer cells. Thus, miR-29a may be a potential target for the patients who acquired ADR-resistance during the treatment of breast cancer. |
| Literature | GENERIF_SUMMARY | PTEN and PIK3CA were shown to be upregulated by 89 and 76% (western blot), respectively, in the pemetrexed-resistant cell line compared with control cells. |
| Literature | GENERIF_SUMMARY | Inhibition of protooncogenic microRNA-181a may suppress proliferation and invasion and promote apoptosis of cervical cancer cells through the PTEN/Akt/FOXO1 pathway. |
| Literature | GENERIF_SUMMARY | TGF-beta1 has a role in stimulating the migration of type II endometrial cancer cells by activating SMAD (inducing SMAD 2/3 |

| Main Category | Category | Terms |
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| | | phosphorylation and knocking down SMAD4) and ERK 1/2, which down-regulates PTEN |
| Literature | GENERIF_SUMMARY | miR-29a may regulate the radiosensitivity of intestinal cell lines by targeting the PTEN gene |
| Literature | GENERIF_SUMMARY | Findings indicate that PTEN and its downstream PI3K/AKT signaling targets were modulated by JQ1 compound. |
| Literature | GENERIF_SUMMARY | our results provide a bridge between PRL-3 and PTEN; PRL-3 decreased the expression of PTEN as well as increased the level of PTEN phosphorylation and inactivated it, consequently activating the PI3K/Akt signaling pathway, and upregulating MMP-2/MMP-9 expression to promote gastric cancer cell peritoneal metastasis. |
| Literature | GENERIF_SUMMARY | The silencing of PTEN by siRNA increased the proliferation and promoted cell invasion of Tca8113 cells (squamous carcinoma). PTEN gene silencing may accelerate the EMT in Tca8113 cells. |
| Literature | GENERIF_SUMMARY | These results suggest that miR-21 could promote TGF-beta1-induced epithelial-mesenchymal transition in gastric cancer cells through up-regulating PTEN expression |
| Literature | GENERIF_SUMMARY | Together these results suggest that miR-26a modulates the proliferation and migration of keratinocytes via regulating PTEN/AKT signaling pathway. |
| Literature | GENERIF_SUMMARY | The detection of PTEN hypermethylation could be an early tumorigenesis marker for breast cancer patients. |
| Literature | GENERIF_SUMMARY | our findings support the notion that PTEN is one of the primary targets of PML/RARA in Acute promyelocitic leukemia |
| Literature | GENERIF_SUMMARY | Data show that miR-4534 exerts its oncogenic effects partly by downregulating the tumor suppressor PTEN gene. |
| Literature | GENERIF_SUMMARY | study demonstrated that miR-141-3p contributed to an acquired chemo-resistance through PTEN modulation both in vitro and in vivo. |
| Literature | GENERIF_SUMMARY | Bmi-1 binds to PTEN promoter and directly inhibits PTEN and thereafter activates AKT |
| Literature | GENERIF_SUMMARY | Targeting of PTEN by miR-486-5p observed in endothelial cells |
| Literature | GENERIF_SUMMARY | Intra-tumour activation of the PTEN/Akt/COX-2 pathway modulates colorectal cancer |

| Main Category | Category | Terms |
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| | | (CRC) progression and invasive capacity. |
| Literature | GENERIF_SUMMARY | findings provide evidence that rs701848, rs2735343 and rs112025902 polymorphisms in the PTEN gene may be correlated with the risk of depression and depressive symptoms in the Chinese population. |
| Literature | GENERIF_SUMMARY | PTEN gene silence disorganizes the PI3K/ AKT/ PTEN/mTOR signaling transduction pathway in laryngeal squamous cell carcinoma, affecting potentially the response rates to specific targeted therapeutic agents and the prognosis in those patients |
| Literature | GENERIF_SUMMARY | PTEN loss in a Gleason score 3 + 4 = 7 biopsy is independently associated with an increased risk of nonorgan confined disease at prostatectomy. It adds to the preoperative parameters commonly used to predict pathological stage |
| Literature | GENERIF_SUMMARY | The inhibitory effects of ODAM on colorectal cancer cell growth were associated with PTEN elevation and PI3K/AKT signaling inactivation. |
| Literature | GENERIF_SUMMARY | miR-9 expression was negatively associated with PTEN expression in laryngocarcinoma cells, modulating cell proliferation and metastasis. |
| Literature | GENERIF_SUMMARY | Which was correlated with the inhibition of PETN/PI3K/Akt level. |
| Literature | GENERIF_SUMMARY | High expression of PTEN is associated with Adriamycin resistance in breast cancer. |
| Literature | GENERIF_SUMMARY | miR-130a could alleviate human coronary artery endothelial cells (HCAECs) injury and inflammatory responses by down-regulating PTEN and activating PI3K/Akt/eNOS signaling pathway. |
| Literature | GENERIF_SUMMARY | Mutations in PTEN gene is associated with T-cell acute lymphoblastic leukemia. |
| Literature | GENERIF_SUMMARY | Absolute exclusivity between SPINK1 overexpression and homozygous PTEN deletion in localized PCA. |
| Literature | GENERIF_SUMMARY | Nuclear PTEN interferes with binding of Ku70 at double-strand breaks through post-translational poly(ADP-ribosyl)ation. |
| Literature | GENERIF_SUMMARY | PTEN inhibits the occurrence and development of endometriosis by regulating angiogenesis and the apoptosis and cell cycle of endometrial cells. |

| Main Category | Category | Terms |
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| Literature | GENERIF_SUMMARY | PTEN/Akt/FOXO1 pathway plays an important role in adriamycin-resistance of breast cancer. |
| Literature | GENERIF_SUMMARY | pregulation of PTEN and LKB1 in concert with negative or low levels of activated Akt, mTOR and S6 indicates that PI3K/Akt/mTOR pathway may not play a significant role in pathogenesis of leiomyoma. |
| Literature | GENERIF_SUMMARY | Marsdenia tenacissimae extract inhibits the proliferation and induces the apoptosis of human acute T cell leukemia cells by inactivating PI3K/AKT/mTOR signaling pathway via PTEN enhancement. |
| Literature | GENERIF_SUMMARY | Frameshift mutation of PTEN is associated with acute T-lymphoblastic leukemia. |
| Literature | GENERIF_SUMMARY | Four gene signature (PTEN, PIK3C2A, ITPA and BCL3) is an independent prognostic factors of both overall survival and disease-free survival in clear-cell renal-cell carcinoma. |
| Literature | GENERIF_SUMMARY | miRNA92a targets KLF2 and PTEN signaling to promote human T follicular helper precursors in type 1 diabetes mellitus islet autoimmunity. |
| Literature | GENERIF_SUMMARY | PTEN downregulates the PI3K/AKT/hTERT pathway, thereby suppressing the growth of lung adenocarcinoma cells. |
| Literature | GENERIF_SUMMARY | Case Report |
| Literature | GENERIF_SUMMARY | this study shows that PTEN dephosphorylates AKT to prevent the expression of GLUT1 on plasmamembrane and to limit glucose consumption in cancer cells |
| Literature | GENERIF_SUMMARY | PTENp1 was aberrantly expressed in oral squamous cell carcinoma. There was a positive correlation between the expression levels of PTENp1 and PTEN. Further, we showed that PTENp1 acted as a competing endogenous RNA that protects PTEN transcripts from being inhibited by miR-21, and consequently inhibited proliferation and colony formation and triggered S-G2/M cell cycle arrest through the AKT pathway. |
| Literature | GENERIF_SUMMARY | In particular, AQP8 affected VEGF-induced redox signaling by increasing the sulfenation of the tumor suppressor PTEN, which resulted in its inactivation and, in turn, caused Akt activation |
| Literature | GENERIF_SUMMARY | PI3Kbeta associated with N-cadherin and PIK3CB depletion selectively disrupted N-cadherin cell adhesions in PTEN-mutant |

| Main Category | Category | Terms |
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| | | bladder carcinoma. |
| Literature | GENERIF_SUMMARY | PTEN/PI3K/AKT proteins are related to the clinicopathological features and prognosis of breast cancer with axillary lymph node metastasis. |
| Literature | GENERIF_SUMMARY | Studies indicate that frequently altered genes in type I carcinomas are TP53, K-Ras, PTEN, and ss-catenin. |
| Literature | GENERIF_SUMMARY | SHP2 and PTEN deficiencies were detected in liver cancer patients with poor prognosis. These data depict a mechanism of hepato-oncogenesis and suggest a potential therapeutic strategy. |
| Literature | GENERIF_SUMMARY | Results show that PTEN expression is down-regulated by mir-205 in esophageal squamous cell carcinoma promoting radiation resistance. |
| Literature | GENERIF_SUMMARY | this study identified PTEN as a previously unrecognized substrate of the ISGylation post-translational modification pathway. The deconjugase USP18 serves as a novel regulator of PTEN stability. This indicates inhibition of ISGylation is therapeutically relevant in cancers. |
| Literature | GENERIF_SUMMARY | PTEN gene expression in acute myeloid leukemia. |
| Literature | GENERIF_SUMMARY | the present study was designed to explore whether PTEN was involved in APN-induced osteogenic differentiation of human bone marrow mesenchymal stem cells, and the related mechanisms. |
| Literature | GENERIF_SUMMARY | PTEN downregulation initiates the signal pathway leading to ROS generation in the progress of replicative skin aging. |
| Literature | GENERIF_SUMMARY | Results show that loss of PTEN is frequently detected in triple-negative BCBM patients and associated with poor prognosis. The findings suggest that PTEN loss promotes a feedback loop between tumor cells and glial cells, which might contribute to disease progression. |
| Literature | GENERIF_SUMMARY | Results showed that PTEN expression is low in nasopharyngeal carcinoma (NPC) cells and revealed that MiR-200c suppresses PTEN expression by targeting its 3'UTR. |
| Literature | GENERIF_SUMMARY | Suggest CBX7 is an important tumor suppressor that negatively modulates PTEN/Akt signaling during pancreatic tumorigenesis. |
| Literature | GENERIF_SUMMARY | Both upregulated miR-21 and |

| Main Category | Category | Terms |
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| | | downregulated PTEN expression have a possible correlation with the aggressive progression and poor prognosis of WT, which suggests that upregulated miR-21 and downregulated PTEN expression may be valuable markers of tumor progression and indicators of the prognosis of WT. |
| Literature | GENERIF_SUMMARY | Results demonstrated that a genome-wide screen of miRNAs and targeting mRNAs revealed the negatively regulatory effect of miR-130b-3p on PTEN in bladder cancer (BC) by PI3K and integrin 1/FAK signaling pathways. |
| Literature | GENERIF_SUMMARY | Taken together, the authors presented here a novel cross-talk between miR-181a and PTEN which was raised by hepatitis B virus X protein, and this shined a new line in hepatitis B virus-related hepato-carcinogenesis. |
| Literature | GENERIF_SUMMARY | role of PTEN as a suppressor of prostate cancer and association of its loss with aggressive disease are well established [review] |
| Literature | GENERIF_SUMMARY | PTEN, a well-defined tumor suppressor was identified to be a target gene of miR-495. Rescue experiments showed that enforced expression of PTEN impaired miR-495-mediated bladder cancer proliferation and invasion. |
| Literature | GENERIF_SUMMARY | Pten protein involvement in breast cancer cells apoptosis |
| Literature | GENERIF_SUMMARY | that the presence of NRAS, PTEN and AKT1 gene mutations may have an influence on the occurrence of CNS metastases in patients with SqCC |
| Literature | GENERIF_SUMMARY | These findings indicate that PTEN not only targets tumor cells themselves by impacting cell behaviors, but also blocks osteoclast-mediated bone destruction, leading to interruption of the vicious cycle during osteosarcomagenesis |
| Literature | GENERIF_SUMMARY | Transcript level expression of PTEN was analyzed by reverse transcriptase PCR and quantitative PCR in 20 GBC and 20 GSD tissue samples. |
| Literature | GENERIF_SUMMARY | we demonstrate that HOTAIR downregulates miR-29b expression and attenuates its control on epigenetic regulation, leading to enhanced PTEN methylation, which contributes to the progression of liver fibrosis. |
| Literature | GENERIF_SUMMARY | MicroRNA miR-155-5p promoted |

| Main Category | Category | Terms |
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| | | proliferation, invasion and migration, but inhibited apoptosis in hepatocellular carcinoma (HCC) by directly targeting the 3'-UTR of PTEN through the PI3K/Akt pathway. |
| Literature | GENERIF_SUMMARY | miR-205 functions as an oncogenic miRNA by directly binding to SMAD4 and PTEN, providing a novel target for the molecular treatment of ovarian cancer. |
| Literature | GENERIF_SUMMARY | partial knockdown of Pten rescues the toxicity observed in the NSC34 (G4C2)102 cellular gain-of-toxic function model of C9ORF72-ALS. Our data indicate that PTEN may provide a potential therapeutic target to ameliorate toxic effects of the (G4C2)n repeat. |
| Literature | GENERIF_SUMMARY | our study reveals a novel mechanism of crosstalk amongst SDHD, PTEN and autophagy pathways and their potential roles in thyroid carcinogenesis |
| Literature | GENERIF_SUMMARY | In PTEN-deficient prostate and breast cancers, CHD1 depletion profoundly and specifically suppressed cell proliferation, cell survival and tumorigenic potential. |
| Literature | GENERIF_SUMMARY | Loss of PTEN is associated with prostate tumors. |
| Literature | GENERIF_SUMMARY | Data show that acquired immunodeficiency syndrome (AIDS) related diffuse large B-cell lymphoma (AR-DLBCL) frequently contain phosphorylated 70 ribosomal S6 protein-kinase (p70S6K), a main downstream effector of the mTOR serine-threonine kinases (mTOR) pathway, and the presence of p70S6K is positively correlated with phosphorylated phosphatase and tensin homolog (pPTEN). |
| Literature | GENERIF_SUMMARY | PTEN role in regulating phosphatidylinositol-3-kinase signaling in the triple-negative breast cancer cells. |
| Literature | GENERIF_SUMMARY | high intratumoral PTEN gene expression was associated with poor prognosis in patients with locally advanced breast cancers harboring wild-type TP53 |
| Literature | GENERIF_SUMMARY | Findings suggest that PTEN and DNMT3B possess common regulation features as well as certain ethnic differences in expression between Han women and Uygur women. |
| Literature | GENERIF_SUMMARY | high-expression CFTR plays an important role in the development of Ph+ leukemia through the HDAC2-mediated PTEN pathway |

| Main Category | Category | Terms |
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| Literature | GENERIF_SUMMARY | The authors demonstrated a new mechanism of miR-26a that it performed converse roles on oncogenicity in different gastric cancer cells; miR-26a and PTEN can be used as combined biomarkers in patients with gastric cancer to predict tumor progression and prognoses. |
| Literature | GENERIF_SUMMARY | Her-2-positive metastatic breast cancers, PTEN positivity was significantly associated with progressive disease, but not with PFS or OS |
| Literature | GENERIF_SUMMARY | IPO11 provides protection in another way, the researchers found. The transport receptor also shuttles UBE2E1, an enzyme necessary for PTEN ubiquitination, into the nucleus. Thanks to IPO11, normal cells keep all of their UBE2E1 in the nucleus, allowing PTEN to continue functioning in the cytoplasm. In cells lacking IPO11, UBE2E1 remains in the cytoplasm and promotes the destruction of PTEN. |
| Literature | GENERIF_SUMMARY | CD164 expression may have affected the proliferation and apoptosis of human glioma cells via the PTEN/phosphoinositide 3-kinase/AKT pathway. |
| Literature | GENERIF_SUMMARY | Findings provide a molecular mechanism for cancer-associated PTEN defects in bran neoplasms. |
| Literature | GENERIF_SUMMARY | showed that PTEN induced miR-26b expression by regulating the differential expression of Ikaros isoforms that are transcriptional regulators of miR-26b |
| Literature | GENERIF_SUMMARY | Out of 171 EGFR mutated tumor-patient cases, 20 were sequenced using at least a limited comprehensive genomic profiling platform. 50% harbored concurrent TP53 mutation, 10% PIK3CA mutation, 5% PTEN mutation, among others. |
| Literature | GENERIF_SUMMARY | Taken together, recent discovery of various PTEN proteoforms, including their property to dimerize and multimerize indicates that we have a lot to learn about biological and pathophysiological functions of PTEN, which till date had remained obscure. [review] |
| Literature | GENERIF_SUMMARY | PARK2 inactivation connects energy and oxidative stress to Akt activation via redox-mediated inactivation of PTEN by S-nitrosylation to support cell survival under conditions of energy deprivation. |
| Literature | GENERIF_SUMMARY | When using Pifithrin-alpha (one of p53 inhibitors), the level of p53-dependent |

| Main Category | Category | Terms |
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| | | apoptosis was significantly lower than untreated, which suggested that p53 was possibly involved in PTEN-dependent apoptosis |
| Literature | GENERIF_SUMMARY | findings shed new light on the STAT1/miR-181a/PTEN pathway in colorectal cancer and add new insight regarding the carcinogenesis of colorectal cancer. |
| Literature | GENERIF_SUMMARY | Disruption of PTEN proteinisoform PTENbeta (PTENbeta) alters rDNA transcription and promotes ribosomal biogenesis. |
| Literature | GENERIF_SUMMARY | Study showed for the first time that the suppression of rheumatiod arthritis fibroblast-like synoviocyte was mediated by phosphatase and tensin homolog involving survivin silencing. |
| Literature | GENERIF_SUMMARY | The first preclinical study evaluates the response to CDK4/6 inhibition in endometrial malignancies driven by PTEN deficiency, and it reveals an important role of cyclin D-CDK4/6 activity in their development. |
| Literature | GENERIF_SUMMARY | This review will summarize the biological roles of phosphatidylinositol 3-kinase regulatory subunit 1 and phosphatase and tensin homolog in breast cancer, with an emphasis on recent findings and the potential of phosphatidylinositol 3-kinase regulatory subunit 1 and phosphatase and tensin homolog as a therapeutic target for breast cancer therapy. |
| Literature | GENERIF_SUMMARY | MiR-181a can promote hepatocellular carcinoma metastasis by targeting PTEN, which is regulated by lncRNA XIST. |
| Literature | GENERIF_SUMMARY | Our data indicated that the combination of Ad-PTEN and LY294002 effectively suppressed the malignant growth of human glioma cells in vitro and in tumor xenografts, suggesting a promising new approach for glioma gene therapy that warrants further investigation |
| Literature | GENERIF_SUMMARY | we demonstrate that PKCepsilon cooperates with the loss of the tumor suppressor Pten for the development of prostate cancer in a mouse model. Mechanistic analysis revealed that PKCe overexpression and Pten loss individually and synergistically upregulate the production of the chemokine CXCL13, which involves the transcriptional activation of the CXCL13 gene |

| Main Category | Category | Terms |
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| Literature | GENERIF_SUMMARY | Down-regulation of miR-155 significantly inhibits proliferation, migration, invasion and promotes apoptosis through PTEN singaling pathway in psoriasis cells. |
| Literature | GENERIF_SUMMARY | PTEN loss as a frequent event in breast cancer that is closely associated with progression and poor prognosis. PTEN may be a promising, useful biomarker for predicting clinical outcomes in women with breast cancer. |
| Literature | GENERIF_SUMMARY | Coexpression of SALL4 with HDAC1 and/or HDAC2 was associated with PTEN underexpression and a poor prognosis in hepatocellular carcinoma. |
| Literature | GENERIF_SUMMARY | This study suggested that PTEN expression was negatively correlated with the miR-24 level in tongue squamous cell carcinoma. The loss of PTEN expression may serve as a predictor of unfavorable prognosis for tongue squamous cell carcinoma patients |
| Literature | GENERIF_SUMMARY | Low Expressions of PTEN is associated with metastasis of esophageal squamous cell carcinoma. |
| Literature | GENERIF_SUMMARY | PTEN rs701848 and AKT1 rs2494752 polymorphisms might be a candidate pharmacogenomic factor to assess the susceptibility of and response and prognosis prediction for interindividualized CE(A)F chemotherapy in breast cancer patients. |
| Literature | GENERIF_SUMMARY | In the present study, we identified that the adaptor protein Numb, which is demonstrated to be a novel binding partner of NEDD4-1, plays important roles in controlling PTEN ubiquitination through regulating NEDD4-1 activity and the association between PTEN and NEDD4-1. |
| Literature | GENERIF_SUMMARY | inhibition of miR21 suppressed cell proliferation and colony formation through regulating the PTEN/AKT pathway, and improved PTX sensitivity in cervical cancer cells. |
| Literature | GENERIF_SUMMARY | our data suggest that microRNA-200c plays an oncogenic role in nasopharyngeal carcinoma by targeting PTEN |
| Literature | GENERIF_SUMMARY | results showed that upregulated lncARSR promotes doxorubicin resistance in HCC via modulating PTEN-PI3K/Akt pathway, and implied that lncARSR may serve as a promising prognostic biomarker and therapeutic target for HCC chemo-resistance. |

| Main Category | Category | Terms |
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| Literature | GENERIF_SUMMARY | our findings demonstrated that dysregulation of miR-19a plays critical roles in the osteosarcoma stem cells, at least in part via targeting the phosphatase and tensin homolog (PTEN) deleted on chromosome 10. Knockdown of miR-19a may represent a potential strategy for the osteosarcoma treatment. |
| Literature | GENERIF_SUMMARY | Data show that phosphatase and tensin homolog (PTEN) interacts with death domain associated protein (DAXX) and, in turn PTEN directly regulates oncogene expression by modulating DAXX-histone H3.3 (H3.3) association on the chromatin. |
| Literature | GENERIF_SUMMARY | Antibody therapy targeted at AFP-producing tumor cells showed an inhibitory effect on the PI3K/AKT pathway via the liberation, restoration and functional stabilization of PTEN. PTEN simultaneously induced both P53 activation and intracellular translocation of GLUT1, since these are closely associated with PTEN. |
| Literature | GENERIF_SUMMARY | The authors conclude that the PTEN-Akt1-mTOR pathway does not appear to play a central role in mitochondrial efficiency in ocular hypertension. |
| Literature | GENERIF_SUMMARY | PTEN loss and p27 loss differ among morphologic patterns of prostate cancer. |
| Literature | GENERIF_SUMMARY | This study for the first time demonstrated that HDAC8 activity determines susceptibility to cell cycle arrest induced by Anthrax Lethal Toxin, through regulating the PI3K-PTEN-AKT signaling axis. |
| Literature | GENERIF_SUMMARY | This study reveals the presence of a PTEN Alu insertion hotspot involved in Cowden syndrome, and suggests that undetected PTEN pathogenic variants could contribute to CS. |
| Literature | GENERIF_SUMMARY | reductive PTEN mediated by miR-106b and miR-93 promoted cell progression through PI3K/Akt pathway in breast cancer. |
| Literature | GENERIF_SUMMARY | PTEN knockdown potentiated Akt phosphorylation. |
| Literature | GENERIF_SUMMARY | Over-expression of miR-21 promotes aggressive behavior of Wilms' tumor by targeting PTEN. |
| Literature | GENERIF_SUMMARY | High PTEN expression is associated with non-small cell lung cancer metastasis. |
| Literature | GENERIF_SUMMARY | PTEN loss has a negative prognostic impact in both adult and paediatric patients. The negative clinical outcome of PTEN loss is, |

| Main Category | Category | Terms |
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| | | however, limited to cases harbouring large genomic deletions, indicating that a detailed analysis of the type of abnormality is indispensable to refine risk stratification and treatment. |
| Literature | GENERIF_SUMMARY | Findings indicated that Mxd1 mediates hypoxia-induced cisplatin-resistance in osteosarcoma cells through its regulation of PTEN expression. |
| Literature | GENERIF_SUMMARY | The expression levels of Bcl11a and Mdm2, Pten in B-ALL patients with CR were decreased significantly when compared with the healthy control (P |
| Literature | GENERIF_SUMMARY | PTEN inhibits replicative senescence-induced MMP-1 expression by regulating NOX4-mediated reactive oxygen species in human dermal fibroblasts. |
| Literature | GENERIF_SUMMARY | TRPC1 regulated HIF1alpha levels in PTEN-deficient MDA-MB-468 and HCC1569 breast cancer cell lines. This regulation arises from effects on the constitutive translation of HIF1alpha under normoxic conditions via an Akt-dependent pathway. |
| Literature | GENERIF_SUMMARY | Western blotting and quantitative real-time PCR demonstrated that miR-214 negatively regulated PTEN in radioresistance ovarian cancer cell lines and ovarian cancer tissues. |
| Literature | GENERIF_SUMMARY | ERK5 associates with CKII to play essential roles in GPIb-IX-mediated platelet activation via the PTEN/PI3K/Akt pathway. |
| Literature | GENERIF_SUMMARY | overexpression of ciRS-7 blocked the miR-7-induced tumor suppression in MGC-803 and HGC-27 cells and led to a more aggressive oncogenic phenotype, via antagonizing miR-7-mediated PTEN/PI3K/AKT pathway. |
| Literature | GENERIF_SUMMARY | PTEN counteracts FBXL2 to promote IP3R3- and Ca(2+)-mediated apoptosis limiting tumour growth |
| Literature | GENERIF_SUMMARY | PHTS has an autosomal dominant mode of inheritance and is characterized by heterozygous germline pathogenic variants in the PTEN gene located on chromosome 10 at position q23.31. The PI3K/AKT/mTOR pathway, which plays a central role in cell-cycle processes including cell proliferation, is negatively regulated by the PTEN gene |
| Literature | GENERIF_SUMMARY | In perspective of the crucial role PTEN plays in balancing proliferation and apoptosis, we propose PTEN as a valuable marker in the diagnosis, assessment of tumor grade, and disease stage in hepatocellular carcinoma patients |

| Main Category | Category | Terms |
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| Literature | GENERIF_SUMMARY | PTEN-long tumor suppressor can repress cancer glial cells proliferation. |
| Literature | GENERIF_SUMMARY | we identified PTEN as a direct and functional target of miR-193a-3p, at least partly responsible for promoting tumor effect of miR-193a-3p in renal cell carcinoma. Taken together, the findings indicated for the first time that miR-193a-3p functions as a tumor-promoting microRNA by directly targeting PTEN in renal cell carcinoma. |
| Literature | GENERIF_SUMMARY | These studies provide the first direct evidence for both the involvement of the FGFR1 V561M mutation and PTEN inactivation in the development of resistance in leukemias overexpressing chimeric FGFR1. These studies also provide a potential strategy to treat leukemias and lymphomas driven by FGFR1 activation that become resistant to FGFR1 inhibitors |
| Literature | GENERIF_SUMMARY | This meta-analysis suggests that PTEN expression is of limited value in predicting the prognosis of patients with RCC for OS and PFS via immunohistochemistry staining analysis; and that for DSS, low PTEN expression is significantly associated with an unfavorable outcome |
| Literature | GENERIF_SUMMARY | Our observations of these PTEN intronic variants should contribute to the determination of pathogenicity of PTEN intronic variants and aid in genetic counseling |
| Literature | GENERIF_SUMMARY | Hypoxia-induced upregulation of miR-214 was found to promote pulmonary hypertension development by targeting PTEN in pulmonary artery smooth muscle cells. |
| Literature | GENERIF_SUMMARY | GAS5/miR-21/PTEN affects the sensitivity of NSCLC to DDP-based therapy. |
| Literature | GENERIF_SUMMARY | PTEN expression levels in brain metastases are lower compared with that of primary site. Nuclear PTEN expression was significantly downregulated in various brain metastases. |
| Literature | GENERIF_SUMMARY | A two-hit loss involving VHL predicted for clear-cell renal cell carcinoma(ccRCC ) and a better prognosis, whereas mutations in PTEN, TP53, or RELN predicted for Sarcomatoid clear-cell renal cell carcinoma and worse prognosis. |
| Literature | GENERIF_SUMMARY | MiRNA-26b may play a key role in cell growth and death of laryngeal carcinoma |

| Main Category | Category | Terms |
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| | | through ULK2 and the PTEN/AKT pathway. |
| Literature | GENERIF_SUMMARY | findings therefore indicate that miR582 promotes colorectal cancer progression by decreasing PTEN expression. |
| Literature | GENERIF_SUMMARY | Our results demonstrate that NEDD4 may promote the acquired resistance of non-small-cell lung cancer cells to erlotinib by decreasing phosphatase and tensin homolog deleted on chromosome 10 protein expression. Targeted decrease in NEDD4 expression may be a potential therapeutic strategy for tyrosine kinase inhibitor-resistant non-small-cell lung cancer. |
| Literature | GENERIF_SUMMARY | Silencing CRKL expression in PTEN-null human cancer cells leads to a decrease in p110beta-dependent PI3K signaling and cell proliferation. |
| Literature | GENERIF_SUMMARY | Results suggest that PTEN may regulate chemotherapy sensitivity of nasal-type natural killer/T-cell (NK/T-cell) lymphoma and contribute to clinical outcomes, indicate PTEN to be a potential target anti-tumor therapeutics for NK/T-cell lymphoma. |
| Literature | GENERIF_SUMMARY | upregualtion of PTENP1 could attenuate BC cell growth and metastasis, and induce apoptosis, possibly because it has a competing endogenous RNA role in reverse regulation of miRNA-19b, and then activation of PTEN, as well as modulation of PTEN/AKT/p53 signaling pathway |
| Literature | GENERIF_SUMMARY | The results show that PTEN controls multicellular assembly through a membrane-associated regulatory protein complex composed of beta-Arrestin1, ARHGAP21 and Cdc42. |
| Literature | GENERIF_SUMMARY | PIK3CA and/or ERBB family mutations were detected in 23 (31.1%) tumour samples tested, whereas PTEN expression was low in 31.1% of cases tested |
| Literature | GENERIF_SUMMARY | Data suggest that low levels of PTEN and KLLN are associated with both sporadic papillary thyroid carcinoma and multi-nodular goiter (as compared to control subjects); however, PTEN and KLLN should not be considered as circulating biomarkers for differential diagnosis between malignancy and benignity in indeterminate thyroid nodules. (PTEN = phosphatase and tensin homolog; KLLN = DNA replication inhibitor killin) |
| Literature | GENERIF_SUMMARY | PTEN promotes DNA repair through Rad51- |

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| | | dependnent homologous recombination. |
| Literature | GENERIF_SUMMARY | We applied the Dual-luciferase Reporter assay to expose the direct interaction between miR-93 and PTEN, which suggested that miR-93 contributes to inducing EMT and drug resistance of breast cancer cells by targeting PTEN |
| Literature | GENERIF_SUMMARY | These results therefore indicate that EGCG targets PML/RARalpha oncoprotein for degradation and potentiates differentiation of promyelocytic leukemia cells in combination with ATRA via PTEN. |
| Literature | GENERIF_SUMMARY | Our results indicated that miR-144 can regulate the proliferation and apoptosis of trophoblastic cells through targeting the PTEN signaling pathway. |
| Literature | GENERIF_SUMMARY | We identified a novel PTEN missense mutation, c.1A>G (p.M1V), in a 16-year-old girl. The patient with this mutation presented with Autism Spectrum Disorder, intellectual disability, language delay, extreme macrocephaly. |
| Literature | GENERIF_SUMMARY | The reducing system of PTEN was comprised of NADPH, thioredoxin reductase (TrxR1), and thioredoxin (Trx). |
| Literature | GENERIF_SUMMARY | PTEN-Long, a translation isoform of PTEN, functions in a cell non-autonomous manner. This study demonstrates that intracellular overexpression of PTEN-Long inhibits HCV replication. |
| Literature | GENERIF_SUMMARY | Our results suggested that BetA was able to enhance radiosensitization at least partially by downregulating Sp1 and upregulating PTEN through inducing Sp1 sumoylation. BetA is suggested to be a promising drug for increasing radiosensitization in oral squamous cell carcinoma radiotherapy. |
| Literature | GENERIF_SUMMARY | The aim of this review is to shed light on the role of four different phosphatases (PTEN, PP2A, CDC25 and DUSP1) in five different solid tumors (breast cancer, lung cancer, pancreatic cancer, prostate cancer and ovarian cancer), in order to better understand the most frequent and aggressive primary cancer of the central nervous system, glioblastoma. |
| Literature | GENERIF_SUMMARY | These results suggest that the activation of NF-kappabeta by MMA(III) may participate in UROtsa cells malignant transformation through the negative regulation of PTEN expression involving p50 homodimers-mediated chromatin remodeling around the |

| Main Category | Category | Terms |
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| | | PTEN promoter. |
| Literature | GENERIF_SUMMARY | miR-221/222 play a critical role in the propagation of breast cancer stem cells and tumor growth possibly through targeting PTEN, which in turn activating the Akt/NF-kappaB/COX-2 pathway. |
| Literature | GENERIF_SUMMARY | USP10 directly interacted with and stabilized PTEN via deubiquitination. |
| Literature | GENERIF_SUMMARY | CNS disease was associated with enhanced expression of cytoplasmic and membranous ITGA10 and nuclear PTEN (P |
| Literature | GENERIF_SUMMARY | Bioinformatics analysis demonstrated that the 3'UTR of PTEN mRNA was targeted by hsa-miR-142-5p which regulates its expression triggering cancer stem cell-like properties of cutaneous squamous cell carcinoma. |
| Literature | GENERIF_SUMMARY | The results of this work suggest that PTEN allelic loss is an important mechanism in the late stage of the development of oral potentially malignant lesions into oral cancer |
| Literature | GENERIF_SUMMARY | High PTEN expression is associated with Non-Small Cell Lung Cancer. |
| Literature | GENERIF_SUMMARY | Meta-analysis demonstrates that PTEN loss is of particular importance for predicting breast cancer aggressiveness and poor prognosis. PTEN is a potential drug target for the development of individualized treatment in breast cancer patients. |
| Literature | GENERIF_SUMMARY | Cells with mutated PTEN showed over-activation of the PI3K/mTOR pathway. These cells were more sensitive to PARP inhibition compared to PTEN wild-type cells. In addition, PI3K inhibitor treatment reduced RAD51 foci formation in PTEN mutated cells, and sensitized these cells to PARP inhibitor. |
| Literature | GENERIF_SUMMARY | MiRNA-20a may have great potential as therapeutic target for coronary artery disease, since its participation can induce alteration of functional genes as well as PTEN. |
| Literature | GENERIF_SUMMARY | Exosomes derived from cisplatin-resistant OSCC cells transferred miR-21 to oral squamous cell carcinoma (OSCC) parental cells and induced cisplatin resistance by targeting phosphatase and tensin homolog (PTEN) and programmed cell death 4 (PDCD4). |
| Literature | GENERIF_SUMMARY | Star-PAP-specific polyadenylation sites |

| Main Category | Category | Terms |
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| | | usage regulates the expression of the eukaryotic translation initiation factor EIF4A1, the tumor suppressor gene PTEN and the long non-coding RNA NEAT1. |
| Literature | GENERIF_SUMMARY | Long non-coding RNA 657 suppresses hepatocellular carcinoma cell growth by acting as a molecular sponge of miR-106a-5p to regulate PTEN expression |
| Literature | GENERIF_SUMMARY | Results indicate miR-130a as an oncogenic miRNA that targets PTEN to drive malignant cell survival and tumor growth. |
| Literature | GENERIF_SUMMARY | miR-132 promotes the proliferation, invasion and migration of human pancreatic cancer by inhibition of PTEN |
| Literature | GENERIF_SUMMARY | miR-198 may promote proliferation and contribute to systemic lupus erythematosus progression by targeting PTEN. |
| Literature | GENERIF_SUMMARY | Study proposes a new mechanism by which loss of PTEN and consequent activation of the PI3K-AKT-mTORC1-S6K1 signalling pathway impairs DNA repair by downregulation of MRE11. |
| Literature | GENERIF_SUMMARY | This study identifies novel mechanistic biomarkers of PI3Kb inhibition in PTEN-null tumors supporting the concept that targeting PI3Kb may exploit a metabolic dependency that contributes to therapeutic benefit in inducing cell stress |
| Literature | GENERIF_SUMMARY | Well-differentiated laryngeal squamous cell carcinomas express moderate to high protein levels of PTEN, an evidence of normal gene function, whereas loss of its expression correlates with a progressive tumor dedifferentiation. |
| Literature | GENERIF_SUMMARY | might induce cell invasion and migration via regulating AKT/GSK-3beta/beta-catenin signaling pathway, playing a vital role in the progression of gastric cancer |
| Literature | GENERIF_SUMMARY | Results showed that PTEN expression was positively associated with the expression of RBM38 in breast cancer tissues and breast cancer cells. Moreover, RBM38 stabilized PTEN transcript to enhance its expression via binding to multiple AU/U- rich elements in 3'-untranslated region of PTEN transcript. |
| Literature | GENERIF_SUMMARY | Suggest a role for PI(3,4)P2 in the phenotype caused by loss-of-function mutations or deletions in PTEN in breast/ prostate tumor cells. |
| Literature | GENERIF_SUMMARY | the antiviral function of PTEN-L may open an avenue for the use of PTEN-L in antiviral |

| Main Category | Category | Terms |
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| | | therapy, particularly in patients with PTEN-deficient tumors. |
| Literature | GENERIF_SUMMARY | Authors have identified a novel mechanism by which P53 was inactivated via SIRT3 in PTEN-deficient cells. This may shed light on the mechanisms underlying the malignancy of PTEN-deficient NSCLC. |
| Literature | GENERIF_SUMMARY | PTEN-4A preferentially localized to the nucleus where it suppressed E2F1-mediated transcription of cell cycle genes. |
| Literature | GENERIF_SUMMARY | When considered together (43 cases), 1/25 cases (4%) with a PIK3CA mutation and/or low PTEN expression levels had a pathologic complete response (pCR)compared to 7/18 cases (39%) with wild-type PI3KCA and high PTEN expression levels (p = 0.006). |
| Literature | GENERIF_SUMMARY | Deregulated expression of miR-107 inhibits metastasis of pancreatic ductal adenocarcinoma through inhibition PI3K/Akt signaling via caveolin-1 and PTEN. |
| Literature | GENERIF_SUMMARY | These results suggest that the miR-130b inhibitor suppressed glioma cell proliferation and invasion via the PTEN/AKT pathway. Therefore, miR130b is suggested to be an effective therapeutic target for glioma. |
| Literature | GENERIF_SUMMARY | miR222 may act as an oncomir in NPC by targeting PTEN in nasopharyngeal carcinoma. |
| Literature | GENERIF_SUMMARY | PTEN as a mechanistically diverse regulator of endocytic trafficking, combining versatile phosphatase- dependent functions with a phosphatase-independent role that ensures homeostasis of signaling receptors in the cell. |
| Literature | GENERIF_SUMMARY | In prostate tumor tissue microarrays, loss of PTEN phosphohydrolase (PTEN) correlates with increased tyrosine kinase 6 PTK6 tyrosine 342 (PY342) phosphorylation and poor outcome. |
| Literature | GENERIF_SUMMARY | Studied association of phosphatase and tensin homolog (PTEN) expression and glucose homeostasis in nutritionally supplemented, hospitalized, critically ill trauma patients. Found PTEN expression has no influence on glucose homeostasis in young, nondiabetic, critically ill trauma patients. |
| Literature | GENERIF_SUMMARY | Results show that PTEN and WNT5A expression are directly repressed by miR-26b which promotes colorectal cancer metastasis. |

| Main Category | Category | Terms |
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| Literature | GENERIF_SUMMARY | Loss of PTEN and ARID1A expression is highly specific for malignancy in effusion pathology |
| Literature | GENERIF_SUMMARY | MTSS1 is stabilized by the protein phosphatase activity of the tumor suppressor PTEN. Our data show that PTEN loss in PDAC cells results in both increased metastatic potential and decreased MTSS1 expression. Furthermore, we show that ectopic MTSS1 expression rescues this effect. |
| Literature | GENERIF_SUMMARY | All B cell subsets from ITP patients, excluding memory B cells, showed lower PTEN expression than cells from controls and correlated with platelet count. IL-21, anti-IgM, and CD40L induced PTEN expression. PTEN expression, regulation, and function all contribute to the B cell hyper-responsiveness associated with ITP. |
| Literature | GENERIF_SUMMARY | Inhibiting miR-21 modulates biological functions of gastric cancer cells via PTEN/PI3K/mTOR pathway. |
| Literature | GENERIF_SUMMARY | PTEN mutation is associated with Riley Ruvalcaba syndrome and diffuse testicular microlithiasis and precocious puberty. |
| Literature | GENERIF_SUMMARY | Decreased PTEN expression in the colonic mucosa might contribute to the pathogenesis of microscopic colitis. |
| Literature | GENERIF_SUMMARY | Knockdown of MTDH expression also upregulated PTEN and suppressed pAKT protein expression in Caki2 cells. In conclusion, the present study is the first, to the best of our knowledge, to provide evidence suggesting that miRNA30a5p suppresses tumor human renal cancer cell proliferation via the MTDH/PTEN/AKT pathway |
| Literature | GENERIF_SUMMARY | a certain degree of mitochondrial oxidative activity, with some difference for PTEN-wild type SF767 cells respect to PTEN-deleted A172 and U87MG characterized by a loss-of-function point mutation of PTEN. |
| Literature | GENERIF_SUMMARY | Integrative Genomic Analysis of Coincident Cancer Foci Implicates CTNNB1 and PTEN Alterations in Ductal Prostate Cancer. |
| Literature | GENERIF_SUMMARY | Ectopic PTEN overexpression in bladder cancer cells blocked the Akt signal pathway which attenuated cell growth via upregualtion of BTG2 gene expression, while reverse effect was found in PTEN-knockdown cells. PTEN activity inhibitor treatment decreased BTG2 expression in |

| Main Category | Category | Terms |
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| | | Highly differentiated bladder cancer cells. Results suggested that BTG2 functioned as a bladder cancer tumor suppressor gene and was induced by PTEN. |
| Literature | GENERIF_SUMMARY | CFTR attaches tumor suppressor PTEN to the membrane and promotes anti Pseudomonas aeruginosa immunity. |
| Literature | GENERIF_SUMMARY | PTEN expression is inversely correlated with miR-28 expression levels in gastric cancer tissues. |
| Literature | GENERIF_SUMMARY | To evaluate the association between the five SNPs (mTOR rs2295080 and rs2536, AKT1 rs2494750 and rs2494752, pTEN rs701848) and cancer risk. |
| Literature | GENERIF_SUMMARY | PTEN expression is regulated by STRAP protein in the hepatocellular carcinoma. |
| Literature | GENERIF_SUMMARY | A novel information on the susceptibility of PTEN to the inflammatory oxidant HOCl and its effects on the structure and activity of the protein is provided. |
| Literature | GENERIF_SUMMARY | PTEN loss is associated with castration-resistant prostate cancer. |
| Literature | GENERIF_SUMMARY | Pten expression in the hepatocellular carcinoma and its role in the hepatocarcinogenesis. |
| Literature | GENERIF_SUMMARY | We conclude that miR-486-5p stimulates cell proliferation, migration, and invasion through inhibition of PTEN expression and activation of the oncogenic PI3K/Akt pathway in cervical cancer. Our findings implicate serum miR-486-5p as a novel molecular biomarker that may provide effective approaches to both diagnosis and treatment in cervical cancer |
| Literature | GENERIF_SUMMARY | Low PTEN expression is associated with osteosarcoma. |
| Literature | GENERIF_SUMMARY | miR-21 may promote salivary adenoid cystic carcinoma progression via PDCD4 and PTEN down-regulation and Bcl-2 up-regulation. |
| Literature | GENERIF_SUMMARY | DBP(n) are non-toxic to the cells and have a weak overall demethylation effect on genomic DNA. DBP(n) demethylate the promoter regions of the tumor suppressor genes PTEN and RARB. DBP(n) promotes expression of the genes RARB, PTEN, CDKN2A, RUNX3, Apaf-1 and APC ""silent"" in the MCF-7 because of the hypermethylation of their promoter regions" |
| Literature | GENERIF_SUMMARY | Ezh2 regulates PTEN/AKT signaling by |

| Main Category | Category | Terms |
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| | | directly binding to the promoter regions of PTEN in Gastric Cancer. |
| Literature | GENERIF_SUMMARY | Knockdown of endogenous PTEN enhanced cell migration in DU145 cells, whereas PTEN overexpression reduced migration in PC3 cells and reduced phosphorylation of AKT in response to TGF-beta. |
| Literature | GENERIF_SUMMARY | Low PTEN expression is associated with thyroid cancer progression. |
| Literature | GENERIF_SUMMARY | Podocyte-specific knockin of PTEN protected the kidney against hyperglycemia and was partly attributed to an improvement in autophagy and motility and the inhibition of apoptosis. |
| Literature | GENERIF_SUMMARY | Loss of PTEN increased cell migration, invasion, and upregulated WNT4, a key regulator of Mullerian duct development during embryogenesis. Further investigation revealed that WNT4 was required for increased migration and colonization of the ovary by PTEN-deficient oviductal cells in a beta-catenin independent manner. |
| Literature | GENERIF_SUMMARY | decreased PTEN expression was associated with higher number of recurrence episodes in non-invasive low-grade papillary urothelial carcinoma of the bladder |
| Literature | GENERIF_SUMMARY | miR-21-5p mediates apoptosis by targeting PTEN and TIMP3. |
| Literature | GENERIF_SUMMARY | ATR signalling adversely impact survival in PTEN-deficient breast cancers. ATR inhibition is synthetically lethal in PTEN-deficient triple-negative breast cancer (TNBC) cells |
| Literature | GENERIF_SUMMARY | The decreased PTEN was associated with poorer survival outcomes of patients with kidney cancer and PTEN acts as a tumor suppressor in tumorigeneses and progression in kidney cancer. |
| Literature | GENERIF_SUMMARY | Taken together, these data suggest that melatonin-MT1 receptor complexes represent a potential target for the treatment of glioma. |
| Literature | GENERIF_SUMMARY | mechanism study shows that STX3 binds to PTEN and increases PTEN ubiquitination and degradation, thus leading to activation of the PI3K-Akt-mTOR signaling. Therefore, STX3 promotes the growth of breast cancer cells by regulating the PTEN-PI3K-Akt-mTOR signaling. |
| Literature | GENERIF_SUMMARY | The overexpression of PTEN concomitant with Livin gene silencing was confirmed as a |

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| | | feasible and effective in vitro and in vivo gene modulation method, which may represent a potential therapeutic strategy for the treatment of Gastric Cancer. |
| Literature | GENERIF_SUMMARY | Intimal PTEN and vascular smooth muscle cell (SMC) contractile protein expression was reduced compared with the media in arteries with nonatherosclerotic and atherosclerotic hyperplasia or complex plaque. Study propose that PTEN is an antiinflammatory, antifibrotic target that functions to maintain SMC differentiation. SMC loss of PTEN results in pathological vascular remodeling of human arteries. |
| Literature | GENERIF_SUMMARY | Decrease of PTEN via increase of miR-181a may be important contributor to endometrial cancer in non-obese patients. |
| Literature | GENERIF_SUMMARY | miR-181 may be a novel and important regulator of cisplatin-resistant non-small cell lung cancer by serving a role in the regulation of apoptosis, as an established rate-limiting miRNA target. |
| Literature | GENERIF_SUMMARY | study found that CKS2 knockdown induced PTEN up-regulation and may associate with P53 pathway activation |
| Literature | GENERIF_SUMMARY | The results of the present study indicate that the expression of miRNA23a may regulate acute myocardial infarction (AMI) through targeting PTEN in patients and in vitro, and PTEN/miRNA23a may therefore be potential targets for the clinical treatment of AMI. |
| Literature | GENERIF_SUMMARY | data implicate PTEN in maintaining genomic stability by delaying G2/M-phase progression of damaged cells, thus allowing time for double strand break repair by homologous recombination |
| Literature | GENERIF_SUMMARY | The present study confirmed that pAURKA is important in the development of gastric adenocarcinoma and revealed a novel functional link between PTEN, AURKA and pAURKA activation |
| Literature | GENERIF_SUMMARY | miR367 was revealed to bind directly to phosphatase and tensin homolog (PTEN) mRNA and regulate the expression of the PTEN protein |
| Literature | GENERIF_SUMMARY | ATM is a promising target for PTEN-defective breast cancer. |
| Literature | GENERIF_SUMMARY | Our findings show that SCD1 promotes metastasis of CRC cells through MUFA production and suppressing PTEN in response to glucose, which may be a novel |

| Main Category | Category | Terms |
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| | | mechanism for diabetes-induced CRC metastasis. |
| Literature | GENERIF_SUMMARY | LASP1 interacted with PTEN and decreased the expression of PTEN in nasopharyngeal carcinoma. |
| Literature | GENERIF_SUMMARY | Study demonstrates that electrophilic 15d-PGJ2 directly binds to cysteine 136 of PTEN and provides new insight into PTEN loss in cancer progression associated with chronic inflammation. These observations suggest that 15d-PGJ2 can undergo nucleophilic addition to PTEN, presumably at Cys(136), thereby inactivating this tumor suppressor protein with concomitant Akt activation. |
| Literature | GENERIF_SUMMARY | PTEN role in the resistance to BRAF inhibitors in melanoma. |
| Literature | GENERIF_SUMMARY | Confirmation studies (step 4) provided additional evidence that NUAK1 and STK11 have PTEN-SSL patterns of activity. Consistent with PTEN-SSL status, inhibition of the NUAK1 protein kinase by the small molecule drug HTH-01-015 selectively impaired viability in multiple PTEN-deficient breast cancer cell lines, while mutations affecting STK11 and PTEN were largely mutually exclusive across large pan-cancer data sets. |
| Literature | GENERIF_SUMMARY | Results provide evidence that PIK3CA mutation and PTEN loss coexist in prostate cancer and can cooperate in vivo to accelerate tumorigenesis and facilitate castration-resistant prostate cancer. |
| Literature | GENERIF_SUMMARY | This work demonstrates a novel function for PTEN during cell polarization in controlling apical membrane size and identifies PTPL1 as a critical apical membrane anchor for PTEN in this process. |
| Literature | GENERIF_SUMMARY | CK1alpha bound the C-terminal tail of PTEN and enhanced both PTEN stability and activity by competitively antagonizing NEDD4-1-induced PTEN polyubiquitination and abrogating PTEN phosphorylation, thereby inhibiting AKT activity and activating FOXO3a-induced transcription of Atg7, to suppress lung tumor growth. |
| Literature | GENERIF_SUMMARY | The human PTEN gene encodes a lipid phosphatase with specificity towards phosphatidylinositol (3,4,5) triphosphate (PIP3). |
| Literature | GENERIF_SUMMARY | miR-130b was upregulated in the lupus nephritis group, compared with that in the control group. PTEN was identified as a |

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| | | virtual target of miR-130b, and there was a negative regulatory association between miR-130b and PTEN. miR-130b and PTEN interfered with the viability and apoptosis of mesangial cells. |
| Literature | GENERIF_SUMMARY | it is highly conceivable that the subset of PTEN-negative mucinous tumors may still have a relatively good prognosis due to a different/PTEN-independent protective pathway or mechanism. |
| Literature | GENERIF_SUMMARY | a statistically significant association between PTEN loss and the triple negative breast cancers was found in African American women |
| Literature | GENERIF_SUMMARY | the abundance of phosphatase and tension homolog deleted on chromosome ten (PTEN), a negative downstream target of SP1, was increased with the ectopic miR-150-3p Collectively, these results suggested that miR-150-3p suppressed the growth of glioma cells partially via regulating SP1 and possibly PTEN. |
| Literature | GENERIF_SUMMARY | Upregulation of plasma miR-93-5p expression significantly increases the risk of esophageal cancer and is associated with poor prognosis. miR-93-5p transferred by exosomes promotes the proliferation of recipient esophageal cancer cells and affects the expression of PTEN and its downstream proteins p21 and cyclin D1. |
| Literature | GENERIF_SUMMARY | We conducted a 4-year multicenter prospective study of incident patients with features of Cowden/Cowden-like (CS/CS-like) and Bannayan-Riley-Ruvalcaba syndromes (BRRS) without PTEN mutations. Pathogenic or likely pathogenic cancer susceptibility gene alterations were found in 7 of the 87 (8%) CS/CS-like and BRRS patients and included MUTYH, RET, TSC2, BRCA1, BRCA2, ERCC2 and HRAS. |
| Literature | GENERIF_SUMMARY | Pathogenic missense mutations in the N-terminus of PTEN lead to accumulation of the mutant protein in the cell nucleus. |
| Literature | GENERIF_SUMMARY | Chemerin upregulated expression and phosphatase activity of PTEN by interfering with PTEN-CMKLR1 interaction, leading to weakened ubiquitination of PTEN and decreased p-Akt (Ser473) level, which was responsible for suppressed migration, invasion and metastasis of HCC cells. |
| Literature | GENERIF_SUMMARY | inflammatory state of adipose tissue is not affected by the anti-inflammatory response of the A2a-adenosine system and miR-221/ |

| Main Category | Category | Terms |
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| | | PTEN |
| Literature | GENERIF_SUMMARY | Low PTEN mRNA expression was associated with down-regulation of a group of genes involved in immune responses and B-cell development. |
| Literature | GENERIF_SUMMARY | several p85a mutations found in human cancers may deregulate PTEN and/or Rab5 regulated pathways to contribute to oncogenesis. We also engineered several experimental mutations within the p85a BH domain and identified L191 and V263 as important for both binding and regulation of Rab5 activity |
| Literature | GENERIF_SUMMARY | High Notch-1 and low PTEN mRNA expression may predict poorer overall survival. |
| Literature | GENERIF_SUMMARY | The test results of diffusion-weighted imaging and magnetic resonance spectroscopy of patients with glioblastoma multiforme (GBM) can accurately reflect the inactivation or mutation of PTEN and p53 |
| Literature | GENERIF_SUMMARY | These results demonstrate that SPAG6 silencing induces PTEN expression to regulate apoptosis though the PI3K/AKT pathway, indicating that SPAG6 may be a potential therapeutic target for myelodysplastic syndromes. |
| Literature | GENERIF_SUMMARY | The inhibition of PTEN also reduced the cancer effects of CD4+ T cells on non-small cell lung cancer (NSCLC) cell lines following miR-142-5p downregulation. Therefore, our study demonstrated that miR-142-5p regulated CD4+ T cells in human NSCLC through PD-L1 expression via the PTEN pathway. |
| Literature | GENERIF_SUMMARY | we first identified that YAP1 can modulate tumour-associated MDSC induction through suppressing PTEN expression to activate P-AKT, P-p65 and COX-2 and subsequently induce the secretion of GM-CSF, which is associated with the expansion of MDSCs in CRC |
| Literature | GENERIF_SUMMARY | Study identified PTEN as a direct target of miR-19a in ovarian cancer (OC) cells. PTEN expression is inversely correlated with the expression of miR-19a in OC tissues; its overexpression attenuated the promotion effect of miR-19a on the growth of OC cells. |
| Literature | GENERIF_SUMMARY | PTEN and IL10RB mRNA levels were reduced in the human colonic mucosa of pediatric ulcerative colitis compared with non-inflammatory bowel disease subjects. |

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| | | Loss of Pten function increases the levels of colitogenic bacteria in the gut, thereby inducing deleterious colitis in an Il10-deficient condition. |
| Literature | GENERIF_SUMMARY | A translational variant of PTEN has been identified and termed PTEN-Long (PTEN-L). The additional 173 amino acids (PTEN-L leader) at the N-terminal constitute a potential signal peptide. Differing from canonical PTEN, PTEN-L is secreted into the extracellular fluid and re-enters recipient cells, playing the similar roles as PTEN in vivo and in vitro. |
| Literature | GENERIF_SUMMARY | findings showed that PTEN inhibits HBV replication as well as HBV HCV co-replication. SREBP-1 is involved in HBV HCV replication inhibition by PTEN |
| Literature | GENERIF_SUMMARY | Results showed that SIX1 was overexpressed in osteosarcoma tissues, blood samples and cell lines, whereas PTEN expression was reduced. |
| Literature | GENERIF_SUMMARY | in silico analysis revealed PTEN to be the downstream target of miR-21, which was further confirmed by expression analysis. |
| Literature | GENERIF_SUMMARY | These results demonstrated that shikonin inhibits CML proliferation and migration and induces apoptosis by the PTEN/PI3K/AKT pathway, revealing the effects of shikonin therapy on CML. |
| Literature | GENERIF_SUMMARY | Our studies identify a previously undefined function of ERG to restrict lineage plasticity and maintain antiandrogen sensitivity in PTEN/TP53-altered prostate cancer. Our findings suggest ERG fusion as a biomarker to guide treatment of PTEN/TP53-altered, RB1-intact prostate cancer. |
| Literature | GENERIF_SUMMARY | miR23b3p and PTEN interfered with the viability and apoptosis of smooth muscle cells. |
| Literature | GENERIF_SUMMARY | The findings revealed that delayed endothelial replicative senescence caused by the inhibition of MTORC1 activation could be modulated by miR-107 via its influence on PTEN. |
| Literature | GENERIF_SUMMARY | MiR-221 together with proteins MDR1 and ABCG2 was upregulated in Cisplatin-resistant A549 lung cancer cells. Anti-miR-221 inhibits proliferation and induces senescence in lung cancer cells. PTEN/Akt pathway axis was identified as a target of drug resistance induced by miR-221. |
| Literature | GENERIF_SUMMARY | Circulating exosomal miR-301a-3p levels |

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| | | positively associated with depth of invasion, lymph node metastasis, late TNM stage, and poor prognosis of pancreatic cancer. Hypoxic exosomal miR-301a-3p induced the M2 polarization of macrophages via activation of the PTEN/PI3Kgamma signaling pathway. |
| Literature | GENERIF_SUMMARY | Nuclear phosphatase and tensin homolog protein (PTEN) expression was observed in approximately half of endometrial adenocarcinoma (EndoCA) patient tumors. |
| Literature | GENERIF_SUMMARY | TERT could induce thyroid carcinoma cell proliferation mainly through the PTEN/AKT signaling pathway. |
| Literature | GENERIF_SUMMARY | Functionally, Ataxin-3 overexpression promoted cell proliferation, and Ataxin-3 knockdown inhibited cell proliferation in testicular cancer cell. In addition, up-regulation of Ataxin-3 inhibited the expression of PTEN and activated the AKT/mTOR pathway. |
| Literature | GENERIF_SUMMARY | Results showed that MiR-374b was highly expressed, while PTEN was downregulated in the GIST tissues. The levels of miR-374b, PI3K, AKT and PTEN were related to tumor diameter and pathological stage. Additionally, miR-374b increased the mRNA and protein levels of PI3K, Akt, MMP2, MMP9, P53 and cyclinD1, suggesting that miR-374b activates PI3K/Akt signaling pathway in GIST-T1 cells. |
| Literature | GENERIF_SUMMARY | These results suggest that miR214 mediates vascular inflammation and apoptosis via PTEN expression. |
| Literature | GENERIF_SUMMARY | MicroRNA-1297 contributes to the progression of human cervical carcinoma through PTEN |
| Literature | GENERIF_SUMMARY | results point out that lncRNA DUXAP8 is overexpressed in bladder cancer tissues, which can promote the progression of bladder cancer through inhibiting PTEN. |
| Literature | GENERIF_SUMMARY | Nuclear phosphatase and tensin homologue on chromosome ten protein (PTEN) interacts with the splicing machinery, spliceosome, to regulate its assembly and pre-mRNA splicing. |
| Literature | GENERIF_SUMMARY | Taken together, these findings demonstrate a novel function of PTEN-L as a protein phosphatase for ubiquitin, which counteracts PINK1-mediated ubiquitin phosphorylation leading to blockage of the feedforward mechanisms in mitophagy |

| Main Category | Category | Terms |
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| | | induction and eventual suppression of mitophagy. |
| Literature | GENERIF_SUMMARY | These findings suggest that PTEN inhibition promotes post-ischemic angiogenesis in HUVECs after exposure to oxygen-glucose deprivation and this enhancing effect might be achieved through activation of the Akt signal cascade. |
| Literature | GENERIF_SUMMARY | PTEN genomic aberrations and the biologically consequential PTEN inactivation contribute to adverse therapy response in T-ALL patients |
| Literature | GENERIF_SUMMARY | t is relevant to investigate the potential additive effect of the overexpression of GAS1 and PTEN on tumor growth. In particular, we employed secreted forms of both GAS1 (tGAS1) and PTEN (PTEN-LONG, or PTEN-L) and tested their combined effect on glioma cells. |
| Literature | GENERIF_SUMMARY | PTEN is a prognostically significant molecular marker in renal cell carcinoma. |
| Literature | GENERIF_SUMMARY | PTEN splice variant PTEN-Delta shows similar function as the tumor suppressor PTEN itself |
| Literature | GENERIF_SUMMARY | SIRT6 inhibited proliferation, migration, and invasion of colon cancer cells by up-regulating PTEN expression and down-regulating AKT1 expression. |
| Literature | GENERIF_SUMMARY | A negative correlation between HRD1 and PTEN expression. |
| Literature | GENERIF_SUMMARY | PTEN is dispensable for myeloid leukemia cells in response to MAPK inhibitors. |
| Literature | GENERIF_SUMMARY | PTENalpha functions as a mitochondrial quality controller that maintains mitochondrial function and cardiac homeostasis. |
| Literature | GENERIF_SUMMARY | A126 in the active site and TI167/168 in the TI loop are essential for PTEN's substrate specificity. |
| Literature | GENERIF_SUMMARY | The role of PTEN in the process is potentially significant providing a suppressor balance to the p53/ survivin complex. |
| Literature | GENERIF_SUMMARY | Low PTEN expression is associated with thyroid cancer progression. |
| Literature | GENERIF_SUMMARY | Low PTEN expression is associated with multiple myeloma. |
| Literature | GENERIF_SUMMARY | The data indicate that diagnostic or |

| Main Category | Category | Terms |
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| | | therapeutic chest radiation may predispose patients with decreased stromal PTEN expression to secondary breast cancer, and that prophylactic EGFR inhibition may reduce this risk. |
| Literature | GENERIF_SUMMARY | study illustrated a novel signaling cascade of LINC00641/miR-197-3p/KLF10/PTEN/PI3K/AKT pathway regulating bladder cancer development |
| Literature | GENERIF_SUMMARY | PDCD4 and PTEN were the functional targets of miR-21. |
| Literature | GENERIF_SUMMARY | Studies have indicated that in breast cancer, PTEN undergo mutations. There is a functional and mechanistic link between the BMI-1 oncoprotein and tumor suppressor PTEN in the development and progression of breast cancer. [review] |
| Literature | GENERIF_SUMMARY | The results implied that both squamous and glandular components of adenosquamous carcinoma of the lung may share the same oncogenic driver pathway for carcinogenesis, which is reflected in the patterns of expression of PTEN and PD-L1. |
| Literature | GENERIF_SUMMARY | Data indicate a significant prognostic role for assessing transcriptional regulator ERG (ERG) and phosphatase and tensin homolog protein (PTEN) in men with prostate cancer. |
| Literature | GENERIF_SUMMARY | A patient with rare tumors that have a germline mutation first hit in PTEN in common. |
| Literature | GENERIF_SUMMARY | We demonstrated that expression of PTEN and miR-718 were significantly correlated in patients with gastric cancer. Low expression of PTEN and high level of miR-718 were notably associated with a lower 5-year overall survival rate. Both PTEN and miR-718 were identified as prognostic factors of gastric cancer. |
| Literature | GENERIF_SUMMARY | the expression of PTEN and miR-144 was inversely correlated in metastatic breast cancer cell lines. |
| Literature | GENERIF_SUMMARY | Study suggested that there may be a regulatory loop between miR21 and PTEN, and that miR21 inhibition affected the proliferative, invasive and apoptotic abilities of oral squamous cell carcinoma (OSCC) cells. miR-21 expression was observed in 80.0% OSCC tissues and in 30.0% of normal tissues. By contrast, PTEN expression exhibited an opposite trend in OSCC tissues 37.1%, and normal tissues 80.0%. |
| Literature | GENERIF_SUMMARY | our results elucidate the role of the |

| Main Category | Category | Terms |
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| | | LINC00312-miR-21-PTEN axis in colorectal cancer (CRC) cell proliferation and tumour progression and may lead to new lncRNA-based diagnostics or therapeutics for CRC |
| Literature | GENERIF_SUMMARY | PTEN lipid phosphatase inactivation abolished the MOB1-LATS1/2 interaction, decreased YAP phosphorylation and finally promoted YAP nuclear translocation, which enhanced the synergistic effect of YAP-TEAD, thus inducing cell proliferation and migration. |
| Literature | GENERIF_SUMMARY | findings provided a comprehensive mutation profiling of AKT1, PIK3CA, PTEN and TP53 genes in Chinese breast cancer patients, which have potential implications in clinical management. |
| Literature | GENERIF_SUMMARY | we provide a review on current understandings of the regulation of PTEN by ncRNAs, which could contribute to the development of novel approaches to the diseases with abnormal expression of PTEN. |
| Literature | GENERIF_SUMMARY | NIS has a role in thyroid cancer cells that involves PTEN signaling |
| Literature | GENERIF_SUMMARY | it was suggested that Cantharidin (CTD)be used as a novel antiproliferation agent of human melanoma via targeting the miR21PTEN signaling pathway. |
| Literature | GENERIF_SUMMARY | In addition, there was a negative correlation between PTEN mRNA and the miR328 expression levels. In addition, higher miR328 expression levels, and lower PTEN mRNA and protein expression levels, were detected in cisplatinresistant A549 (A549rCDDP) cells when compared with in their parental cells |
| Literature | GENERIF_SUMMARY | In this study, we used the Ion Personal Genome Machine (PGM) and Ion Torrent Ampliseq Cancer panel to sequence hotspot regions from PIK3CA, AKT and PTEN genes to identify genetic mutations in 39 samples of TNBC subtype from Moroccan patients and to correlate the results with clinical-pathologic data |
| Literature | GENERIF_SUMMARY | Low PTEN expression is associated with Epithelial-mesenchymal transition in gastric carcinoma. |
| Literature | GENERIF_SUMMARY | The IRIS-driven metastatic mechanism results from IRIS-dependent suppression of phosphatase and tensin homolog (PTEN) transcription, which in turn perturbs the PI3K/AKT/GSK-3beta pathway leading to |

| Main Category | Category | Terms |
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| | | prolyl hydroxylase-independent HIF-1alpha stabilization and activation in a normoxic environment. |
| Literature | GENERIF_SUMMARY | PTEN loss (heterozygous or homozygous) was significantly associated with shorter time to biochemical recurrence in clinically localized prostate cancer. |
| Literature | GENERIF_SUMMARY | LncRNA HORAIRM1 suppressed the PI3K/AKT pathway in gastric cancer (GC) and inhibited the progression of GC by serving as a competing endogenous RNA of miR-17-5p, mediating the expression of PTEN. |
| Literature | GENERIF_SUMMARY | LncRNA MEG3 and PTEN were down-regulated in ovarian cancer cells. LncRNA MEG3 regulated the downstream gene PTEN in ovarian cancer cells to prohibit cell proliferation, promote apoptosis and block cell cycle progression. |
| Literature | GENERIF_SUMMARY | This study aims to investigate the expression level of mir-let7b-3p and mir-548, which are involved in PTEN expression in tissue samples of prostate cancer patients versus benign prostate hyperplasia and normal adjacent tissue |
| Literature | GENERIF_SUMMARY | Clinically meaningful differences in PFS were seen based on PTEN expression by IHC, lost in >50% of patients. |
| Literature | GENERIF_SUMMARY | Bioinformatic studies showed that these two miRNAs target PTEN and DUSP14 tumor suppressor genes. Quantitative Real-time PCR confirmed the overexpression of the miRNAs and downregulation of their targets. Luciferase assay confirmed that the miRNAs target PTEN and DUSP14. |
| Literature | GENERIF_SUMMARY | Results show that PTEN depletion induces cellular senescence via the p53-p21 signaling pathway without triggering a DNA damage response |
| Literature | GENERIF_SUMMARY | Low expression of microRNA-21 attenuated cell infiltration of trophoblasts via direct regulating PTEN expression in preeclampsia. |
| Literature | GENERIF_SUMMARY | Results indicated that PTEN was a downstream target of microRNA-200a and their interaction was greatly involved in the development of ovarian cancer. |
| Literature | GENERIF_SUMMARY | Down- regulated miR-328 has a tumor-suppressive effect, preventing hepatocellular carcinoma cell metastasis and proliferation by targeting PTEN. |

| Main Category | Category | Terms |
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| Literature | GENERIF_SUMMARY | Highly expressed FAL1 can promote proliferation and migration of endothelial cells through activating PTEN/AKT signaling pathway in diabetic atherosclerosis. |
| Literature | GENERIF_SUMMARY | The HOTAIR/miR-17-5p/PTEN axis could be regarded as the potential treatment targets for gastric cancer. |
| Literature | GENERIF_SUMMARY | The miR-10a could promote GCT development via synergistically regulating PTEN, Akt, and Wnt pathways. |
| Literature | GENERIF_SUMMARY | ARL4C stabilized by AKT/mTOR pathway promotes the invasiveness of PTEN-deficient glioblastoma cells. |
| Literature | GENERIF_SUMMARY | Compared to PTEN-positive BCs, PTEN-negative BCs were significantly more frequently associated with lobular tumor histology (p |
| Literature | GENERIF_SUMMARY | The CD28 and PTEN gene pair were found to play important roles in the occurrence and development of cervical cancer. |
| Literature | GENERIF_SUMMARY | The inhibition of PTEN reduces the function of miR494 in glioma cell growth through the Akt/mTOR pathway; on the contrary, the promotion of PTEN promotes the function of miR494 in glioma cancer cell proliferation through the Akt/mTOR pathway. |
| Literature | GENERIF_SUMMARY | It is a tumor suppressive gene regulated by several biomarkers including a small single-stranded molecule, microRNA26b (miR-26b). |
| Literature | GENERIF_SUMMARY | study suggested for the first time that PTEN was independent prognostic immune biomarkers in classical Hodgkin's lymphoma |
| Literature | GENERIF_SUMMARY | Low PTEN expression is associated with colorectal cancer progression. |
| Literature | GENERIF_SUMMARY | The levels of PTEN were negatively associated with PRL-3 expression. PRL-3 might inhibit the protein level of PTEN through enhancing its phosphorylation level. |
| Literature | GENERIF_SUMMARY | miR-4465 significantly inhibited the expression of PTEN, upregulated phosphorylated AKT, and thereby inhibited autophagy by activating mTOR. |
| Literature | GENERIF_SUMMARY | Absence of detectable CEBPA expression was markedly more frequent in ERG negative (45%) as compared to ERG positive cancers (20%, P |
| Literature | GENERIF_SUMMARY | Low PTEN expression is associated with gastric cancer. |

| Main Category | Category | Terms |
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| Literature | GENERIF_SUMMARY | Low PTEN expression is associated with progression of endometrial cancer. |
| Literature | GENERIF_SUMMARY | High PTEN expression is associated with hepatocellular carcinoma. |
| Literature | GENERIF_SUMMARY | LncRNA TUG1 could compete with PTEN for miR-21 binding. |
| Literature | GENERIF_SUMMARY | we showed that TP73AS1 inhibits CRC cell growth by functioning as a ceRNA (competing endogenous RNAs) to regulate PTEN levels. Our findings provide new insights into the underlying molecular mechanisms of TP73AS1-mediated CRC. |
| Literature | GENERIF_SUMMARY | The present findings confirm that PTEN is a prognostic factor for metastatic head and neck squamous cell carcinoma |
| Literature | GENERIF_SUMMARY | SLC25A22 is highly expressed in osteosarcoma, promoting osteosarcoma cell proliferation and invasion by inhibiting the phosphatase and tensin homolog signaling pathway |
| Literature | GENERIF_SUMMARY | results highlight the synthetic lethal relationship between PTEN and BRG1 and support targeting BRG1 as an effective approach to the treatment of PTEN-deficient prostate cancer. |
| Literature | GENERIF_SUMMARY | In hepatocellular carcinoma cells,CASC11 epigenetically silenced PTEN by binding with EZH2. |
| Literature | GENERIF_SUMMARY | Compared with normal glucose tolerance (NGT) groups, the PTEN mRNA expression was significantly higher in Uyghur patients with mild type 2 diabetes mellitus (T2DM) groups; PTEN protein expression was upregulated in Uyghur patients with mild T2DM groups. PTEN methylation in T2DM patients was significantly lower than that in NGT groups. 2 CpG units demonstrated a significant difference between NGT and Uyghur patients. |
| Literature | GENERIF_SUMMARY | senescent alveolar epithelial cell controlled by the PTEN/NF-kappaB pathway facilitated collagen accumulation in fibroblasts, resulting in lung fibrosis. |
| Literature | GENERIF_SUMMARY | PTEN loss in fibroblasts promotes extracellular matrix deposition and alignment independently from cancer cell presence, and this reorganization regulates cancer cell behavior. Importantly, stromal PTEN negatively correlated with collagen alignment and high mammographic density in human breast tissue, suggesting parallel function for PTEN in patients. |

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| Literature | GENERIF_SUMMARY | Deleterious variants are associated with an increased risk of relapse and death in Castration-Sensitive Prostate Cancer . Poorer outcomes are seen with compound gene hits in both early and advanced disease, and this may in part reflect increasing global genomic instability. |
| Literature | GENERIF_SUMMARY | The overexpression of lncRNA FAL1 regulates cell proliferation and cell cycle in gastric cancer cells by inhibiting PTEN. |
| Literature | GENERIF_SUMMARY | WDR66 expression was identified to be inversely associated with PTEN expression and negatively correlated with the overall survival of patients with salivary adenoid cystic carcinoma (SACC). Collectively, the results of the study revealed a novel function of WDR66 in mediating the progression of PTENdeficient SACCs. |
| Literature | GENERIF_SUMMARY | MiR-103/107 regulates cell proliferation and PI3K/AKT signaling partially through PTEN dependent mechanism. |
| Literature | GENERIF_SUMMARY | GAS5 suppresses proliferation, migration, invasion, and epithelial-mesenchymal transition in OSCC through the miR-21/PTEN. |
| Literature | GENERIF_SUMMARY | it can be concluded that estrogen improves the miR-200c level by combining with ER, PTENP1 and PTEN could be inhibited by miR-200c, and then activate the PI3K-AKT pathway. |
| Literature | GENERIF_SUMMARY | YY1-regulated LINC00152 promotes triple negative breast cancer progression by affecting the stability of PTEN protein. |
| Literature | GENERIF_SUMMARY | Findings support that NQO1 displays a paradoxical role in mediating GBM growth in response to tumor suppressor PTEN. |
| Literature | GENERIF_SUMMARY | present study showed 25% somatic mutational frequency in exonic region 5 of the PTEN gene. This may indicate its role in the pathogenesis of ameloblastoma |
| Literature | GENERIF_SUMMARY | Through regulating the homeostasis of CD4(+) T cells in a PD1/PTEN/AKT signalling pathway. |
| Literature | GENERIF_SUMMARY | SIRT4 was induced and cooperated with IDE to degrade PTEN. |
| Literature | GENERIF_SUMMARY | MiR-182-5p exhaustion blocked cell proliferation and invasion by regulating PTEN expression, providing a novel therapeutic avenue for treatment of breast cancer. |

| Main Category | Category | Terms |
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| Literature | GENERIF_SUMMARY | PTEN mutations affect innate immune cell function in the PTEN hamartoma tumor syndrome patients. |
| Literature | GENERIF_SUMMARY | Rpn10 directly promoted PTEN degradation. |
| Literature | GENERIF_SUMMARY | PTEN expression provides differential therapeutic strategies for north Indian breast cancer. |
| Literature | GENERIF_SUMMARY | HBx promoted the proliferation of HCC cell and reduced the apoptosis of HCC cells through the SNHG20/PTEN signalling pathway |
| Literature | GENERIF_SUMMARY | results uncover that miR-21 targets PTEN/Akt signaling pathway and regulates cell proliferation, migration and apoptosis in human gastric cancer cells. |
| Literature | GENERIF_SUMMARY | GFRA2 was upregulated in human neuroblastoma cells and tissues, and promoted neuroblastoma cell proliferation through interacting with the tumor suppressor PTEN. |
| Literature | GENERIF_SUMMARY | NF-kappaB-Induced Upregulation of miR-548as-3p Increases Invasion of NSCLC by Targeting PTEN |
| Literature | GENERIF_SUMMARY | In this case-control study we evaluated the association of genetic variants in PTEN-L with chronic myeloid leukemia risk and therapy response in the Argentine population. The allele A of SNP rs12573787 was found to be associated with chronic myeloid leukemia risk OR (95% CI) 1.71 (1.11-2.63) p=0.016, which resulted consistent by multivariate analysis adjusted by gender and age. |
| Literature | GENERIF_SUMMARY | Oncogenic Y68 frame shift mutation of PTEN represents a mechanism of docetaxel resistance in endometrial cancer cell lines. |
| Literature | GENERIF_SUMMARY | The inhibition of microRNA-21-5p increased PTEN and reduced mTOR signaling and remodeling. |
| Literature | GENERIF_SUMMARY | the expression profiles of ZHX2 and PTEN were positively correlated in hepatocellular carcinoma tissues |
| Literature | GENERIF_SUMMARY | Single Cell Profiling Reveals PTEN Overexpression in Influenza-Specific B cells in Aging HIV-infected individuals on Anti-retroviral Therapy. |
| Literature | GENERIF_SUMMARY | EZH2 silencing and PTEN overexpression significantly abrogated the SOX2-OT overexpression-mediated promotion of laryngeal squamous cell carcinoma (LSCC) |

| Main Category | Category | Terms |
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| | | cell malignant behavior. Collectively, our findings demonstrate that SOX2-OT inhibits PTEN expression to facilitate LSCC development through EZH2-mediated H3K27me3 |
| Literature | GENERIF_SUMMARY | Authors revealed that MTA2 promoted PDAC cell proliferation, migration, and invasion in vitro and PDAC tumor growth in vivo by downregulation of PTEN. |
| Literature | GENERIF_SUMMARY | We identified that PTEN loss and increased MET copy number may predict an unfavorable survival in mutated-EGFR non-small-cell lung cancer patients. |
| Literature | GENERIF_SUMMARY | Data shows that pY240-PTEN mediates resistance to DNA damage through enhancement of the DNA repair process. This particular phosphorylation-related function of PTEN is separate from its canonical function of opposing AKT pathway activation. |
| Literature | GENERIF_SUMMARY | MiR-144-3p promotes the progression of NPC by directly targeting PTEN. |
| Literature | GENERIF_SUMMARY | This study establishes that GRA16 is a HAUSP inhibitor that targets the nuclear localization of PTEN and induces the anticancer effect in a p53-dependent manner. The efficacy of GRA16 could be newly highlighted in hepatocellular carcinoma (HCC) treatment in a p53-dependent manner |
| Literature | GENERIF_SUMMARY | These results indicated that LINC00702 modulated the expression of PTEN gene by acting as a ceRNA for miR-510 in non-small cell lung cancer. |
| Literature | GENERIF_SUMMARY | MiR-183-5p is required for NSCLC development through the suppressing PTEN. |
| Literature | GENERIF_SUMMARY | The PTEN expression was increased in the CAD or ICAD group compared to the control group (CAD vs. controls P |
| Literature | GENERIF_SUMMARY | YAP1 promotes cell growth and inhibits cell apoptosis of breast cancer (BC) through the phosphatase and tensin homolog deleted on chromosome 10-AKT signaling pathway, and thus suggest that YAP1 might serve as a new target for inhibiting BC progression |
| Literature | GENERIF_SUMMARY | tis study shows that PTEN inactivation induces epithelial-mesenchymal transition and metastasis by intranuclear translocation of beta-catenin and snail/slug in non-small cell lung carcinoma cells |

| Main Category | Category | Terms |
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| Literature | GENERIF_SUMMARY | PTEN arginine methylation modulates pre-mRNA alternative splicing and influences diverse physiologic processes. |
| Literature | GENERIF_SUMMARY | Low PTEN expression is associated with Pulmonary arterial hypertension. |
| Literature | GENERIF_SUMMARY | The effect of decitabine on the expression and methylation of the PPP1CA, BTG2, and PTEN in association with changes in miR-125b, miR-17, and miR-181b in NALM6 cell line. |
| Literature | GENERIF_SUMMARY | FDPS plays an oncogenic role in PTEN-deficient Prostate cancer through GTPase/AKT axis. Identifying mevalonate pathway proteins could serve as a therapeutic target in PTEN dysregulated tumors. |
| Literature | GENERIF_SUMMARY | The BRD4-KDM5C-PTEN may be a new oncogenic pathway in CRPC development. |
| Literature | GENERIF_SUMMARY | Study found PTEN as a direct target of miR-96. There is an inverse correlation between the expression level of PTEN protein and miR-96 level in neck squamous cell carcinoma (HNSCC) cells. PTEN is a mediator of chemo-radiotherapy treatment for HNSCC cells. |
| Literature | GENERIF_SUMMARY | Low levels of PDZK1 in gastric cancer specimens lead to increase proliferation of gastric cancer cells via phosphorylation of PTEN at the S380/T382/T383 cluster and constitutively activation of PI3K/AKT signaling, which results in poor prognosis of gastric cancer patients. |
| Literature | GENERIF_SUMMARY | Activation of the oncogenic miR-21-5p promotes HCV replication and steatosis induced by the viral core 3a protein. |
| Literature | GENERIF_SUMMARY | Authors found PTEN to be a direct target of miR-371. The overexpression or knockdown of PTEN exhibited the opposite effects from those of miR-371 on cell proliferation and migration. Study demonstrates that miR-371 promotes proliferation and metastasis in HCC by targeting PTEN. |
| Literature | GENERIF_SUMMARY | these results confirm PTEN as a direct target of miR-301a in HeLa cells and suggest that miR-301a/PTEN pathway contributes to the development and progression of cervical cancer |
| Literature | GENERIF_SUMMARY | phosphatase and tensin homolog (PTEN) was recognized as a target of miR-21, and STAT3 inhibition restored AngII-induced reduction in PTEN. Similarly, the STAT3/miR-21 axis was shown to mediate AngII-provoked angiogenesis in vivo, which was |

| Main Category | Category | Terms |
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| | | demonstrated by using the appropriate inhibitors. |
| Literature | GENERIF_SUMMARY | Bioactive miR-19b-3p transmitted to cancer cells by cancer stem cells exosomes induced EMT via repressing the expression of PTEN. |
| Literature | GENERIF_SUMMARY | these data support a critical role for NF-kappaB-dependent upregulation of miR-181b, which further inhibited PTEN expression and promoted the cell proliferation of osteosarcoma cell lines. |
| Literature | GENERIF_SUMMARY | phosphatase and tensin homolog (PTEN) was a direct target of miR-186-5p verified by luciferase reporter assay. Overexpression of PTEN significantly suppressed LUAD cells to proliferate, migrate and invade. This study demonstrated that miR-186-5p promotes the proliferation, migration and invasion of LUAD cells by targeting PTEN. |
| Literature | GENERIF_SUMMARY | miR-202-5p was high-expressed in breast cancer tissues and cells, targeted and downregulated PTEN expression. DOX-resistance of breast cancer cells was enhanced by miR-202-5p through PTEN/PI3K/Akt signaling |
| Literature | GENERIF_SUMMARY | Findings imply that PTEN deficiency is linked to an immunosuppressive state in PCa with distinct changes in the frequency of immune cell types in tumors from different metastatic sites. |
| Literature | GENERIF_SUMMARY | Results found that the increase of GSK3beta in chemoresistant breast cancer cells then phosphorylates PTEN at Thr366, leading to a loss of PTEN activity. This loss of PTEN function results in activation of Akt leading to increased migratory and invasive properties and the development of chemoresistance, probably contributing to cancer progression. |
| Literature | GENERIF_SUMMARY | RhoB inhibits the proliferation, invasion, EMT, and PTEN/AKT signal pathway in breast tumor cells. |
| Literature | GENERIF_SUMMARY | LncRNA LEF1-AS1 regulates the migration and proliferation of vascular smooth muscle cells by targeting miR-544a/PTEN axis. |
| Literature | GENERIF_SUMMARY | PTEN-loss independently conferred higher risk for relapse in Breast Cancer. |
| Literature | GENERIF_SUMMARY | MiR-363-3p was significantly highly expressed in osteoporotic samples. Mechanistically, miR-363-3p promotes osteoclastogenesis and inhibits osteogenic differentiation by targeting PTEN and |

| Main Category | Category | Terms |
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| | | therefore activating PI3K/AKT signaling pathway. MiR-363-3p was activated by its upstream transcription activator MYB |
| Literature | GENERIF_SUMMARY | These results demonstrated that overexpression of TUSC8 could inhibit the invasion and migration of cervical cancer cells by upregulating PTEN via miR-641. |
| Literature | GENERIF_SUMMARY | Our study demonstrated that miR-21 could promote proliferation and inhibit apoptosis of hepatic stellate cells, and its mechanism may be related to PTEN/PI3K/AKT pathway. |
| Literature | GENERIF_SUMMARY | athogenic PTEN variant c.956_959delCTTT p.(Thr319LysfsTer24) was detected in his thyroid tissue. |
| Literature | GENERIF_SUMMARY | HOXC13-AS, which is significantly upregulated in breast cancers, promoted cell proliferation through the suppressed miR-497-5p and further upregulated PTEN. |
| Literature | GENERIF_SUMMARY | Assessment of epigenetic alterations and in silico analysis of mutation affecting PTEN expression among Indian cervical cancer patients. |
| Literature | GENERIF_SUMMARY | our findings elucidate that miR-4286 promotes the tumorigenesis of NSCLC by interacting with PTEN. This miR-4286-mediated upregulation of PTEN might lead to new therapeutic strategies for NSCLC. |
| Literature | GENERIF_SUMMARY | Expression of the PTEN/FOXO3a/PLZF signalling pathway in pancreatic cancer and its significance in tumourigenesis and progression. |
| Literature | GENERIF_SUMMARY | Among traditional serrated pathway-associated colorectal carcinomas with aberrant PTEN expression, 8 harbored PTEN somatic mutations, which were characterized by random distribution without hotspot clustering, 12 had promoter hypermethylation, and 14 had deleted alleles. |
| Literature | GENERIF_SUMMARY | Positive association has been found between 32-bp deletion polymorphism, promoter methylation of PTEN, and hepatitis C virus induced hepatocellular carcinoma. |
| Literature | GENERIF_SUMMARY | Study reveal that in mesenchymal stem cells derived from human bone marrow (BM-MSCs), but not in dental pulp (DP-MSCs), DNMT3B DNA methyltransferase and G9a histone methyltransferase establish DNA methylation and H3K9Me2 on the PTEN promoter, respectively. Higher PTEN expression in DP-MSCs than in BM-MSCs is responsible for the lineage commitment and |

| Main Category | Category | Terms |
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| | | tumorigenesis differences in both cells. |
| Literature | GENERIF_SUMMARY | prostate cancer with PTEN loss demonstrates a strong correlation with known unfavorable histologic features, particularly intraductal carcinoma of the prostate |
| Literature | GENERIF_SUMMARY | PTEN plays a key role in hepatocarcinogenesis and reduction of PTEN expression is related to increased expression of CD133, EpCAM and CK19 and hepatocellular carcinoma prognosis and recurrence. |
| Literature | GENERIF_SUMMARY | Our observations demonstrate an imperative role of PTEN in global transcriptional regulation that will provide a new direction to understand PTEN-associated pathologies and its management. |
| Literature | GENERIF_SUMMARY | NBAT-1 overexpression remarkably upregulated PTEN expression in gastric cancer (GC) cells. Low expression of NBAT-1 promotes GC development by downregulating PTEN expression. |
| Literature | GENERIF_SUMMARY | The findings of this study suggested L-THP increased the sensitivity of ovarian cancer cells to cisplatin via modulating miR-93/PTEN/AKT pathway in A2780/DDP ovarian cancer cell line. |
| Literature | GENERIF_SUMMARY | MiR-21 promotes tumor cell proliferation and migration by targeting inhibition of PTEN expression. |
| Literature | GENERIF_SUMMARY | TFF3 expression was detected in 131/226 (57.9%) RP samples and 148/318 (46.5%) of TURP cases. In general, TFF3 positivity was less frequently observed with advanced Gleason Groups. TFF3 expression was also assessed in relation to PTEN expression. Only 15-16% of TFF3-expressed cases were present in association with complete loss of PTEN expression in the TURP and localized cohorts, respectively. |
| Literature | GENERIF_SUMMARY | Results found that PTEN is a direct target of miR-188 which expression negatively correlated with PTEN expression in gastric cancer (GC) tumors. Further data suggested that suppression of PTEN is necessary for the pro-metastatic property of miR-188 upregulation in GC cells. |
| Literature | GENERIF_SUMMARY | PTEN improve renal fibrosis in vitro and in vivo through inhibiting FAK/AKT signaling pathway. |
| Literature | GENERIF_SUMMARY | our findings indicate that TP53 changes and CDKN2A deletions tend to coexist in LGG, |

| Main Category | Category | Terms |
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| | | suggesting that they are early events in progression of these tumors, as well as CDKN2A and PTEN deletions, since a significant association was observed, with CDKN2A changes preceding PTEN deletions, that are involved with malignant progression of gliomas. |
| Literature | GENERIF_SUMMARY | Study confirms that PTEN was the downstream candidate gene of miR-205. Its expression is downregulated by mir-205 in nasopharyngeal carcinoma. |
| Literature | GENERIF_SUMMARY | Nuclear phosphatase and tensin homologue deleted on chromosome ten (PTEN), which by interacting with the transcription machinery in the context of chromatin exerts an additional layer of regulatory control on RNA polymerase II (RNAPII)-mediated transcription. |
| Literature | GENERIF_SUMMARY | In PTEN-null glioblastoma multiforme xenograft mouse models, inhibition of LOX reduces macrophage infiltration and tumor growth. |
| Literature | GENERIF_SUMMARY | PTENP1/miR-20a/PTEN axis involved in the malignant behaviors of BC cells, illuminating the possible mechanism mediated by PTEN via PI3K/Akt pathway. Targeting PTENP1/miR-20a/PTEN may provide a potential diagnosis and treatment strategy for BC. |
| Literature | GENERIF_SUMMARY | Analysis of Site-Specific Phosphorylation of PTEN by Using Enzyme-Catalyzed Expressed Protein Ligation. |
| Literature | GENERIF_SUMMARY | KRAS exon 3 and PTEN exon 7 mutations had no significant impact on overall survival of small-cell lung cancer patients. |
| Literature | GENERIF_SUMMARY | STXBP5-AS1 might function as a ceRNA to drive CC cells proliferation and invasion via regulating miR-96-5p/PTEN axis. |
| Literature | GENERIF_SUMMARY | The prostate cancer suppressors PTEN and NKX3.1 interact and loss of PTEN is responsible, at least in part, for progressive loss of NKX3.1 that occurs during tumor progression. |
| Literature | GENERIF_SUMMARY | Study reports that DNA double-strand breaks (DSBs) promote NSD2-mediated dimethylation of PTEN at K349, which is read by the tudor domain of 53BP1 to recruit PTEN into DNA-damage sites to govern the timely repair of DSBs in part through dephosphorylating gammaH2AX. |
| Literature | GENERIF_SUMMARY | miR-32 negatively modulates PTEN, thereby promoting M2 macrophage transformation through PI3K/AKT signaling, enhancing |

| Main Category | Category | Terms |
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| | | glioma proliferation and migration abilities. |
| Literature | GENERIF_SUMMARY | MiR-26a-5p enhances cells proliferation, invasion, and apoptosis resistance of fibroblast-like synoviocytes in rheumatoid arthritis by regulating PTEN/PI3K/AKT pathway. |
| Literature | GENERIF_SUMMARY | Results found that found that miR-130b and miR-494 could both bind to the 3'-UTR region of PTEN to inhibit its expression which promotes migration, invasion, and proliferation of bladder cancer cells. |
| Literature | GENERIF_SUMMARY | CELF2 suppresses non-small cell lung carcinoma growth by inhibiting the PREX2-PTEN interaction. |
| Literature | GENERIF_SUMMARY | Overexpression of miR-19a enhanced the proliferation and migration of airway smooth muscle (ASM) cells, whereas inhibition of miR-19a decreased the proliferation and migration of ASM cells. Western blot analysis demonstrated that miR-19a negatively regulated PTEN expression and positively regulated p-AKT expression. |
| Literature | GENERIF_SUMMARY | RSRC1 knockdown promoted the proliferation and migration of gastric cancer cells. In addition, the knockdown of RSRC1 decreased the expression of phosphatase and tensin homolog deleted on chromosome 10 (PTEN), a potent tumor suppressor gene controlling cellular growth and viability. |
| Literature | GENERIF_SUMMARY | Association between PTEN and clinical-pathological features of osteosarcoma. |
| Literature | GENERIF_SUMMARY | the combined effect of PTEN and CFTR dysfunction stimulates mitochondrial activity, resulting in excessive release of succinate and reactive oxygen species. |
| Literature | GENERIF_SUMMARY | MEG3 interacted with miR-494 to repress bladder cancer progression through targeting PTEN. |
| Literature | GENERIF_SUMMARY | Autism Spectrum Disorder Associated with Germline Heterozygous PTEN Mutations. |
| Literature | GENERIF_SUMMARY | Studied association of prognosis and survival in gastric cancer patients with regards to phosphatase and tensin homolog (PTEN), epidermal growth factor receptor (EGFR), erb-b2 receptor tyrosine kinase 2 (HER-2), and insulin like growth factor 1 receptor (IGF-1R) expression. |
| Literature | GENERIF_SUMMARY | Long noncoding RNA Linc01296 promotes hepatocellular carcinoma development through regulation of the miR-26a/PTEN |

| Main Category | Category | Terms |
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| | | axis. |
| Literature | GENERIF_SUMMARY | aberrant expression of miR-498 changed the expression levels of phosphatase and tensin homolog and p-AKT in LNCaP and DU-145 PCa cells. |
| Literature | GENERIF_SUMMARY | PTEN and BMI1 help mediate NEUROGENIN-3-induced cell cycle arrest |
| Literature | GENERIF_SUMMARY | LncRNA GHET1 promotes osteoblast proliferation and differentiation by inhibiting PTEN. |
| Literature | GENERIF_SUMMARY | PLCE1 inhibits apoptosis of non-small cell lung cancer via promoting PTEN methylation. |
| Literature | GENERIF_SUMMARY | PTEN, an established target of miR367 in skin melanoma (SM) cell lines, was positively regulated by Linc00961, and its inhibition reversed the inhibitory effects of Linc00961 on the proliferation and invasion of SM cell lines. |
| Literature | GENERIF_SUMMARY | Daurinol Attenuates Autoimmune Arthritis via Stabilization of Nrp1-PTEN-Foxp3 Signaling in Regulatory T Cells. |
| Literature | GENERIF_SUMMARY | Further study of underlying mechanisms demonstrated that DMBT1 combined with PTEN which could stabilize PTEN protein, resulting in inhibiting the activation of PI3K/AKT signaling pathway |
| Literature | GENERIF_SUMMARY | Low PTEN expression detected by fluorescence immunohistochemistry in primary prostate cancer is an independent prognostic biomarker for metastatic disease and death after definitive therapy. |
| Literature | GENERIF_SUMMARY | Findings show that miR-200b and miR-200c suppress the proliferation of granulosa-like tumor cell line (KGN) cells by targeting PTEN protein tyrosine phosphatase (PTEN). |
| Literature | GENERIF_SUMMARY | Dual-luciferase reporter gene assay confirmed the binding site between microRNA-19a and PTEN. PTEN expression was negatively regulated by microRNA-19a both at mRNA and protein levels. |
| Literature | GENERIF_SUMMARY | PTEN in Autism and Neurodevelopmental Disorders. |
| Literature | GENERIF_SUMMARY | Increased expression of miR-205-5p and PTEN in gastric cancer tissues were correlated with tumor stage. In SGC-7901 cells, miR-205-5p mRNA expression in the miR-inhibitor and miR-inhibitor+si-PTEN groups was significantly lower than that in the NC group (P |

| Main Category | Category | Terms |
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| Literature | GENERIF_SUMMARY | Promoter hypermethylation of PTEN is associated with Endometrial Cancer. |
| Literature | GENERIF_SUMMARY | The Impact of Genetic Variants on PTEN Molecular Functions and Cellular Phenotypes. |
| Literature | GENERIF_SUMMARY | the anticancer mechanism of PTEN in glioblastoma is dependent on the activation of Drp1-related mitochondrial division via Akt pathway modulation. |
| Literature | GENERIF_SUMMARY | MicroRNA-26a regulates cerebral ischemia injury through targeting PTEN. |
| Literature | GENERIF_SUMMARY | MiR-647 promotes proliferation and migration of ox-LDL-treated vascular smooth muscle cells through regulating PTEN/PI3K/AKT pathway. |
| Literature | GENERIF_SUMMARY | The protein phosphatase activity of PTEN dephosphorylates and inhibits autophosphorylated PGK1, thereby inhibiting glycolysis, ATP production, and brain tumor cell proliferation. |
| Literature | GENERIF_SUMMARY | MicroRNA-483-3p Promotes Proliferation, Migration, and Invasion and Induces Chemoresistance of Wilms' Tumor Cells. |
| Literature | GENERIF_SUMMARY | CXCL12 regulates PTEN expression and is associated with the proliferation and invasion of colon cancer cells via PI3k/Akt signaling |
| Literature | GENERIF_SUMMARY | Precise Immunodetection of PTEN Protein in Human Neoplasia. |
| Literature | GENERIF_SUMMARY | The Role of PTEN in Innate and Adaptive Immunity. |
| Literature | GENERIF_SUMMARY | PTEN loss regulates alveolar epithelial cell senescence in pulmonary fibrosis depending on Akt activation. |
| Literature | GENERIF_SUMMARY | Patients with PTEN-deficient tumors may represent a subpopulation potentially amenable to treatment. |
| Literature | GENERIF_SUMMARY | Targeted multiplex proteomics for molecular prescreening and biomarker discovery in metastatic colorectal cancer. |
| Literature | GENERIF_SUMMARY | LncRNA MAGI2-AS3 suppresses the proliferation and invasion of non-small cell lung carcinoma through miRNA-23a-3p/PTEN axis. |
| Literature | GENERIF_SUMMARY | the present study revealed that PTEN knockdown resulted in MAD2 degradation via the ubiquitinproteasomal pathway, while restoration of MAD2 expression partially |

| Main Category | Category | Terms |
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| | | ameliorated the mitotic defects induced by PTEN loss. |
| Literature | GENERIF_SUMMARY | Downregulation of PTEN and phosphorylated AKT is associated with refractory acute myeloid leukemia (AML) and resistance to idarubicin and cytarabine treatment. |
| Literature | GENERIF_SUMMARY | Data revealed that PTEN expression was markedly lower in vulvar squamous cell carcinoma (VSCC) tissues. Its expression is directly regulated by miR-4712. |
| Literature | GENERIF_SUMMARY | Dual-Specific Protein and Lipid Phosphatase PTEN and Its Biological Functions. |
| Literature | GENERIF_SUMMARY | High mRNA Expression of PTEN is associated with Breast Cancer. |
| Literature | GENERIF_SUMMARY | Inhibition of microRNA-21-5p reduces keloid fibroblast autophagy and migration by targeting PTEN after electron beam irradiation. |
| Literature | GENERIF_SUMMARY | PTEN in Chromatin Remodeling. |
| Literature | GENERIF_SUMMARY | PTEN loss activates a functional AKT/CXCR4 signaling axis to potentiate tumor growth and lung metastasis in human osteosarcoma cells. |
| Literature | GENERIF_SUMMARY | Results provide evidence that PTEN upregulates TAGLN in human bladder carcinoma cells. |
| Literature | GENERIF_SUMMARY | PTEN Loss Mediates Clinical Cross-Resistance to CDK4/6 and PI3Kalpha Inhibitors in Breast Cancer. |
| Literature | GENERIF_SUMMARY | These analyses revealed a distinct neuropsychological profile associated with mutations in PTEN suggesting primary disruption of frontal lobe systems. Cognitive deficits in PTEN-ASD (autism spectrum disorder) are more severe than those in PTEN-no ASD and extend to other areas of neurobehavioral function, specifically, adaptive behavior and sensory deficits. |
| Literature | GENERIF_SUMMARY | Immunohistochemically Detected Expression of ATRX, TSC2, and PTEN Predicts Clinical Outcomes in Patients With Grade 1 and 2 Pancreatic Neuroendocrine Tumors. |
| Literature | GENERIF_SUMMARY | miR-298 plays a pivotal role in colon cancer invasiveness by targeting PTEN. |
| Literature | GENERIF_SUMMARY | MiR-205 promotes endothelial progenitor cell angiogenesis and deep vein thrombosis recanalization and resolution by targeting |

| Main Category | Category | Terms |
|---|---|---|
| | | PTEN to regulate Akt/autophagy pathway and MMP2 expression. |
| Literature | GENERIF_SUMMARY | Structural Mechanisms of PTEN Regulation. |
| Literature | GENERIF_SUMMARY | Hypermethylation of FOXA1 and allelic loss of PTEN drive squamous differentiation and promote heterogeneity in bladder cancer. |
| Literature | GENERIF_SUMMARY | miR4255p expression was increased in breast cancer (BC) tissues and cell lines, and was associated with tumor size, clinical stage, lymph node metastasis, distant metastasis and poor overall survival in patients with BC. Knockdown of miR4255p in BC cell lines inhibited proliferation and migration. PTEN was identified as a downstream target gene of miR4255p. |
| Literature | GENERIF_SUMMARY | LncRNA PCAT6 promotes occurrence and development of ovarian cancer by inhibiting PTEN. |
| Literature | GENERIF_SUMMARY | PTEN promotes intervertebral disc degeneration by regulating nucleus pulposus cell behaviors. |
| Literature | GENERIF_SUMMARY | WWP2 promotes proliferation of gastric cancer cells in a PTEN-dependent manner. |
| Literature | GENERIF_SUMMARY | MIR-19a overexpression enhanced PI3K/AKT signaling, which was attenuated by enforced expression of PTEN in HPASMCs. More importantly, knockdown of miR-19 attenuated the chronic hypoxia-induced PAH. |
| Literature | GENERIF_SUMMARY | USP9X promotes tumorigenesis and FBXW11 suppresses tumorigenesis through PTENalpha/beta. |
| Literature | GENERIF_SUMMARY | The Mechanisms Underlying PTEN Loss in Human Tumors Suggest Potential Therapeutic Opportunities. |
| Literature | GENERIF_SUMMARY | GAS5 functioned as a tumor suppressor role in HCC through regulation of miR-21-PTEN singling pathway. |
| Literature | GENERIF_SUMMARY | LINC00470 associated with PTEN mRNA and suppressed its stability through interaction with the N(6)-methyladenosine (m(6)A) writer METTL3 in gastric cancer. |
| Literature | GENERIF_SUMMARY | Circular RNA circGRAMD1B inhibits gastric cancer progression by sponging miR-130a-3p and regulating PTEN and p21 expression. |
| Literature | GENERIF_SUMMARY | We provide evidence that the allelic loss of PTEN is present in premalignant oral lesions and oral squamous cell carcinoma (OSCC), |

| Main Category | Category | Terms |
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| | | however the LOH of PTEN does not seems to influence its protein expression |
| Literature | GENERIF_SUMMARY | AKT, which mediates Ser432-USP14 phosphorylation, was required for IRIF formation by USP14. Similar to USP14 block, AKT inhibition rescued the activity of NHEJ-DDR proteins in autophagy- and PTEN-deficient cells. These findings reveal a novel negative PTEN/Akt-dependent regulation of NHEJ by |
| Literature | GENERIF_SUMMARY | PTEN and AKT1 Variations in Childhood T-Cell Acute Lymphoblastic Leukemia |
| Literature | GENERIF_SUMMARY | Statin Use Is Associated with Lower Risk of PTEN-Null and Lethal Prostate Cancer. |
| Literature | GENERIF_SUMMARY | The potential roles of several oncogenic pathways (phosphatase and tensin homolog (PTEN) and phosphatidylinositol 3-kinase (PI3K)) in relation to HER2 status. |
| Literature | GENERIF_SUMMARY | findings suggested that PTEN mutation in prostate cancer may induce changes in a variety of genes and pathways and affect disease progression, suggesting the significance of PTEN mutation in individualized treatment of prostate cancer |
| Literature | GENERIF_SUMMARY | hUC-MSCs secreted exosomes inhibit the glioma cell progression through PTENP1/miR-10a-5p/PTEN pathway. |
| Literature | GENERIF_SUMMARY | Resistance to BTK inhibition by ibrutinib can be overcome by preventing FOXO3a nuclear export and PI3K/AKT activation in B-cell lymphoid malignancies. |
| Literature | GENERIF_SUMMARY | Biomarker May Predict Cancer Versus Autism Risk in Pten Hamartoma Tumor Syndrome |
| Literature | GENERIF_SUMMARY | PTEN expression in U251 glioma cells enhances their sensitivity to ionizing radiation by suppressing DNA repair capacity. |
| Literature | GENERIF_SUMMARY | FOXF2 aggravates the progression of non-small cell lung cancer through targeting lncRNA H19 to downregulate PTEN. |
| Literature | GENERIF_SUMMARY | Relationship between polycomb-group protein BMI-1 and phosphatases regulating AKT phosphorylation level in endometrial cancer. |
| Literature | GENERIF_SUMMARY | Study showed that in Ishikawa endometrial carcinoma cells, downregulation of PTEN was associated with the expression of the CDH1 and CDH2 genes and upregulated expression of the cell membrane |

| Main Category | Category | Terms |
|---|---|---|
| | | glycoprotein, CD133, which are associated with epithelial-mesenchymal transition (EMT) in malignancy. |
| Literature | GENERIF_SUMMARY | The Complex Landscape of PTEN mRNA Regulation. |
| Literature | GENERIF_SUMMARY | Identification and validation of mRNA 3'untranslated regions of DNMT3B and TET3 as novel competing endogenous RNAs of the tumor suppressor PTEN. |
| Literature | GENERIF_SUMMARY | Significantly differential expression of PTEN with corresponding changes in JunB/FosB (subunits of AP-1) and NF-kappaB activity after LPS stimulation. |
| Literature | GENERIF_SUMMARY | Circular RNA_LARP4 Sponges miR-1323 and Hampers Progression of Esophageal Squamous Cell Carcinoma Through Modulating PTEN/PI3K/AKT Pathway. |
| Literature | GENERIF_SUMMARY | Considering both the dismal prognosis and the characteristic mutation profiles revealed by WES analysis, the associated mediastinal GCT and AMKL harboring TP53 and PTEN mutations represent a unique biological entity. |
| Literature | GENERIF_SUMMARY | MicroRNA-221 promotes breast cancer resistance to adriamycin via modulation of PTEN/Akt/mTOR signaling. |
| Literature | GENERIF_SUMMARY | REVIEW |
| Literature | GENERIF_SUMMARY | Protein phosphatase Mg(2+) /Mn(2+) dependent-1A and PTEN deregulation in renal fibrosis |
| Literature | GENERIF_SUMMARY | Role and mechanism of PTEN in Burkitt's lymphoma. |
| Literature | GENERIF_SUMMARY | Crosstalk between PTEN/PI3K/Akt Signalling and DNA Damage in the Oocyte |
| Literature | GENERIF_SUMMARY | serum exosomal miR-106b level was significantly increased in lung cancer patients and positively associated with TNM stages and lymph node metastasis. Furthermore, exosomal miR-106b enhanced the migrated and invasive ability of lung cancer cells by regulating PTEN. |
| Literature | GENERIF_SUMMARY | miR-885 mediated cardioprotection against hypoxia/reoxygenation-induced apoptosis in human cardiomyocytes via inhibition of PTEN and BCL2L11 and modulation of AKT/mTOR signaling. |
| Literature | GENERIF_SUMMARY | Genome sequencing analysis of a family with a child displaying severe abdominal distention and recurrent hypoglycemia. |

| Main Category | Category | Terms |
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| Literature | GENERIF_SUMMARY | LINC-PINT alleviates lung cancer progression via sponging miR-543 and inducing PTEN. |
| Literature | GENERIF_SUMMARY | Western blot analyses demonstrated that while the expression of p53 and PTEN was increased, the p-Akt levels were decreased in COUP-TFII-SNU-C4 cells. |
| Literature | GENERIF_SUMMARY | Lowintensity ultrasound enhances the antitumor effects of doxorubicin on hepatocellular carcinoma cells through the ROSmiR21PTEN axis. |
| Literature | GENERIF_SUMMARY | Long noncoding RNA GAS5 increases the radiosensitivity of A549 cells through interaction with the miR21/PTEN/Akt axis. |
| Literature | GENERIF_SUMMARY | Overexpression of tripartite motif containing 26 inhibits non-small cell lung cancer cell growth by suppressing PI3K/AKT signaling. |
| Literature | GENERIF_SUMMARY | A Sox2 |
| Literature | GENERIF_SUMMARY | PTEN Expression as a Complementary Biomarker for Mismatch Repair Testing in Breast Cancer. |
| Literature | GENERIF_SUMMARY | MiR-214 Mediates Cell Proliferation and Apoptosis of Nasopharyngeal Carcinoma Through Targeting Both WWOX and PTEN. |
| Literature | GENERIF_SUMMARY | Risk Stratification of Prostate Cancer Through Quantitative Assessment of PTEN Loss (qPTEN). |
| Literature | GENERIF_SUMMARY | PTEN Loss and Cyclin A2 Upregulation Define a PI3K/AKT Pathway Activation in Helicobacter pylori-induced MALT and DLBCL Gastric Lymphoma With Features of MALT. |
| Literature | GENERIF_SUMMARY | PTEN promoter methylation is found in a number of ameloblastomas but not significantly correlated with loss of PTEN expression. Genetic or epigenetic mechanisms other than PTEN promoter methylation may contribute to PTEN inactivation in ameloblastoma tumor cells |
| Literature | GENERIF_SUMMARY | MiR-130a regulating the biological function of colon cancer by targeting inhibition of PTEN. |
| Literature | GENERIF_SUMMARY | LncRNA SNHG5 regulates cell apoptosis and inflammation by miR-132/PTEN axis in COPD. |
| Literature | GENERIF_SUMMARY | Autism-associated PTEN missense mutation leads to enhanced nuclear localization and neurite outgrowth in an induced pluripotent stem cell line. |

| Main Category | Category | Terms |
|---|---|---|
| Literature | GENERIF_SUMMARY | Clinical and molecular aspects of PTEN mutations in 10 pediatric patients. |
| Literature | GENERIF_SUMMARY | Targeted inhibition of ACK1 can inhibit the proliferation of hepatocellular carcinoma cells through the PTEN/AKT/mTOR pathway. |
| Literature | GENERIF_SUMMARY | Brain morphological analysis in PTEN hamartoma tumor syndrome. |
| Literature | GENERIF_SUMMARY | MicroRNA-744 promotes proliferation of osteosarcoma cells by targeting PTEN. |
| Literature | GENERIF_SUMMARY | PTEN inhibitor VO-OHpic attenuates GC-associated endothelial progenitor cell dysfunction and osteonecrosis of the femoral head via activating Nrf2 signaling and inhibiting mitochondrial apoptosis pathway. |
| Literature | GENERIF_SUMMARY | The germline mutation of the PTEN gene is not closely related to the occurrence of breast cancer in the Chinese population. |
| Literature | GENERIF_SUMMARY | PTEN may be a biomarker and therapeutic target in patients with metastatic renal cell carcinoma . |
| Literature | GENERIF_SUMMARY | Exosome-transmitted lncRNA UFC1 promotes non-small-cell lung cancer progression by EZH2-mediated epigenetic silencing of PTEN expression. |
| Literature | GENERIF_SUMMARY | knocking out either the ADAMTSL3 or PTEN genes promoted either the proliferation or metastasis of hepatocellular carcinoma cells, respectively |
| Literature | GENERIF_SUMMARY | MiR-20a acted as a ceRNA of lncRNA PTENPL and promoted bladder cancer cell proliferation and migration by regulating PDCD4. |
| Literature | GENERIF_SUMMARY | Enhancing PTEN suppressor gene expression in pancreatic carcinoma. |
| Literature | GENERIF_SUMMARY | PTEN mutation |
| Literature | GENERIF_SUMMARY | Inhibition of PTEN Gene Expression by Small Interfering RNA on PI3K/Akt/FoxO3a Signaling Pathway in Human Nasopharyngeal Carcinoma. |
| Literature | GENERIF_SUMMARY | Long Noncoding RNA XIST/miR-17/PTEN Axis Modulates the Proliferation and Apoptosis of Vascular Smooth Muscle Cells to Affect Stanford Type A Aortic Dissection. |
| Literature | GENERIF_SUMMARY | Diagnostic significance of DNA methylation of PTEN and DAPK in thyroid tumors. |

| Main Category | Category | Terms |
|---|---|---|
| Literature | GENERIF_SUMMARY | The Combination of Transient Receptor Potential Vanilloid Type 1 (TRPV1) and Phosphatase and Tension Homolog (PTEN) is an Effective Prognostic Biomarker in Cervical Cancer. |
| Literature | GENERIF_SUMMARY | PTEN and PHLPP crosstalk in cancer cells and in TGFbeta-activated stem cells. |
| Literature | GENERIF_SUMMARY | Sperm miR-26a-5p and its target PTEN transcripts content in men with unexplained infertility. |
| Literature | GENERIF_SUMMARY | Replication Study |
| Literature | GENERIF_SUMMARY | lncRNA LA16c313D11.11 modulates the development of endometrial cancer by binding to and inhibiting microRNA2055p function and indirectly increasing PTEN activity. |
| Literature | GENERIF_SUMMARY | LncRNA SNHG4 regulates miR-10a/PTEN to inhibit the proliferation of acute myeloid leukemia cells. |
| Literature | GENERIF_SUMMARY | miR21/PTEN pathway mediates the cardioprotection of geniposide against oxidized lowdensity lipoproteininduced endothelial injury via suppressing oxidative stress and inflammatory response. |
| Literature | GENERIF_SUMMARY | Linc00702 inhibits cell growth and metastasis through regulating PTEN in colorectal cancer. |
| Literature | GENERIF_SUMMARY | YY1-mediated PTEN dephosphorylation antagonizes IR-induced DNA repair contributing to tongue squamous cell carcinoma radiosensitization. |
| Literature | GENERIF_SUMMARY | Prognostic Significance of Transient Receptor Potential Vanilloid Type 1 (TRPV1) and Phosphatase and Tension Homolog (PTEN) in Epithelial Ovarian Cancer. |
| Literature | GENERIF_SUMMARY | Multi-model functionalization of disease-associated PTEN missense mutations identifies multiple molecular mechanisms underlying protein dysfunction. |
| Literature | GENERIF_SUMMARY | Overexpression of METTL3 attenuates high-glucose induced RPE cell pyroptosis by regulating miR-25-3p/PTEN/Akt signaling cascade through DGCR8. |
| Literature | GENERIF_SUMMARY | hsa-miR-20b-5p and hsa-miR-363-3p Affect Expression of PTEN and BIM Tumor Suppressor Genes and Modulate Survival of T-ALL Cells In Vitro. |
| Literature | GENERIF_SUMMARY | HBV-Induced Increased N6 Methyladenosine Modification of PTEN RNA |

| Main Category | Category | Terms |
|---|---|---|
| | | Affects Innate Immunity and Contributes to HCC. |
| Literature | GENERIF_SUMMARY | lncRNA TP73-AS1 Regulates miR-21/PTEN Axis to Affect Cell Proliferation in Acute Myeloid Leukemia. |
| Literature | GENERIF_SUMMARY | Curcumin inhibits proliferation of hepatocellular carcinoma cells through down regulation of DJ-1. |
| Literature | GENERIF_SUMMARY | MiR-519d-3p in Trophoblastic Cells |
| Literature | GENERIF_SUMMARY | Circular RNA hsa_circ_0008285 inhibits colorectal cancer cell proliferation and migration via the miR-382-5p/PTEN axis. |
| Literature | GENERIF_SUMMARY | PTEN and FOXO3 expression in the prenatal and postnatal human ovary. |
| Literature | GENERIF_SUMMARY | Pathway-Specific Genome Editing of PI3K/mTOR Tumor Suppressor Genes Reveals that PTEN Loss Contributes to Cetuximab Resistance in Head and Neck Cancer. |
| Literature | GENERIF_SUMMARY | An Integrated Deep-Mutational-Scanning Approach Provides Clinical Insights on PTEN Genotype-Phenotype Relationships. |
| Literature | GENERIF_SUMMARY | Combined p53- and PTEN-deficiency activates expression of mesenchyme homeobox 1 (MEOX1) required for growth of triple-negative breast cancer. |
| Literature | GENERIF_SUMMARY | The association of diabetes with risk of prostate cancer defined by clinical and molecular features. |
| Literature | GENERIF_SUMMARY | The interaction of interleukin-8 and PTEN inactivation promotes the malignant progression of head and neck squamous cell carcinoma via the STAT3 pathway. |
| Literature | GENERIF_SUMMARY | Single-cell profiling of pediatric T-cell acute lymphoblastic leukemia |
| Literature | GENERIF_SUMMARY | Comprehensive Analysis of PTEN in Primary Cutaneous Melanoma. |
| Literature | GENERIF_SUMMARY | PTENP1 is a ceRNA for PTEN |
| Literature | GENERIF_SUMMARY | MicroRNA-17-5p regulates the growth, migration and invasion of the human osteosarcoma cells by modulating the expression of PTEN. |
| Literature | GENERIF_SUMMARY | miR-410 induces both epithelial-mesenchymal transition and radioresistance through activation of the PI3K/mTOR pathway in non-small cell lung cancer. |
| Literature | GENERIF_SUMMARY | Regulating tumor suppressor genes |

| Main Category | Category | Terms |
|---|---|---|
| Literature | GENERIF_SUMMARY | Association of phosphatase and tension homologue deleted on chromosome ten polymorphism rs1903858, but not serum levels with the risk of non-small-cell lung cancer |
| Literature | GENERIF_SUMMARY | Integrative analysis of miRNAs-mRNAs reveals that miR-182 up-regulation contributes to proliferation and invasion of nasopharyngeal carcinoma by targeting PTEN. |
| Literature | GENERIF_SUMMARY | Correlation Between Single Nucleotide Polymorphisms of an miRNA Binding Site in the 3'UTR of PTEN and Risk of Cervical Cancer Among the Han Chinese. |
| Literature | GENERIF_SUMMARY | Clinical and pathological associations of PTEN expression in ovarian cancer |
| Literature | GENERIF_SUMMARY | Pseudogene PTENP1 sponges miR-214 to regulate the expression of PTEN to modulate osteoclast differentiation and attenuate osteoporosis. |
| Literature | GENERIF_SUMMARY | PTEN hamartoma tumour syndrome |
| Literature | GENERIF_SUMMARY | A new human adipocyte model with PTEN haploinsufficiency. |
| Literature | GENERIF_SUMMARY | Association of genetic variations in phosphatase and tensin homolog (PTEN) gene with polycystic ovary syndrome in South Indian women |
| Literature | GENERIF_SUMMARY | MiRNA-620 promotes TGF-beta1-induced proliferation of airway smooth muscle cell through controlling PTEN/AKT signaling pathway. |
| Literature | GENERIF_SUMMARY | Elevated V-ATPase Activity Following PTEN Loss Is Required for Enhanced Oncogenic Signaling in Breast Cancer. |
| Literature | GENERIF_SUMMARY | New germline mutations in non-BRCA genes among breast cancer women of Mongoloid origin. |
| Literature | GENERIF_SUMMARY | Chemerin Reactivates PTEN and Suppresses PD-L1 in Tumor Cells via Modulation of a Novel CMKLR1-mediated Signaling Cascade. |
| Literature | GENERIF_SUMMARY | The Role of circRNA-SETD2/miR-519a/PTEN Axis in Fetal Birth Weight through Regulating Trophoblast Proliferation. |
| Literature | GENERIF_SUMMARY | The long noncoding RNA CASC7 inhibits growth and invasion of nonsmall cell lung cancer cells through phosphatase and tensin homolog upregulation via sequestration of miR92a. |

| Main Category | Category | Terms |
|---|---|---|
| Literature | GENERIF_SUMMARY | MicroRNA-92b acts as an oncogene by targeting PTEN/AKT in NSCLC. |
| Literature | GENERIF_SUMMARY | PTEN and DNA-PK determine sensitivity and recovery in response to WEE1 inhibition in human breast cancer. |
| Literature | GENERIF_SUMMARY | Circ100284 promotes invasion and migration of osteosarcoma cells by down-regulating PTEN and EMP1. |
| Literature | GENERIF_SUMMARY | PTEN and ERG expression in MRI-ultrasound guided fusion biopsy correlated with radical prostatectomy findings in men with prostate cancer. |
| Literature | GENERIF_SUMMARY | Circ-ADAM9 targeting PTEN and ATG7 promotes autophagy and apoptosis of diabetic endothelial progenitor cells by sponging mir-20a-5p. |
| Literature | GENERIF_SUMMARY | Long non-coding RNA AWPPH enhances malignant phenotypes in nasopharyngeal carcinoma via silencing PTEN through interacting with LSD1 and EZH2. |
| Literature | GENERIF_SUMMARY | PTEN dephosphorylates Abi1 to promote epithelial morphogenesis. |
| Literature | GENERIF_SUMMARY | Overexpression of miR-217-5p protects against oxygen-glucose deprivation/ reperfusion-induced neuronal injury via inhibition of PTEN. |
| Literature | GENERIF_SUMMARY | LncRNA CTBP1-AS2 regulates miR-216a/ PTEN to suppress ovarian cancer cell proliferation. |
| Literature | GENERIF_SUMMARY | Role of phosphatase and tensin homolog in pathogenesis of ameloblastoma |
| Literature | GENERIF_SUMMARY | PTEN Function at the Interface between Cancer and Tumor Microenvironment |
| Literature | GENERIF_SUMMARY | PTEN suppresses epithelial-mesenchymal transition and cancer stem cell activity by downregulating Abi1. |
| Literature | GENERIF_SUMMARY | Metabolic stress regulates genome-wide transcription in a PTEN-dependent manner. |
| Literature | GENERIF_SUMMARY | miR-188-5p inhibits apoptosis of neuronal cells during oxygen-glucose deprivation (OGD)-induced stroke by suppressing PTEN. |
| Literature | GENERIF_SUMMARY | Prevalence of APC and PTEN Alterations in Urachal Cancer. |
| Literature | GENERIF_SUMMARY | LncRNA AY343892 inhibits breast cancer development by positively regulating BRCA1-mediated transcription of PTEN. |

| Main Category | Category | Terms |
|---|---|---|
| Literature | GENERIF_SUMMARY | [Down-regulation of HOTAIR Reverses the Resistance of Gefitinib in HCC827 Cells by Increasing the Expression of PTEN]. |
| Literature | GENERIF_SUMMARY | miR-520h Stimulates Drug Resistance to Paclitaxel by Targeting the OTUD3-PTEN Axis in Breast Cancer. |
| Literature | GENERIF_SUMMARY | The loss of PTEN expression and microsatellite stability (MSS) were predictors of unfavorable prognosis in gastric cancer (GC). |
| Literature | GENERIF_SUMMARY | LncRNA IUR downregulates miR-144 to regulate PTEN in nasopharyngeal carcinoma. |
| Literature | GENERIF_SUMMARY | CircPSMC3 alleviates the symptoms of PCOS by sponging miR-296-3p and regulating PTEN expression. |
| Literature | GENERIF_SUMMARY | Long Non-Coding RNA (lncRNA) NEAT1 Aggravates Cerebral Ischemia-Reperfusion Injury by Suppressing the Inhibitory Effect of miR-214 on PTEN. |
| Literature | GENERIF_SUMMARY | Enforced C-Src Activation Causes Compartmental Dysregulation of PI3K and PTEN Molecules in Lipid Rafts of Tongue Squamous Carcinoma Cells by Attenuating Rac1-Akt-GLUT-1-Mediated Sphingolipid Synthesis. |
| Literature | GENERIF_SUMMARY | PTEN in osteosarcoma |
| Literature | GENERIF_SUMMARY | Loss of PTEN expression is associated with PI3K pathway-dependent metabolic reprogramming in hepatocellular carcinoma. |
| Literature | GENERIF_SUMMARY | INNO-406 inhibits the growth of chronic myeloid leukemia and promotes its apoptosis via targeting PTEN. |
| Literature | GENERIF_SUMMARY | Enhancing PTEN suppressor gene expression in pancreatic carcinoma. |
| Literature | GENERIF_SUMMARY | SHIP2 inhibition alters redox-induced PI3K/AKT and MAP kinase pathways via PTEN over-activation in cervical cancer cells. |
| Literature | GENERIF_SUMMARY | Circular RNA Circ_0000442 acts as a sponge of MiR-148b-3p to suppress breast cancer via PTEN/PI3K/Akt signaling pathway. |
| Literature | GENERIF_SUMMARY | Nuclear Heparanase Regulates Chromatin Remodeling, Gene Expression and PTEN Tumor Suppressor Function. |
| Literature | GENERIF_SUMMARY | Expression level of NEAT1 differentiates benign and malignant thyroid nodules by regulating NEAT1/miR9/PTEN and NEAT1/ |

| Main Category | Category | Terms |
|---|---|---|
| | | miR124/PDCD6 signalling. |
| Literature | GENERIF_SUMMARY | miR1835p attenuates cerebral ischemia injury by negatively regulating PTEN. |
| Literature | GENERIF_SUMMARY | High-metastatic cancer cells derived exosomal miR92a-3p promotes epithelial-mesenchymal transition and metastasis of low-metastatic cancer cells by regulating PTEN/Akt pathway in hepatocellular carcinoma. |
| Literature | GENERIF_SUMMARY | miR-4516 inhibits the apoptosis of RB tumor cells by targeting the PTEN/AKT signaling pathway. |
| Literature | GENERIF_SUMMARY | Negative cross talk between LIMK2 and PTEN promotes castration resistant prostate cancer pathogenesis in cells and in vivo. |
| Literature | GENERIF_SUMMARY | METTL3-mediated maturation of miR-126-5p promotes ovarian cancer progression via PTEN-mediated PI3K/Akt/mTOR pathway. |
| Literature | GENERIF_SUMMARY | Circulating miR-155, let-7c, miR-21, and PTEN levels in differential diagnosis and prognosis of idiopathic granulomatous mastitis and breast cancer. |
| Literature | GENERIF_SUMMARY | Targeting the histone demethylase PHF8-mediated PKCalpha-Src-PTEN axis in HER2-negative gastric cancer. |
| Literature | GENERIF_SUMMARY | Upregulation of DJ-1 expression in melanoma regulates PTEN/AKT pathway for cell survival and migration. |
| Literature | GENERIF_SUMMARY | Polymicrogyria is Associated With Pathogenic Variants in PTEN. |
| Literature | GENERIF_SUMMARY | PTEN status determines chemosensitivity to proteasome inhibition in cholangiocarcinoma. |
| Literature | GENERIF_SUMMARY | [Overexpression of miR-382-5p Mediated Downregulation of PTEN on Malignant Biological Behavior of Human Glioma U251 Cells]. |
| Literature | GENERIF_SUMMARY | LncRNA GAS5 regulates epithelial-mesenchymal transition and viability of glioma cells by targeting microRNA-106b and regulating PTEN expression. |
| Literature | GENERIF_SUMMARY | Restoration of circPSMC3 sensitizes gefitinib-resistant esophageal squamous cell carcinoma cells to gefitinib by regulating miR-10a-5p/PTEN axis. |
| Literature | GENERIF_SUMMARY | Divergent Roles of PI3K Isoforms in PTEN- |

| Main Category | Category | Terms |
|---|---|---|
| | | Deficient Glioblastomas. |
| Literature | GENERIF_SUMMARY | SET8 suppression mediates high glucose-induced vascular endothelial inflammation via the upregulation of PTEN. |
| Literature | GENERIF_SUMMARY | Yin Yang 1-induced LINC00667 up-regulates pyruvate dehydrogenase kinase 1 to promote proliferation, migration and invasion of cholangiocarcinoma cells by sponging miR-200c-3p. |
| Literature | GENERIF_SUMMARY | HBV down-regulates PTEN expression via Nrf2/GSK3beta signaling pathway."", trans ""HBVNrf2/GSK3betaPTEN." |
| Literature | GENERIF_SUMMARY | MiR-130b can suppress proliferation of glioma cells through targeting PTEN to regulate AKT pathway. |
| Literature | GENERIF_SUMMARY | Significance of P16(INK4A) Expression and PTEN Loss of Heterozygosity in Human Papilloma Virus-related Oral Squamous Cell Carcinoma. |
| Literature | GENERIF_SUMMARY | CircRNA_0092516 regulates chondrocyte proliferation and apoptosis in osteoarthritis through the miR-337-3p/PTEN axis. |
| Literature | GENERIF_SUMMARY | Long non-coding RNA GAS5 overexpression inhibits M2-like polarization of tumour-associated macrophages in SMCC-7721 cells by promoting PTEN expression. |
| Literature | GENERIF_SUMMARY | Upregulated microRNA-126 induces apoptosis of dental pulp stem cell via mediating PTEN-regulated Akt activation. |
| Literature | GENERIF_SUMMARY | A Single Amino Acid Residue Regulates PTEN-Binding and Stability of the Spinal Muscular Atrophy Protein SMN. |
| Literature | GENERIF_SUMMARY | BAP1 suppresses prostate cancer progression by deubiquitinating and stabilizing PTEN. |
| Literature | GENERIF_SUMMARY | Genetic ablation of FASN attenuates the invasive potential of prostate cancer driven by Pten loss. |
| Literature | GENERIF_SUMMARY | LINC00861 inhibits the progression of cervical cancer cells by functioning as a ceRNA for miR513b5p and regulating the PTEN/AKT/mTOR signaling pathway. |
| Literature | GENERIF_SUMMARY | Molecular dynamic study on PTEN frameshift mutations in breast cancer provide c2 domain as a potential biomarker. |
| Literature | GENERIF_SUMMARY | Senescence Reprogramming by TIMP1 Deficiency Promotes Prostate Cancer Metastasis. |

| Main Category | Category | Terms |
|---|---|---|
| Literature | GENERIF_SUMMARY | PAG1 stimulates proliferation and metastasis of nasopharyngeal carcinoma through downregulating PTEN. |
| Literature | GENERIF_SUMMARY | PTEN deficiency facilitates the therapeutic vulnerability to proteasome inhibitor bortezomib in gallbladder cancer. |
| Literature | GENERIF_SUMMARY | WDHD1 is essential for the survival of PTEN-inactive triple-negative breast cancer. |
| Literature | GENERIF_SUMMARY | Dysregulation of Transcription Factor Activity During Formation of Cancer-Associated Fibroblasts. |
| Literature | GENERIF_SUMMARY | Oncogenic activation of PI3K-AKT-mTOR signaling suppresses ferroptosis via SREBP-mediated lipogenesis. |
| Literature | GENERIF_SUMMARY | High-risk HPV infection modulates the promoter hypermethylation of APC, SFRP1, and PTEN in cervical cancer patients of North India. |
| Literature | GENERIF_SUMMARY | PTEN, ATM, IDH1 mutations and MAPK pathway activation as modulators of PFS and OS in patients treated by first line EGFR TKI, an ancillary study of the French Cooperative Thoracic Intergroup (IFCT) Biomarkers France project. |
| Literature | GENERIF_SUMMARY | Activation of PAR2 by tissue factor induces the release of the PTEN from MAGI proteins and regulates PTEN and Akt activities. |
| Literature | GENERIF_SUMMARY | Circulatory miRNA-155, miRNA-21 target PTEN expression and activity as a factor in breast cancer development. |
| Literature | GENERIF_SUMMARY | Phosphorylation and Driver Mutations in PI3Kalpha and PTEN Autoinhibition. |
| Literature | GENERIF_SUMMARY | LncRNA ANCR promotes glioma cells invasion, migration, proliferation and inhibits apoptosis via interacting with EZH2 and repressing PTEN expression. |
| Literature | GENERIF_SUMMARY | Neddylation of PTEN regulates its nuclear import and promotes tumor development. |
| Literature | GENERIF_SUMMARY | New PTEN mutation identified in a patient with rare bilateral choroidal ganglioneuroma. |
| Literature | GENERIF_SUMMARY | Upregulation of PI3K/AKT/PTEN pathway is correlated with glucose and glutamine metabolic dysfunction during tamoxifen resistance development in MCF-7 cells. |
| Literature | GENERIF_SUMMARY | Pten-NOLC1 fusion promotes cancers involving MET and EGFR signalings. |

| Main Category | Category | Terms |
|---|---|---|
| Literature | GENERIF_SUMMARY | Over expression of PTEN induces apoptosis and prevents cell proliferation in breast cancer cells. |
| Literature | GENERIF_SUMMARY | Long non-coding RNA MEG3 promotes autophagy and apoptosis of nasopharyngeal carcinoma cells via PTEN up-regulation by binding to microRNA-21. |
| Literature | GENERIF_SUMMARY | Does obesity modify the expression of cyclin D1 and pten in endometrial polyps in postmenopausal women? |
| Literature | GENERIF_SUMMARY | Interactomic affinity profiling by holdup assay |
| Literature | GENERIF_SUMMARY | Correlation Between Onco-suppressors PTEN and NM23 and Clinical Outcome in Patients With T1 Breast Cancer. |
| Literature | GENERIF_SUMMARY | Lnc-GAN1 expression is associated with good survival and suppresses tumor progression by sponging mir-26a-5p to activate PTEN signaling in non-small cell lung cancer. |
| Literature | GENERIF_SUMMARY | Long noncoding RNA FER1L4 inhibits cell proliferation and promotes cell apoptosis via the PTEN/AKT/p53 signaling pathway in lung cancer. |
| Literature | GENERIF_SUMMARY | NF1 and PTEN gene polymorphisms and the susceptibility to soft tissue sarcomas in a Chinese population |
| Literature | GENERIF_SUMMARY | Downregulated exosomal microRNA-148b-3p in cancer associated fibroblasts enhance chemosensitivity of bladder cancer cells by downregulating the Wnt/beta-catenin pathway and upregulating PTEN. |
| Literature | GENERIF_SUMMARY | Multiple signaling pathways are essential for synapse formation induced by synaptic adhesion molecules. |
| Literature | GENERIF_SUMMARY | Downregulation of microRNA253p inhibits the proliferation and promotes the apoptosis of multiple myeloma cells via targeting the PTEN/PI3K/AKT signaling pathway. |
| Literature | GENERIF_SUMMARY | Circular RNA circ_0001287 inhibits the proliferation, metastasis, and radiosensitivity of non-small cell lung cancer cells by sponging microRNA miR-21 and up-regulating phosphatase and tensin homolog expression. |
| Literature | GENERIF_SUMMARY | Long noncoding RNA MEG3 inhibits cell migration and invasion of nonsmall cell lung cancer cells by regulating the miR215p/ |

| Main Category | Category | Terms |
|---|---|---|
| | | PTEN axis. |
| Literature | GENERIF_SUMMARY | Cross-level analysis of molecular and neurobehavioral function in a prospective series of patients with germline heterozygous PTEN mutations with and without autism. |
| Literature | GENERIF_SUMMARY | HOTAIR suppresses PTEN via DNMT3b and confers drug resistance in acute myeloid leukemia. |
| Literature | GENERIF_SUMMARY | PTEN is required for the migration and invasion of Ras-transformed MDCK cells. |
| Literature | GENERIF_SUMMARY | LncRNA-SNHG16 promotes proliferation and migration of acute myeloid leukemia cells via PTEN/PI3K/AKT axis through suppressing CELF2 protein. |
| Literature | GENERIF_SUMMARY | Brief Report |
| Literature | GENERIF_SUMMARY | A global analysis of the reconstitution of PTEN function by translational readthrough of PTEN pathogenic premature termination codons. |
| Literature | GENERIF_SUMMARY | Regulation of PTEN translation by PI3K signaling maintains pathway homeostasis. |
| Literature | GENERIF_SUMMARY | Association of Prediagnostic Blood Metabolomics with Prostate Cancer Defined by ERG or PTEN Molecular Subtypes. |
| Literature | GENERIF_SUMMARY | E3 ubiquitin ligase RNF126 affects bladder cancer progression through regulation of PTEN stability. |
| Literature | GENERIF_SUMMARY | PTEN/AKT upregulation of TMSB10 contributes to lung cancer cell growth and predicts poor survival of the patients. |
| Literature | GENERIF_SUMMARY | Diethylhexyl phthalate (DEHP) regulates the proliferation and chemosensitivity of esophageal squamous cell carcinoma cells via regulation of PTEN. |
| Literature | GENERIF_SUMMARY | Altered expression of serum miR-106a, miR-19b, miR-17, and PTEN in patients with idiopathic membranous nephropathy. |
| Literature | GENERIF_SUMMARY | NOTCH and EZH2 collaborate to repress PTEN expression in breast cancer. |
| Literature | GENERIF_SUMMARY | Circular RNA hsa_circ_0072309 inhibits the proliferation, invasion and migration of gastric cancer cells via inhibition of PI3K/AKT signaling by activating PPARgamma/PTEN signaling. |
| Literature | GENERIF_SUMMARY | miR2245p regulates the proliferation, migration and invasion of pancreatic |

| Main Category | Category | Terms |
|---|---|---|
| | | mucinous cystadenocarcinoma by targeting PTEN. |
| Literature | GENERIF_SUMMARY | PBK/TOPK promotes chemoresistance to oxaliplatin in hepatocellular carcinoma cells by regulating PTEN. |
| Literature | GENERIF_SUMMARY | Understanding the Origins of Loss of Protein Function by Analyzing the Effects of Thousands of Variants on Activity and Abundance. |
| Literature | GENERIF_SUMMARY | Inhibition Sequence of miR-205 Hinders the Cell Proliferation and Migration of Lung Cancer Cells by Regulating PETN-Mediated PI3K/AKT Signal Pathway. |
| Literature | GENERIF_SUMMARY | MiR-193a-3p Promotes Fracture Healing via Targeting PTEN Gene. |
| Literature | GENERIF_SUMMARY | Silencing PTEN in the fallopian tube promotes enrichment of cancer stem cell-like function through loss of PAX2. |
| Literature | GENERIF_SUMMARY | LncRNA PlncRNA-1 accelerates the progression of prostate cancer by regulating PTEN/Akt axis. |
| Literature | GENERIF_SUMMARY | PTEN loss correlates with T cell exclusion across human cancers. |
| Literature | GENERIF_SUMMARY | Glioblastomas with primitive neuronal component harbor a distinct methylation and copy-number profile with inactivation of TP53, PTEN, and RB1. |
| Literature | GENERIF_SUMMARY | Interference with lncRNA NEAT1 promotes the proliferation, migration, and invasion of trophoblasts by upregulating miR-411-5p and inhibiting PTEN expression. |
| Literature | GENERIF_SUMMARY | PTEN mutations predict benefit from tumor treating fields (TTFields) therapy in patients with recurrent glioblastoma. |
| Literature | GENERIF_SUMMARY | PTEN Regulates Mitochondrial Biogenesis via the AKT/GSK-3beta/PGC-1alpha Pathway in Autism. |
| Literature | GENERIF_SUMMARY | PTEN is a predictive biomarker of trastuzumab resistance and prognostic factor in HER2-overexpressing gastroesophageal adenocarcinoma. |
| Literature | GENERIF_SUMMARY | EPHA3 Contributes to Epigenetic Suppression of PTEN in Radioresistant Head and Neck Cancer. |
| Literature | GENERIF_SUMMARY | A Low Tumor Mutational Burden and PTEN Mutations Are Predictors of a Negative Response to PD-1 Blockade in MSI-H/dMMR Gastrointestinal Tumors. |

| Main Category | Category | Terms |
|---|---|---|
| Literature | GENERIF_SUMMARY | Depleting long noncoding RNA HOTAIR attenuates chronic myelocytic leukemia progression by binding to DNA methyltransferase 1 and inhibiting PTEN gene promoter methylation. |
| Literature | GENERIF_SUMMARY | Structural analysis of the PTEN |
| Literature | GENERIF_SUMMARY | Glycogen synthase kinase 3beta promotes osteosarcoma invasion and migration via regulating PTEN and phosphorylation of focal adhesion kinase. |
| Literature | GENERIF_SUMMARY | Effect of miR-126 on the Proliferation and Migration of Vascular Smooth Muscle Cells in Aortic Aneurysm Mice Under PI3K/AKT/mTOR Signaling Pathway. |
| Literature | GENERIF_SUMMARY | EGR1/2 Inhibits Papillary Thyroid Carcinoma Cell Growth by Suppressing the Expression of PTEN and BAX. |
| Literature | GENERIF_SUMMARY | Distinct roles of tumor associated mutations in collective cell migration. |
| Literature | GENERIF_SUMMARY | MiRNA-26a inhibits myocardial infarction-induced apoptosis by targeting PTEN via JAK/STAT pathways. |
| Literature | GENERIF_SUMMARY | Notch3 inhibits cell proliferation and tumorigenesis and predicts better prognosis in breast cancer through transactivating PTEN. |
| Literature | GENERIF_SUMMARY | Distribution of genetic alterations in high-risk early-stage cervical cancer patients treated with postoperative radiation therapy. |
| Literature | GENERIF_SUMMARY | PTEN somatic mutations contribute to spectrum of cerebral overgrowth. |
| Literature | GENERIF_SUMMARY | Abnormal Expression of microRNA-296-3p in Type 2 Diabetes Patients and its Role in Pancreatic beta-Cells Function by Targeting Tensin Homolog Deleted on Chromosome Ten. |
| Literature | GENERIF_SUMMARY | AQP3-mediated H2 O2 uptake inhibits LUAD autophagy by inactivating PTEN. |
| Literature | GENERIF_SUMMARY | CircRAB3B suppresses proliferation, motility, cell cycle progression and promotes the apoptosis of IL-22-induced keratinocytes depending on the regulation of miR-1228-3p/PTEN axis in psoriasis. |
| Literature | GENERIF_SUMMARY | Prognostic value of PTEN in de novo diagnosed metastatic prostate cancer. |
| Literature | GENERIF_SUMMARY | MicroRNA-9 inhibits proliferation and progression in retinoblastoma cells by targeting PTEN. |

| Main Category | Category | Terms |
|---|---|---|
| Literature | GENERIF_SUMMARY | Integrated multi-omics analyses on patient-derived CRC organoids highlight altered molecular pathways in colorectal cancer progression involving PTEN. |
| Literature | GENERIF_SUMMARY | """"Thyroid nodular disease and PTEN mutation in a multicentre series of children with PTEN hamartoma tumor syndrome (PHTS)""""." |
| Literature | GENERIF_SUMMARY | AKT1 and PTEN show the highest affinities among phosphoinositide binding proteins for the second messengers PtdIns(3,4,5)P3 and PtdIns(3,4)P2. |
| Literature | GENERIF_SUMMARY | PTEN Is Activated by the Addition of Cetuximab to Paclitaxel in Oral Squamous Cell Carcinoma. |
| Literature | GENERIF_SUMMARY | Molecular mechanism of microRNA-26a regulation of phosphatase and tensin homolog gene in condyloma acuminatum and penile squamous cell carcinoma. |
| Literature | GENERIF_SUMMARY | lncRNA NEAT1 aggravates sepsis-induced lung injury by regulating the miR-27a/PTEN axis. |
| Literature | GENERIF_SUMMARY | Mesenchymal stem cells-derived exosomes modulate vascular endothelial injury via miR-144-5p/PTEN in intracranial aneurysm. |
| Literature | GENERIF_SUMMARY | CLIC1 promotes the progression of cervical cancer through the PTEN/PI3K/AKT pathway. |
| Literature | GENERIF_SUMMARY | PTEN suppresses tumorigenesis by directly dephosphorylating Akt. |
| Literature | GENERIF_SUMMARY | Plasma Cell-Free DNA Profiling of PTEN-PI3K-AKT Pathway Aberrations in Metastatic Castration-Resistant Prostate Cancer. |
| Literature | GENERIF_SUMMARY | MicroRNA-30b-5p promotes the proliferation and migration of human airway smooth muscle cells induced by platelet-derived growth factor by targeting phosphatase and tensin homolog deleted on chromosome ten. |
| Literature | GENERIF_SUMMARY | Prevalence and clinical/molecular characteristics of PTEN mutations in Turkish children with autism spectrum disorders and macrocephaly. |
| Literature | GENERIF_SUMMARY | PTEN regulates adipose progenitor cell growth, differentiation, and replicative aging. |
| Literature | GENERIF_SUMMARY | Circular RNA circSETD3 hampers cell growth, migration, and stem cell properties in bladder cancer through sponging miR-641 to upregulate PTEN. |

| Main Category | Category | Terms |
|---|---|---|
| Literature | GENERIF_SUMMARY | Differential impact of tumor suppressor gene (TP53, PTEN, RB1) alterations and treatment outcomes in metastatic, hormone-sensitive prostate cancer. |
| Literature | GENERIF_SUMMARY | Synovial mesenchymal stem cell-derived extracellular vesicles containing microRN555A-26a-5p ameliorate cartilage damage of osteoarthritis. |
| Literature | GENERIF_SUMMARY | MicroRNA-103a-3p Promotes Cell Proliferation and Invasion in Non-Small-Cell Lung Cancer Cells through Akt Pathway by Targeting PTEN. |
| Literature | GENERIF_SUMMARY | Transcriptional landscape of PTEN loss in primary prostate cancer. |
| Literature | GENERIF_SUMMARY | Piwi-interacting RNA-651 promotes cell proliferation and migration and inhibits apoptosis in breast cancer by facilitating DNMT1-mediated PTEN promoter methylation. |
| Literature | GENERIF_SUMMARY | Non-canonical function of nuclear PTEN and its implication on tumorigenesis. |
| Literature | GENERIF_SUMMARY | A high CD8 to FOXP3 ratio in the tumor stroma and expression of PTEN in tumor cells are associated with improved survival in non-metastatic triple-negative breast carcinoma. |
| Literature | GENERIF_SUMMARY | Expression of activator proteins of SHH/GLI and PI3K/Akt/mTORC1 signaling pathways in human gliomas is associated with high grade tumors. |
| Literature | GENERIF_SUMMARY | Association of PIK3CA mutation and PTEN loss with expression of CD274 (PD-L1) in colorectal carcinoma. |
| Literature | GENERIF_SUMMARY | Overexpression of lncRNA-NEF regulates the miR-155/PTEN axis to inhibit adipogenesis and promote osteogenesis. |
| Literature | GENERIF_SUMMARY | Role of lncRNA XIST/microRNA-19/PTEN network in autophagy of nucleus pulposus cells in intervertebral disc degeneration via the PI3K/Akt signaling pathway. |
| Literature | GENERIF_SUMMARY | Aberrant PTEN, PIK3CA, pMAPK, and TP53 expression in human scalp and face angiosarcoma. |
| Literature | GENERIF_SUMMARY | MiR-508-3p promotes proliferation and inhibits apoptosis of middle ear cholesteatoma cells by targeting PTEN/PI3K/AKT pathway. |
| Literature | GENERIF_SUMMARY | The deubiquitinase USP10 restores PTEN activity and inhibits non-small cell lung |

| Main Category | Category | Terms |
|---|---|---|
| | | cancer cell proliferation. |
| Literature | GENERIF_SUMMARY | Toward better characterization of restricted and repetitive behaviors in individuals with germline heterozygous PTEN mutations. |
| Literature | GENERIF_SUMMARY | New evidence on tumor suppressor activity of PTEN and KLLN in papillary thyroid carcinoma. |
| Literature | GENERIF_SUMMARY | RNA-binding protein RBM38 inhibits colorectal cancer progression by partly and competitively binding to PTEN 3'UTR with miR-92a-3p. |
| Literature | GENERIF_SUMMARY | Modeling PTEN overexpression-induced microcephaly in human brain organoids. |
| Literature | GENERIF_SUMMARY | Long noncoding RNA HAND2AS1/miR106a/PTEN axis resensitizes cisplatinresistant ovarian cells to cisplatin treatment. |
| Literature | GENERIF_SUMMARY | A scalable Drosophila assay for clinical interpretation of human PTEN variants in suppression of PI3K/AKT induced cellular proliferation. |
| Literature | GENERIF_SUMMARY | Enhanced immortalization, HUWE1 mutations and other biological drivers of breast invasive carcinoma in Black/African American patients. |
| Literature | GENERIF_SUMMARY | Cancer derived exosomes induce macrophages immunosuppressive polarization to promote bladder cancer progression. |
| Literature | GENERIF_SUMMARY | USP52 inhibits cell proliferation by stabilizing PTEN protein in non-small cell lung cancer. |
| Literature | GENERIF_SUMMARY | MiR-200c promotes papillary thyroid cancer cell proliferation, migration, and invasion by downregulating PTEN. |
| Literature | GENERIF_SUMMARY | Reliable Identification of Endometrial Precancers Through Combined Pax2, beta-Catenin, and Pten Immunohistochemistry. |
| Literature | GENERIF_SUMMARY | Alteration of PPAR-GAMMA (PPARG; PPARgamma) and PTEN gene expression in acute myeloid leukemia patients and the promising anticancer effects of PPARgamma stimulation using pioglitazone on AML cells. |
| Literature | GENERIF_SUMMARY | PTEN regulates proliferation and osteogenesis of dental pulp cells and adipogenesis of human adipose-derived stem cells. |
| Literature | GENERIF_SUMMARY | Circular RNA hsa_circ_0000730 restrains |

| Main Category | Category | Terms |
|---|---|---|
| | | cell proliferation, migration, and invasion in cervical cancer through miR-942-5p/PTEN axis. |
| Literature | GENERIF_SUMMARY | PI3K/PTEN/AKT Signaling Pathways in Germ Cell Development and Their Involvement in Germ Cell Tumors and Ovarian Dysfunctions. |
| Literature | GENERIF_SUMMARY | Exosomal microRNA-125a-3p from human adipose-derived mesenchymal stem cells promotes angiogenesis of wound healing through inhibiting PTEN. |
| Literature | GENERIF_SUMMARY | PTENalpha is responsible for protection of brain against oxidative stress during aging. |
| Literature | GENERIF_SUMMARY | Circular RNA ITCH attenuates the progression of nasopharyngeal carcinoma by inducing PTEN upregulation via miR-214. |
| Literature | GENERIF_SUMMARY | PTEN mutations in autism spectrum disorder and congenital hydrocephalus |
| Literature | GENERIF_SUMMARY | The structural basis of PTEN regulation by multi-site phosphorylation. |
| Literature | GENERIF_SUMMARY | Red flags for early recognition of adult patients with PTEN Hamartoma Tumour Syndrome. |
| Literature | GENERIF_SUMMARY | The E3 Ubiquitin Ligase NEDD4-1 Mediates Temozolomide-Resistant Glioblastoma through PTEN Attenuation and Redox Imbalance in Nrf2-HO-1 Axis. |
| Literature | GENERIF_SUMMARY | Anti-oncogenic PTEN induces ovarian cancer cell senescence by targeting P21. |
| Literature | GENERIF_SUMMARY | Integrating thousands of PTEN variant activity and abundance measurements reveals variant subgroups and new dominant negatives in cancers. |
| Literature | GENERIF_SUMMARY | YB-1 Oncoprotein Controls PI3K/Akt Pathway by Reducing Pten Protein Level. |
| Literature | GENERIF_SUMMARY | [Methylation detection of phosphatase and tensin homolog deleted on chromosome ten gene promoter in hepatocellular carcinoma samples by next-generation sequencing]. |
| Literature | GENERIF_SUMMARY | RNA-binding protein RBM24 represses colorectal tumourigenesis by stabilising PTEN mRNA. |
| Literature | GENERIF_SUMMARY | Conditional ERK3 overexpression cooperates with PTEN deletion to promote lung adenocarcinoma formation in mice. |
| Literature | GENERIF_SUMMARY | USP22-mediated deubiquitination of PTEN inhibits pancreatic cancer progression by |

| Main Category | Category | Terms |
|---|---|---|
| | | inducing p21 expression. |
| Literature | GENERIF_SUMMARY | LncRNA MIR31HG is induced by tocilizumab and ameliorates rheumatoid arthritis fibroblast-like synoviocyte-mediated inflammation via miR-214-PTEN-AKT signaling pathway. |
| Literature | GENERIF_SUMMARY | PTEN alleviates maladaptive repair of renal tubular epithelial cells by restoring CHMP2A-mediated phagosome closure. |
| Literature | GENERIF_SUMMARY | Closing the gap |
| Literature | GENERIF_SUMMARY | Fluctuations in AKT and PTEN Activity Are Linked by the E3 Ubiquitin Ligase cCBL. |
| Literature | GENERIF_SUMMARY | Long noncoding RNA NEAT1 inhibits the acetylation of PTEN through the miR-524-5p /HDAC1 axis to promote the proliferation and invasion of laryngeal cancer cells. |
| Literature | GENERIF_SUMMARY | CircFUNDC1 knockdown alleviates oxygen-glucose deprivation-induced human brain microvascular endothelial cell injuries by inhibiting PTEN via miR-375. |
| Literature | GENERIF_SUMMARY | Abnormal hsa_circ_0003948 expression affects chronic periodontitis development by regulating miR-144-3p/NR2F2/PTEN signaling. |
| Literature | GENERIF_SUMMARY | Impaired social cognition and fine dexterity in patients with Cowden syndrome associated with germline PTEN variants. |
| Literature | GENERIF_SUMMARY | The Novel Phosphatase Domain Mutations Q171R and Y65S Switch PTEN from Tumor Suppressor to Oncogene. |
| Literature | GENERIF_SUMMARY | Expression of obesity- and type-2 diabetes-associated genes in omental adipose tissue of individuals with obesity. |
| Literature | GENERIF_SUMMARY | G-protein coupled receptor 34 regulates the proliferation and growth of LS174T cells through differential expression of PI3K subunits and PTEN. |
| Literature | GENERIF_SUMMARY | Functional impact and targetability of PI3KCA, GNAS, and PTEN mutations in a spindle cell rhabdomyosarcoma with MYOD1 L122R mutation. |
| Literature | GENERIF_SUMMARY | A mass spectrometric method for in-depth profiling of phosphoinositide regioisomers and their disease-associated regulation. |
| Literature | GENERIF_SUMMARY | LncRNA HCG11 represses ovarian cancer cell growth via AKT signaling pathway. |

| Main Category | Category | Terms |
|---|---|---|
| Literature | GENERIF_SUMMARY | MicroRNA-625-3p improved proliferation and involved chemotherapy resistance via targeting PTEN in high grade ovarian serous carcinoma. |
| Literature | GENERIF_SUMMARY | Collision tumors revealed by prospectively assessing subtype-defining molecular alterations in 904 individual prostate cancer foci. |
| Literature | GENERIF_SUMMARY | DNA damage induced cellular senescence and it's PTEN-armed exosomes-the warriors against prostate carcinoma cells. |
| Literature | GENERIF_SUMMARY | Novel long non-coding RNA lncAMPC downregulates PTEN via miR-214 to promote nasopharyngeal carcinoma progression. |
| Literature | GENERIF_SUMMARY | Immunoexpression of PTEN, HER2/neu, and Ki-67 in endoscopic gastric carcinoma biopsies; their correlation with histological subtypes and one-year survival. |
| Literature | GENERIF_SUMMARY | PTEN Regulates Dendritic Arborization by Decreasing Microtubule Polymerization Rate. |
| Literature | GENERIF_SUMMARY | Regulation of PTEN and ovarian cancer progression by an E3 ubiquitin ligase RBCK1. |
| Literature | GENERIF_SUMMARY | HBX suppresses PTEN to promote the malignant progression of hepatocellular carcinoma through mi-R155 activation. |
| Literature | GENERIF_SUMMARY | Non-phosphorylatable cyclin D1 mutant potentiates endometrial hyperplasia and drives carcinoma with Pten loss. |
| Literature | GENERIF_SUMMARY | HMGB1/PTEN/PI3K axis participates in the peripheral immune cell differentiation in two representative TCM syndromes of chronic hepatitis B patients. |
| Literature | GENERIF_SUMMARY | Considerations on diagnosis and surveillance measures of PTEN hamartoma tumor syndrome |
| Literature | GENERIF_SUMMARY | PTPIP51 inhibits non-small-cell lung cancer by promoting PTEN-mediated EGFR degradation. |
| Literature | GENERIF_SUMMARY | miR-96-5p targets PTEN to mediate sunitinib resistance in clear cell renal cell carcinoma. |
| Literature | GENERIF_SUMMARY | METTL14-mediated N6-methyladenosine modification of Pten mRNA inhibits tumour progression in clear-cell renal cell carcinoma. |
| Literature | GENERIF_SUMMARY | Significance of a PTEN Mutational Status- |

| Main Category | Category | Terms |
|---|---|---|
| | | Associated Gene Signature in the Progression and Prognosis of Endometrial Carcinoma. |
| Literature | GENERIF_SUMMARY | PTEN alterations in sporadic and BRCA1-associated triple negative breast carcinomas. |
| Literature | GENERIF_SUMMARY | Diverse Macrophages Constituted the Glioma Microenvironment and Influenced by PTEN Status. |
| Literature | GENERIF_SUMMARY | Validation of a computational phenotype for finding patients eligible for genetic testing for pathogenic PTEN variants across three centers. |
| Literature | GENERIF_SUMMARY | Comprehensive characterization of PTEN mutational profile in a series of 34,129 colorectal cancers. |
| Literature | GENERIF_SUMMARY | Sp1 transcription factor represses transcription of phosphatase and tensin homolog to aggravate lung injury in mice with type 2 diabetes mellitus-pulmonary tuberculosis. |
| Literature | GENERIF_SUMMARY | Identification and Somatic Characterization of the Germline PTEN Promoter Variant rs34149102 in a Family with Gastrointestinal and Breast Tumors. |
| Literature | GENERIF_SUMMARY | PTEN in cancer associated fibroblasts. |
| Literature | GENERIF_SUMMARY | KBTBD7 promotes non-small cell lung carcinoma progression by enhancing ubiquitin-dependent degradation of PTEN. |
| Literature | GENERIF_SUMMARY | Hsa_circ_0000345 inhibits cell proliferation, migration and invasion of nasopharyngeal carcinoma cells via miR-513a-3p/PTEN axis. |
| Literature | GENERIF_SUMMARY | Identification of distinct genomic features reveals frequent somatic AHNAK and PTEN mutations predominantly in primary malignant melanoma presenting in the ureter. |
| Literature | GENERIF_SUMMARY | Decreased Expressions of CK1alpha and PTEN in Sinonasal Inverted Papilloma. |
| Literature | GENERIF_SUMMARY | PIK3CA mutations-mediated downregulation of circLHFPL2 inhibits colorectal cancer progression via upregulating PTEN. |
| Literature | GENERIF_SUMMARY | Epigenetic Silencing of PTEN and Epi-Transcriptional Silencing of MDM2 Underlied Progression to Secondary Acute Myeloid Leukemia in Myelodysplastic Syndrome Treated with Hypomethylating Agents. |

| Main Category | Category | Terms |
|---|---|---|
| Literature | GENERIF_SUMMARY | Deguelin Attenuates Non-Small-Cell Lung Cancer Cell Metastasis by Upregulating PTEN/KLF4/EMT Signaling Pathway. |
| Literature | GENERIF_SUMMARY | The Relationship and Expression of miR-451a, miR-25-3p and PTEN in Early Peritoneal Endometriotic Lesions and Their Modulation In Vitro. |
| Literature | GENERIF_SUMMARY | The Role of PTEN-L in Modulating PINK1-Parkin-Mediated Mitophagy. |
| Literature | GENERIF_SUMMARY | PTEN is required for human Treg suppression of costimulation in vitro. |
| Literature | GENERIF_SUMMARY | Phosphatase and tensin homolog (PTEN) variants and epilepsy |
| Literature | GENERIF_SUMMARY | PTEN |
| Literature | GENERIF_SUMMARY | PTEN and LKB1 are differentially required in Gli1-expressing mesenchymal cells to suppress gastrointestinal polyposis. |
| Literature | GENERIF_SUMMARY | Non-canonical EphA2 activation underpins PTEN-mediated metastatic migration and poor clinical outcome in prostate cancer. |
| Literature | GENERIF_SUMMARY | The interaction of hsa_circ_0002594 and eIF4A3 promotes T-helper 2 cell differentiation by the regulation of PTEN. |
| Literature | GENERIF_SUMMARY | Circ_0008768 Suppresses the Pancreatic Cancer Progression via miR-330- 3p/PTEN Axis. |
| Literature | GENERIF_SUMMARY | Loss of PTEN Expression, PIK3CA Mutations, and Breast Cancer Survival in the Nurses' Health Studies. |
| Literature | GENERIF_SUMMARY | Circular RNA hsa_circ_0000317 inhibits non-small cell lung cancer progression through regulating microRNA-494-3p/phosphatase and tensin homolog deleted on chromosome 10 axis. |
| Literature | GENERIF_SUMMARY | Exome sequencing reveals a distinct somatic genomic landscape in breast cancer from women with germline PTEN variants. |
| Literature | GENERIF_SUMMARY | Loss of KLK4 |
| Literature | GENERIF_SUMMARY | METTL3 promotes prostatic hyperplasia by regulating PTEN expression in an m(6)A-YTHDF2-dependent manner. |
| Literature | GENERIF_SUMMARY | Gastrointestinal manifestations in PTEN hamartoma tumor syndrome. |
| Literature | GENERIF_SUMMARY | Elevated Notch-1 expression promotes the lymph node metastasis of gastric cancer |

| Main Category | Category | Terms |
|---|---|---|
| | | and the Notch-1-PTEN-ERK1/2 signalling axis promotes the progression of gastric cancer. |
| Literature | GENERIF_SUMMARY | Both phosphorylation and phosphatase activity of PTEN are required to prevent replication fork progression during stress by inducing heterochromatin. |
| Literature | GENERIF_SUMMARY | Investigation of the expression levels of CDH1, FHIT, PTEN, and TTPAL genes in colorectal tumors. |
| Literature | GENERIF_SUMMARY | MicroRNA-382-5p Promotes Oral Squamous Cell Carcinoma Development and Progression by Negatively Regulating PTEN Expression. |
| Literature | GENERIF_SUMMARY | Linear ubiquitination of PTEN impairs its function to promote prostate cancer progression. |
| Literature | GENERIF_SUMMARY | Phosphatase and tensin homologue determine inflammatory status by differentially regulating the expression of Akt1 and Akt2 in macrophage alternative polarization of periodontitis. |
| Literature | GENERIF_SUMMARY | Genotype-phenotype associations in a large PTEN Hamartoma Tumor Syndrome (PHTS) patient cohort. |
| Literature | GENERIF_SUMMARY | Smurf1 Suppression Enhances Temozolomide Chemosensitivity in Glioblastoma by Facilitating PTEN Nuclear Translocation. |
| Literature | GENERIF_SUMMARY | PTEN directs developmental and metabolic signaling for innate-like T cell fate and tissue homeostasis. |
| Literature | GENERIF_SUMMARY | PTEN regulates invasiveness in pancreatic neuroendocrine tumors through DUSP19-mediated VEGFR3 dephosphorylation. |
| Literature | GENERIF_SUMMARY | PTEN Loss Enhances Error-Prone DSB Processing and Tumor Cell Radiosensitivity by Suppressing RAD51 Expression and Homologous Recombination. |
| Literature | GENERIF_SUMMARY | [Investigation of the role of polymorphic loci RS2299941, RS1903858, RS10490920, RS2735343 of the PTEN gene in patients with prostate cancer]. |
| Literature | GENERIF_SUMMARY | The equilibrium of tumor suppression |
| Literature | GENERIF_SUMMARY | PTEN overexpression and nuclear beta-catenin stabilization promote morular differentiation through induction of epithelial-mesenchymal transition and cancer stem cell-like properties in |

| Main Category | Category | Terms |
|---|---|---|
| | | endometrial carcinoma. |
| Literature | GENERIF_SUMMARY | CircSTAM inhibits migration and invasion of trophoblast cells by regulating miR-148a-5p/PTEN axis. |
| Literature | GENERIF_SUMMARY | Role of Tumor Suppressor PTEN and Its Regulation in Malignant Transformation of Endometrium. |
| Literature | GENERIF_SUMMARY | Functional analysis of PTEN variants of unknown significance from PHTS patients unveils complex patterns of PTEN biological activity in disease. |
| Literature | GENERIF_SUMMARY | BAP1 and PTEN mutations shape the immunological landscape of clear cell renal cell carcinoma and reveal the intertumoral heterogeneity of T cell suppression |
| Literature | GENERIF_SUMMARY | Silencing of RhoC induces macrophage M1 polarization to inhibit migration and invasion in colon cancer via regulating the PTEN/FOXO1 pathway. |
| Literature | GENERIF_SUMMARY | Molecular dynamic simulation and functional analysis of pathogenic PTEN mutations in glioblastoma. |
| Literature | GENERIF_SUMMARY | PTEN mutations as predictive marker for the high-grade endometrial cancer development in slovak women. |
| Literature | GENERIF_SUMMARY | PTEN rs701848 Polymorphism is Associated with Trastuzumab Resistance in HER2-positive Metastatic Breast Cancer and Predicts Progression-free Survival. |
| Literature | GENERIF_SUMMARY | Inactivation of PTEN and ZFHX3 in Mammary Epithelial Cells Alters Patterns of Collective Cell Migration. |
| Literature | GENERIF_SUMMARY | Cancer Risk Associated With PTEN Pathogenic Variants Identified Using Multigene Hereditary Cancer Panel Testing. |
| Literature | GENERIF_SUMMARY | JARID2 regulates epithelial mesenchymal transition through the PTEN/AKT signalling pathways in ovarian endometriosis. |
| Literature | GENERIF_SUMMARY | Foxo3a-Mediated DNMT3B Impedes Cervical Cancer Cell Proliferation and Migration Capacities through Suppressing PTEN Promoter Methylation. |
| Literature | GENERIF_SUMMARY | TET1-mediated microRNA-188-5p promoter hydroxymethylation regulates PTEN/PI3K/AKT signaling pathway in acute myeloid leukemia cells. |
| Literature | GENERIF_SUMMARY | Vascular adaptation to cancer beyond angiogenesis |

| Main Category | Category | Terms |
| --- | --- | --- |
| Literature | GENERIF_SUMMARY | LncRNA GABPB1-IT1 inhibits the tumorigenesis of renal cancer via the miR-21/PTEN axis. |
| Literature | GENERIF_SUMMARY | The PTEN-associated immune prognostic signature reveals the landscape of the tumor microenvironment in glioblastoma. |
| Literature | GENERIF_SUMMARY | PTEN Deficiency in Tubo-Ovarian High-Grade Serous Carcinoma is Associated with Poor Progression-Free Survival and is Mutually Exclusive with CCNE1 Amplification. |
| Literature | GENERIF_SUMMARY | Microarray expression results of VEGF, YAP1 and PTEN immunostains in preeclampsia cases. |
| Literature | GENERIF_SUMMARY | Nuclear PTEN's Functions in Suppressing Tumorigenesis |
| Literature | GENERIF_SUMMARY | The Rate of Phosphatase and Tensin (PTEN) Gene Expression Loss in Prostate Cancer and its Link to Tumor Upgrading. |
| Literature | GENERIF_SUMMARY | Mitochondrial Hydrogen Peroxide Activates PTEN and Inactivates Akt Leading to Autophagy Inhibition-Dependent Cell Death in Neuronal Models of Parkinson's Disease. |
| Literature | GENERIF_SUMMARY | Multiplatform Analysis of Intratumoral PTEN Heterogeneity in Melanoma. |
| Literature | GENERIF_SUMMARY | miR-19-3p Targets PTEN to Regulate Cervical Cancer Cell Proliferation, Invasion, and Autophagy. |
| Literature | GENERIF_SUMMARY | PTEN mutation predicts unfavorable fertility preserving treatment outcome in the young patients with endometrioid endometrial cancer and atypical hyperplasia. |
| Literature | GENERIF_SUMMARY | Phospho PTEN mediated dephosphorylation of mitotic kinase PLK1 and Aurora Kinase A prevents aneuploidy and preserves genomic stability. |
| Literature | GENERIF_SUMMARY | Pan-cancer genomic analysis shows hemizygous PTEN loss tumors are associated with immune evasion and poor outcome. |
| Literature | GENERIF_SUMMARY | Hsa_circ_0001535 inhibits the proliferation and migration of ovarian cancer by sponging miR-593-3p, upregulating PTEN expression. |
| Literature | GENERIF_SUMMARY | Hsa_circITGA4/ miR-1468/EGFR/ PTEN a Master Regulators Axis in Glioblastoma Development and Progression. |
| Literature | GENERIF_SUMMARY | AZD8186 in Combination With Paclitaxel in Patients With Advanced Gastric Cancer |

| Main Category | Category | Terms |
|---|---|---|
| Literature | GENERIF_SUMMARY | Overexpression of the PTEN Gene in Myocardial Tissues of Coronary Bypass Surgery Patients. |
| Literature | GENERIF_SUMMARY | Quadra-Stable Dynamics of p53 and PTEN in the DNA Damage Response. |
| Literature | GENERIF_SUMMARY | YTHDC1 positively regulates PTEN expression and plays a critical role in cisplatin resistance of bladder cancer. |
| Literature | GENERIF_SUMMARY | Longitudinal Analysis of Cancer Risk in Children and Adults With Germline PTEN Variants. |
| Literature | GENERIF_SUMMARY | Autism-specific PTEN p.Ile135Leu variant and an autism genetic background combine to dysregulate cortical neurogenesis. |
| Literature | GENERIF_SUMMARY | Ubiquitin ligase MARCH8 promotes the malignant progression of hepatocellular carcinoma through PTEN ubiquitination and degradation. |
| Literature | GENERIF_SUMMARY | lncNALT knockdown ameliorates hypertensive retinopathy via PTEN/PI3K/AKT pathway. |
| Literature | GENERIF_SUMMARY | The impact of PIK3CA mutations and PTEN expression on the effect of neoadjuvant therapy for postmenopausal luminal breast cancer patients. |
| Literature | GENERIF_SUMMARY | MicroRNAs as important players in regulating cancer through PTEN/PI3K/AKT signalling pathways. |
| Literature | GENERIF_SUMMARY | Spatiotemporal Coordination of RPE Cell Quality by Extracellular Vesicle miR-494-3p Via Competitive Interplays With SIRT3 or PTEN. |
| Literature | GENERIF_SUMMARY | Localized Endometrial Proliferations of Pregnancy are Clonal Glandular Outgrowths Characterized by PTEN Loss and PIK3CA Pathogenic Variants. |
| Literature | GENERIF_SUMMARY | miR-181a targets PTEN to mediate the neuronal injury caused by oxygen-glucose deprivation and reoxygenation. |
| Literature | GENERIF_SUMMARY | Mir-183 functions as an oncogene via decreasing PTEN in breast cancer cells. |
| Literature | GENERIF_SUMMARY | Circular RNA hsa_circ_0084054 promotes the progression of periodontitis with diabetes via the miR-508-3p/PTEN axis. |
| Literature | GENERIF_SUMMARY | PTEN loss is not a determinant of time to castration-resistance following androgen-deprivation therapy in prostate cancer |

| Main Category | Category | Terms |
|---|---|---|
| Literature | GENERIF_SUMMARY | HNRNPA2B1-mediated m[6]A modification of lncRNA MEG3 facilitates tumorigenesis and metastasis of non-small cell lung cancer by regulating miR-21-5p/PTEN axis. |
| Literature | GENERIF_SUMMARY | PTEN Loss Confers Resistance to Anti-PD-1 Therapy in Non-Small Cell Lung Cancer by Increasing Tumor Infiltration of Regulatory T Cells. |
| Literature | GENERIF_SUMMARY | PTEN promoter methylation and expression in endometrial cancer tissues. |
| Literature | GENERIF_SUMMARY | Renal Neoplasia Occurring in Patients With PTEN Hamartoma Tumor Syndrome |
| Literature | GENERIF_SUMMARY | RETN gene polymorphisms interact with alcohol dependence in association with depression. |
| Literature | GENERIF_SUMMARY | Nicotinamide Adenine Dinucleotide Phosphate Oxidase Promotes Glioblastoma Radiation Resistance in a Phosphate and Tensin Homolog-Dependent Manner. |
| Literature | GENERIF_SUMMARY | Lower PTEN may be associated with CD8+ T cell exhaustion in diffuse large B-cell lymphoma. |
| Literature | GENERIF_SUMMARY | p-mTOR, p-ERK and PTEN Expression in Tumor Biopsies and Organoids as Predictive Biomarkers for Patients with HPV Negative Head and Neck Cancer. |
| Literature | GENERIF_SUMMARY | GAS5 silencing attenuates hypoxia-induced cardiomyocytes injury by targeting miR-21/PTEN. |
| Literature | GENERIF_SUMMARY | ELK4 Promotes Cell Cycle Progression and Stem Cell-like Characteristics in HPV-associated Cervical Cancer by Regulating the FBXO22/PTEN Axis. |
| Literature | GENERIF_SUMMARY | miR-138 from ADSC Exo accelerates wound healing by targeting SIRT1/PTEN pathway to promote angiogenesis and fibrosis. |
| Literature | GENERIF_SUMMARY | Long non coding RNA, C8orf49, a novel diagnostic and prognostic biomarker, enhances PTEN/FZD4-mediated cell growth and metastasis by sponging miR-1323 in endometriosis. |
| Literature | GENERIF_SUMMARY | Colorectal cancer tumor cell-derived exosomal miR-203a-3p promotes CRC metastasis by targeting PTEN-induced macrophage polarization. |
| Literature | GENERIF_SUMMARY | PTEN deficiency exposes a requirement for an ARF GTPase module for integrin-dependent invasion in ovarian cancer. |

| Main Category | Category | Terms |
|---|---|---|
| Literature | GENERIF_SUMMARY | Pharmacological Inhibition of PTEN Rescues Dopaminergic Neurons by Attenuating Apoptotic and Neuroinflammatory Signaling Events. |
| Literature | GENERIF_SUMMARY | Cross-Comparison of Inflammatory Skin Disease Transcriptomics Identifies PTEN as a Pathogenic Disease Classifier in Cutaneous Lupus Erythematosus. |
| Literature | GENERIF_SUMMARY | Variants in PTEN Are Associated With a Diverse Spectrum of Cortical Dysplasia. |
| Literature | GENERIF_SUMMARY | MYC Deregulation and PTEN Loss Model Tumor and Stromal Heterogeneity of Aggressive Triple-Negative Breast Cancer. |
| Literature | GENERIF_SUMMARY | Reciprocal negative feedback regulation of ATF6alpha and PTEN promotes prostate cancer progression. |
| Literature | GENERIF_SUMMARY | Spinster homolog 2 reduces malignancies of glioblastoma via PTEN/PI3K/AKT pathway. |
| Literature | GENERIF_SUMMARY | FOXO3 and PTEN expression in the ovary of girls with extra-gonadal cancer with or without chemotherapy treatment prior to cryopreservation. |
| Literature | GENERIF_SUMMARY | Abrogation of ATR function preferentially augments cisplatin-induced cytotoxicity in PTEN-deficient breast cancer cells. |
| Literature | GENERIF_SUMMARY | A critical role of the endothelial S-phase kinase-associated protein 2/phosphatase and tensin homologue axis in angiogenesis and psoriasis. |
| Literature | GENERIF_SUMMARY | lncRNA SNHG9 enhances liver cancer stem cell self-renewal and tumorigenicity by negatively regulating PTEN expression via recruiting EZH2. |
| Literature | GENERIF_SUMMARY | CLDN6 inhibits colorectal cancer proliferation dependent on restraining p53 ubiquitination via ZO-1/PTEN axis. |
| Literature | GENERIF_SUMMARY | PTEN and soluble epoxide hydrolase in intestinal cell differentiation. |
| Literature | GENERIF_SUMMARY | The SNHG10-miR-495-3p-PTEN axis is involved in sevoflurane-mediated protective effects in cardiomyocytes against hypoxia/reoxygenation injury. |
| Literature | GENERIF_SUMMARY | A New Frameshift Mutation of PTEN Gene Associated with Cowden Syndrome-Case Report and Brief Review of the Literature. |
| Literature | GENERIF_SUMMARY | Concurrent PTEN and PDGFRB Alterations Characterize Storiform Collagenoma. |

| Main Category | Category | Terms |
|---|---|---|
| Literature | GENERIF_SUMMARY | Cancer and Autism |
| Literature | GENERIF_SUMMARY | PINK1-PTEN axis promotes metastasis and chemoresistance in ovarian cancer via non-canonical pathway. |
| Literature | GENERIF_SUMMARY | Extracellular vesicle-encapsulated microRNA-296-3p from cancer-associated fibroblasts promotes ovarian cancer development through regulation of the PTEN/AKT and SOCS6/STAT3 pathways. |
| Literature | GENERIF_SUMMARY | PTEN-mediated FOXO signaling affects autophagy, migration and invasion of rheumatoid arthritis fibroblast-like synoviocytes. |
| Literature | GENERIF_SUMMARY | Germline PTEN genotype-dependent phenotypic divergence during the early neural developmental process of forebrain organoids. |
| Literature | GENERIF_SUMMARY | Overexpression of HMGB1 in hepatocytes accelerates PTEN inactivation-induced liver cancer. |
| Literature | GENERIF_SUMMARY | PTEN expression can be used as a switch between senescence and apoptosis in breast cancer cells according to a logical model of the G2/M checkpoint. |
| Literature | GENERIF_SUMMARY | PROGNOSTIC IMPLICATIONS OF PD-L1 EXPRESSION AND LOSS OF PTEN IN PATIENTS WITH RHABDOMYOSARCOMA, EWING'S SARCOMA AND OSTEOSARCOMA. |
| Literature | GENERIF_SUMMARY | Investigating the effects of PTEN mutations on cGAS-STING pathway in glioblastoma tumours. |
| Literature | GENERIF_SUMMARY | Coexpression of TP53, BIM, and PTEN Enhances the Therapeutic Efficacy of Non-Small-Cell Lung Cancer. |
| Literature | GENERIF_SUMMARY | Evaluation of phosphatase and tensin homologue (PTEN) expression in gastric cancer and its relationship with histopathological findings. |
| Literature | GENERIF_SUMMARY | A Circular RNA Derived from the Pumilio 1 Gene Could Regulate PTEN in Human Cumulus Cells. |
| Literature | GENERIF_SUMMARY | Anti-tumor effect of PTEN and its effect on inhibition of the bone tumor through the CD47-SIRPalpha signaling pathway. |
| Literature | GENERIF_SUMMARY | Correlation of PTEN signaling pathway and miRNA in breast cancer. |
| Literature | GENERIF_SUMMARY | PTEN hinders the formation of scars by regulating the levels of proteins in the |

| Main Category | Category | Terms |
|---|---|---|
| | | extracellular matrix and promoting the apoptosis of dermal fibroblasts through Bcl-xL. |
| Literature | GENERIF_SUMMARY | PTEN Mediates the Silencing of Unintegrated HIV-1 DNA. |
| Literature | GENERIF_SUMMARY | PTEN hamartoma tumor syndrome |
| Literature | GENERIF_SUMMARY | PTEN deficiency potentiates HBV-associated liver cancer development through augmented GP73/GOLM1. |
| Literature | GENERIF_SUMMARY | MiR-29a-3p mediates phosphatase and tensin homolog and inhibits osteoarthritis progression. |
| Literature | GENERIF_SUMMARY | PTEN decreases NR2F1 expression to inhibit ciliogenesis during EGFR[L858R]-induced lung cancer progression. |
| Literature | GENERIF_SUMMARY | Hsa_circ_0004872 alleviates meningioma progression by sponging miR-190a-3p/PTEN signaling. |
| Literature | GENERIF_SUMMARY | Cerebellar phenotypes in germline PTEN mutation carriers. |
| Literature | GENERIF_SUMMARY | Thyroid nodules with DICER1 mutation or PTEN alteration |
| Literature | GENERIF_SUMMARY | Cancer-associated fibroblasts drive colorectal cancer cell progression through exosomal miR-20a-5p-mediated targeting of PTEN and stimulating interleukin-6 production. |
| Literature | GENERIF_SUMMARY | SNHG4-mediated PTEN destabilization confers oxaliplatin resistance in colorectal cancer cells by inhibiting ferroptosis. |
| Literature | GENERIF_SUMMARY | Combined deletion of MEN1, ATRX and PTEN triggers development of high-grade pancreatic neuroendocrine tumors in mice. |
| Literature | GENERIF_SUMMARY | Therapeutic role of PTEN in tissue regeneration for management of neurological disorders |
| Literature | GENERIF_SUMMARY | FBXO32-mediated degradation of PTEN promotes lung adenocarcinoma progression. |
| Literature | GENERIF_SUMMARY | Effect of stability of PTEN on hepatocellular carcinoma. |
| Literature | GENERIF_SUMMARY | miR-30a-5p attenuates hypoxia/reoxygenation-induced cardiomyocyte apoptosis by regulating PTEN protein expression and activating PI3K/Akt signaling pathway. |

| Main Category | Category | Terms |
|---|---|---|
| Literature | GENERIF_SUMMARY | The NTE domain of PTENalpha/beta promotes cancer progression by interacting with WDR5 via its SSSRRSS motif. |
| Literature | GENERIF_SUMMARY | Hsa_circ_0007590/PTBP1 complex reprograms glucose metabolism by reducing the stability of m[6]A-modified PTEN mRNA in pancreatic ductal adenocarcinoma. |
| Literature | GENERIF_SUMMARY | A miRNA-21-Mediated PTEN/Akt/NF-kappaB Axis Promotes Chronic Obstructive Pulmonary Disease Pathogenesis. |
| Literature | GENERIF_SUMMARY | Rs12039395 Variant Influences the Expression of hsa-miR-181a-5p and PTEN Toward Colorectal Cancer Risk. |
| Literature | GENERIF_SUMMARY | [Genetic analysis of a Chinese pedigree affected with Cowden syndrome due to variant of PTEN gene]. |
| Literature | GENERIF_SUMMARY | CBLL1 promotes endometrial stromal cell senescence via inhibiting PTEN in recurrent spontaneous abortion. |
| Literature | GENERIF_SUMMARY | The Role of PTEN in Chemoresistance Mediated by the HIF-1alpha/YY1 Axis in Pediatric Acute Lymphoblastic Leukemia. |
| Literature | GENERIF_SUMMARY | RNA-binding protein GIGYF2 orchestrates hepatic insulin resistance through STAU1/PTEN-mediated disruption of the PI3K/AKT signaling cascade. |
| Literature | GENERIF_SUMMARY | NHEJ is promoted by the phosphorylation and phosphatase activity of PTEN via regulation of DNA-PKcs. |
| Literature | GENERIF_SUMMARY | HOTAIR Promotes the Hyperactivation of PI3K/Akt and Wnt/beta-Catenin Signaling Pathways via PTEN Hypermethylation in Cervical Cancer. |
| Literature | GENERIF_SUMMARY | A retrospective analysis suggests PTEN expression is associated with favorable clinicopathological features of breast cancer. |
| Literature | GENERIF_SUMMARY | Decoding PTEN |
| Pathways | KEGG_PATHWAY | Inositol phosphate metabolism |
| Pathways | KEGG_PATHWAY | Metabolic pathways |
| Pathways | KEGG_PATHWAY | EGFR tyrosine kinase inhibitor resistance |
| Pathways | KEGG_PATHWAY | FoxO signaling pathway |
| Pathways | KEGG_PATHWAY | Phosphatidylinositol signaling system |
| Pathways | KEGG_PATHWAY | Sphingolipid signaling pathway |

| Main Category | Category | Terms |
|---|---|---|
| Pathways | KEGG_PATHWAY | p53 signaling pathway |
| Pathways | KEGG_PATHWAY | Autophagy - animal |
| Pathways | KEGG_PATHWAY | mTOR signaling pathway |
| Pathways | KEGG_PATHWAY | PI3K-Akt signaling pathway |
| Pathways | KEGG_PATHWAY | Cellular senescence |
| Pathways | KEGG_PATHWAY | Focal adhesion |
| Pathways | KEGG_PATHWAY | Insulin resistance |
| Pathways | KEGG_PATHWAY | Human papillomavirus infection |
| Pathways | KEGG_PATHWAY | Human T-cell leukemia virus 1 infection |
| Pathways | KEGG_PATHWAY | Pathways in cancer |
| Pathways | KEGG_PATHWAY | MicroRNAs in cancer |
| Pathways | KEGG_PATHWAY | Chemical carcinogenesis - reactive oxygen species |
| Pathways | KEGG_PATHWAY | Endometrial cancer |
| Pathways | KEGG_PATHWAY | Glioma |
| Pathways | KEGG_PATHWAY | Prostate cancer |
| Pathways | KEGG_PATHWAY | Melanoma |
| Pathways | KEGG_PATHWAY | Small cell lung cancer |
| Pathways | KEGG_PATHWAY | Breast cancer |
| Pathways | KEGG_PATHWAY | Hepatocellular carcinoma |
| Pathways | KEGG_PATHWAY | Central carbon metabolism in cancer |
| Pathways | KEGG_PATHWAY | PD-L1 expression and PD-1 checkpoint pathway in cancer |
| Pathways | KEGG_PATHWAY | Diabetic cardiomyopathy |
| Pathways | REACTOME_PATHWAY | PIP3 activates AKT signaling |
| Pathways | REACTOME_PATHWAY | Adaptive Immune System |
| Pathways | REACTOME_PATHWAY | Metabolism |
| Pathways | REACTOME_PATHWAY | Inositol phosphate metabolism |
| Pathways | REACTOME_PATHWAY | PI Metabolism |
| Pathways | REACTOME_PATHWAY | Phospholipid metabolism |
| Pathways | REACTOME_PATHWAY | Signal Transduction |
| Pathways | REACTOME_PATHWAY | Disease |
| Pathways | REACTOME_PATHWAY | Synthesis of PIPs at the plasma membrane |

| Main Category | Category | Terms |
|---|---|---|
| Pathways | REACTOME_PATHWAY | Immune System |
| Pathways | REACTOME_PATHWAY | Synthesis of IP3 and IP4 in the cytosol |
| Pathways | REACTOME_PATHWAY | Negative regulation of the PI3K/AKT network |
| Pathways | REACTOME_PATHWAY | TCR signaling |
| Pathways | REACTOME_PATHWAY | Downstream TCR signaling |
| Pathways | REACTOME_PATHWAY | Generic Transcription Pathway |
| Pathways | REACTOME_PATHWAY | PI3K/AKT Signaling in Cancer |
| Pathways | REACTOME_PATHWAY | Transcriptional Regulation by TP53 |
| Pathways | REACTOME_PATHWAY | Metabolism of proteins |
| Pathways | REACTOME_PATHWAY | Metabolism of lipids |
| Pathways | REACTOME_PATHWAY | TP53 Regulates Metabolic Genes |
| Pathways | REACTOME_PATHWAY | Diseases of signal transduction by growth factor receptors and second messengers |
| Pathways | REACTOME_PATHWAY | PTEN Loss of Function in Cancer |
| Pathways | REACTOME_PATHWAY | Deubiquitination |
| Pathways | REACTOME_PATHWAY | Ub-specific processing proteases |
| Pathways | REACTOME_PATHWAY | Ovarian tumor domain proteases |
| Pathways | REACTOME_PATHWAY | Post-translational protein modification |
| Pathways | REACTOME_PATHWAY | PTEN Regulation |
| Pathways | REACTOME_PATHWAY | RNA Polymerase II Transcription |
| Pathways | REACTOME_PATHWAY | Gene expression (Transcription) |
| Pathways | REACTOME_PATHWAY | Regulation of PTEN mRNA translation |
| Pathways | REACTOME_PATHWAY | Regulation of PTEN localization |
| Pathways | REACTOME_PATHWAY | Regulation of PTEN stability and activity |
| Pathways | REACTOME_PATHWAY | Transcriptional Regulation by MECP2 |
| Pathways | REACTOME_PATHWAY | Intracellular signaling by second messengers |
| Pathways | WIKIPATHWAYS | Androgen receptor signaling |
| Pathways | WIKIPATHWAYS | Integrated cancer pathway |
| Pathways | WIKIPATHWAYS | Integrated breast cancer pathway |
| Pathways | WIKIPATHWAYS | Leptin signaling |
| Pathways | WIKIPATHWAYS | Microtubule cytoskeleton regulation |
| Pathways | WIKIPATHWAYS | Glioblastoma signaling |

| Main Category | Category | Terms |
|---|---|---|
| Pathways | WIKIPATHWAYS | Androgen receptor network in prostate cancer |
| Pathways | WIKIPATHWAYS | T cell receptor and co stimulatory signaling |
| Pathways | WIKIPATHWAYS | Focal adhesion |
| Pathways | WIKIPATHWAYS | Hepatocyte growth factor receptor signaling |
| Pathways | WIKIPATHWAYS | Copper homeostasis |
| Pathways | WIKIPATHWAYS | MECP2 and associated Rett syndrome |
| Pathways | WIKIPATHWAYS | Circadian rhythm genes |
| Pathways | WIKIPATHWAYS | Pathways affected in adenoid cystic carcinoma |
| Pathways | WIKIPATHWAYS | Hematopoietic stem cell gene regulation by GABP alpha beta complex |
| Pathways | WIKIPATHWAYS | PI3K AKT mTOR signaling and therapeutic opportunities in prostate cancer |
| Pathways | WIKIPATHWAYS | Factors and pathways affecting insulin like growth factor IGF1 Akt signaling |
| Pathways | WIKIPATHWAYS | RIOK1 and RIOK2 in EGFR and PI3K mediated tumorigenesis |
| Pathways | WIKIPATHWAYS | Embryonic stem cell pluripotency pathways |
| Pathways | WIKIPATHWAYS | Focal adhesion PI3K Akt mTOR signaling |
| Pathways | WIKIPATHWAYS | miRNA regulation of p53 pathway in prostate cancer |
| Pathways | WIKIPATHWAYS | Clear cell renal cell carcinoma pathways |
| Pathways | WIKIPATHWAYS | Endometrial cancer |
| Pathways | WIKIPATHWAYS | PI3K Akt signaling |
| Pathways | WIKIPATHWAYS | Somatroph axis GH and its relationship to dietary restriction and aging |
| Pathways | WIKIPATHWAYS | Breast cancer pathway |
| Pathways | WIKIPATHWAYS | Extracellular vesicles in the crosstalk of cardiac cells |
| Pathways | WIKIPATHWAYS | EGF EGFR signaling |
| Pathways | WIKIPATHWAYS | Synaptic signaling associated with autism spectrum disorder |
| Pathways | WIKIPATHWAYS | Fragile X syndrome |
| Pathways | WIKIPATHWAYS | Small cell lung cancer |
| Pathways | WIKIPATHWAYS | Head and neck squamous cell carcinoma |
| Pathways | WIKIPATHWAYS | Melanoma |

| Main Category | Category | Terms |
|---|---|---|
| Pathways | WIKIPATHWAYS | IL18 signaling |
| Pathways | WIKIPATHWAYS | EGFR tyrosine kinase inhibitor resistance |
| Pathways | WIKIPATHWAYS | Insulin signaling |
| Pathways | WIKIPATHWAYS | mBDNF and proBDNF regulation of GABA neurotransmission |
| Pathways | WIKIPATHWAYS | 16p11 2 proximal deletion syndrome |
| Pathways | WIKIPATHWAYS | p53 transcriptional gene network |
| Pathways | WIKIPATHWAYS | Phosphoinositides metabolism |
| Pathways | WIKIPATHWAYS | Pleural mesothelioma |
| Pathways | WIKIPATHWAYS | T cell activation SARS CoV 2 |
| Pathways | WIKIPATHWAYS | Urotensin II mediated signaling |
| Pathways | WIKIPATHWAYS | Targeted agents in triple negative breast cancer |
| Pathways | WIKIPATHWAYS | TROP2 regulatory signaling |
| Pathways | WIKIPATHWAYS | Macrophage stimulating protein MSP signaling |
| Pathways | WIKIPATHWAYS | Smith Magenis and Potocki Lupski syndrome copy number variation |
| Pathways | WIKIPATHWAYS | Phosphatidyl inositol phosphate pathway |
| Pathways | WIKIPATHWAYS | ADHD and autism ASD pathways |
| Pathways | WIKIPATHWAYS | Cancer pathways |
| Pathways | WIKIPATHWAYS | Fanconi anemia |
| Pathways | WIKIPATHWAYS | Effect of omega 3 PUFA on Huntington 39 s disease pathways |
| Pathways | WIKIPATHWAYS | Hallmark of cancer sustaining proliferative signaling |
| Pathways | WIKIPATHWAYS | Molecular pathway for oxidative stress |
| Pathways | WIKIPATHWAYS | Tyrosine kinase inhibitors in glioblastoma |
| Pathways | WIKIPATHWAYS | Senescence and autophagy in cancer |
| Pathways | WIKIPATHWAYS | DNA damage response only ATM dependent |
| Protein Domains | INTERPRO | Tyr_Pase_dom |
| Protein Domains | INTERPRO | Tyr_Pase_cat |
| Protein Domains | INTERPRO | Tensin_C2-dom |

| Main Category | Category | Terms |
|---|---|---|
| Protein Domains | INTERPRO | Tyr_Pase_AS |
| Protein Domains | INTERPRO | Bifunc_PIno_P3_Pase/Pase_PTEN |
| Protein Domains | INTERPRO | Prot-tyrosine_phosphatase-like |
| Protein Domains | INTERPRO | Tensin_phosphatase |
| Protein Domains | INTERPRO | C2_domain_sf |
| Protein Domains | INTERPRO | PTP_PTEN |
| Protein Domains | INTERPRO | Dual-spec_lipid-protein_phosph |
| Protein Domains | PIR_SUPERFAMILY | PTEN |
| Protein Domains | SMART | PTPc_motif |
| Protein Domains | SMART | PTPlike_phytase |
| Protein Domains | SMART | PTEN_C2 |